

Introduction

A “Robot” Definition First

KEY QUESTION: What is a robot?

One of the official and most complete def:

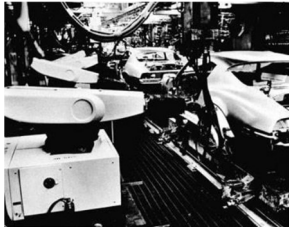
“A robot is an **autonomous system** that exists and interact in the **physical** world, can **sense** its environment and can **act** on it to achieve some **goals**”.

Going through each of the parts of this definition we see that a robot is:

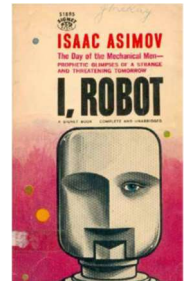
- “autonomous”, so not a teleoperated system
- is not a digital agent since it can interact with the physical world
- it is equipped with sensor that allow to sense its environment, actuators to perform actions and with a controller to achieve its goals.



The origins of modern robotics



Two key figures are Isaac Asimov (science fiction author of “I, Robot”) and Joseph Engelberger, an engineer who designed and developed the first industrial robot, the Unimate, installed in 1960 in the manufacturing plant of General Motors in New Jersey (USA). Thus, 1960 has been worldwide recognized as the official birth of robotics.



Since then, robotics has grown exponentially and the theories and techniques for robot control, fabrication and sensing represent an incredible wealth of knowledge. Robotics technologies are today very solid and robust, in the **accurate, fast, and reliable control** of robot motion.

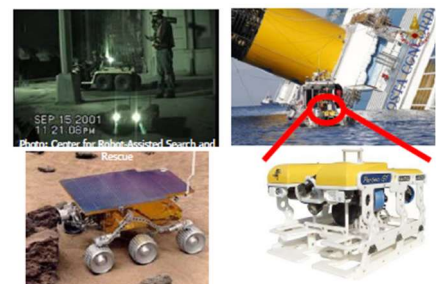
From Structured to Unstructured Environments



So far, we have very advanced robotic systems exploiting all the advances in control techniques, sensing and actuation but only within “**structured environment**”, e.g., manufacturing plants. Such scenario is characterized by a very rigid and fixed disposition of the elements, everything is planned and going exactly as it should, no unknown variables.

Such environment is externally bounded by a cage separating the area that is safe for the workers from the one not accessible during robotics operations. This cage is also limiting the operability of the robots, whose maximum ability can be exploited only within this limited space.

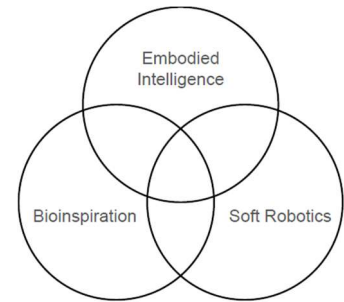
If we would like to transition to the so called “**service robotics**”, where robots able to interact with the real world, full of unknowns and uncertainties that can happen without the robot preparation, there is an increase need in development of theories and techniques for robot perception and action control. Here we deal with the so called “**Unstructured Environment**”. Examples, pushed to the extreme are the one of search and rescue, ship wrecks. These are the one where we would really like to take advantage of robotics but we are still struggling with since robots are not yet able to cope with the uncertainties common of such environments.



How to leave the cage?

IDEA: Recipe made of 3 pillars (that will be the core focus of the course)

- **Embodied Intelligence**
- **Bioinspiration**
- **Soft Robotics**



BIOINSPIRATION:



In scenarios such as rescue ones, where there is the risk of interaction with living beings that may be endangered and that are full of uncertainties and irregularities, we are still forced to involve living beings, e.g., humans, animals, since the abilities that robots have are not yet enough optimized to be able to cope with such environments. Taking this as a starting point, since humans are in this case the best solution so far, the idea is to **learn something from nature** and living beings to improve the capabilities of robots to deal with such scenario.

But why robots are still not so advanced to fully match the requirements needed to substitute humans in such scenario, despite the good performances of the “humanoid robots”? The main limitation is due to the fact they are too simple or rather too complex?

Studying living organisms let us to understand that despite being very complex organisms, they seem to behave in a very smart and efficient, implementing their intelligence in very simple ways. So the actual aim is to understand the mechanism (nature secret) making possible to implement intelligent behavior through a simplified control in complex systems.

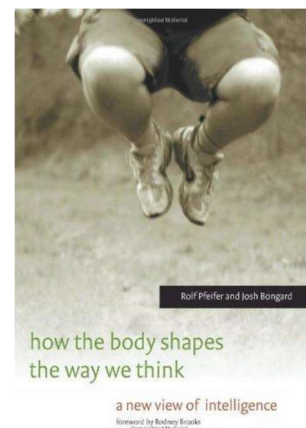
KEY TAKEAWAY (Nature Lesson): shift from the idea of designing complex robots complexifying also the control part and focus instead on **simplifying principles** for control and behavior.

EMBODIED INTELLIGENCE – the modern view of AI:

Embodied Intelligence is a relatively new way of looking at AI, that is shifting the **classical approach** seeing as the main and only processor of any kind of info, and center of intelligence the brain and central processing of an agent to a **modern approach**. Here the focus is more on interaction with the environment, and it sees cognition as an emerging behavior from interaction of the system with the environment.

According to one of the pioneers of the so called “Embodied Intelligence”, Rolf Pfeifer (author of “*How the body shapes the way we think: a new view of intelligence*”) the key concepts are:

- The nature of the human mind is largely determined by the form of the human body.
- Adaptive behavior is not just control and computation, but it emerges from the complex and dynamic interaction between:
 - **Morphology** (acting as a computing element)
 - **sensory-motor control**
 - **environment** (new part – not usually considered when dealing with AI)



These are the three pillars to always take into account when dealing with the study of intelligence.

KEY TAKEAWAY: The important revolution introduced is the idea of embodiment, i.e., the fact that intelligence is associated also to the body of the agent and not only related to the brain.

Morphological Computation

The concept of Morphological Computation is one of the main pillars of soft robotics theory and is related to the idea that morphology on its own can act as a computing element. Since a brain to interact with the environment needs a body (embodiment), then any kind of info coming from environment and that should reach the brain have to pass through the body.

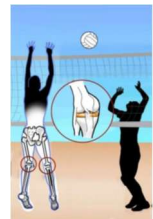
In this way we are considering the body not just as a wire connecting environment to brain but as a complex system that while transferring the info also transforms them, a processor that preprocess the info before reaching the brain. So the brain is no longer the only center of processing.

The characteristics of a body that allows such preprocessing of info are:

- Shape as body structure, that specifies the behavioral response of the agent (e.g., s-shaped spinal cord acting as a spring to stabilize instead of straight)
- Arrangement of motor, perceptive and processing units of the body (e.g., specific arrangement of neurons forming the circuitry involved in the preprocess of visual info before reaching brain)
- Mechanical properties allowing emergent behaviors and adaptive interaction with the environment (different materials and tissue that have specific role in simplifying the control of movements e.g., cartilages in the knee that automatically adapt their stiffness to absorb the impact energy when jumping and reaching the ground back)

Pushing this part to the extreme, to grasp the simplifying principles of morphological computation we need to:

- Exploit the mechanical feedback from the interaction with the environment
- Use the **compliance** of the human body (fact that 90% of animal/human body is made of soft tissue) instead of fighting it.



KEY POINT: in traditional robotics based on the use of hard materials and metals, compliance is seen as something to avoid at all costs, since related to singularities and instabilities. While in soft robotics compliance, if well exploited, might be an added value.

These concepts lead to the development of new **body ware**, i.e., new ways of developing the hardware parts of the body that can be made also of **soft materials**.

SOFT ROBOTICS

Why using robots made of soft materials can be useful?

- 1) **Richer Behavior:** Soft materials, along with compliance and non-linearity in general are added value since they enable building robots characterized by a richer behavior (even though they are harder to manage and design).

NOTE: In traditional robotics the final attempt is to always linearize everything, in order to deal with predictable deviations from such linear relations. An example is the linear domain defined by the strain vs stress plot of the deformation of a metal. Within this linear domain the mechanical behavior is predictable just by using the Young's modulus. Instead with materials such as elastomers (e.g., silicones, rubber), that are polymers having a very low elastic modulus, so that can deform a lot even

with small forces, whose linear part is very short, the Young's modulus is no longer useful to describe their elastic behavior.

However, since our final aim is to use nature as a model, and being nature non linear we need to reject the constraint of rigid and linear bodies.

- 2) **Safer:** elastomers-based platforms, being compliant and deformable have intrinsic safety with respect to metal-based ones, with less impact than hard robots.
- 3) **Mostly programmable:** related to the fact that we have different material combinations and behaviors that can be programmed on the base of the arrangement, mechanical and geometrical properties of the elastomers used. They are also very dexterous.

There are also some key drawbacks so far that are limiting the widespread use of soft materials:

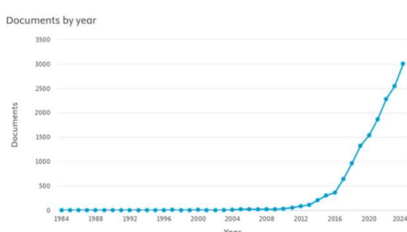
- 1) **Hard to manage:** Robots based on this kind of materials are hard to manage. Although there are many combinations that can be used, the specific way to design a soft robot is still uneasy.
- 2) **Missing References:** the difficulty lies also in the lack of references. In fact, every kind of tools related to robotics that have been developed so far cannot be used in soft robotics, at least without strong modification. This is because we are unhinging the whole base of traditional robotics.

FOCUS (D-H convention): Traditional robotics is strongly founded on the “**Denavit Hartenberg robotic convention**”, that is no longer valid for soft robotics. In fact, the main concept of the a priori division between joints and links when dealing with soft robots made of elastomers that can deform at any point it is impossible.

The D-H convention: Method used in robotics to describe in a simplified way (using the minimum number of parameters) the behavior of every kind of traditional robot represented by a chain of links and joints each with a specific length and d.o.f., e.g., manipulator. Especially to analyze its kinematic and dynamic. It is based on the use of tables of parameters that summarize all the characteristics of joints and links.

- 3) **Variable Stiffness needed:** especially when having the need to apply higher forces, the use of soft materials may be a limitation. An example are deformable surgical tools developed for endoscopy that are able to interact safely with soft tissue and have a high intrinsic dexterity, but when reaching the surgical target and having to execute the operation, the advantage of being deformable in interacting with tissues becomes a disadvantage. To solve this issue many labs are working on variable stiffness technologies that make the soft materials become hard to enable the possibility of interaction with environment in a more effective way. So, softness in this way becomes hard on demand and locally, enabling the possibility of interaction with environment with higher forces. This of course increases the system complexity, burden and bulkiness.

Soft Robotics so far



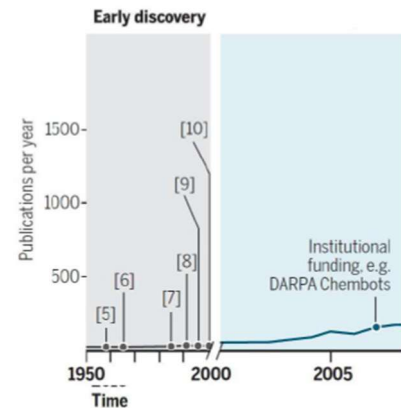
Starting from 2009/10 there has been a strong increase in the interest on soft robotics, that is still continuing, leading to an increase not only in the number but also in the quality of the publications on this topic not only on the number of articles.

Soft Robotics: early discoveries

Till 2000 there were some early but isolated soft robotics publications. An example is the work of Professor K. Suzumori in Japan on the development of a Flexible Microactuator based on elastomers and powered by pneumatic (published on ICRA).

Past 2004/05 there has been a combination of events worldwide that made the actual start of soft robotics:

- In the US the defense agency, DARPA started funding some pioneering projects aiming at developing so-called “Chembots”, based on a combination of material science (functionalization of materials) and microfluidics. Here the core idea was the one of using soft materials to develop robots.
- At the same time, in 2008 by chance appeared in the IEEE Robotics and Automation Magazine an issue called “Soft Robotics”. However, the “soft robotic” concept introduced there was mainly based on hard robotics (metal based), where the softness was introduced by the control algorithm able to change the impedance of the actuator allowing to control the stiffness.



TWO “SOFT ROBOTICS” WAVES

- 1ST VIEW: use of soft materials in robotics → robots made of soft materials that undergo large deformations during interaction, soft actuators and soft components. With control partially embedded in the robot morphology and mechanical properties.
- 2nd VIEW: variable impedance actuators and stiffness control → highly flexible (hyper-redundant) robots with high number of rigid links and multi-axis joints.

The definition “war”

These two completely different trends on how to implement soft robots led to controversy in the working definition of soft robotics.

Some definitions, took advantage from the concept of soft matter (key words in material science to identify with low elastic modulus) to define “soft matter robotics”. (Wang L., Iida F.). Other definition was of “soft bodied robots” (Kim S., Laschi C., Trimmer B.), in analogy with soft bodied animals. Or “Robots built with soft materials” (Laschi C., and Cianchetti M.).

Within this field, some research groups started to question about the fact that systems made of soft materials couldn’t be considered as robots. To fight this thought, Mike Tolley and Daniela Rus from MIT published in Nature the article “*Design, fabrication and control of soft robots*” in which they defined soft robots as:

“systems that are capable of autonomous behavior, and that are primarily composed of materials with moduli in the range of that of soft biological materials”

By comparing the Yung’s Modulus of biological and artificial material they identified as “soft” all the materials having moduli comparable with those of well-known biological materials (skin, cartilage, etc.), or less than around 1 GPa, exhibiting considerable compliance under normal loading.

This definition war was concluded in 2014 through the Coordination action on Soft Robotics funded by the European Commission (RoboSoft), where the community proposed and agreed on the following definition of Soft Robotics:

“Soft robot/devices that can actively interact with the environment and can undergo “large” deformations relying on inherent or structural compliance”

Where the term “large” deformation is taken from material science that defines a very specific range for such deformations.

NEW ELEMENT: Such definition goes beyond only looking at the Young’s Modulus to define a “soft material”, but considers instead **compliance** as directly proportional to the length of the body and inversely to the Young’s modulus and the cross section.

$$k^{-1} \propto \frac{L}{EA}$$

So in addition to the compliance associated to high or low values of E the definition considers also factors like the **cross section** and **length** that can contribute too. In fact, a very long and thin object can still be easy to bend and thus soft, independently from the modulus. In this way also metals can be considered soft provided their length and cross section.

Such definition, considers both **inherent material compliance** associated to the Young’s Modulus and **structural compliance** that depends only on geometrical features. In this way it merged the two waves of soft robotics.

Soft Robotics: the crossroad

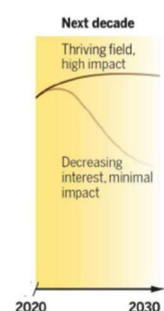
During the same years (starting from 2009) not only in the US but also around the world there were important contributions to soft robotics’ spread.

- An example are the European funding Projects like “Octopus Project”, “RoboSoft”, “Stiff-Flop”, that were explicitly dedicated to the birth and growth of the soft robotic discipline.
- At the same occurred the expiration of the FDM patent that is the base technology of most additive manufacturing techniques.
- The birth of some spin offs companies and organization of workshops (e.g., the first one organized in Monte Verita that combined soft robotics and morphological computation).
- In 2012 the IEEE technical committee on Soft Robotics was founded.
- In 2014 the soft Robotics journal was founded and in 2015 the first Soft Robotics weeks were organized starting spreading the fundamentals of developing robots based on soft materials.
- Finally in 2018 the first soft robotics conference was organized, and it is still happening on yearly base.

What about the next decades? Is Soft robotics going to have an increasing impact or will it decrease? This was defined in the article “Hard questions for Soft Robotics” published by M. Tolley in 2021. It suggested that while the innovation in this field attracts a lot of interest from stakeholders and users, this interest can decrease if the intuition behind the idea of soft robots is not concretely put into practice and fails to meet public’s expectations.

The authors identified three levels of contribution, from:

- **Level 0:** minimal contribution, limited advancement to the field in the second decade. With new systems, similar performance that is achieved in a different manner.
- **Level 1:** contribution only within the field of soft robotics. Leads to advances in soft robotics but limited impact on other research fields or technology domains.
- **Level 2:** best level where contribution to robotics and engineering are more broadly. Not only moving forward soft robotics but also advancing other scientific fields and application



domains more broadly. This is the ideal case In which with soft robotics we are enabling new possibilities and solving problems that in traditional robotics are not possible.

Level 2 (transversal) contributions from the Lab of Prof. Cianchetti:

From enabling technologies for soft actuators, smart compliant sensors and flexible mechanisms to full soft robotic systems based on soft/flexible materials



Embodiment & Morphological Computation

Introduction to embodiment

DEF (given by Pfeifer et al., 2007) of Embodiment:

"the behavior of any system is not merely the outcome of an internal control structure (such as the central nervous system). A system's behavior is also affected by the ecological niche in which the system is physically embedded, by its morphology (the shape of its body and limbs, as well as the type and placement of sensors and effectors), and by the material properties of the elements composing the morphology."

The **trivial** implication of such def is that intelligence requires a body. But the **deeper** implications are that the physical body of the agent actually has a role in processing the information, and then that embodiment can be used for cognition.

Before going into detail there are two key concepts that we need to cover for the “embodied turn”:

- Synthetic Methodology: the potential of the synthetic approach to studying intelligence: “understanding by building”
- Frame of reference Problem: too often, we automatically attribute intelligence to a certain behavior, without realizing that the mechanism that brings it about may be surprisingly simple.

SYNTHETIC METHODOLOGY:

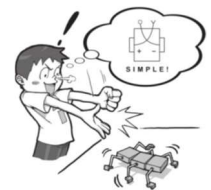
Shift from analytic approach, i.e., based on trying to **reproduce** the same **result** to Synthetic methodology, aimed at explaining **why the result came about**.



Analytic Approach → The analytic approach is universally applied in all empirical sciences. Typically, experiments are performed on an existing system, a human, a desert ant, or a brain region, and the results are analyzed in various ways. Often the goal is to develop a model, obtained by fitting the data that must

be able to predict the outcome of future experiments.

Synthetic Approach → Different approach very useful in the context of soft robotics. While still being based on empirical experiments, it changes the perspective: instead of starting from the observation of the result, it suggest to start by building an artificial system that is implementing the most important aspects of the natural system and then use it to understand the behavior. Shift in focus from the replication of the correct experimental results, i.e., the correct output, to the reproduction of the internal mechanisms that have led to the particular results => “**understanding by building**”



!! Understanding by building is a powerful tool since by building the physical system and looking at how it interacts with the environment we are able to make the cognition emerge directly **!!**

FRAME OF REFERENCE PROBLEM:

The tendence of humans to see intelligence as a projection of their intelligence in everything they see was called by the Oxford ethologist David McFarland "anthropomorphization: the incurable disease.". another example is contained in the Herb Simon's seminal book "The Sciences of the Artificial" (the ant walking along a beach: looking at how ants are able to find their path over long

distances, reach food and other tasks, one might think that it is the result of a very efficient path planning. Instead it arises from instantaneous decision based on very simple rules.)

KEY POINT: The point is that agents can exploit physical laws even if they are not aware of them. And Intelligence in this sense is not so much a property of an agent/brain/evolution, but rather resides in the eye of who observes the exploitation. → sometimes looking just at the result lead to missing the simplicity behind it and focus only on its complexity, and this leads in turn to project models that are very complex while in reality based on very simple behaviours..

The **frame of reference problem** refers to the fact that our reference system whenever we see something is always biased towards us as human, so we tend to “anthropomorphize” instead of actually seeing things for what they are. Sometimes overcomplicating things that are essentially simpler.

Traditional AI

KEY QUESTION: What is intelligence? And is the extreme high ability of processing info of traditional AI actually intelligence?

In 1921, the Journal of Educational Psychology asked 14 experts for definitions. The results were 14 different definitions. In fact, intelligence is difficult to define and understand, thus one approach could be the synthetic one, i.e., try to build an **artificial intelligent system** in order to understand intelligence.

This opened the field of traditional AI, whose modern definition is “*the study and design of intelligent agents*”.

Success of traditional AI: in the classical/traditional approach what really matters is the **algorithm** or the **program** that run on the computer and not the hardware (so the body or brain) on which it runs, that is instead irrelevant.

→ **KEY SUCCESS REASON of COMPUTING:** abstract algorithms/programs that function independently of the specifics of a particular hardware.

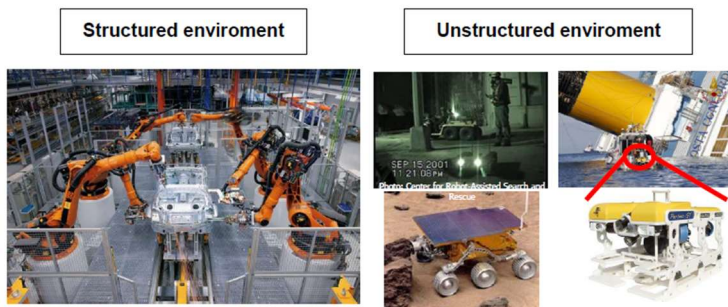
This is at the base of several artificial intelligence systems such as expert systems (medical and technical diagnosis, configuration of complex computer systems, portfolio management), search engines on the internet that use clever machine learning algorithms, text-processing systems (use algorithms to try infer user intentions from the context of what has been done earlier), data-mining systems, natural language processing.



However, is this high capability of processing info in real time and in huge amount, really intelligence? So can we conclude that **human intelligence** is information processing, i.e., computing? The original intention of artificial intelligence was not to develop clever algorithms, but to understand natural human intelligence. It is now generally agreed that the classical approach while being successful in implementing algorithms that can well support human beings has failed to deepen our understanding of many intelligent processes.

Problem of traditional AI: Natural forms of intelligence require a **direct interaction** with the real world. This constitutes the main limitation of traditional AI, that instead is not able to do so in the deeper sense, and putting wheels on computers won't solve this quest.

KEY TAKEAWAY: In order to understand what intelligence is we cannot just consider the **brain** on its own but also how the **body** and **brain** of an intelligent agent **interact with the real world**. (Notice how the three main pillars of embodied intelligence appear)



This view that insert in the loop also the interaction with the real world represents the shifts from the industrial robotics, mainly linked to idea of a **structured environment** to modern service robotics based on theories and techniques for robot perception and action control, that uses as setting **unstructured environment**. Thus, all the uncertainties

that we have in the environment with this new view are not seen as a limitation but are actually part of the recipe we need to make cognition emerge.

Real World VS Virtual World

The comparison we can make between what we have in an environment that is fully digital, without any interaction with the real world and the actual real world, can be associated to the one between playing chess and soccer:

- **Chess:** formal game, settled in a virtual world that has discrete, clearly defined states, board positions, operations, and legal moves (everything is well defined and with specific limitations and rules). Everything is known by just looking at the board, so at any time we have complete information of what is happening. It is also a static game, where opponents cannot do anything while thinking about the next move during each turn.
- **Football:** non formal game. It takes place in the real world with no uniquely defined states, it is a continuous and never repeating sequence of actions (No pre defined moves and freedom of operation). Information is always incomplete, players cannot have a complete spectrum of what is happening at any time (position of opponents, ball position and rotation, wind and so on). And it involves the time pressure.

This distinction mirrors the differences between condition in virtual VS real world.

The symbol-grounding problem

One last pint, that is important to cover before going deep in detail in the more formal description of how to use embodied intelligence is the so called “symbol-grounding problem”.

Whenever we deal with Traditional AI that is based on Algorithms, the bottleneck that we are not always aware is that Algorithms are instructed to answer to user quests by using symbols whose real meaning are completely unaware of.

Symbols are typically defined in a purely syntactic way by how they relate to other symbols and how they are processed by some interpreter. The relation of symbols (e.g., in database applications) to the outside world is never discussed; it is assumed as somehow given, with the (typically implicit) assumption that designers and potential users know what the symbols mean.

Symbol-grounding problem: computer systems are not able on their own to establish the appropriate relations between the used symbols and elements of the outside world. They are not autonomous but instead need the presence in the loop of a human interpreter. Therefore

this mapping is "grounded" in the human's experience of his or her interaction with the real world.

However, once we remove the human interpreter from the loop, as in the case of autonomous agents, we have to take into account that the system needs to interact with the environment on its own.

KEY TAKEAWY: In order to develop a system that is able to implement intelligence that emerges from interaction between brain, body and the environment the robot must be designed as a “**complete agent**” behaving in real world

Note: we use the term agent whenever we do not want to make a distinction between humans, animals, or robots, i.e., when what we say applies to all three.

Complete Agents

Another open topic is how to test or measure the completeness of the agent and its intelligent capabilities? In the history of artificial intelligence research there have been many attempts to measure intelligence of humans. However, most of laboratory studies have proved to be useless, since people were tested on tasks that were not only somewhat artificial but also unusually difficult for humans: subjects are asked, for example, to remember long lists of numbers or to read text upside down.

In this scenario, in 1982, Toda, M. in “Man, Robot, and Society” proposed that “*if we are to learn something relevant about intelligence — something that holds true in real-world behavior — we need to study complete systems, i.e., systems that have to act and perform tasks autonomously in the real world*”.

KEY TAKEAWAY: even if there are no specific experimental protocols or tasks to measure and test intelligence, we can exclude everything that is not aiming at evaluating the ability of the agent to act and perform task autonomously.

Properties of a complete agent:

We now define a list of properties of a complete agent that can allow us to understand what a complete agent should look like.

1) **A complete agent is subject to the laws of physics.**

Walking requires energy, friction, and gravity in order to work.

Because the agent is embodied (must have a body), it is a physical system (biological or not) and thus subject to the laws of physics from which it cannot possibly escape; it must comply with them. If an agent jumps up in the air, gravity will inevitably pull it back to the ground.

2) **A complete agent generates sensory stimulation.**

!! This property refers to the part of the interaction with environment that has a consequence internally, i.e., what the agent perceives while interacting with the environment) !!

For ex. when we walk, we generate sensory stimulation, whether we like it or not:

- when we move, objects seem to flow past us (this is known as optic flow);
- by moving we induce wind that we then sense with our skin and our hair;

- walking also produces pressure patterns on our feet (feedbacks while walking);
- and we can feel the regular flexing and relaxing of our muscles as our legs move (proprioception).

3) A complete agent affects its environment.

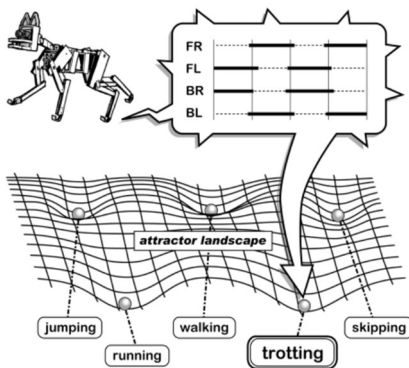
!! This property is related to how the agent have an impact on the environment while interacting with it, i.e., the consequences on the external world of the interaction!!

- When we walk across a lawn, the grass is crushed underfoot;
- when we breathe, we blow air into the environment;
- when we walk and burn energy, we dissipate energy as heat the environment;
- when we drink from a cup, we reduce the amount of liquid in the glass;
- when we drop a cup, it breaks;
- when we talk, we put pressure waves out into the air;
- when we sit down in a chair, it squeaks, and the cushion is squashed.

4) Agents tend to settle into attractor states.

FOCUS (Theory of chaos – Non linear dynamic systems):

When studying complete agents, it is useful to think of them as **nonlinear dynamic systems**, a concept from **chaos theory**. Such systems are always changing and active, but over time they tend to **settle into stable patterns of behavior** called **attractor states**. An attractor state is a condition in which the system naturally stabilizes, even though it remains dynamic. If small disturbances occur, the system tends to return to that same stable pattern rather than changing completely.



A helpful way to visualize this is through an **attractor landscape**. Imagine a surface where the horizontal plane represents all the possible states an agent could be in (for example, combinations of variables like position and momentum, often called the **phase space**). The vertical dimension represents the **energy required to maintain each state**. Some states require more energy, while others require less. The states that require **less energy appear as valleys** in this landscape. Because these states are more efficient, the system naturally tends to fall into and remain in these valleys. These valleys correspond to **attractor states**.

In the context of agents, the “energy” involved is often better understood as **computational effort**. Attractor states therefore reduce the amount of computation needed to maintain a behavior, making the system simpler and more efficient to control.

EX 1 (Horse Gait): A common example is the **gaits of a horse**. A horse can walk, trot, canter, or gallop. Each of these locomotion modes can be seen as an attractor state. While the horse is moving, it typically remains in one gait because it is a stable and energy-efficient pattern. If a small disturbance occurs—like uneven ground—the horse may briefly deviate from the exact movement pattern but will quickly return to the same gait. The horse only leaves that attractor state when enough energy is applied to transition to another gait, such as moving from a trot to a gallop. This transition requires temporarily climbing out of one “valley” in the attractor landscape and moving into another.

EX 2 (Central Pattern Generators): This idea also appears in biological systems such as **Central Pattern Generators (CPGs)** that are neural circuits that produces **rhythmic movements**

automatically, such as walking or breathing, without requiring continuous control from the brain. It is an example of a **nonlinear dynamic system** because it naturally stabilizes into repeating patterns of activity. Located mainly in the **spinal cord or peripheral nervous system**, CPGs generate self-sustaining oscillations that remain stable even when small disturbances occur, thanks to their interaction with the body and the environment. This allows the control of locomotion to be **partly offloaded from the central nervous system to local circuits and the body**, making movement more efficient and reducing the need for conscious control.

KEY POINTS:

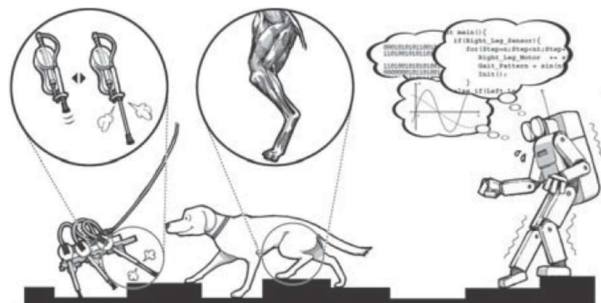
- Attractor states are important because they make behavior **stable, robust, and energy-efficient**. By exploiting the processing abilities of the mechanical parts of the body the agent is able to simplify the control, reducing the computational burden of the central processor /brain that just has to supervise and trigger the behaviour.
- Attractor states into which the agent settles are always the result of the interaction of three systems: agent's body, its brain (or control system) and its environment.

5) Complete agents perform morphological computation.

FOCUS (Morphological Computation):

- **Morphological computation** refers to the idea that an agent's **body itself performs part of the information processing**, reducing the amount of work that must be done by the brain. Instead of the brain handling all computations internally, the **physical morphology** – muscles, joints, biomechanics and body dynamics - of the body actively **preprocess information** as it interacts with the environment.

EX 1 (elasticity of human leg): An example is the fact that the human leg's muscles and tendons are elastic so that the knee, when the leg impacts the ground while running, performs small adaptive movements without neural control. The control is supplied by the muscle tendon system itself, which is part of the morphology of the agent.



EX 2 (Stabilization during walking): A clear example comes from an experiment that measured **stability during walking**. Researchers placed two cameras on a person's body: one near the **center of mass** (around the torso) and one on the **head**. Intuitively, one might expect the center of mass to be the most stable point during locomotion. However, when the researchers analyzed the **optic flow**—how stable the visual image appeared from each camera—they found the opposite: the center of mass moved and vibrated significantly, while the **head remained much more stable**.

This stability is not primarily due to active control by the brain, but to the **mechanical and structural properties of the body**. Through evolution, the human body has developed ways to naturally stabilize the **eyes and head** during movement, because stable vision is crucial for perception. As a result, the body itself performs part of the task of stabilizing visual input.

These five properties that we can use to describe a complete agent are simply unavoidable consequences of embodiment.

!! Systems that are **not complete** hardly ever possess all of these properties **!!**

- For example, a **vision system** consisting of a fixed camera and a desktop computer (that might be seen as an intelligent system) is not a complete agent since: does not generate sensory stimulation because it cannot produce behavior and it influences the environment only by emitting heat and light from the computer screen. Moreover, it does not perform morphological computation and does not have physical attractor states that could be useful to the system.

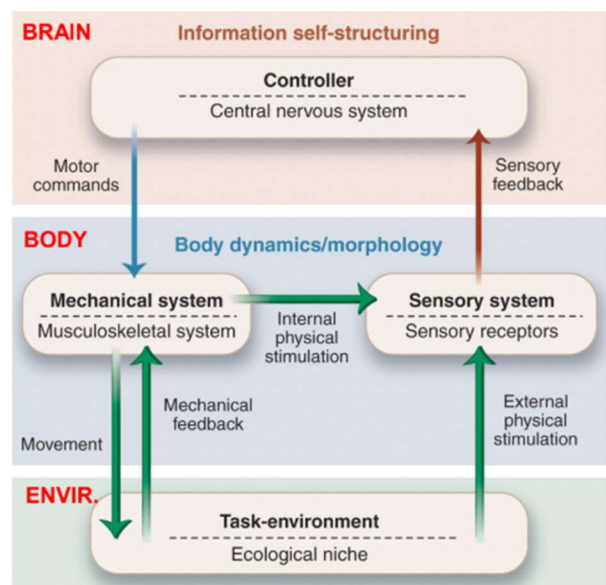
General Scheme of an Agent

We will now see examples of agents implementing some of the characteristics seen above to underline their power. To do so we use the diagram reporting the general scheme used to describe an agent (biological or artificial). Made of three main blocks:

- **Brain:** include the controller/ Central nervous system that generates instructions called motor commands
- **Body:** with its dynamics and morphology that include on one side the mechanical/ musculoskeletal system and on the other the sensory system/receptor (perceptive part)
- **Environment:** that can be task-environment or ecological niche (in case of natural system). It encompasses all forces such gravity and friction that act on the agent.

All the blocks are connected in specific ways.

- Controller generate instructions called **motor commands** that are sent to the mechanical system.
- The Mechanical system that is engaged with the sensory system through the **internal physical stimuli** (i.e., forces and torques developed in the muscles and joint-supporting ligaments). But is also interacting with the environment via **movements**.
- The task-environment sends back to the mechanical system **mechanical feedbacks** (reactions that the agent's mechanical system experiences after acting on the environment)
- The sensory systems has inputs both from environment via **external physical stimulation** (i.e., exteroception, so effects of the environment on the agent's sensory system) and from the mechanical system via internal physical stimulation, i.e., proprioception. It closes the loop by sending back info to the controller via **sensory feedbacks**.



I. Locomotion case studies

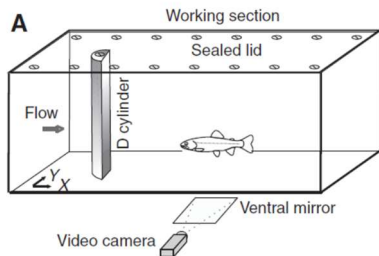
The fact that moving from one place to another, or locomotion, requires a body, comes as no surprise. However, it has been treated predominantly as a control problem by many; the body playing the part of a mere tool that has to be commanded appropriately.

On contrary: shaping the body morphology and thereby the dynamics that result from the interaction with the environment can lead to stable and efficient locomotion, requiring very little control.

KEY TAKEAWAY: locomotion does not always require complex active control. Instead, much of the work can be done by the **interaction between body structure and environmental forces.**

a) **Brain-less and power-less creatures** (no brain or power putting energy in the locomotion)

EX 1: Euthanized trout



In **James C. Liao's experiment**, a trout was placed in a water tank with a flowing current and a cylinder that created vortices (swirling currents) behind it. Liao observed three conditions: a live trout swimming in free flow, a live trout swimming behind the cylinder, and a **euthanized (dead) trout** placed behind the cylinder. In free flow, the live trout had to actively move its body to counter the current. Behind the cylinder, however, the vortices caused the fish's body to oscillate naturally, allowing it to maintain position with much less effort. Surprisingly, even the dead trout oscillated and moved in a similar way when placed in the vortex flow.

This showed that the **shape, flexibility, and mechanical properties of the fish's body interact with the water flow to generate movement**, meaning that locomotion can partly emerge from **body–environment dynamics** rather than active neural control. The key conclusion is that the environment and the body can do much of the work—hence the idea that “*the water swims the fish.*”

EX 2: Passive Dynamic Walker



The **Passive Dynamic Walker**, introduced by **Tad McGeer (1990)**, is a simple mechanical system that demonstrates how locomotion can emerge purely from body mechanics and environmental interaction. The device consists only of **passive mechanical components**—such as legs, joints, and sometimes springs—carefully tuned in terms of **leg length, mass distribution, and foot shape**. It contains **no motors, sensors, or computer control**, making it effectively “brainless.”

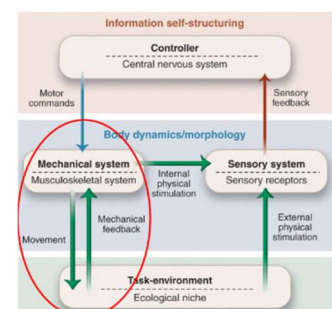
When placed on a **slight incline**, gravity becomes the only energy source, and the walker begins to move downhill with a natural walking motion. The movement emerges from the interaction between the walker's **mechanical structure and gravity**, showing that locomotion can arise without active control or power. The stability of this motion comes from the system settling into a **limit cycle**, a type of attractor in nonlinear dynamic systems where the movement repeats in a stable rhythmic pattern. This example illustrates how:

→ **Locomotion can be realized through pure, but carefully tuned mechanics only.**

THEORETICAL SCHEME: For the category of brainless and powerless creatures, the general scheme of the behavior doesn't include the controller block and sensory system (since they have no sensors). So such systems to implement locomotion just need a mechanical system and the interaction with the environment, i.e., movement and mechanical feedback. And such behaviour is also self stabilizing since no control is in the loop.

b) **Brainless and powered creatures**

EX 1: Crazy Bird





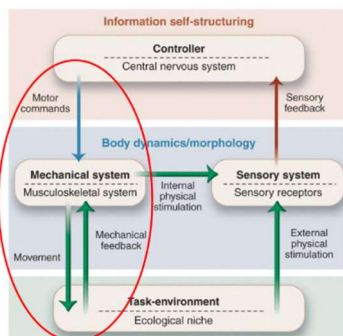
The **Crazy Bird robot**, built using **LEGO Mindstorms**, demonstrates how complex behavior can emerge from very simple control when the body and environment do most of the work. The robot has **two motors and no sensors**, meaning it is still effectively **brainless**, although it is powered. Both motors rotate at the **same speed and in the same direction**, but they start with a **phase delay** between them (for example 0° , 90° , 180° , or 270°). The legs are mechanically linked to the rotating wheels, so their movement interacts with the surface of the table. Because of this setup, the robot's behavior emerges from the **interaction between the simple mechanical structure, the phase difference between the motors, and the environment**. Despite the extremely simple control, the robot is able to **explore the entire surface of the table**.

→ the controller does not need to precisely control each joint or movement; it only needs to induce a global bias, i.e., set a **global parameter (the phase delay)**, while the body–environment dynamics generate the complex motion.

EX 2: Self-Stabilization

The theory of nonlinear dynamical systems is often employed to analyze the phenomena involved in the mechanical (and also neural) aspects of locomotion. The walker is an example of a nonlinear dynamical system and walking patterns (which are periodic motions) correspond to limit cycles. **Limit cycles** in a nonlinear system can display attractive behavior, i.e. nearby trajectories are 'pulled' toward the limit cycle.

A familiar example of **limit-cycle stability** can be seen in a **moving bicycle**. If a bicycle is pushed and given enough speed, it can continue moving forward and remain upright **without anyone actively controlling it**. This happens because of the bicycle's **dynamic properties**—such as its geometry, momentum, and interaction with the ground—which naturally stabilize the motion while it is moving.

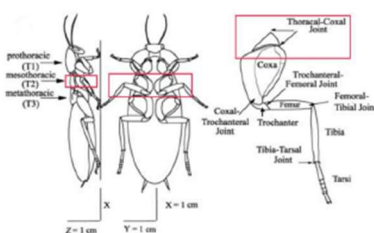


As long as the bicycle maintains sufficient momentum, it tends to remain balanced. Even if small perturbations occur, such as a slight push or uneven ground, the system briefly deviates but then naturally returns to its stable motion.

THEORETICAL SCHEME: it is close to the previous case but this time given the presence of some power we have the addition of motor command (push of the bike or motors). Just a single motor command is able to stabilize the locomotion developed in the closed loop.

c) Simple control exploiting body dynamics and interaction with the environment

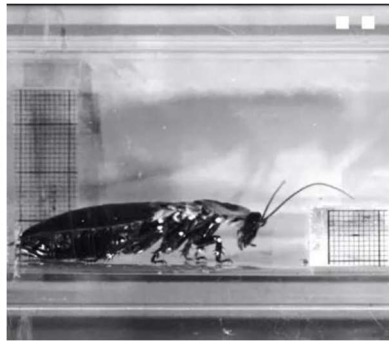
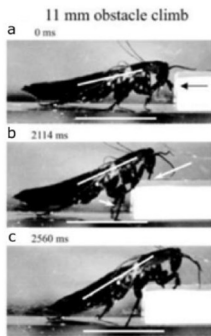
EX1: Cockroaches climbing over obstacles



Cockroaches provide a striking example of how complex behavior can emerge from the interaction between the **body, local neural circuits, and minimal brain control**. Despite having a very complex body—with three thoracic segments, six legs, and more than **seven degrees of freedom per leg**—only about **250 neurons descend from the brain to the body**. This is surprisingly small considering the complexity of the locomotion they perform.

→ **KEY POINT:** Now, how is it possible to control all these degrees of freedom, and such complexity with such a small number of descending neurons?

The reason this works is that most of the control of walking is not handled by the brain but by **local neural circuits called Central Pattern Generators (CPGs)** located in the thoracic ganglia. These circuits produce **stable, rhythmic locomotion patterns** that coordinate the legs automatically, allowing the cockroach to walk efficiently without constant input from the brain.



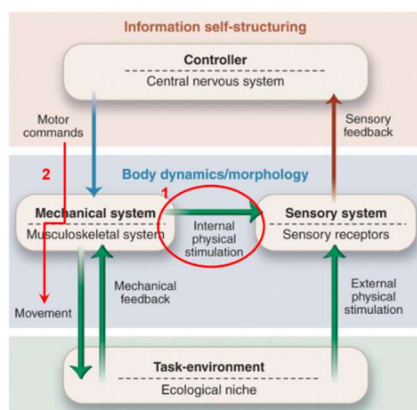
The brain becomes involved mainly when the locomotion pattern needs to **adapt to a new situation**, such as encountering an obstacle. Cockroaches detect obstacles with their **antennae**, which provide information about their presence and approximate height. Instead of completely taking over control of the legs—which would require far more neural resources—the brain makes a **simple morphological adjustment**: it changes

the configuration of the shoulder joint by **lifting the front part of the thorax**. The legs continue executing the same rhythmic walking pattern generated by the local circuits, but because the body posture has changed, the interaction between the legs and the environment also changes. This allows the cockroach to **climb over obstacles higher than its body** while maintaining the same stable locomotion pattern.

This strategy illustrates an efficient division of labor between the **brain, the body, and decentralized neural control**.

✓ **CHEAP SOLUTION:** The brain provides only a minimal intervention—modifying body configuration—while the local neural circuits maintain the rhythmic movement.

By delegating much of the control to **morphology and peripheral neural systems**, the cockroach solves complex locomotion tasks with very few neurons (CHEAP SOLUTION). This approach exploits the **inherent stability of local feedback loops and body mechanics**, showing how intelligent behavior can arise through cooperation between neural control, physical structure, and environmental interaction.



THEORETICAL SCHEME:

In this case we have the internal physical stimulation, which is what makes the peripheral nervous system to synchronize the legs and coordinate them to guarantee stabilization. And then we have the motor commands, i.e., the central nervous system, that only act on the mechanical system to generate movement and change the way the system interacts with the environment.

II. Grasping Case Studies

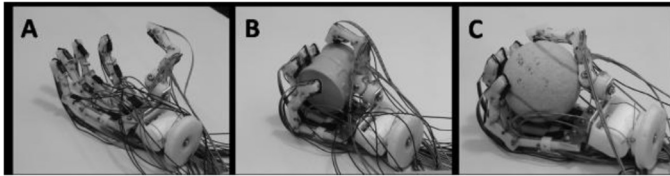
Morphology and materials contribute to grasping behavior. Hand joint structure, muscle mechanics, and the distribution and density of bone to joint movements and muscle recruitment during manipulative behavior are all important variables.

Illustration of ‘cheap grasping’, i.e. grasping that is stable and reliable, yet requires little control.

Cheap Grasping

EX 1: Robotic Yokoi Hand

A first example is the **Yokoi robotic hand** developed in Japan. This hand has **18 degrees of freedom (DOF)** and is tendon-driven, but it is built using **elastic, flexible, and deformable materials**: the tendons are elastic, the fingertips are deformable, and soft material is placed between the fingers.

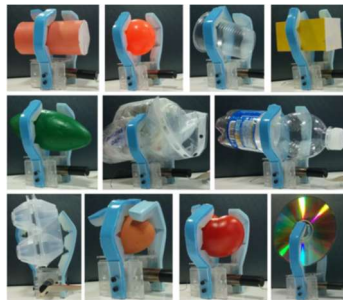
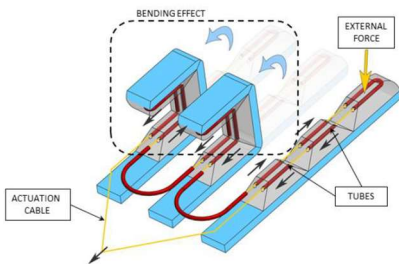


UNDERACTUATION: The hand uses an **underactuated mechanism**, meaning that it has more joints than actuators. For instance, each finger may have **three joints but only one actuator**, typically a cable that runs along

the finger. When the cable is pulled, the finger bends as a whole rather than each joint being controlled individually. As a result, the fingers **automatically adapt to the shape of the object** they encounter. Instead of precisely controlling every joint—as in traditional robotics, where the object’s shape and position must be known—the hand simply performs a **global “close” command** the interaction between the hand’s morphology and the object determines the final grasp configuration.

KEY POINT: This design exploits elasticity, deformability, and underactuation to achieve **automatic shape adaptation with very simple control**.

EX 2: SSSA Soft Robotic Gripper



A similar principle is used in the **SSSA soft robotic gripper**, a system developed in later research. This gripper has nine passive degrees of freedom and only one active degree of freedom, driven by a single motor and a cable that runs through all three fingers. The fingers are made entirely

from **soft, deformable materials**, allowing them to adapt naturally to objects. The design introduces both **intra-digital underactuation** (within each finger) and **inter-digital underactuation** (between different fingers). Because the same tendon controls all fingers, if one finger is blocked by contact with an object, the others can continue moving and close independently. In this way, one motor can control three fingers that adapt automatically to the object’s shape and contact points.

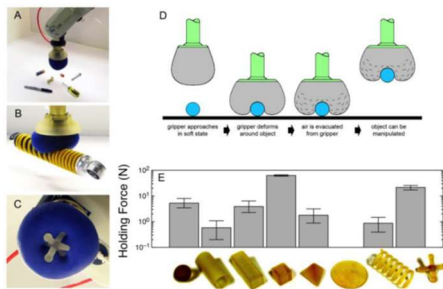
KEY POINT: As in the Yokoi hand, grasping emerges from the interaction between **morphology, material flexibility, and the environment**, rather than from precise joint-level control.

The main limitation of this approach is that increasing the number of fingers can lead to **higher friction in the tendon system**, which reduces efficiency.

TAKEAWAY: Illustrate how **embodiment and morphological design** can simplify manipulation tasks. By embedding adaptability in the **structure and materials of the hand**, complex grasping behaviors can be achieved with **minimal sensing, computation, and control**.

EX 3: Universal Gripper

Another example of **cheap design in grasping** is the **universal gripper**, introduced around 2010–2011. Instead of using multiple fingers, this device consists of a **soft elastic membrane filled with granular material** (such as coffee grounds).



How it works: When the membrane is pressed onto an object, the grains inside behave like a fluid and **flow around the object**, allowing the gripper to conform to almost any shape. Once the membrane has adapted to the object, a **vacuum pump removes the air inside the bag**, causing the grains to undergo a **jamming transition**. In this state, the granular material becomes rigid, locking the shape of the membrane and firmly holding the object.

NOTE: The key principle behind this technology—the **jamming transition of granular materials**—is widely used in soft robotics as a method to create **variable stiffness mechanisms**.

The grasp is stabilized through several physical mechanisms:

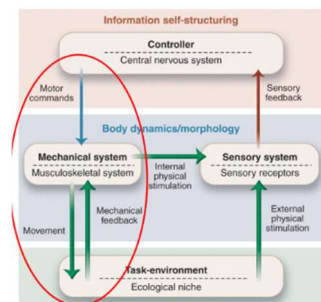
- **geometric interlocking** between the membrane and the object's surface
- **friction generated by normal forces**
- sometimes an additional **suction effect** if the membrane seals part of the object's surface.

When the vacuum is released and air flows back into the membrane, the granular material becomes loose again and the object is automatically released.

This design allows the gripper to grasp a **wide variety of objects without precise sensing or control**, since the adaptation happens through the material's interaction with the object. However, it has limitations: it struggles with **very sharp objects** that could puncture the membrane, **very flat surfaces**, and **very soft objects** (like a sponge) that deform instead of deforming the gripper.

KEY POINT: The gripper conforms to arbitrary shapes passively, that is without any sensory feedback, thanks to its morphological and mechanical properties only.

THEORETICAL SCHEME: In this case, we have a fully mechanical system without sensing, without control. We only have a motor command (e.g., the vacuum in the case of the universal gripper, the single degree of freedom in the hand). And then the rest of the job is directly an effect of the interaction between the mechanical system and the environment.



III. Implications of Embodiment: Information self-structuring

In embodied systems, the **brain and the body should be seen as collaborators rather than competitors**. While mechanical structure can perform part of the “intelligence” of the system, more complex tasks still require control and decision-making. For a controller (or brain) to generate appropriate motor commands, it must receive **relevant** and **well-structured** information about the interaction between the agent and the environment. Embodiment plays a key role in this process because the **physical structure of the body shapes the information that reaches the brain**.

This happens mainly in two ways:

- through the **morphology of the sensory apparatus**

- through the **active generation of information via sensorimotor coordination**

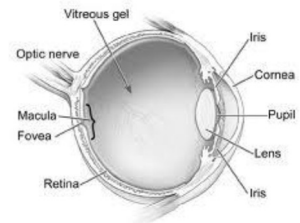
a) Morphology

EX 1: Human Eye vs Insect Eye

Human eye

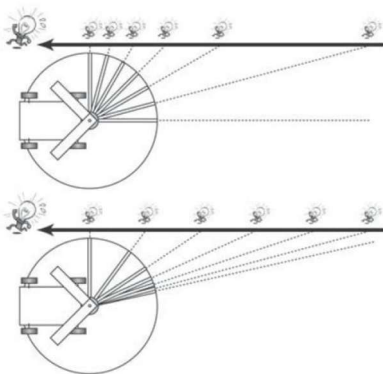
The retina of a human eye is a variable resolution sensor because the distribution of photoreceptors is not uniform. Two types of receptors are present: **cones**, responsible for high-acuity and color vision, and **rods**, which are more sensitive to light.

- Cones are densely concentrated in the **fovea**, the center of the retina, allowing us to see fine details and colors when we look directly at an object.
- In contrast, rods are more prevalent in the **peripheral regions**, where they provide greater sensitivity to light but lower visual acuity.



This non-homogeneous distribution allows the visual system to achieve **both high-resolution central vision and a wide field of view** using a limited number of receptors. It also explains why, in very dark conditions, we sometimes see objects more clearly by **looking slightly away from them**, allowing the more light-sensitive peripheral receptors to detect them.

Insect Eye



Another example comes from the **compound eye of the housefly**, where the arrangement of visual receptors is optimized for detecting motion, which is crucial for survival. The facets of the eye are **not evenly spaced**; they are **denser toward the front of the head** and more widely spaced toward the sides. When a fly moves forward and an object in the environment remains stationary, the object is detected sequentially by different receptors at **regular time intervals**. However, if the object is actually moving—such as a predator approaching—the timing between successive receptor activations changes. This change in timing provides a **simple and reliable cue for motion detection**, allowing the fly to react quickly and escape. If the receptors were distributed uniformly, this timing-based detection would be more difficult to exploit.

Researchers have even reproduced this principle using a **robotic platform and evolutionary algorithms** to optimize sensor placement. The experiment showed that the most efficient configuration for detecting motion corresponded closely to the **non-uniform sensor distribution found in real fly eyes**. In this way, the morphology of the sensory system performs a form of **morphological computation**, simplifying perception by structuring the information before it reaches the brain or controller.

KEY POINT: Certain aims can be solved by proper morphological arrangement of the sensors without changing anything inside the neural controller.

b) Sensory-motor Coordination

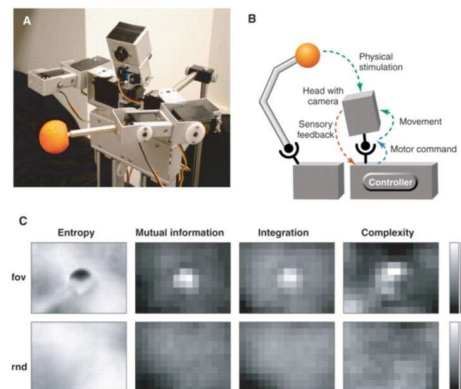
Sensory information is not simply passively received but often **actively generated through movement**. As John Dewey argued:

“We begin not with a sensory stimulus, but with a sensory-motor coordination [...] In a certain sense it is the movement which is primary, and the sensation which is secondary, the movement of the body, head, and eye muscles determining the quality of what is experienced. In other words, the real beginning is with the act of seeing; it is looking, and not a sensation of light”

KEY POINT: movement comes first, and sensation follows from it. In this sense, perception arises from **the act of looking**, where the movements of the body, head, and eyes shape what is perceived.

The visual perception was demonstrated with a robot equipped with **a camera and a robotic arm holding a ball**. Researchers conducted three experiments to analyze how different sensorimotor relationships affect the information obtained from visual input.

- Fov: In the first condition, the sensory feedback is exploited by the controller of the robot head with the camera to actively **track the moving ball**, keeping it centered in the field of view.
- Rnd: In the second condition, the ball moved in the same way, but the **camera remained fixed**, so the ball moved across the visual scene without coordinated camera motion (sensory-motor coupling disrupted).
- (third with two eyes): In the third condition, the robot used **two cameras**: one tracked the ball and kept it centered, while the second camera replicated the same movements but did not actually focus on the ball (it followed the same motion without vergence).

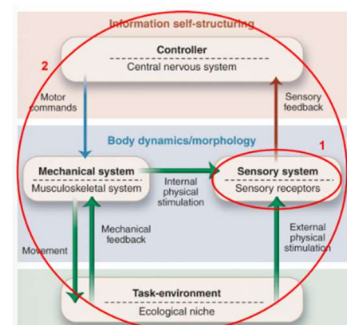


The researchers measured the **information content (entropy)** in the visual signals produced in each case. The results showed that only when the camera **actively tracked the ball and kept it in the center of the visual field** did the sensory input have a low enough entropy to produce meaningful information. In the other cases, although the ball was still visible in the scene, the information structure was poor and did not clearly separate the ball from the background.

This experiment demonstrates that **useful information emerges from the interaction between perception and action**. By actively coordinating movement with sensing, the system structures the sensory input so that relevant information stands out from the environment.

KEY POINT: The fact that we are moving makes the info we are interested on to emerge from the scene. So, Morphology and active perception cannot be considered in isolation.

THEORETICAL SCHEME: in this case we have a sensory system, then there is the morphology of the sensory apparatus, i.e., the way it is made physically, and then there is the active generation of information through sensory motor coordination, so in this case the entire system is used.



How to design a complete agent?

Designing a **complete agent** for the real world is challenging because real environments are complex and unpredictable. Unlike formal systems, we cannot rely on a clean, axiomatic theory that provides exact rules for design. Instead, researchers such as **Rolf Pfeifer** have proposed a set of **design**

heuristics—practical guidelines that help engineers think about how intelligent systems should be built. These principles are not strict laws but **interdependent guidelines** that both support the design of agents and help explain how intelligent behavior emerges.

Principle 1: **The three-constituents principle**

The first principle states that designing an agent always requires considering **three fundamental components**:

- **Environment (ecological niche)**: the context in which the agent will operate
- **Task**: the behaviors or goals the agent must achieve
- **Body**: the physical structure of the agent that enables interaction with the environment

These three elements must be designed **together**, because behavior emerges from their interaction. The body should not be treated merely as a structure that carries the controller or “brain,” but as an **active component that shapes perception, action, and computation**.

SCAFFOLDING: An important aspect related to this principle is **scaffolding**, which refers to modifying the environment to simplify tasks. Humans frequently do this: when working on a desk, we rearrange objects so they are easier to access. Instead of repeatedly reaching far for a pen, we move it closer. Surprisingly, most robots do not yet exploit this strategy. They typically adapt themselves to the environment without trying to **restructure the environment to make tasks easier**, even though humans constantly do so in everyday life.

Principle 2: **The complete agent principle**

The second principle emphasizes that agents must be designed as **complete systems operating in real environments**. This means considering not only the internal mechanisms of the agent but also how it interacts with the environment. In practice, this creates challenges for engineers because the real world cannot be perfectly reproduced in simulations.

KEY ISSUE (Reality Gap): This leads to the well-known **reality gap**: the difference between how a system behaves in a simulated environment and how it behaves in the real world. Real environments contain **geometric, material, and interaction nonlinearities** that are extremely difficult to model accurately. While simulations such as finite element models or simplified dynamic models can help during development, designers must always remain aware that a gap between simulation and reality will exist.

Principle 3: **Cheap Design**

The third principle is **cheap design**, which encourages designers to exploit the **properties of the body and the environment** instead of relying solely on complex control systems. By carefully designing the morphology, materials, and mechanics of the agent—and by understanding the characteristics of the ecological niche—many behaviors can emerge naturally from the interaction between the agent and the environment. When this happens, the control system becomes simpler, and the overall design becomes **more efficient and less computationally demanding**.

!!REMARK!! In this sense, “cheap” does not mean low quality; rather, it means achieving functionality **through intelligent exploitation of physical and environmental properties** rather than through heavy computation or complex control.

Principle 4: **Redundancy**

Another important design principle is **redundancy**, which refers to:

→ the **partial overlap of functionality between different subsystems**.

Intelligent agents should be designed so that their subsystems rely on **different physical processes**, while still being able to perform similar functions if necessary. The goal is not simple duplication, but **adaptability and resilience**. If two components perform the same function using the **same physical mechanism**, they are likely to fail under the same conditions. Instead, redundancy should involve **different mechanisms achieving overlapping outcomes**.

EXAMPLE (**car door locking system**): Modern cars often use **electronic locking**, where pressing a button unlocks the doors. However, if the battery fails or the electronic system stops working, drivers can still access the vehicle using a **physical key**. Both systems perform the same function—unlocking the door—but rely on **different physical principles**. This overlap ensures that the system remains functional even when one mechanism fails.

Principle 5: **Sensory-Motor Coordination**

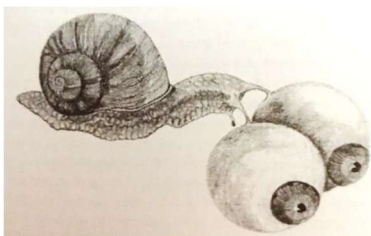
The fifth principle is **sensory–motor coordination**, which emphasizes that perception and action are tightly linked. Instead of passively receiving sensory input, an agent can

→ **actively generate structured sensory information through movement**, a process known as **active perception**. Importantly, this coordination always occurs with respect to a **specific goal or intention**.

EXAMPLE: For example, consider a robot trying to identify an object such as a fruit on a table. Rather than analyzing the entire visual scene, the robot can **move its arm or hand** in the environment. As the hand moves, it creates motion in the visual field. Since the robot’s sensory system can focus on **moving elements in the scene**, the moving hand and the object it interacts with become easier to isolate. When the hand touches the apple, the apple also moves, making it straightforward to identify it as the target object. By **actively moving what it wants to perceive**, the robot simplifies the perception task and processes only the most relevant information.

Principle 6: **Ecological Balance**

The sixth principle is **ecological balance**, which states that there must be an appropriate **match between the complexity of the agent and the demands of its environment and tasks**. In particular, the **sensory, motor, and neural systems** of an agent should be balanced, and tasks should be distributed appropriately among **morphology, materials, control mechanisms, and environmental interaction**.



EXAMPLE (**Snail with Human Eyes**): A well-known illustration of this idea comes from **Richard Dawkins’ example of a snail with human eyes**. Even if a snail had highly advanced visual sensors capable of detecting predators quickly, this ability would provide little benefit if the snail lacked the **motor capabilities to escape rapidly**. In such a case, the complexity of the sensory system would

be wasted because the rest of the organism could not effectively use the information. Ecological balance therefore emphasizes that:

→ **all components of the system must be designed at compatible levels of complexity**, ensuring that perception, action, and control work together effectively.

Principle 7: **Parallel, Loosely Coupled Processes**

The seventh principle concerns **how intelligent systems process information**.

FOCUS (Sense-Think-Act Model): Traditional robotics often follows the **sense–think–act model**: first the system collects sensory information, then processes it internally, and finally produces an action. While this sequential model works in some controlled situations, it is often **inadequate for real-world environments** for two main reasons:

1. First, it assumes a **one-way flow of information**, where sensory input always precedes action. However, perception can also be **actively generated through movement**, meaning sensing and acting are tightly intertwined.
2. Second, the strict hierarchical structure of this model makes it **too slow for real-time interaction**. Each step must finish before the next begins, similar to a computer program waiting for a subroutine to complete before continuing.

Instead, intelligent behavior is better understood as emerging from **many parallel and asynchronous processes** that operate simultaneously. These processes are **loosely coupled**, meaning they are not strictly organized in a hierarchy but still influence each other through **physical interaction within the body and with the environment**.

For example, many physical processes in the body—such as muscle dynamics, mechanical interactions, and environmental forces—occur **in parallel without central coordination**. Their coupling arises naturally from embodiment rather than from a centralized control structure.

→ As a result, intelligent behavior emerges from the **interaction of many concurrent processes**, rather than from a rigid sequential pipeline.

Principle 8: **Value**

The eighth principle concerns the role of **value systems** in guiding behavior. Intelligent agents must possess some notion of **what is beneficial or desirable for them**, which influences how they choose actions in different situations. A value system therefore represents a **set of assumptions about what is good for the agent**, helping determine which behaviors are preferable in a given context.

ISSUE: However, the concept of value is also **one of the most difficult principles to define precisely**. In biology, psychology, robotics, and artificial intelligence, there is still **no consensus on how value systems should be implemented**.

Humans naturally develop such systems through experience. For instance, when reaching for an object like a remote control, there are many possible ways to grasp it, but we instinctively choose the method that is **most efficient, stable, or comfortable**, because we have learned through experience which actions work best. This evaluation assigns a **higher value to certain actions compared to others**.

For artificial agents, implementing such value-based decision making is challenging. Robots typically lack an intuitive understanding of their own **body dynamics, efficiency, and stability**, which humans gradually learn through interaction with the world.

→ while incorporating value systems into artificial agents is recognized as important for guiding intelligent behavior, **defining and implementing them remains an open research problem**.

Bioinspiration and Biomimetics

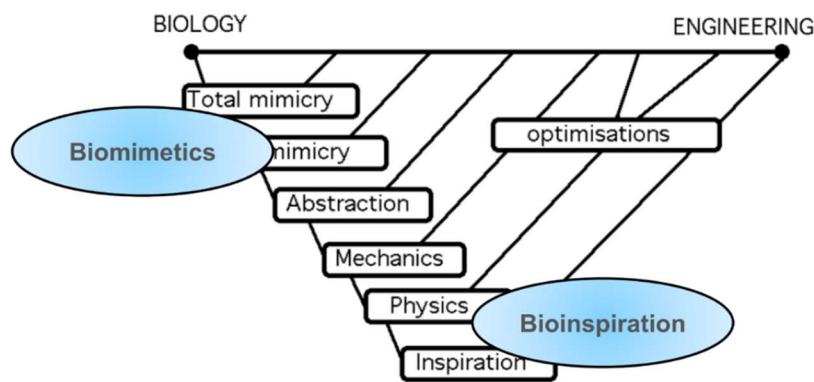
The field of bioinspiration and biomimetics is based on the idea that natural biological systems can provide inspiration for the development of new technologies. By studying biological systems and understanding their underlying principles, researchers can create innovative designs that mimic or improve upon the functions of these systems. It seems straightforward to look at nature to take the inspiration and steel some design principles.

How biomimetics starts? The pioneers of biomimetic were Dr. Schmitt and his wife: by studying natural biological processes and inventing methods and machines that duplicated those actions. The word ‘biomimetics’ was used for the first time by Schmitt in 1957, and used again in 1969 in the title of his paper. Dr. Schmitt and his wife had a dual carrier: Viola was a biologist, that studied neurons and how they work, while Otto tried to replicate the behaviour of a neuron. In fact, his doctoral research was an attempt to produce a physical device that explicitly mimicked the electrical action of a nerve. This new science became known as ‘Biomimetics’ or ‘Mimicry of nature’.

However, before them other scientists took inspiration from nature, for example Leonardo Da Vinci. Leonardo Da Vinci was unsatisfied by the description of the human anatomy reported by his contemporaries and their predecessors. Initially, Leonardo approached the study of the human being and his anatomy in a superficial manner with the aim to be able to reproduce in a realistic way the human body in its paintings. With the passage of time, he developed a passion for the muscular physiology and mechanics, particularly the motion. In his treatise, to better explain the movement, he proposed to replace the muscles and tendons with ropes and wires. Furthermore, Leonardo was determined to understand the full skeleton and each individual bone, as he believed that they were the bearing elements of the ‘human machine’. Leonardo accurately described the curvature of the spine and the structure of the vertebrae. In 1480, he started to study flight: in particular his interest to flight was to understand if human capabilities match the motor abilities with the requirements for the flight. He built a ‘flapping wind mechanism’ to understand if the human power was enough to lift its own body weight (flapping the wings), and it turned out it was not. He tried to develop other machines to allow humans to fly, such as the flying machine or mechanical wings, without success. After some years, in 1500, he resumed his investigation on flight but with focus on the bird’s flight strategies. The project of the flying machine becomes a radical attempt of nature mimicry. He focused on flapping flight or active flight, in which propulsion is obtained by the beating of the wings. He studied the biological system and then modelled and implemented biomimetic machines.

More recently, the formal start of biomimetics as a scientific field could be associated with the second half of the 20th century, in fact by searching for the papers with terms such as ‘biomim*’ or ‘bioinspir*’ their number of publication per year has risen significantly over the last two decades, and this field is still growing today.

However, nature is not necessarily optimal for what we want. Organism that are capable of surviving are not optimal for their technical performance. The unique law of nature is to optimize organism/animals to survive long enough to reproduce. For this reason, we do not have to copy nature, but we have to understand it and take inspiration from it. Furthermore, we can not copy nature because it is way too complex. We should carefully choose nature’s behaviour of interest and extract the underlying key principle at a level of description that is actually possible to implement.



The extraction of key principles can be made in several levels: following the proposed model by Vincent (image), we see that on the right side we have only Biology, that means total mimicry, then at the following level there is the partial mimicry, then abstraction, then looking at the mechanics, then at physics and then only take inspiration.

Vincent, J. V. F.; Pellegrino, S. (Ed.) Stealing Ideas from Nature in Deployable structures Springer, Vienna, 2001, 51-58

The terms biomimetic and bioinspiration are sometimes used as synonyms, actually they are not. Biomimetics is more related to the approaches that are related to mimicry, while bioinspiration is more related to looking at something that can inspire to create new engineering through optimization.

Bioinspiration design approach (application-driven)

Application-driven approach means that we start with an engineering problem. The methodology for applying bioinspiration to design challenges is based on 4 points:

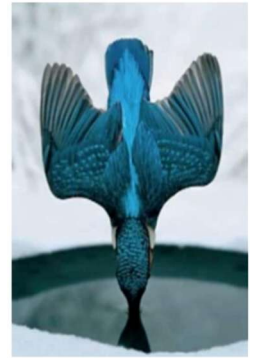
1. Define the problem: This involves identifying the desired outcome and any constraints or limitations that need to be considered. By clearly defining the problem, the designer can focus their research and analysis on finding relevant biological systems that may provide inspiration. This point is fundamental because identifying in a wrong way the problem can not lead to find any solution.
2. Research and Analyse Biological Systems. This involves observing and studying the structures, functions, and behaviours of natural systems and identifying any features or processes that may be relevant to the design problem. This step requires a deep understanding of biology and ecology, as well as the ability to observe and analyse complex natural systems.
3. Synthesize Information. This involves distilling the insights gained from the natural systems into a set of general design principles that can be applied to the design problem. This step requires creativity and an ability to make connections between seemingly unrelated fields.
4. Apply Insights to Create a Solution. This may involve developing technologies, new products, or processes that are inspired by natural systems, or it may involve modifying existing designs to incorporate biomimetic features. This step requires a deep understanding of the design problem and the ability to apply the principles learned from the natural systems in a practical way.

Examples of bioinspiration approach:

Shinkansen bullet train

At the beginning of the development of these trains, train's designers were faced with a huge problem related to the fact that when the train enters through a tunnel at a very high speed, they create a very loud sound. The sonic boom was caused by the compressed air which generates a sound wave that

made the loud sound exiting the tunnel. Furthermore, the air reduces the speed. The problem was like diving through water. Looking at nature, engineers notice that the kingfisher was able to enter the water creating very minimal disturbances to the surface. After studying the animal, they understood that was due to the unique shape of their peak: the peak is long, narrow and extreme line, it steadily increased in diameter from the tip to the end and this reduce the impact with the water; the water flowing past the peak rather than being pushed in front. The peak reassembles two triangles with rounded edges together forming a squashed diamond shape. So the engineering copied the design and solve the problem of the sonic boom.



1. Define the Problem: Develop a high-speed train (more than 300 km/h) that does not create a sonic boom in tunnels and without increasing energy consumption.

2. Research and Analyse Biological Systems: Analysis of the anatomy of the kingfisher, a bird that is known for its ability to dive into water without creating a splash. The bird's beak has a unique shape that reduces the turbulence of the water, allowing it to enter the water with minimal disturbance.



3. Synthesize Information: Create a new shape for the train front that resembles the beak of a kingfisher, with a sharp, tapered tip and a gently sloping back end. This shape helps to reduce the air pressure in front of the train, reducing the amount of sonic boom it creates.

4. Apply Insights to Create a Solution: The new design is applied to the train and tested it in real-world conditions. The new design proved to be highly effective, reducing the sonic boom of the train by as much as 30%.

The bionic car

In 2005 DaimlerChrysler has developed a new high fuel efficiency concept vehicle based on the body shape of the boxfish (*Ostracion meleagris*), a common cube-shaped fish found in tropical marine habitats. The bionic car will offer 20 percent lower fuel consumption and up to 80 percent lower nitrogen oxide emissions according to a release from DaimlerChrysler.



Tour Eiffel

At the end of the 19th century Gustave Eiffel was asked to design a tower of iron that was as high as possible. But iron is extremely heavy, so any building would have collapsed under its own weight. The main problem was to design a structure that could redistribute the weight and propagate the load in a correct way. He found the solution by thinking about the trabeculae of human long bones: the bones are not completely full, but the trabeculae create a sort of net. This makes the sustainability capability of the material much lower, but the weight is extremely reduced. He tried to reproduce the same concept and ended up with the design of the Eiffel tower that we know today.

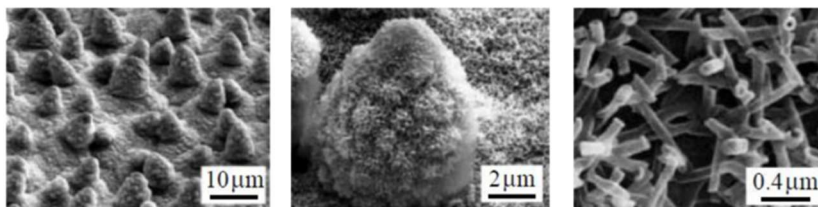
Bioinspiration design process approach (curiosity-driven)

This approach starts with directly by observing nature. The curiosity-driven approach should be seen as a circle and not as a one-way procedure because after the implementation you may need to return to the inspiration to redefine the idea.

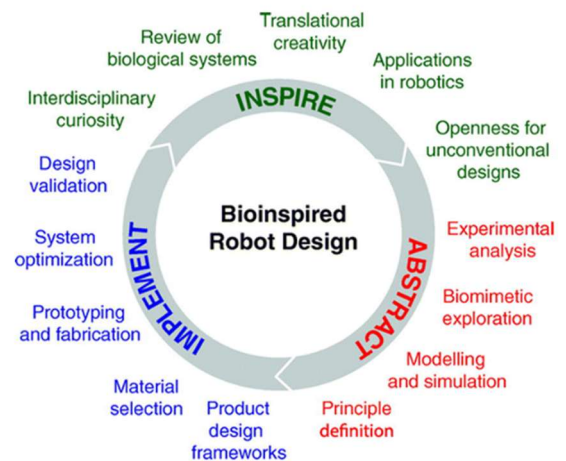
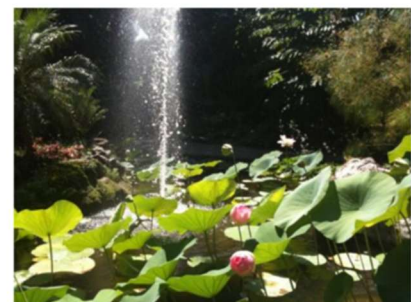
Examples of curiosity-driven approaches:

Lotus Effect

There are some plants whose leaves are water-repellent, such as *Nelumbo nucifera* (lotus) and *Colocasia esculenta*. This came with the observation that when there is water on their surface, the water completely flows away. But the capability that has been observed is that they have the advantage, from a biological point of view, that they are never dirt, and this is very important because if the leaves are dirt the process of energy production (photosynthesis) is hindered. The leaves are known to be superhydrophobic and self-cleaning due to hierarchical roughness (microstructure formed by papillose epidermal cells) and the presence of a hydrophobic coating (epicuticular wax tubules). From a biological point of view, this effect is due to a combination of structures and a kind of wax material on the top. By observing these, some scientists realize that these features could be 'copied' to protect painting, avoiding the surface to become dirt.



Bhushan B. Biomimetics: lessons from nature-an overview, Phil. Trans. R. Soc. A 2009 367, 1445-1486



1. Define the Problem: Developing a non-stick coating that would repel water and other liquids and prevent the adhesion of dirt and other contaminants.
2. Research and Analyze Biological Systems: Studied the surface of the lotus leaf, which is known for its ability to repel water and other liquids. They analysed the structure and chemical composition of the lotus leaf and discovered that its surface is covered by micro and nano-scale bumps that prevent water droplets from sticking to it.
3. Synthesize Information: developed a coating that mimicked the lotus leaf's structure and composition. They used a combination of silicone and fluorine to create a surface that was both water repellent and non-stick.
4. Apply Insights to Create a Solution: Tested the coating on various surfaces and found that it was highly effective at repelling water and other liquids. The coating was able to prevent the adhesion of dirt and other contaminants, making it a highly useful solution for a wide range of applications.

Velcro

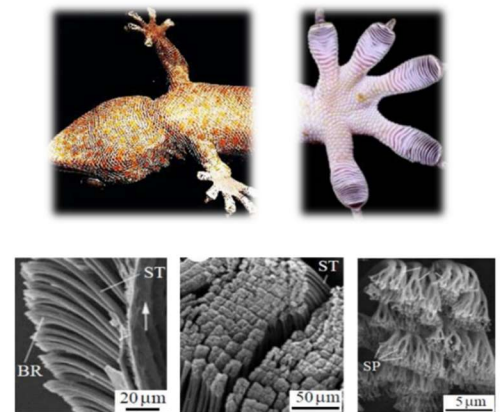
Velcro was invented by George de Mestral, a Swiss engineer, who was inspired by the way that burdock seeds stuck to his clothing and his dog's fur during a walk in the woods. He noticed that the mechanism was reversible, attaching and detaching, but at the same time it was very effective.



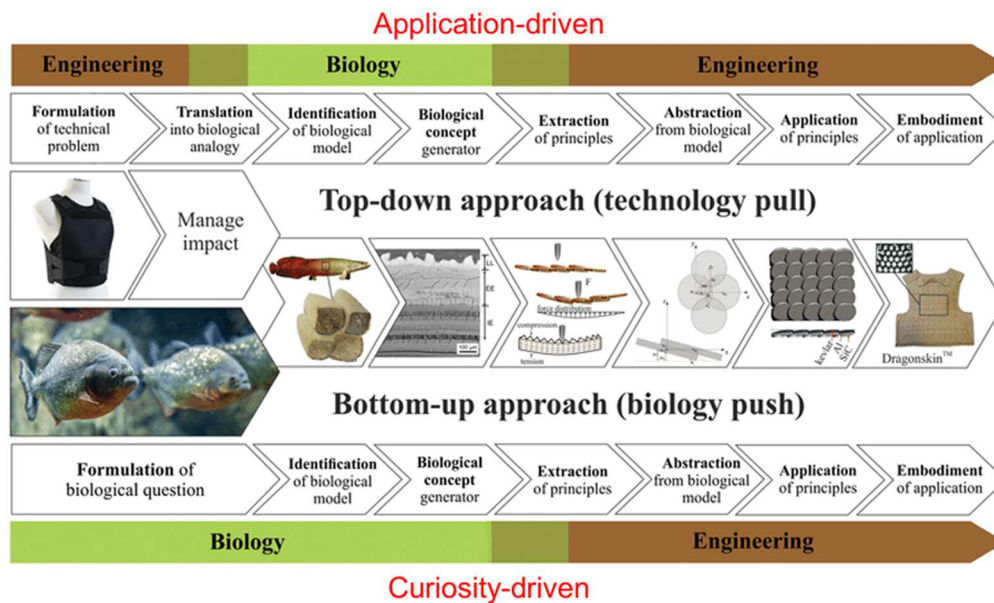
1. Define the Problem: De Mestral observed the way that burdock seeds attached themselves to fabric and fur and identified this as a potential solution for fastening materials together.
2. Research and Analyze Biological Systems: He studied the structure of the burdock seed, which has tiny hooks that attach to loops of fabric or fur. He then applied this concept to the design of a new type of fastener, using synthetic hooks and loops that would attach to each other when pressed together.
3. Synthesize Information: He tested his new fastener design and found that it was more durable and efficient than traditional fastening methods such as buttons and zippers.
4. Apply Insights to Create a Solution: De Mestral Finally patented his invention and began marketing it under the name "Velcro", which is a combination of the French words velours (velvet) and crochet (hook).

Gecko

Gecko is the largest animal that can produce dry adhesion to support its own weight. They can walk upside down without any problems in dry conditions (there exist other largest animal that can sustains their weight but in wet conditions). Researchers studied very in details the fingertips of the gecko and found out that their foot comprise a complex hierarchical structure of branches, lamellae, setae and spatula. They have an extremely small structure, so small that allows to interact with the materials with Van Der Waals forces, very low forces, that summing up are able to sustain the body weight. Some researchers at Stanford tried to replicate this principle in a robot. The robot was able to climb surfaces, also ones that are not so flat. However, they did not replicate the real key secret of the gecko fingertips (nanostructures that use Van Der Waals forces), but they used silicon in a particular shape and orientation to have some advantages into silicon-surface interaction. Furthermore, silicon is not actually completely dry material.



Biomimetics Approaches



Biomimetic approaches can generally be divided into two main strategies: application-driven and curiosity-driven.

In an application-driven approach, the starting point is an engineering problem. First, the problem is analysed and translated into a biological analogy: in other words, researchers look for a natural system that faces a similar functional challenge. Then, a suitable biological model is identified as a source of inspiration. The key principles behind the biological solution are extracted, abstracted from the specific organism or structure, and finally applied to the engineering scenario. Since this method starts from a technical need and then looks at biology for a solution, it is considered a top-down approach.

In a curiosity-driven approach, the process starts from a biological question, such as: *How does this phenomenon occur in nature?* Researchers first investigate the biological mechanism and identify the model organism or natural structure involved. After that, the process is similar: the biological principles are extracted, abstracted, and potentially transferred to an engineering application. Since this approach starts from the observation of nature and may later lead to technological applications, it is considered a bottom-up approach.

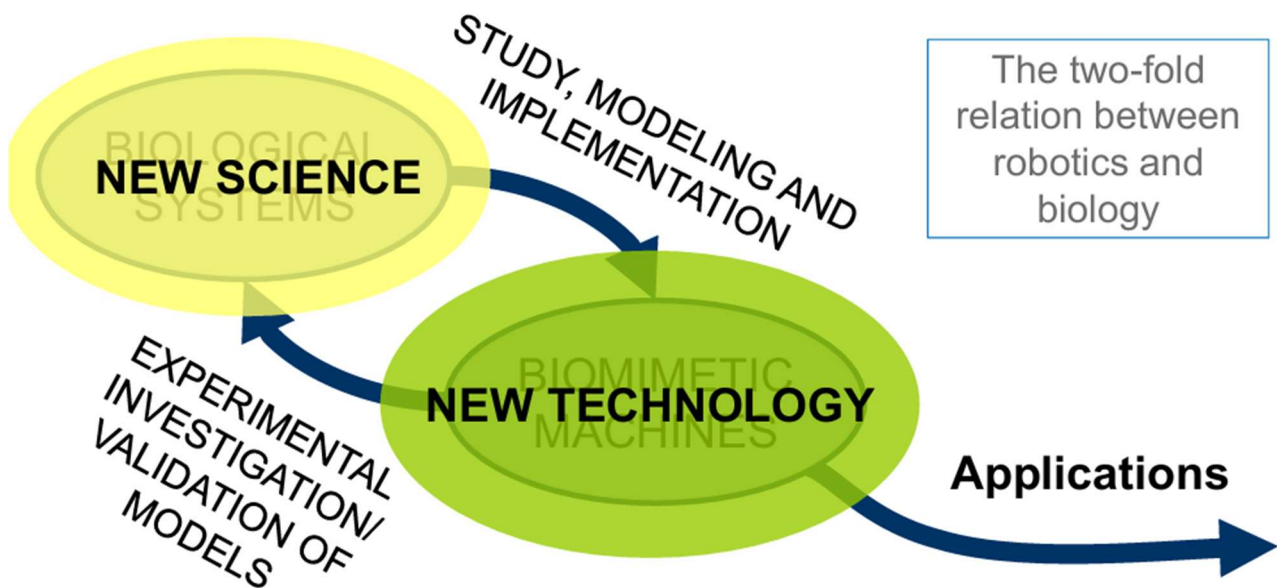
However, a possible limitation of the curiosity-driven approach is the risk of over-engineering. A biological mechanism may be fascinating but translating it into technology is useful only if it provides a real advantage compared to existing solutions. Therefore, innovation occurs only when the biomimetic design improves performance, efficiency, adaptability, sustainability, or solves a problem in a better way than conventional engineering methods.

The two-fold relation between robotics and biology

The relationship between robotics and biology can be considered as a two-way process. On one hand, biological systems can be studied, modelled, and implemented in order to create biomimetic machines. In this case, nature acts as a source of inspiration: biological principles are extracted from

animals, plants, or natural mechanisms and then translated into robotic systems. These robots can then lead to the development of new technologies for real-world applications.

On the other hand, the relationship also works in the opposite direction. Robotics does not only receive inspiration from biology but can also become a powerful tool to better understand biological systems. By building robotic models of natural mechanisms, scientists can test hypotheses, validate biological models, and discover whether the current understanding of a natural phenomenon is complete. In this sense, robotics can contribute to the development of new scientific knowledge.



A clear example is the work carried out in Japan by a researcher, who tried to replicate the locomotion strategy of snakes. He collected the information available at the time about snake locomotion and implemented the main principles into a robotic machine. However, the first model did not work properly. This failure suggested that some important biological information was still missing. After further investigation of real snake movement, he discovered that the snake's body does not remain completely flat during locomotion. Instead, some parts of the body stay in contact with the ground, while other parts are lifted. Without this mechanism, neither the robot nor the real animal would be able to move effectively.

This example shows that robotics can be used not only to imitate nature, but also to investigate it. The attempt to reproduce a biological behaviour in a machine can reveal gaps in scientific knowledge and lead to a deeper understanding of how natural systems actually work.

The octopus as a biological model for soft robotics

The octopus is an extremely interesting biological model, especially from the perspective of soft robotics. From an anatomical point of view, the octopus has only one relatively rigid structure: the ‘beak’, near the head, which is used to break the shells of prey. The rest of the body is mainly composed of soft materials, such as tissues, muscles, and cartilage-like structures. Thanks to this soft composition, the octopus can deform its body and squeeze through extremely small openings, even through holes approximately as large as its eye.

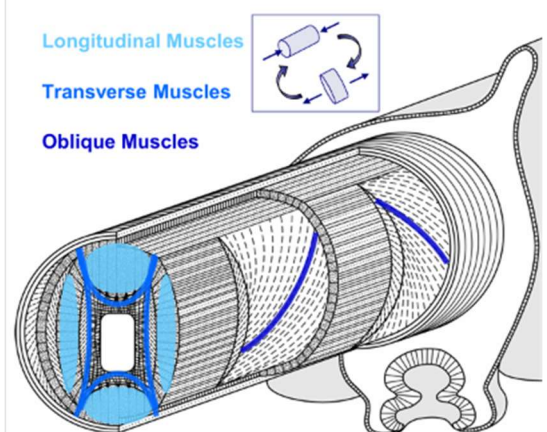
One of the most remarkable features of the octopus is the extraordinary mobility of its arms. The arms can bend, elongate, twist, and move in almost any direction. They are highly dexterous and are used not only for locomotion and swimming, but also for interacting with the environment, exploring objects, and catching prey. From an engineering point of view, this ability to move freely in space gives the octopus arms an extremely high number of degrees of freedom. In fact, since the arms are soft and continuously deformable, they can be considered as having almost infinite passive degrees of freedom. Another important feature is that, although the arms are made of soft materials, they are still able to vary their stiffness and generate significant forces. For example, octopuses are able to manipulate objects and even open jars. This shows that softness does not necessarily mean weakness: the octopus combines flexibility, dexterity, and strength in the same structure.

To better understand the motor capabilities of the octopus, researchers performed non-invasive tests and observed how the animal moves, manipulates objects, and controls its arms. These studies showed that the octopus has unexpected manipulation abilities, despite the absence of rigid bones or joints. Its peculiar motion capabilities are due to a structure called a muscular hydrostat. This biological mechanism is very important for robotics because it shows how movement and force generation can be achieved using soft structures. For this reason, the octopus has inspired the development of soft robotic arms and manipulators that can adapt to complex environments, grasp delicate objects, and move safely around humans.

Arm Morphology

A muscular hydrostat is a biological structure made almost entirely of muscles arranged in a specific way. Since these muscles are mainly composed of water, their volume remains almost constant during contraction. This means that when the muscles contract in one direction, the structure must expand or deform in another direction. For example, if the arm becomes shorter in one direction, it may become wider; if it becomes thinner, it may elongate. A small deformation in diameter (20%), induce a long elongation (80%). The muscular hydrostats serves as a modifiable skeleton and allows the transformation of force into motion. The octopus arm is made by three types of muscles: longitudinal, transverse and oblique muscles.

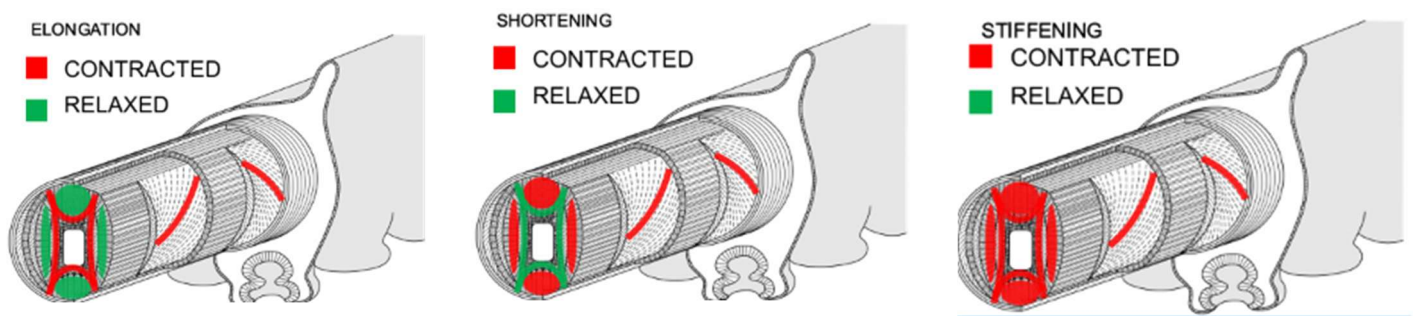
Longitudinal muscles are arranged longitudinally along all the muscle, while the transverse muscles have a radial arrangement. The oblique muscles have a support function and allow rotation (orientation of the arm).



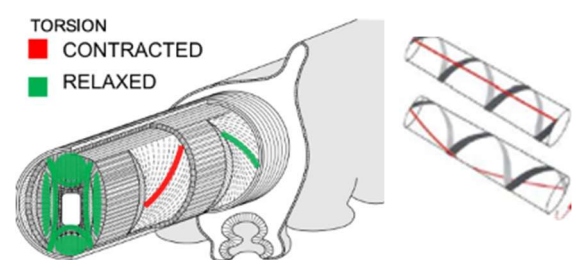
The octopus arm can produce different movements thanks to the coordinated activation of different muscle groups. Elongation is achieved mainly by contracting the transverse and oblique muscles, while the longitudinal muscles remain relaxed. Since the arm has an almost constant volume, contraction in the transverse direction reduces the diameter of the arm and causes it to extend in length. For this reason, elongation can be considered a sort of passive elongation, because it is not produced by directly pulling the arm along its length, but by reducing its cross-sectional area.

Shortening, instead, is achieved by contracting the longitudinal and oblique muscles, while the transverse muscles remain relaxed. In this case, the longitudinal contraction reduces the length of the arm, while the transverse section increases in diameter in order to preserve the overall volume.

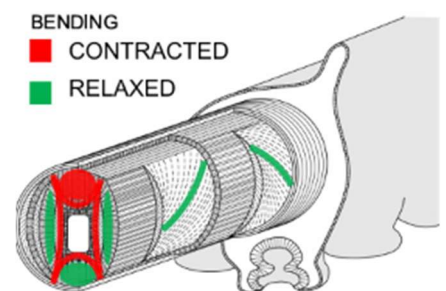
The octopus can also stiffen its arm by contracting all the muscle groups simultaneously. This co-contraction generates internal forces that compensate each other, allowing the arm to maintain both its shape and its volume while increasing its mechanical stiffness. In other words, the arm becomes more resistant to deformation without necessarily producing a visible movement. This ability to vary stiffness is fundamental for interaction with the environment. A soft arm allows safe and adaptive contact with objects, while a stiffer arm is useful when the octopus needs to apply force, manipulate prey, support part of its body, or perform more controlled movements.

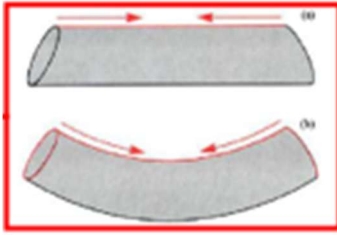


Torsion is achieved by selectively contracting part of the oblique muscles, while the remaining muscles are relaxed. Since the oblique fibers are helically arranged around the arm, their asymmetric contraction generates a twisting movement around the longitudinal axis. This mechanism allows the octopus to rotate the arm and orient its suckers during manipulation tasks.



Bending is a more complex movement compared to elongation, shortening, and torsion. If the octopus contracted only the longitudinal muscles on the side toward which it wants to bend, the movement would not be efficient. Since the arm is entirely soft and does not contain rigid skeletal structures, the contraction of the longitudinal muscles alone would not preserve the shape of the cross-section. Instead of producing a controlled bend, the two ends of the arm segment would simply move closer to each other, causing an undesired deformation, as shown in condition A.





To obtain an effective bending movement, the octopus must also preserve the transverse shape of the arm. This is achieved by activating the transverse muscles, which help maintain the cross-sectional geometry and prevent collapse or excessive deformation. At the same time, the longitudinal muscles on the side of the intended bending are contracted. This coordinated activation produces a controlled curvature of the arm, as shown in condition B.

The Octopus Project

The aim of the Octopus Project, carried out from 2009 to 2013, was to understand the biological principles behind the manipulation abilities, dexterity, and force generation of the octopus, and to use this knowledge to replicate these capabilities in robotic systems.

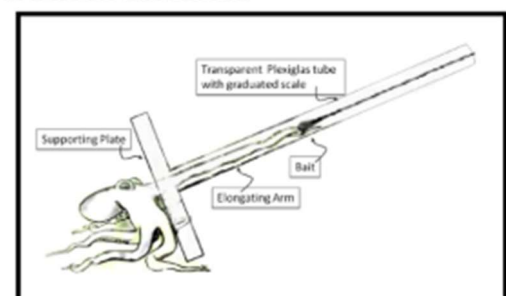
At that time, some important data about the mechanical properties of the octopus arm were still missing, especially information related to force generation, elongation, stiffness variation, and the internal organization of tissues. For this reason, researchers developed specific experimental tools to study the octopus in a non-invasive way and to measure its motor capabilities during natural or semi-natural tasks.

For example, the tool shown in Image A was developed to measure the elongation and force capabilities of a single arm. A tube was inserted into the tank in such a way that the octopus could insert only one arm inside it. A prey was placed at the end of the tube, encouraging the animal to elongate the arm as much as possible in order to reach it. Since the tube had a graduated scale, researchers could measure the maximum elongation of the arm. Once the octopus reached and grasped the prey, the researchers pulled it in order to estimate the force exerted by the arm.



Image A

In addition to these biomechanical tests, anatomical analyses were also performed. Ultrasound imaging was used to study the internal structure of the arm in a non-invasive way, while histological analyses were carried out by laboratories in Naples. These analyses provided information about the structure, geometry, and spatial organization of the biological tissues inside the arm.



The combination of biomechanical, anatomical, and histological data allowed researchers to obtain a more complete understanding of the octopus arm. In particular, the histological analysis provided useful information about the insertion points of the muscles and the percentage distribution of transverse and longitudinal muscle fibers along different regions of the arm. This was fundamental to

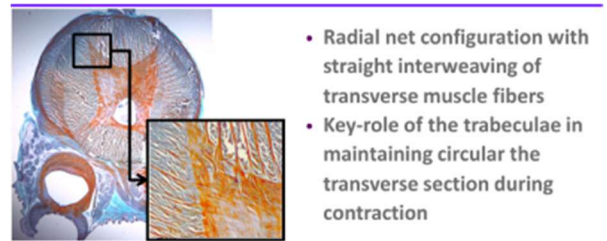
understand how the octopus can combine softness, flexibility, elongation, variable stiffness, and force generation in the same biological structure.

Design of transverse actuators inspired by the octopus muscles arrangement:

In the biological octopus arm, the transverse muscles are not attached only at four discrete points, as it might seem from a simplified radial view. Instead, they are distributed throughout the longitudinal muscle structure. The arm presents a radial network configuration, with transverse muscle fibers interwoven through the longitudinal muscles. This arrangement plays a key role in maintaining the circular shape of the transverse section during contraction.

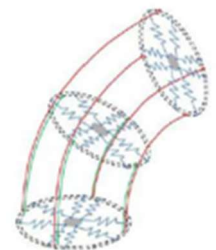
In fact, the first preliminary robotic tests used a simplified configuration based on four arcs, but this solution was not effective. It was too simple and did not preserve the shape of the arm properly during contraction. A more biomimetic radial configuration was then adopted, which proved to be more efficient in reducing the diameter of the artificial arm. In the robotic implementation, this contraction mechanism was reproduced using shape memory alloy springs.

Regarding the longitudinal muscles, in nature they have multiple insertion points along the length of the arm. This allows the octopus to bend the arm at many different points, producing highly flexible and complex movements. However, reproducing a continuous distribution of insertion points in an artificial system was not technically feasible, and this inevitably reduced the efficiency and biological similarity of the robotic model.



For this reason, the researchers adopted a simplified solution: they used 10 insertion points along an arm segment of about 30 cm. Although this configuration is still far from the complexity of the natural octopus arm, it was sufficient to reproduce a similar bending behaviour.

In the scheme, the transverse actuators are represented as springs, because they were made of shape memory alloy springs. The longitudinal muscles, instead, are represented as cables. These cables run along the arm toward the base and are pulled by several motors, allowing the artificial arm to bend in different directions.



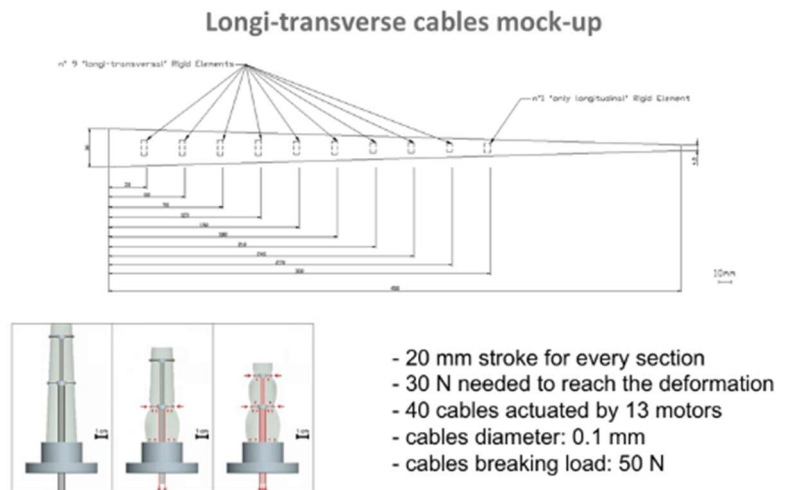
Results on octopus biomechanics:

The octopus can reach 75% of arm elongation corresponding to a 23% of diameter reduction. The pulling force is 50 N when the force application point is at 400 mm from the base and 28 N when the application point is at 200 mm. So the force generation depends on the application point of the force. The octopus has a preferred grasping point, he prefers to interact with the environment with the point that is at the 0.75 of the total arm length, in this way it can generate a high force and use the remaining part of the arm to envelope the object and stabilize the grasp. The contraction time is about 1-2 seconds.

A technological problem encountered into the reproduction of the longitudinal muscles was in the generation of force by pulling some cables that are surrounded in silicon. The interaction between the

cable and the silicon must be managed, otherwise the cable will cut the silicon. The sheaths used was hard silicon or DFE (?).

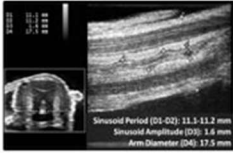
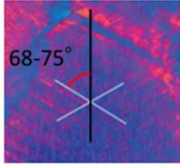
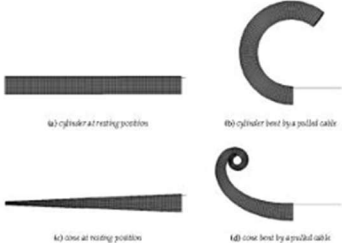
The final Mock-up: the longitudinal muscles were reproduced by cables, which generate high force, while the transversal muscles were reproduced with shape-memory springs. In order to test the capability of stiffness, simultaneously contract transversal and longitudinal muscles, they came up with the idea of a longi-transverse cables: cables run longitudinally from the base and at a certain point turn 90 degrees and emerge transversally. In this way, pulling the cables will reduce the length but also the diameter. This do not work (reduce diameter) continuously as in the octopus, but allow to obtain similar performance (although there is a small deformation).



Combining some basics movements, such as shortening, bending, twisting, elongation, the octopus is able to perform higher level movements. One of the most studied is the reaching movement, which is complex due to the underwater drag. To reduce the drag and avoiding that the drag is on the entire arm, the octopus generate a sort of 'wave': the arm is initially completely bend that then it is bended time by time, in this way the only part exposed to drag is the tip. Furthermore, from a cognitive point of view, the central nervous system, sent information only about the orientation of the base of the arm (which determine the direction of the movement) and the velocity, all the rest is controlled by the peripheral nervous system though a propagating stiffness wave. The stiffness wave is generated by having stiffness locally, by contracting the transverse and longitudinal muscles locally, in series. This allows the propagation of the stiffness wave from the base to the tip. The brain of the octopus is very small, and the 90% of it is dedicated to vision and the rest 10% is for the dexterity of the arm. This is way most of the control of the arms is up to the ganglions that are along the arms. Octopuses face a unique visual challenge: because their eyes are positioned 180 degrees apart, they have a very narrow field of binocular overlap. To compensate for this lack of stereoscopic vision and better perceive depth, they perform a behaviour known as 'head bobbing'. By moving their head up and down, they create motion parallax, allowing their brain to calculate distances based on how objects shift against the background. This biological 'scanning' technique is essential for mapping their surroundings and executing precise strikes on prey. Since this 'scanning' provides only an approximate distance, octopuses compensate by launching multiple arms at once. This strategy creates a wide 'net' that increases the probability of contact, allowing their highly sensitive tactile receptors to take over once the prey is within reach.

A major challenge in creating artificial octopus arms is that electrical wires, much like nervous tissue, are not naturally stretchable. Nature solves this through a brilliant structural adaptation: in a relaxed state, the octopus's nerve cords are arranged in a sinusoidal, wave-like shape. This folding allows the nerves to uncoil and straighten during extreme elongation without being stretched or damaged. By mimicking this sinusoidal geometry, engineers can integrate flexible wiring that follows the arm's

movement, ensuring a constant power supply to the shape-memory actuators even during full extension.

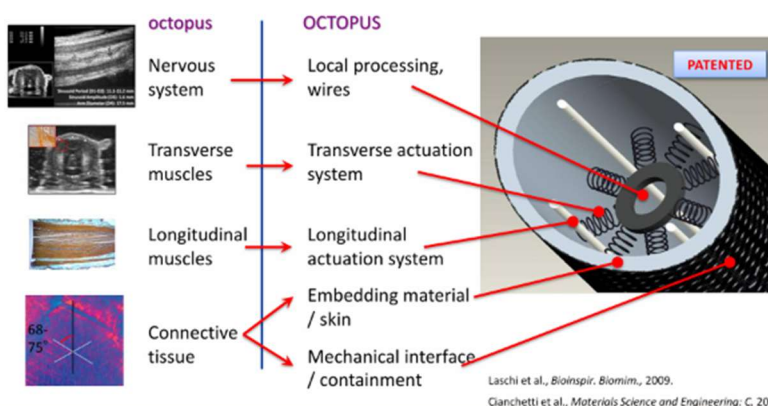
Nerve cord arrangement	Connective tissue	Arm shape
 Sinusoidal arrangement at the arm rest length	 Helicoidally braided arrangement	 (a) cylindrical at resting position (b) cylinder bent by a pulled cable (c) cone at resting position (d) cone bent by a pulled cable

The octopus's arm is conical shape provides a significant mechanical advantage over a constant cylindrical cross-section. In a cylinder, the bending curvature is uniform throughout; however, a conical geometry allows the curvature to decrease along its length. This gradient enables the octopus to adapt its grip to objects of various sizes: the thinner, more flexible tip can curl tightly around small items, while the thicker, more robust base provides the necessary strength and larger curvature to grasp and stabilize bigger objects.

In the artificial octopus, the actuators for the artificial muscular hydrostat were made of an electroactive polymer (dielectric elastomers), which is a soft material with passive deformation capabilities and relatively 'fast' response time. Is based on silicon material and a polymeric film which have two compliant electrodes. When the electrodes are charged the two plates moves closer, squeezing the polymeric field. However, the fabrication procedure was not feasible, at least at the size they needed.

From biology to robotics

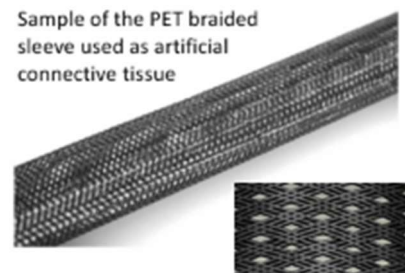
Design and development of an artificial muscular hydrostat



The shape memory alloys have a lot of advantages: lightweight, flexible, high work density, high possibility of miniaturization, simple electronic drivers. The cons were 'solved' with some strategies: the low strain capabilities problem was solved giving the spring shape; the relatively high current and low bandwidth problems with a well-studied cooling system; and the non-linear behaviour and the wide hysteresis with an all-or-nothing activation and antagonist arrangement.

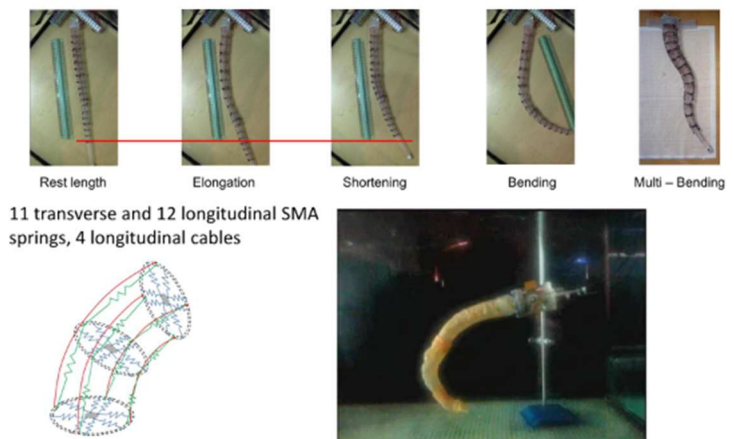
The shape memory alloys spring with a PTFE sheath was coupled with an elastomeric material that provides a restoring force. The SMA in water cannot work for many cycles due to electrolysis, with 'extract' titanium from the alloy, reducing their life cycle.

From literature, the connective tissue in the octopus arm is arranged as a net to wrap the entire arm. The net angle varies from 68° to 75° . This was a source of inspiration to solve the problem of the 'sausage' effect. In order to distribute the deformation both transversally and longitudinally, the deformation should be global and not local. In the configuration seen before, where the cables are emerging transversally the deformation is local (sausage effect). To obtain a global deformation, a PET braided sleeve was used, making the reduction in diameter much more efficient. With the braided sleeve used as connective tissue, the 20% of diameter reduction lead to a 89% of elongation. The sleeve provides a mechanical support and transforms transverse contraction in elongation and propagated the deformation longitudinally.



The first prototype had 12 longitudinal cables and 18 transverse actuators, each one made 8 SMA springs. The second prototype used only SMA springs: the longitudinal cables were substituted with the SMA springs. The SMA springs actuators were directly linked to braided sleeve without rigid parts and the electrical cable interlaced with the yarns of the braided sheath (using the same path of the braided sheath). The SMA springs created a local effect. The third prototype had both cables and SMA springs (4 longitudinal cables, 12 longitudinal SMA springs and 11 transvers SMA springs) because without cables the force generated was low. This set-up allowed to generate enough force to interact with object, thanks to longitudinal and radial deformation. The final prototype of the robot had two frontal arms equipped with SMA springs and cables while the other arms are only used for locomotion.

OCTOPUS arm prototype v3



Advantages and limitations of Bioinspiration and Biomimetics

Advantages of using these kinds of approaches are that:

- **Novel Solutions:** Natural systems have evolved over millions of years to be highly efficient and effective, and bioinspiration and biomimetics offer a way to tap into this vast source of knowledge to develop new solutions to complex problems. Natural systems are a source of inspiration
- **Sustainability:** Bioinspired and biomimetic technologies can be designed to use sustainable and renewable resources, and to minimize negative impacts on local ecosystems and the wider environment. Using same materials of nature, will reduce the negative impact of artificial robots/products.
- **Cost-Effective:** Biomimetic designs can also be cost-effective, as they often make use of existing natural structures and materials that are readily available in the environment.
- **Efficiency:** Natural systems are often highly efficient, and biomimetic designs can help to improve the efficiency of many different technologies, from transportation to energy production to manufacturing.
- **Interdisciplinarity:** Bioinspiration and biomimetics involve the integration of knowledge from many different disciplines, including biology, physics, materials science, and engineering. This interdisciplinary approach can lead to new insights and breakthroughs that might not be possible with a narrower focus

Limitations:

- **Lack of Understanding:** While natural systems are highly effective, we still have much to learn about how they work and how they can be translated into practical designs. This lack of understanding can make it challenging to develop effective biomimetic technologies.
- **Complexity:** Natural systems are often highly complex and difficult to replicate, which can make it challenging to translate biomimetic designs into practical technologies.
- **Compatibility:** Biomimetic designs may not be compatible with existing manufacturing processes or materials, which can make it difficult to bring them to market in a cost-effective way.
- **Patentability:** In some cases, biomimetic designs may not be patentable, as they may be seen as direct copies of natural systems. This can limit the potential for commercialization and may make it difficult to attract investment. Without an 'invention step' there are no patents.

Actuation Technologies: Introduction

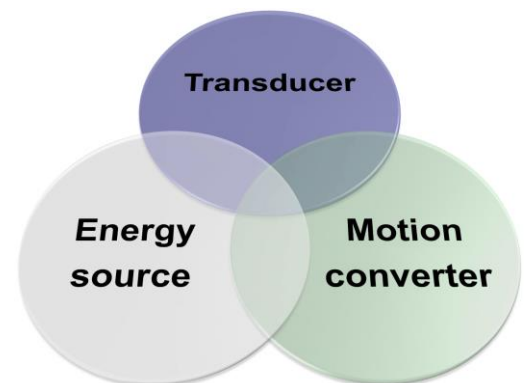
ACTUATOR = system which is able to provide **mechanical action in the requested time**.

This definition is important because sometimes there is a general misunderstanding of what an actuator is. Sometimes actuator is used as synonym of motor for example, but it's not (this is acceptable in general terms, not during the exam because it's able to generate mechanical power). At this level is important to understand the difference and what is the specific definition of an actuator. And actuator is composed of 3 main components:

1. **TRANSDUCER** → It's the only one that is **mandatory** (without a transducer you cannot have an actuator). A transducer is the only physical element that converts energy. An energy, that is usually given by an electrical input, is transformed into mechanical energy. So, in general, it takes a type of input energy and transforms it into mechanical energy.
2. **ENERGY SOURCE** (optional) → It can be integrated inside the system. It is what **provides** the electrical **energy as an input** to be transformed, by the transducer, into mechanical energy
3. **MOTION CONVERTER** (optional) → It's a mechanical component that **transforms mechanical couplings**. So, considering the case that you have a certain energy as an input, for example electric energy, that can be provide with a specific power. So, you can provide energy in a certain amount of time, and this is generally fixed because your energy source is set. But what you can do with a motion converter is to transform what is the coupling at the output in terms of **force and speed**.
 - Direct coupling = Without motion converter you have an input, as electric power, and, due to the capabilities of the transducer, you have a certain speed and force, like rotation of gear. That process does not have conversion.
 - With motion converter = for example gearboxes or levers. These kinds of mechanisms convert this force one F_1 , associated with speed one V_1 , to another coupling that is force two F_2 with speed two V_2 . The power is the same at the beginning and after the motion converter → $F_1 \cdot V_1 = F_2 \cdot V_2$. Then, the power in input is a certain number provided by the product of F_1 and V_1 and you can amplify or reduce the force F_2 , increasing or decreasing V_2 and vice versa. **The product of these 2 is always the same.**

These 3 elements can be combined together to have all kind of possible actuators:

- ❖ Only *transducer* = actuator is called **effector**.
- ❖ *Transducer* + *embedded energy source* = **autonomous effector**
- ❖ *Transducer* + *motion converter* = **motor**
- ❖ *Transducer* + *motion converter* + *embedded energy source* = **autonomous motor**



You can have only these 4 classes because transducer is mandatory, you cannot have an actuator formed just by energy source and motion converter.

If you want to compare actuator performances, maybe to select the most readable for your application, you cannot just compare numbers because actuator is formed by different elements. So, below there are more in detail what are the fundamental characteristic that describes an actuator performance so you will be able also to decompose the actuation system that you are looking for or analyse it. We should be familiar with these definitions. It's important to fix them in our mind.

FUNDAMENTAL CHARACTERISTICS of actuator performance:

1. **STRESS** = typical force exerted on area unit and it's measured in Pascal. $\sigma = \frac{F}{A} \left[\frac{N}{m^2} \right]$
2. **STRAIN** = deformation but normalize for initial length. It's final length minus initial length, divided for initial length (dimensionless). When we have large deformation, not the case of actuators, is sometimes use also the *stretch* that is the ration between final and initial length.
 $\varepsilon = \frac{l_f - l_o}{l_o} \left[\frac{m}{m} \right]$
3. **SPECIFIC WORK** = generated for mass unit of actuator. It's work generated that is divided by density and it's measured in Joule per unit mass. $W_s = \frac{\sigma \varepsilon}{\rho} \left[\frac{Nm m^3}{m^2 m kg} = \frac{J}{kg} \right]$
4. **SPECIFIC POWER** = same as before but we have power and not energy. This is the power per mass unit of the actuator. $P_s = \frac{P}{m} \left[\frac{W}{kg} \right]$
5. **FREQUENCY** = this is the number of cycles that actuators is able to perform in time unit. It's measured in Hertz. $f = \frac{n^o cycles}{s} [Hz]$
6. **POWER DENSITY** = it's different from specific power, that is per mass unit. Power density is related to the volume; it's the ratio between the effective power and the actuator volume. $P_d = \frac{1}{2} \sigma \varepsilon f \left[\frac{Nm}{m^2 m s} = \frac{W}{m^3} \right]$
7. **TRASDUCER EFFICIENCY** = it's the ratio between mechanical work (mechanical output) and energy (energy input) $\eta_t = \frac{W_o}{E_i} \left[\frac{J}{J} \right]$
8. **MOTION CONVERTER EFFICIENCY** = In this case we are consider the motion converter and, in this case both the input and the output are mechanical. Then it's the ration between mechanical energy in output and mechanical energy in input. You can have the motion converter with the previous coupling (F1V1=F2V2) but it's not for free and in between there is always an energy dissipation. For this reason, motion converter efficiency should always be considered. $\eta_{mc} = \frac{E_o}{E_i} \left[\frac{J}{J} \right]$

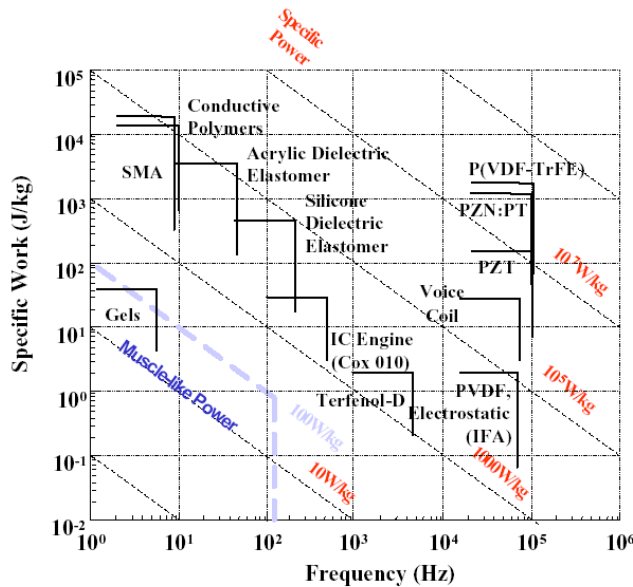
9. **BULK RATIO** = This provides the idea of how much the motion converter impacts the dimension of the actuator because it's the ration between the volume of the motion converter and the transducer. $BR = \frac{V_{mc}}{V_t} \left[\frac{m^3}{m^3} \right]$

So, if you are able to apply these characteristics properly you are able to discriminate and compare actuation technologies, but the problem is that it's not easy to identify all these characteristics on the

same place and to have a direct comparison of technologies. There are some tools and instruments, like graphical representation to do it. In the following example, there is a frequency- specific work graph and in “the third dimension” there is the specific power.

PZT is piezoelectric technologies.

The point is that you can directly compare 2/3 parameters at the time, we do not have something to compare all the characteristics together and, also in the example, are reported covering areas.

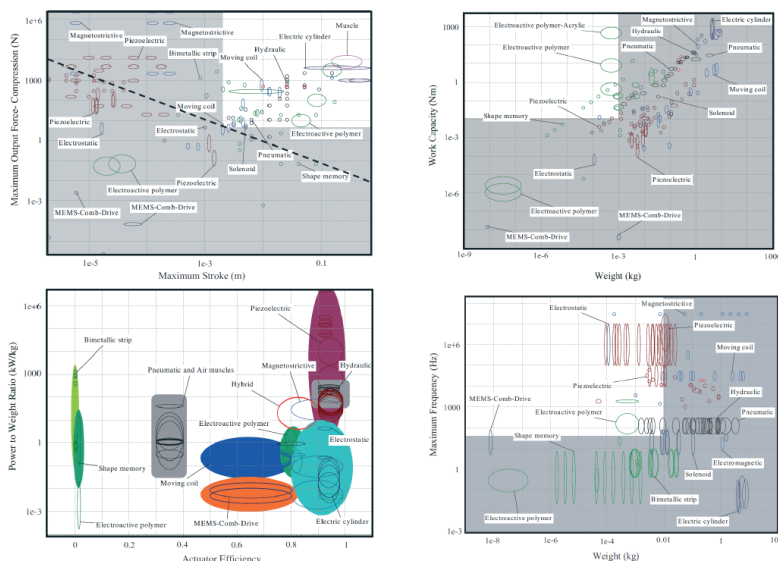


Source: DARPA and SRI International

ASHBY DIAGRAMS:

Towards these coverage areas there are Ashby diagrams that have basically the same aim to represent parameters for an easy and direct comparison. In the first one there are the maximum stroke and the maximum output force compression, while in the second one there is the comparison of weight and **work capacity**.

Ashby diagrams



The work capacity is the product between the force and the displacement, the strokes that are able to generate. The third one compares the efficiency and the power to weight ratio and the last one represents the weight and the maximum frequency. Why are we using these diagrams? Because we should be able to identify what could be the technologies that may be useful for a specific application.

TABLES:

Other possibilities are **tables**; on slide 8 there are some technologies with typical values. Tables can be really different, and you can find the best option for you. Now is reported specific examples:

1. ONLY EFFECTORS:

In this table there are different elements. Remember the comparison should be guided by the specific requirement you have. You cannot compare parameters randomly. In this case if you are looking for the best technologies for just and **effector**. The best way is to use **only power density**.

	Stress (MPa)	Strain	Efficiency	Bandwidth (Hz)	Power (W/cm ³)
Electromagnetic	0.02	0.5	90%	20	0.1
Hydraulic	20	0.5	80%	4	20
Pneumatic	0.7	0.5	90%	20	3.5
Muscle	0.35	0.2	30%	10	0.35
Shape memory	200	0.1	3%	3	30
Electrostrictive	50	0.002	50%	5000	250
Piezoelectric	35	0.002	50%	5000	175
Magnetostrictive	35	0.002	80%	2000	70
Contractile polymer	0.3	0.5	30%	10	0.75
*Single Crystal PZN:PT	300	0.017	90%	5800	15000

In this case you have unlimited energy source, because you considering it outside. In this example also efficiency is not considered. You control only power density

$$P_d = \frac{1}{2} \sigma \epsilon f \left[\frac{Nm}{m^2 m s} = \frac{W}{m^3} \right]$$

In this case there is no dispute, piezoelectric is the

best one, followed by shape memory alloys.

2. AUTONOMOUS EFFECTOR

If you start to consider energy source you start to consider autonomous effectors. In this case power density is not enough. What the power density of the effector you have to consider also the **power density of the battery** and the **transduction efficiency**.

$$(\eta_t = \frac{W_o}{E_i} \left[\frac{J}{J} \right])$$

Autonomous effector

	Stress (MPa)	Strain	Efficiency	Bandwidth (Hz)	Power (W/cm ³)
Electromagnetic	0.02	0.5	90%	20	0.08
Hydraulic	20	0.5	80%	4	16
Pneumatic	0.7	0.5	90%	20	3.15
Muscle	0.35	0.2	30%	10	0.1
Shape memory	200	0.1	3%	3	0.01
Electrostrictive	50	0.002	50%	5000	125
Piezoelectric	35	0.002	50%	5000	0.2
Magnetostrictive	35	0.002	80%	2000	56
Contractile polymer	0.3	0.5	30%	10	0.23
*Single Crystal PZN:PT	300	0.017	90%	5800	13500

in this case, at the end, the main parameter is the Power density of the autonomous effector with the value provided by the formula below.

Piezoelectric materials still in the top of the ranking but shape memory alloys pass to be the third, because in this case you must consider other 2 elements.

$$P_{d,auto,effector} = \frac{P_{d,effector}}{1 - \frac{P_{d,effector}}{\eta_t * P_{d,battery}}}$$

3. MOTOR

Then going to the case of motor. In this case we should consider the power density, while the energy source should be not considered because you have infinite energy. The transducer efficiency is not an issue, but we need to consider how bad is the motion converter. Then

bulk ratio and motion converter efficiency should be considered $\eta_{mc} = \frac{E_o}{E_i} \left[\frac{J}{J} \right]$

	Motion Converter Bulk Ratio	Motion Converter efficiency (%)	Motor Power (W/cm ³)
Electromagnetic	1	80	0.04
Shape Memory	10	50	1.4
Piezoelectric	100	30	0.52

$$P_{d,motor} = \frac{P_{d,effector}}{1 + BR} \eta_{mc}$$

Now in the tables we have is power density comparing motors, the bulk ratio and the motion converter efficiency. In this case electricity usually required very big motion converters, so this makes the overall ranking very much different rather than the previous one. Now the best is shape memory alloys.

4. AUTONOMOUS MOTOR

In this case you have to consider all the previous elements:

- Have limited energy
- Transduction efficiency
- Battery power density
- Bulk ration
- Motion converter efficiency

Now after all of these you can see the result. What makes the differences is the transduction efficiency of shape memory alloys that is dramatically low.

$$P_{d,aut.motor} = \frac{P_{d,effector}}{BR + 1 + \frac{P_{d,effector}}{\eta_t * P_{d,battery}}} \eta_{mc}$$

	Motion Converter Bulk Ratio	Motion Converter efficiency (%)	Auton. Motor Power (W/cm ³)
Electromagnetic 90%	1	80	0.035
Shape Memory 3%	10	50	0.006
Piezoelectric 50%	100	30	0.054

in this case electromagnetic and piezoelectric are in the same order of magnitude and they can be considered for applications. (Question from the class: “why are we

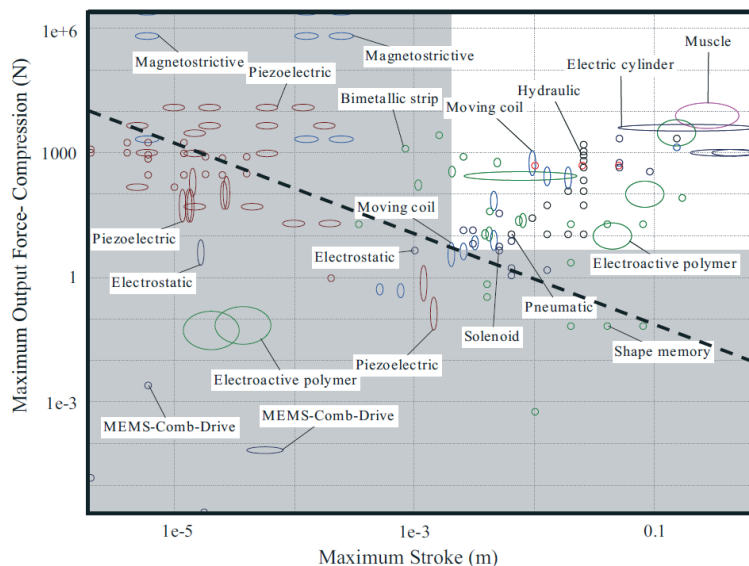
not talking about other examples on the tables?” answer: you can basically move forward also with all of them; the teacher just choose to compare these.)

EXAMPLE WITH SPECIF REQUIREMENTS:

Now we go in a more specific case. We want to create an actuator with these specific characteristics. The force that the actuator can generate is 5N, the stroke, that is basically the displacement, should be at least millimetres, while the operational frequency is 50 Hz. Life as much as possible but not less than how is indicated and also the mass should be light as possible.

Now we start, for example, using Ashby diagrams.

- Let's start with **force and stroke**:



This is a coupling that sets the **work capacity** of an actuator (because the product of these 2 provides the work that an actuator must be able to generate. **WORK CAPACITY = FORCE X STROKE**. In this case $5 \times 0.002\text{m} = 0.01\text{Nm}$).

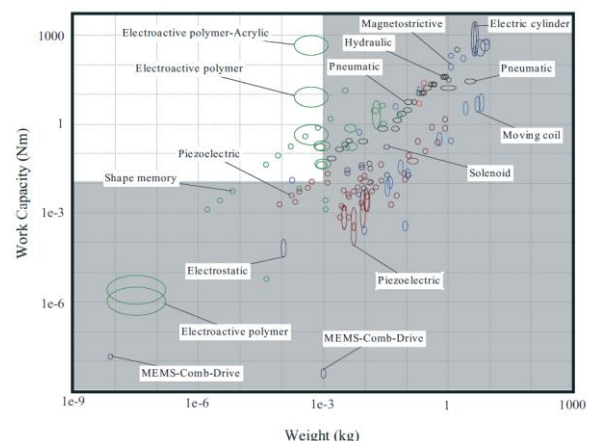
So according to this, you are not forced to restrict the range of technologies with forces with a value over 5 N, but your actuators can be able to produce a force minor of 5 N but with a stroke higher enough to obtain the same work capacity.

The product of 2 has to be higher than, in this specific case 0.01. If you have an actuator that is able to produce a force lower than 5 N, but the work capacity is higher than the targeted level, you are able to add a motion convert, to have an actuator able to produce 5 N and enough stroke to match the value.

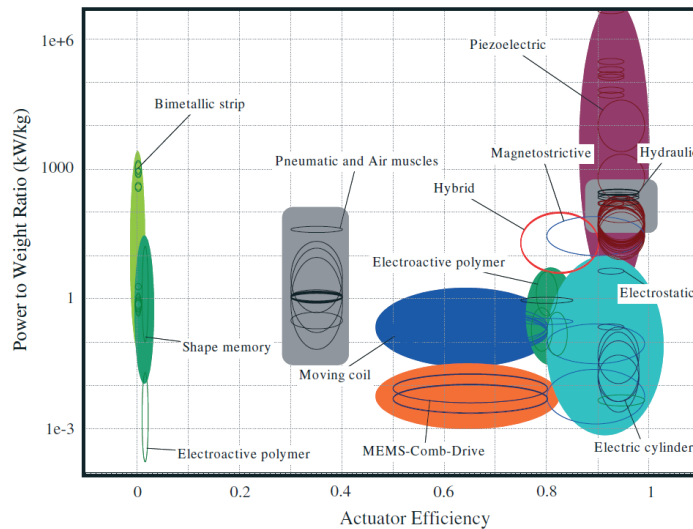
And this is why in the Ashby diagram these values are not restricted to a point but there is actually a **line**, because this line is setting the product that is acceptable for the specific application. Everything above the line is acceptable. From this grey area that is acceptable (you can use motion converter for example), you can restrict the system into a square in which force is more than 5 N and the stroke is more than 2 mm (the white square).

- Let's considering also **the mass**:

Now we have another diagram with weight and work capacity. The work capacity has to be more than 0.01 Nm then only value above this is



acceptable. There isn't a specific number for the weight so you should stay much as possible on the left part of the diagram. The white square is not numerically written but should stay as much as possible on the upper-left side. Now you should start to consider and adding

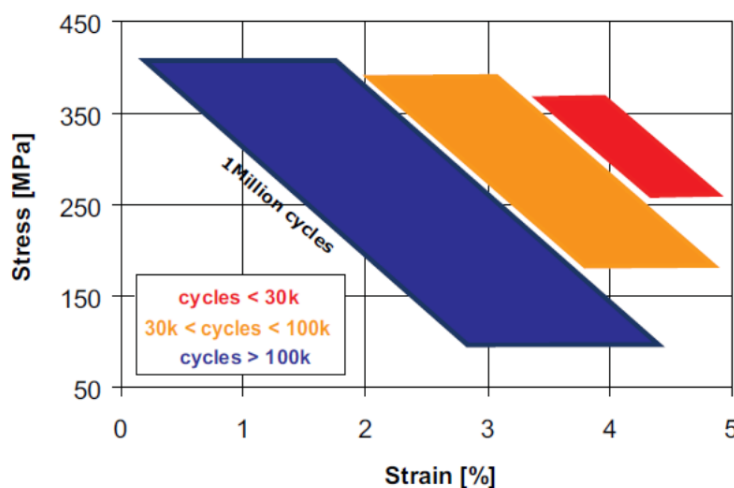


some parameters that are not constrained by requirements like actuator efficiency. By identify technologies that are on the white square and have highest actuator efficiency you can start to narrow down and to a kind of “**trade-off**”. And the same should be done for the frequency. In this case the trade-off is done with the frequency because there is a numerical constrain that is 50 Hz. You cannot move lower than 50 Hz but on the Ashby diagrams

you should stay as much as possible on the left part because for the weight there isn't a specific number.

➤ **For the life:**

For the life usually there are some tables reporting the data sheet of the technologies and some tables report directly the life. Life usually depends a lot on working conditions. On the slide 17 is possible to see an example in which a number of cycle range is set and can be considered valid for this technology. But, other times, especially for innovative technologies you can have information of life expressed in



a different way. For example, shape memory alloys have a number of cycles that depends on the strain and the stress that alloys generate. And this is kind of graph. *On the blue area* there are low stress whit high strain and low strain with high stress. In this case life can be higher than *100 000 cycles*. *On the orange area* there is relative high stress and strain the life is in *intermediate range*. *The red area* is related to a life cycles less than *30000 cycles*.

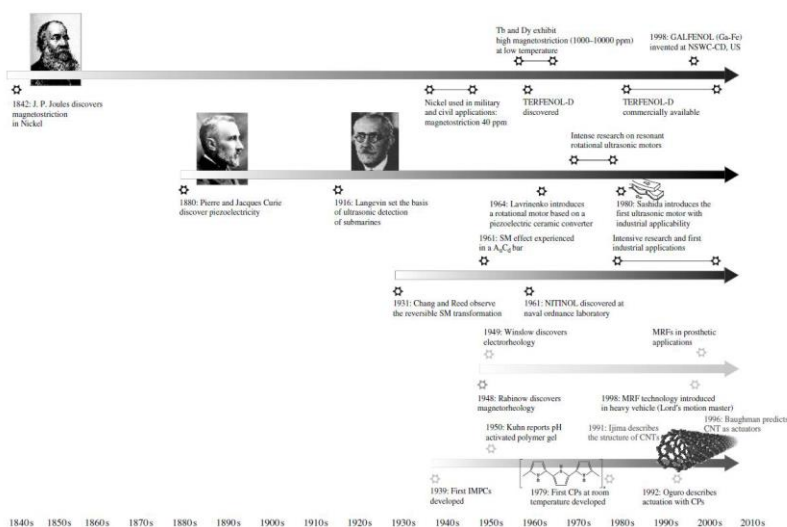
(question from the class of shape memory alloys, answer of the prof: ‘we’ll do 5 hours on shape memory alloys, in general the higher is deformation the higher is the force)

How to have new actuator technologies:

In order to have an actuator and new actuator technologies there is a sort of procedure. There are some steps that must be consolidated before having something that can be used proficiently.

1. **To Master the New Phenomenon** = at the beginning you have to discover the new phenomena that can be associated with a transduction principle to be able to translate energy into mechanical power. Then you must understand it, clarify what are the physical law behind and extract ***the constitutive equations*** that fully describes the phenomenon.
2. **To Master the Material Aspects** = once you understand, you ***have to optimize to enhance the transduction principles through analysing the material aspects***. For example, again shape memory alloys are alloys, so they are made by different kinds of materials. The first time they observed the phenomenon and found the equations, but then they understood that changing the composition of the alloy they were able to enhance transduction of the actuation. (trova la composizione migliore)
3. **To Develop Appropriate Actuation Concepts** = to develop appropriate actuation system. This last step is a little bit more difficult to grasp intuitively and we'll see a very clear example while treating piezoelectric materials. Piezoelectric materials are basically ceramics that can deform thank to application of an electric field. Their deformation is very small. It can be performed also at high frequency but it's really small. Usually, you need a motion converter that is very powerful, so it can transform this high force into a lower force and can increase the deformation. This is the direct exploitation of piezoelectric materials, but more recently some guys came up with the idea to apply or to exploit some resonance effect of piezoelectric materials to use it in ultrasound motors. ***They do not exploit the basic principle but more elaborated actuation concepts***, they use resonance to use ultrasound motors.

Then, once you have respected the points you really have something that can be used. Some



examples are reported on the image. They show how much time it took for some technologies from the discovery to the commercial exploitation. Of course, is not the same for all of them. Piezoelectricity has been discovered in 1880, but ultrasonic detection was 36 years later, and the first use in commercial applications, except for military use, was in 1980.

Magneto technology took a little less. IMPCs (ionic metal-polymer composites), electroactive polymers, are still taking time because they don't have commercial exploitation.

On slide 21 are reported the actuation technologies that the professor would like to show during this course, obviously this depends on the time. For sure we'll cover the first 4 of active actuators, the last 2 are uncertain. For semi-active, for sure the first 3.

Active = active technologies are technologies that can exchange work both positive and negative. So, if you have active system, the system can dissipate or increase the level of energy of the system in which in embedded

- Piezoelectric Materials
- Shape Memory Materials
- Electro-Active Polymers
- Flexible Fluidic Actuators
- Magneto-strictive Actuators
- Supercoiled Polymer Actuators

Semi-active = technologies in which the work exchange is only negative or zero. The actuator inside the system can only dissipate the energy of the system. It's still active but have to be activated to dissipate energy during mechanical interactions.

- Electro Rheological Fluids
- Magneto Rheological Fluids
- Jamming transition phenomenon
- Low melting-point materials



What kind of actuators you need, depends on several things but now we have fundamental characteristics that we can spot, identify and compare. It depends on requirement you have.

“IN EVALUATING THE BEST ACTUATION TECHNOLOGY FOR YOUR APPLICATION, YOU SHOULD CONSIDER ALL THE STEPS NEEDED FOR THE ADAPTATION OF THE ENERGY FROM THE SOURCE TO THE FINAL USE”

In the second part of lesson, we'll do traditional technologies as well, not considering innovative technologies as the only possible application to be used in our career. The course is on innovative technologies, but there is no worse way to apply this knowledge than trying to replicate something that traditional technologies are already doing well. So, the suggestion is: “using innovative technologies and innovative concepts only where you see traditional technologies are not good”



Actuation Technologies: Traditional technologies

This summarizes most of the traditional actuation technologies and their main features

Classification

Actuator			Feature
<i>Electromechanical</i>			
DC motor	Wound field	Separately excited	Speed can be controlled either by the voltage across the armature winding or by varying the field current
		Shunt series	Constant-speed application High starting torque, high acceleration torque, high speeds with light load
		Compound	Low starting torque, good speed regulation Instability at heavy loads
	Permanent magnet	Conventional PM motor	High efficiency, high peak power, and fast response
		Moving-coil PM motor	Higher efficiency and lower inductance than conventional DC motor
		Torque motor	Designed to run for long periods in stalled or a low rpm condition
	Electronic commutation (brushless motor)		Fast response High efficiency, often exceeding 75% Long life, high reliability, no maintenance needed Low radio frequency interference and noise production
AC motor		AC induction motor	The most commonly used motor in industry Simple, rugged, and inexpensive
		AC synchronous motor	Rotor rotates at synchronous speed Very high efficiency over a wide range of speeds and loads Need additional system to start
		Universal motor	Can operate in DC or AC Very high horsepower per pound ratio Relatively short operating life
Stepper motor		Hybrid	Change electrical pulses into mechanical movement Provide accurate positioning without feedback
		Variable reluctance	Low maintenance

Actuator		Feature
<i>Electromagnetic</i>		
Solenoid type devices Electromagnets, relay		Large force, short duration On/off control
<i>Hydraulic and Pneumatic</i>		
Cylinder		Suitable for linear movement
Hydraulic motor	Gear type	Wide speed range
	Vane type	High horsepower output
Air motor	Rotary type	No electric shock hazard
	Reciprocating	Low maintenance
Valves	Directional control valves	

There are **electromechanical group** (formed by DC motors, stepper motors, AC motors), **electromagnetics principles** (solenoid type devices or electromagnets) and **hydraulic and pneumatic technologies** (cylinder hydraulic motors and pumps).

Electromechanical actuators:

These are by far, probably, the most used. This is a case where you have a motor (transducer + motion converter) that converts electrical energy into mechanical motion. They convert energy from **electric domain → to mechanical domain**.

(DC motors, stepper motors and AC motors)

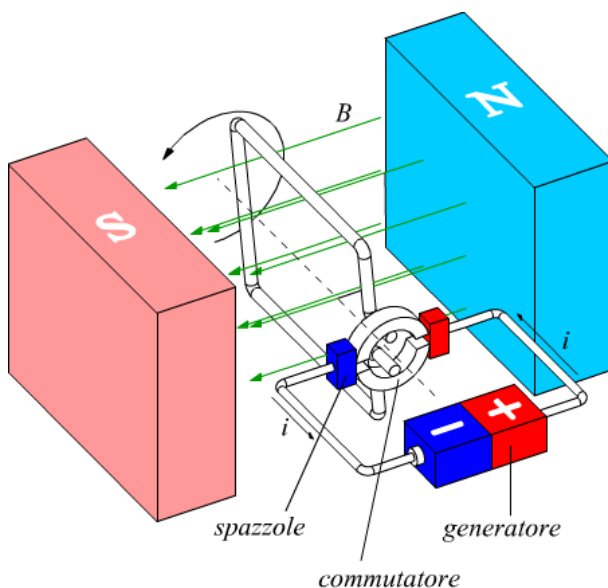
1. DC motors

History:

DC motor has been invented by accident in 1873, because the first that has been invented is the dynamo, that is the reverse principle (idea of store electric energy through mechanical inputs). But the principle is reversible and when Zènobe Gramma just connected one dynamo to another saw that the first one was able to move the second one.

➤ BRUSHED

The principle itself is very simple:



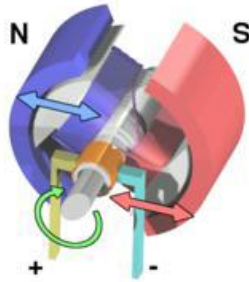
There is an energy source that provides different voltage and the **current is generated inside the coil**. **Outside** the coil there is a permanent magnet that creates a **static magnetic field**. But, when the current flows inside the coil, it generates an electric and magnetic field. The magnetic field that is generated through the current interacts with the external field and **this interaction makes the coil rotate due to align, basically, the two magnetic fields**. The magnetic field that is generated is proportional to the number of coils. In the 'real' inner part of this kind of motors there are many coils bundled together. (the magnetic field is higher in the

reality, than one single coil here).

Why is the disc not continuous (commutatore)?

There are some cuts because it's necessary to maintain the rotation of the DC motors. Why? Imagine that the battery is driving the current inside the coil. The current is generating the magnetic field, and the magnetic field rotates the coil to align the generated magnetic field to the permanent magnetic field. But, once it's aligned, it stops rotating and this is what we don't want. So, the smart solution is this. While rotating, the coil is sliding on **brushes**, that are **where the voltage is brought from the battery**. The 2 cuts, identified before, commutate the direction of the current. When the coil is aligned with the external magnetic field, the cuts on the ring and the brush, so the connector, allow the current to jump from one side of the ring to the other one. So, it's reversing the current that is running inside the coil. **And this automatically maintain the rotation of the coil.**

When the coil is aligning the 2 magnetic fields, the ring stops to move, and the cut are over the brushes. For inertia the ring rotates a little bit, and the brushes touch the part after the cut. Then the part after the cut of the ring touches the brushes. The positive brush, after the aligning, touch the second half of the ring, and the negative brush enters in contact with the part of the ring that previous was connected with the positive part. So, the current enters on the opposite side of the coil. And this is done forever until there is a battery that is powering the ring.



This is the case for just single coil; in the real cases there are a lot of coils.

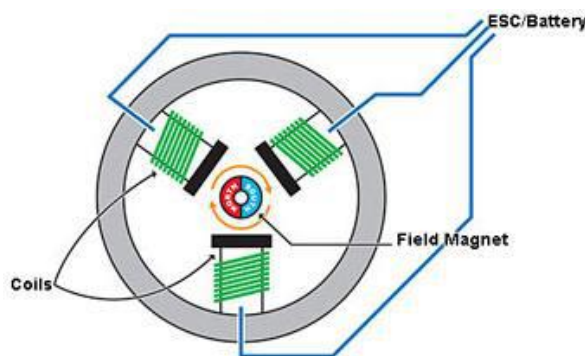
Coils are arranged on a **shaft**. On the image is possible to see the yellow and light blue elements; these are the brushes that are sliding on the orange **ring** that is called **commutator**. The **permanent magnet** on the side is called **stator**, because it does not move. There is the shaft inside, with the coils in which current is running and generating the electromagnetic field. Then, there is the **rotor** that is rotating inside. The battery is connected with the commutator through these branches, just a reminder. Brushes are elements that are sliding continuously on metal structures, metal brushes are sliding on the commutator.

➤ BRUSHLESS

Problems of brushed motors:

Also in Dc motors, there is another version that is called brushless. **Brush** are elements that slide on the motors and bring current into coil, but they have **many limitations**, **they are quite low cost so are used a lot, but limitations are present**. For example, sliding can bring to some damage because it's based on contact pressure. And, also for this reason the maximum rotation is also limited. The rotor, the part that is rotating, is the more subject to inertia and other dynamics. It's extremely heavy because it's lodging the weeding of the coil. Usually in brush-DC motors the rotor is the heavy part. The generation of electromagnetism generates a lot of heat, and it's not really easy to dissipate because it's in the inner part of the actuator.

Brushless structure and principles:



To overcome these limitations the brushless version was created, this is a smart version. Now **in the rotor** there is the **permanent magnets** and **externally there is the electromagnets**. And this resolves the problem of heat dissipation, because the field is external, and also resolves the weight of the rotor because in much lighter. There is not brushes anymore. What is the complex? You need some

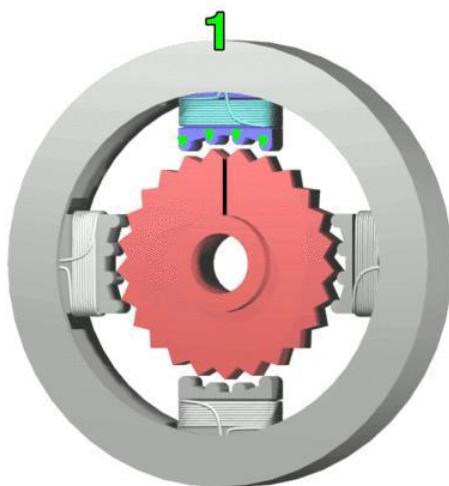
control in the interaction between the generated electromagnetic fields and the permanent magnet. In the previous case the synchronization of the commutation is for

free, because it's directly led by the commutator that is physically cut. In this case there is no control on the generation of the magnetic field that is maintaining the rotation controlling the interaction between the magnetic field of the magnetic magnet and the electro-magnetic field. In this case the 3 electro-magnets should be driven in a way that the permanent magnet is maintaining the rotation by rotating and also the activation of electromagnets. So first you activate one coil, then when the angle is around 60 degrees it starts to decrease and the second coil to increase and also the third coil, while the magnet continues to rotate. And this should be controlled, indeed in this case we need an encoder to set and understand what the rotation value of the paramagnet magnet is. And, on a closed loop, you must also control the activation of the three external paramagnets.

Respect to AC motors both the versions are much simpler, but they required more maintenance compared to AC motors.

2. Steppers Motors

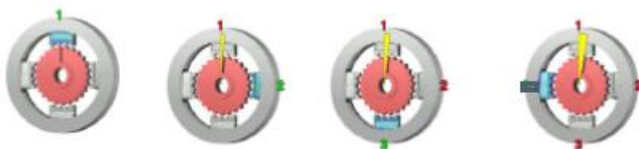
Still in the group of electromechanical actuators, there are steppers motors. This one is one



of the simplest (on image). This one is based on 4 electromagnets, the number can change, but in this case are 4 and they are arranged by angles of 90 degrees. So, in this case the stator is the grey structure, and the rotor is the red structure. The rotation of the rotors is induced by the 4 electromagnets by activating the electromagnetic field and relying on the **PRINCIPLES OF MINIMUM RELUCTANCE**. The red part is paramagnetic, but it's not activated, it's not a permanent magnet. But when you have one of those electromagnets that is activated, the teeth of the wheel are oriented to minimize the gap between

the four teeth of the electromagnet and the teeth of the rotor. The orientation of geometrical features makes sure that every time I activate one of the electromagnets, I can obtain a very small amount of rotation that minimize the reluctance for the one that is activated but increases the reluctance on the others that are not powered.

Step- by-step procedure:



the first electromagnet is activated → there is a certain degree of rotation.

The first one is de-activated and activated the second → there is another degree of rotation.

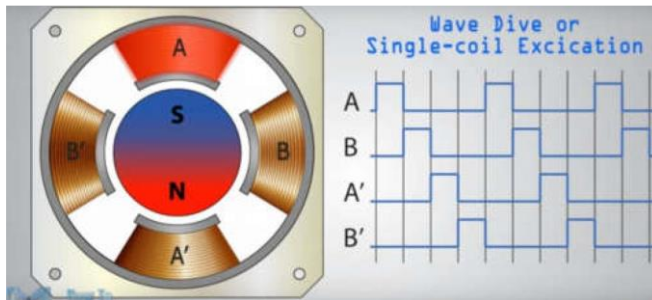
Again, the second is de-activated and the third is activated → another degree of rotation

And the same is for the last one electromagnet. And then, the motor returns to the first situation, re-activating the first and re-starting the process.

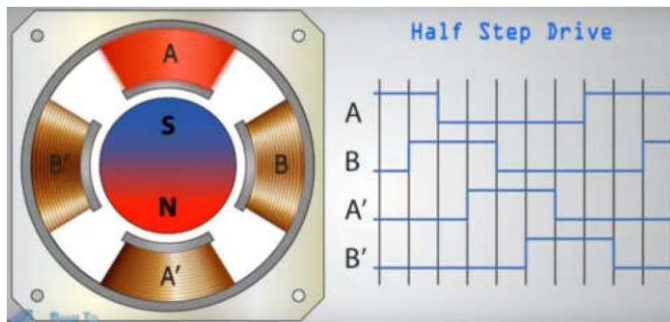
So, in serious activation of the four electromagnets, makes the rotation of one **single tip**. This is really simple and it's called step – by – step procedure.

But these steps can be obtained by others means:

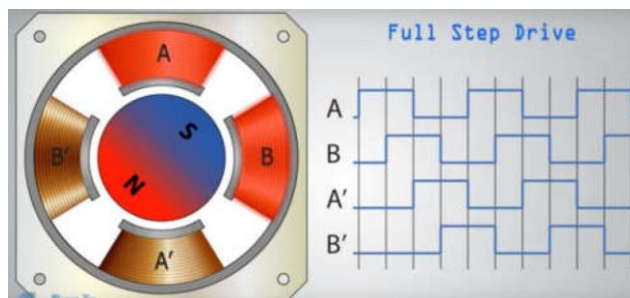
- WAVE DIVE or CO-EXITATION = there are electromagnets and a **permanent magnet**. The activation of the electromagnets is provided by square waves. The four electromagnets are **activated one by one, without any kind of overlapping**.



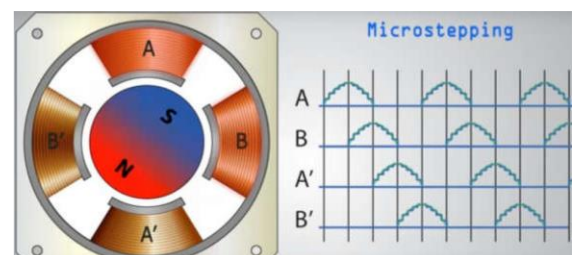
- HALF STEP DRIVE = in this case there is a pair. So, 2 electromagnets are activated at the same time, and the rotation is even more controlled step-by-step.



- FULL STEP DRIVE = in this case this activating of the electromagnets still continuously but *without having any step where only one electromagnet is still activated*.



- MICROSTEPPING = in this case all the electromagnets can be also modulated in amplitude, modifying the value of magnetic field.



3. AC Motors:

The main difference between the previous one is that in the electromagnets the current is alternating and also the magnetic field generated would be variable. (in the first case, when the current was constant, also the magnetic field was constant and there was the problem of orientation because once they were aligned, there wasn't motion anymore)

The alternative current is a feature that we can exploit to have more complex activation concepts.

- SYNCHRONOUS MOTORS:



This is the simplest case. The adjective synchronous says everything. In this case there are:

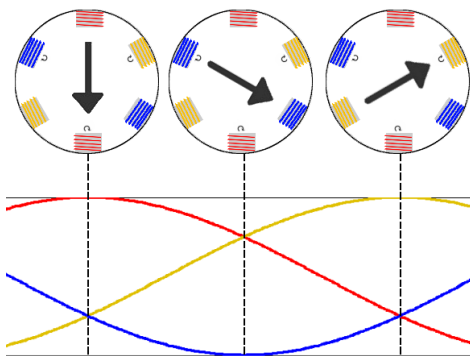
3 electromagnets on the stator.

2 electromagnets on the rotor.

Just by having the coordination of generated magnetic fields, **the inner magnetic field** is generated, and it maintains **always the rotation by the rotation of the external electrically induced magnetic fields**. The external electromagnets produce varying magnetic fields, or magnetic fields with various amplitudes. The same is done internally and the interaction between the 2 magnetic fields maintains the rotation. The important thing is that it's synchronous → the velocity of rotation of the magnetic field (magnetic field produced by alternation of current flow in the 3 electromagnets) is the same of the rotation of the rotor. The magnetic field is rotating at the same speed of the rotor, the phase may be a little different, but the velocity is the same.

(Question from the class: “this velocity of the magnet is the AC frequency?” answer: “yeah is related to AC frequency unless you are using some references, but this is not the case”)

- ASYNCHRONOUS MOTORS:



This is simpler but more complex to grasp. The electromagnets in the stator generate a magnetic field and this magnetic field is modulated for allow rotation. In the inner part, the rotor, there are coils. In the previous case all the coils are powered and the magnetic fields are interacting. But, in this case only the three electromagnets on the stator are powered and the inside coils on the rotator are not powered.

The internal coils, on the rotator, experience a rotating external magnetic field that rotates around them. We know that, if you have a coil immersed in an invaluable magnetic field, there is a *FEM*, and *induced electromotive force*.

The steps are these:

1. There is a rotating magnetic field created by the current flow inside the electromagnet coils located on the stator.
2. This rotating magnetic field induced a FEM on the coils located on the rotator.
3. A current is generated inside the coils on rotator by the external magnetic field.

4. The current inside the coils that are in the rotor, also generates a magnetic field due to the current flow inside the internal coils.

In the end, there are again 2 magnetic fields that are interacting each other, but in a different way. Because in this case, the rotation of magnetic field of the rotor and the rotation of the rotor, is only due to the magnetic field that is induced by the external electromagnets. **This means that they cannot rotate at the same velocity** → because if you consider the extreme case that is also impossible, when the electromagnetic field and the rotor are rotating at the same speed. In this case the internal coils will not experience variable magnetic field anymore, and that means that internally there is not current generated and also magnetic field. This means that the speed will decreasing and while decreasing, the rotor will start experience the variable magnetic field→ at the end the velocity of the rotor will be slightly less of the rotation of external magnetic field. They cannot have the same speed.

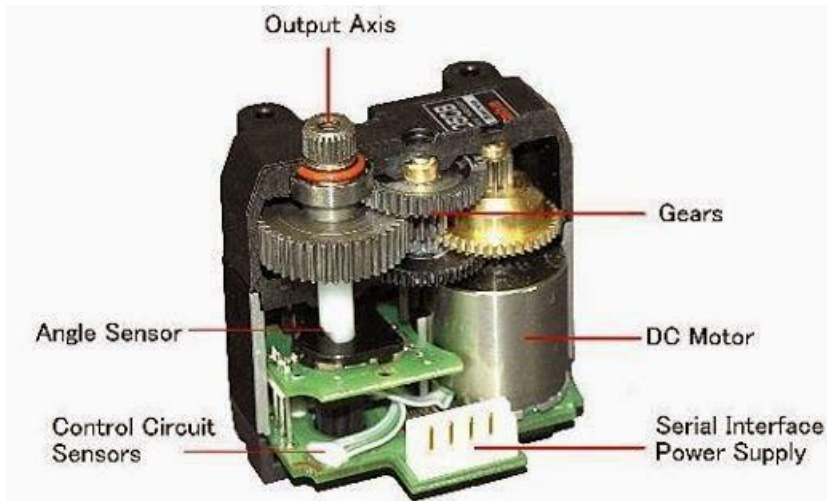
(Video on induction motors explains the same things, plus things:

The difference between synchronous and rotor speed is known as **slip**. Energy loss during motor operation is dissipated like heat)

(Question from the class: “is the speed of the rotor constantly increasing and decreasing or is it constant? as long as the rotation of the magnetic field is constant, it will be constant. If you need to modulate the velocity of the rotation of the rotor you have to change the velocity of the magnetic field in order to increase the velocity of the rotor)

SERVOMOTORS

This is not a class per se. They are extremely useful and very much used, especially at research level because these kinds of components are very simple to use and very versatile. They have a lot of components that are already installed and can easily put researchers in conditions to apply precise forces or deformations to any kind of even simple devices. Indeed, servomotors are very used at prototyping level.



You can customize this for your specific application. However, since a servomotor already includes most of the core components, you'll need for advanced designs, it is an **excellent starting point** for many projects. A servomotor indeed usually relies on **DC, AC or AC induction motors**, but it also has other components:

- Gears
- Motion converter is already inside
- Angle sensor, that is monitoring the rotation of the output axis.
- Small board/ PCB boards to control circuit sensors.
- Interface when usually there are 3 leading cables.

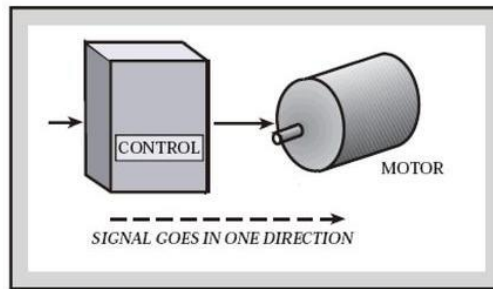
You kind find a lot of different servomotors, some of them have also more sensors inside, for example tension, especially for the torque. The simpler servomotors have only the angle sensor, needed for rotation control. Usually, they are **controlled** by the sensor **on the angle**, so they are controlled in rotation and not in velocity. **Velocity is not something that can be controlled.**

Servomotors have remarkable torque, but they are **limited in rotation**. Some versions can provide continuous rotation but, in most of cases they can provide well controlled rotation but very limited rotation like 90/80 degrees (can be a little bit more or less than 90 degrees). Others can also rotate 360 degrees, but just for one time, that means one complete turn but only one. This characteristic changes by version

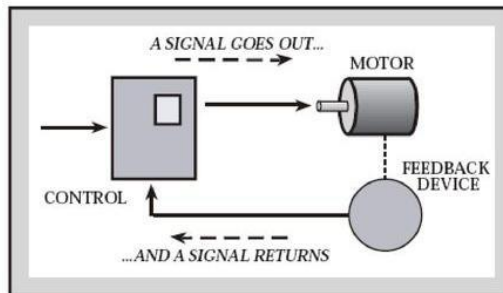
CLOSED LOOP

The peculiarity of servomotors is that it is not an open loop, like motors, it is a **closed loop**. Internally it's able to implement a **control loop** → **closed loop**.

Open-loop

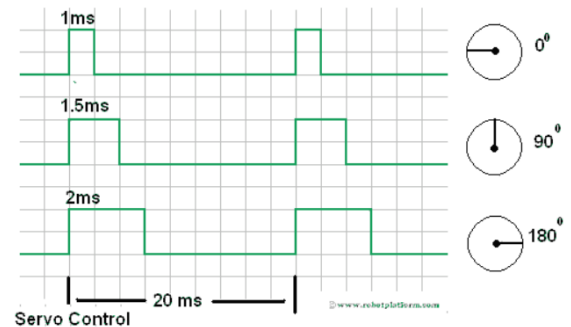


Closed-loop



So, you have the control and the signal for generate rotation. But the rotation can be motoring through the sensor that given back the value that the controller uses to set the motor rotation curve.

How does this loop work?

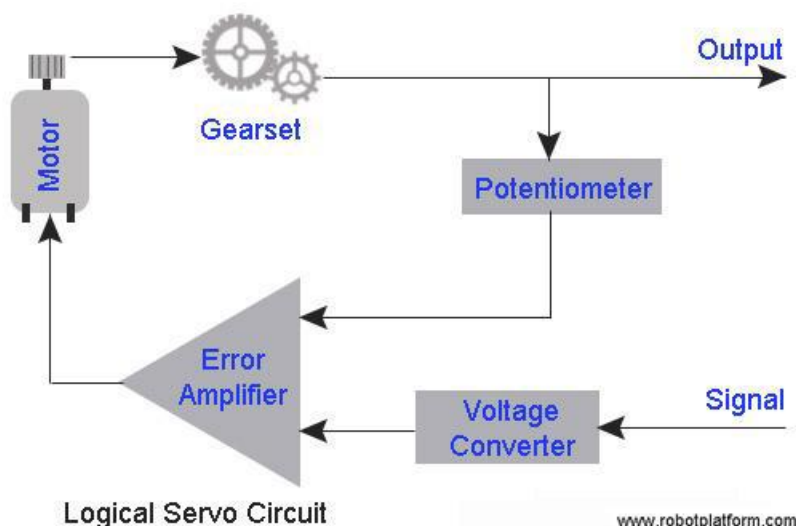


The image on the right shows the kind of signal that servomotor uses in order to control position, so the rotation of the shaft. The signal is a **square wave** called **PWM** signal (pulse width modulation) → these kinds of signals are not modulating the amplitude, so the amplitude is always the same, but the information is related to the width of the signal.

These signals-controlled servomotors. The servomotors expect to have one signal every 20 ms, so the **frequency is already set**. Depending on the width of the signal that it receives every 20 ms the rotation of the shaft can be set:

- if the PWM of the signal is 1 ms → 0 degree
- PWM is 1.5 ms → 90 degrees
- PWM is 2 ms → 180 degrees
- Everything that is in the middle will generate an angle that is corresponding to that signal.

this image shows more in detail how the close loop works:

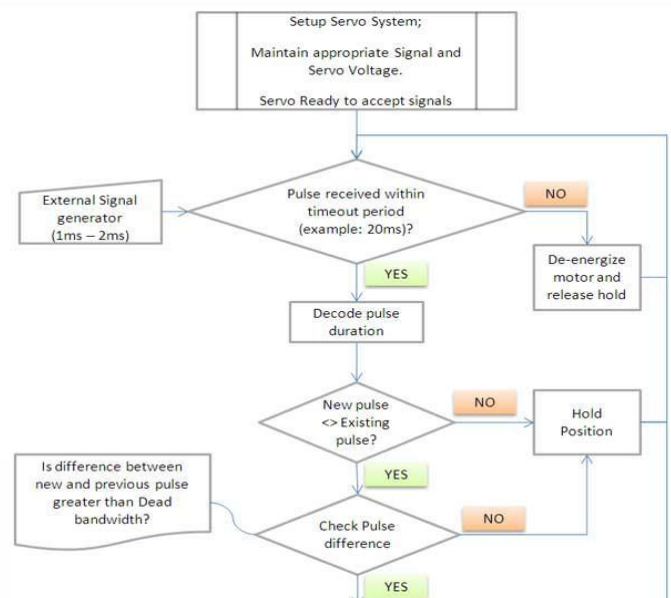


- There is a **signal (input)** that is provided by **the user or by the central processor**.
 - Then there is a **voltage converter** that converts the information in the PWM into a constant voltage value (this is made usually by capacitors that are charged).
 - There is an **error amplifier** that compares the signals: the voltage level after the voltage converter and the voltage level that comes back from the potentiometer.
 - On the base of the output of the amplifier, especially if there is an imbalance between V_p and V_s , this error or difference is amplified due to **activate the motor to rotate the shaft**.
 - There is the **motion converter and the output**.
 - But, before the output, there is **potentiometer** that reads how much is the shaft rotation and gives back in terms of voltage input to the amplifier. And this maintain the rotation until the error between the 2 signals is zero → this means that the rotation of the shaft corresponds to the signal given as an input (the error is zero).
 - The motor is not rotating anymore.
- Also, if the difference is negative is amplified and the rotation is set.
 If $V_p < V_s \rightarrow$ the rotation will be the angle
 If $V_p > V_s \rightarrow$ the rotation includes the angles

FLOWCART OF HOW SERVOMOTOR WORKS:

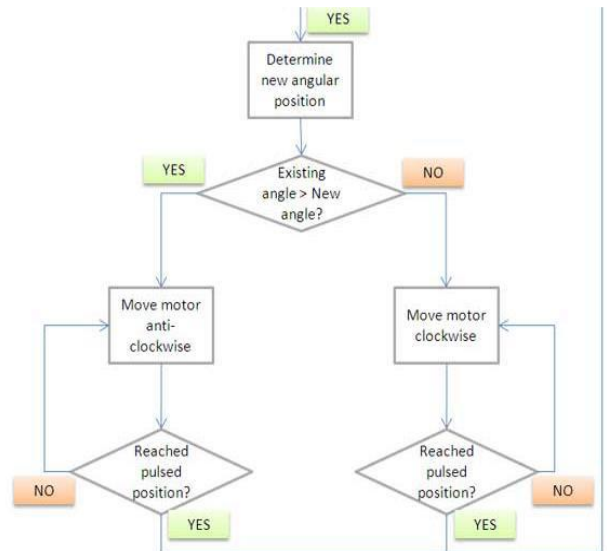
There is the servomotor ready to accept signals, energized by 5 volts. (usually, they are energized by volts of 7.3). in this way it is **ready to accept signals**.

1. First of all: “is the signal in the timeout period?” Because any kind of pulse that is received less than 20 ms, and it takes more than 20 ms is not considered to be accessible. In this case, this is not a signal to activate the servomotor. If the signal generator gives a signal that is within the timeout period of 20 ms, it is acceptable. Otherwise, the energized motor will go back to the condition to be ready to accept signals.
2. If it is in the timeout period, then you need to decode the whole pulse duration.
3. If the new pulse is different from the existing one you go down (yes), otherwise you hold the position and go back to the acceptance of signal state.
4. If the new pulse is different you go to the next step to evaluate the pulse difference.



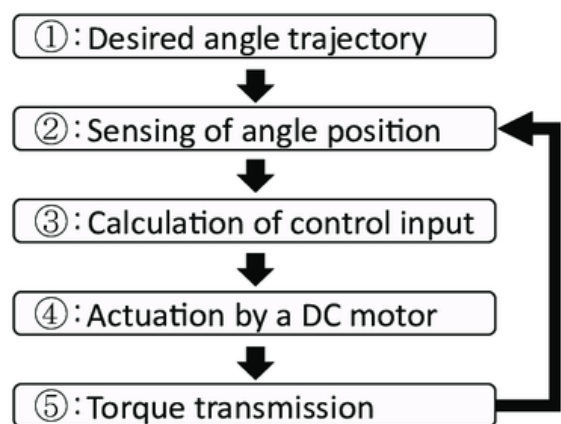
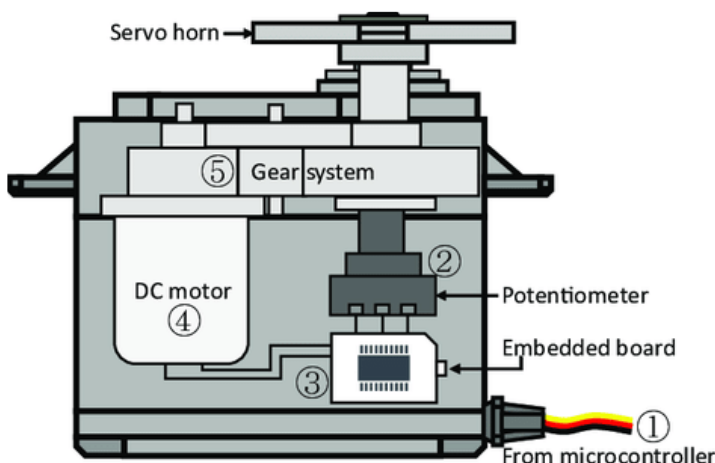
(limitation = comparing the values of voltage provide a continuous value, but if the difference is extremely small, this cannot be decoded into a rotation output. This situation is called **dead bandwidth**, when the difference is not exactly zero but there isn't any conversion into a rotation)

5. If the difference is not enough, you hold position and go back to be ready to accept the new signal.
6. If the difference is high, then you determine the outward position. So now, the servomotor has to evaluate how much the shaft should rotate.
7. If the existing angle $>$ new angle $\rightarrow$ you go to the branch that moves the shaft **anticlockwise**. And it is repeated until the difference is zero, and then you go back to the initial state.
8. If the existing angle $<$ new angle $\rightarrow$ the motor rotates clockwise on the other side until it reaches the positive position and then you can go back to initial state.



This is another image of the servomotor, from which it's possible to take information about colours. Usually there are **3 pins** that are coming from servo motors, and they are **color-coded**, and they are important if you are going to play around with cables separately and not using the standard connectors.

- Ground is usually brown or black
- Power is always red and always in the center of the connector
- The control usually is white, orange or yellow.



(Question about pins $\rightarrow$ answer: in order to make servo motor work these 3 cables are sufficient, because you have the signal, the power and the ground. For servomotors functionality these 3 elements are enough, but there are more complex servo motors that can embed additional sensors that are not used for the internal close loop, but you can use it for external loop. For example, the

motor that also the possibility to monitor the torque of the shaft. this information is acquired directly inside the servomotors, and it is made available externally. So will be another wire for that. That information can be used to modulate the control signal by the external. It's possible to have additional features inside, like the Dynamixel → these kinds of servomotors are very useful when you want additional features without adding to many hardware components in your system. In this case the servo motors are a little bit bigger)

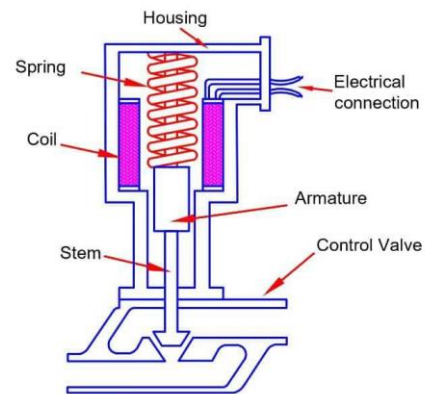
Electromagnetic actuators:

In this case we see the specific case of SOLENOID.

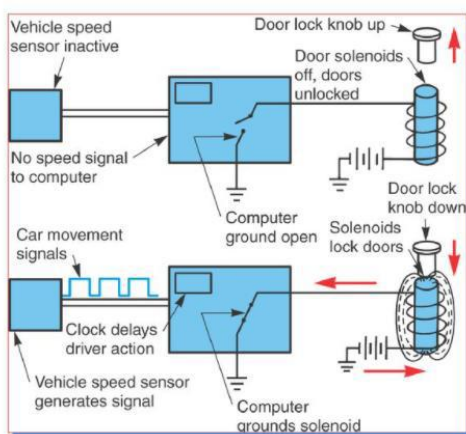
SOLENOIDS:

In this case there are coils that are generating magnetic field that induce an interaction with ferromagnetic materials that compose another part of the system. This is a sort of solenoid-valve. There is:

- **Armature** that can be pushed or pulled inside the coil. And the armature is usually pulled inside when they are energized because of minimum reluctance of material that is inside the coil. The armature is generally pulled inside, minimizing the interaction between the magnetic field generated by the coil and the armature.
- **Spring** is necessary because we obviously want the opposite direction, so the armature can return in initial position. The spring is usually a passive element. As long as the coils are energized, the armature is pulled in between. When the coils are re-energized, the spring will make the armature go back outside the coil and the valve will be closed.



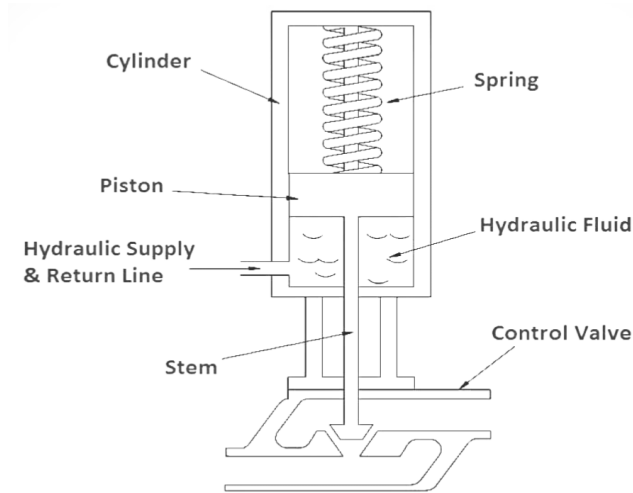
So basically, this is an electromagnet that is maximizing the interaction with the armature. This is what happens in modern automatic door locks in every car. In this case is automatically but, of



course, you can do it manually, but the idea is that there is a sensor that is monitoring the speed of vehicle. If the speed is zero or very low, the computer does not activate any kind of switch. So, the solenoid is not activated, and the loop is open and the knob is out. When the car starts moving, there is a signal that activates the switch of the computer that controls the solenoids. This closes the electric loop, generating the magnetic field that is pulling the knob down, that it closes the doors.

Hydraulic and Pneumatic actuators

Below the 2 cases of hydraulic and pneumatic actuators are reported. The teacher explains both at the same time, because the principle is the same.



the main difference is that hydraulic actuators can count on additional constraint that can be exploited → hydraulic liquids are usually **incompressible**, so they maintain a constant volume + it's also better to control quite precise how actuator is working.

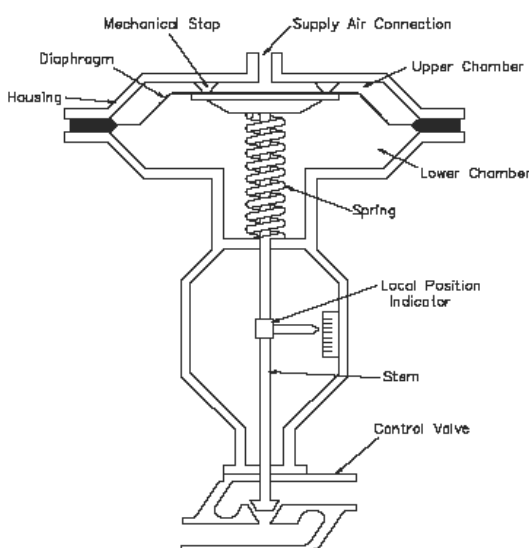
While in pneumatics, gases are used and they are compressible. This has to be considered during control.

Both of them are in the macro group of **fluidic actuators**.

Principle:

There is a cylinder, a rigid metallic structure, and a spring. The chamber is in contact with a piston that can slide inside the cylinder, then there are also a stem and a valve. The idea is: if you use and hydraulic source, with a certain pressure, that can fill the chamber, the piston can be pushed upside, and the spring is compressed and you can open the valve. If the hydraulic supply moves away from the chamber or if the fluid is recalled in the source, the spring reacts to the compression and make the valve closed again.

The exactly the same principle is applied in pneumatic actuators. They have just quite different implementation.



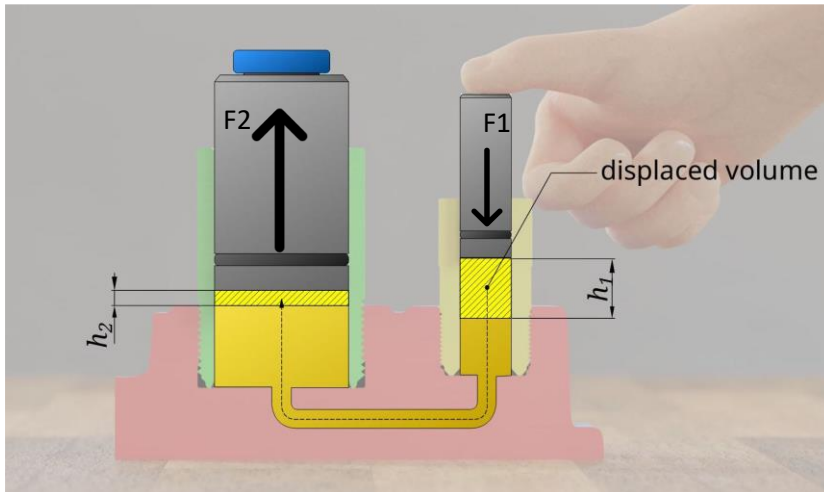
← in this image there are 2 chambers, the upper and the lower one. There is **diaphragm** that can be pushed down or can go back up according to the pressure in the upper chamber. The supply air connection can increase the supply of pressure, and the diaphragm moves up or down to compress the spring and then this is reflected on the stem and then on the valve. There is also an additional part, called the **position indicator**, that is a circle. This provides the measure of how much the stem has been moved. (there is also an additional component to visualize the direction of the air)

This is a very simple application but there are more complex implementations that use the same simple

principle.

PASCAL PRINCIPLE → if you use an incompressible liquid. The applied pressure distributes equally in all parts of the liquid.

This principle is used to make hydraulic actuators like motion converters. It's possible to exploit it by playing with geometries to have a leverage effect in the force that can be exerted in the other side. On the image there is a manually driven system there is F_2 that is the force that you want to apply to a certain object.



- F_2 is directed upward that is due to the pressure that the liquid still has multiplied by the area A_2 . But this pressure is the same that is present on the other side.

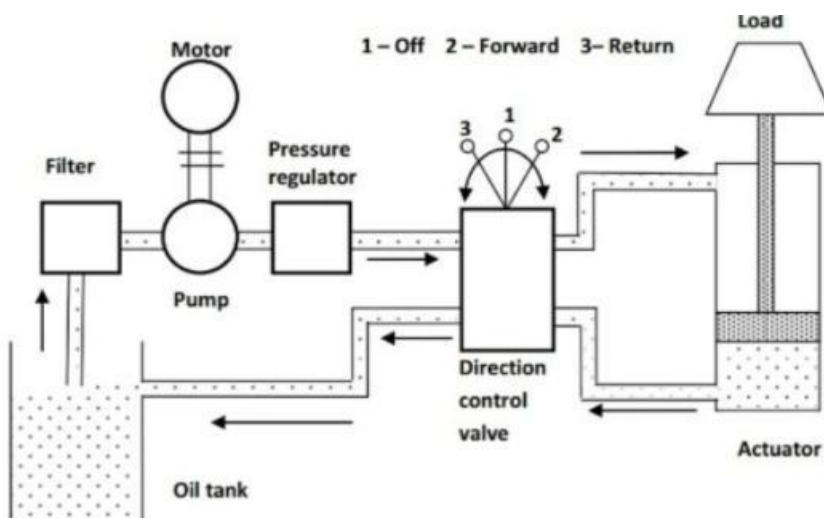
- In this case A_1 is less than A_2 so F_1 is proportional to maintain the same pressure but, since A_1 is smaller also F_1 is smaller.

- So, with A_1 and F_1 there is a pressure that is driven to the other side, when there is a higher area, and then also higher force. There is an amplification of the initial force.
- But this is related to the fact that the liquid has a constant volume, and the reduction of the area leads to an increase of the height. For this reason, the **displacement on the side of the lower force with a lower area, will be much higher** than the other side.

Formulas: $F_2 = pA_2 = F_1A_2/A_1$ and $h_2 = V/A_2 = h_1 = A_1/A_2$

HYDRAULIC SYSTEM:

Now let's see more in detail a **hydraulic system** and its components. There is a tank, containing the liquid, that usually it's oil. Then there is a pump, that leads the fluid in the pressure regulator that is basically the valve that induces a certain pressure to the rest of the system.



After that, there is a component that sends the direction of the motion of the liquid (Direction control valve). And, at the end there is a cylinder that is the actuator itself. There is a piston inside the cylinder, determining 2 chambers. One push the piston down to squeeze liquids out. This is a closed circuit, so the **pressure only sets the position of the position of the piston**, and the piston is connected with

the external load. This is the **effector** that you can connect with the system, and the hydraulic system will control how much you are moving it. If someone invert the direction of the control valve the second chamber will fill and the pistol will go up.

IMPORTANT OBSERVATION:



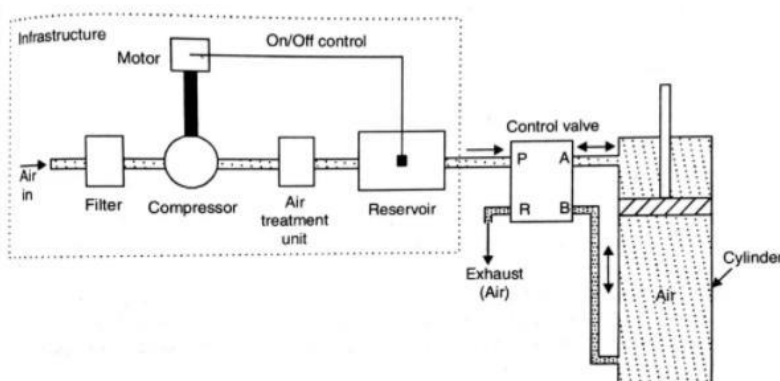
At the beginning of lesson, the teacher said that an actuator is an actuator only if it has transducer. In this system can you spot when the transducer is? The transducer is the physical component that converts the energy input into mechanical energy and it's mandatory and essential. The transducer is the **pump** (in soft robotics 90% of actuators are fluidic actuators in soft robotics and usually they confuse the term hydraulic system and hydraulic actuators but 99% rely on an external pump). Why is this important? If you are considering some characteristic of your system, you can still refer to fluid actuators without, really say that the actuator is outside. But, once you want to stress that you system is portable or more efficient, if you are not considering pump, you are not making a fair comparison!!! If you want to compare actuator technologies, you must consider where the actuators are in systems. YOU ALWAYS HAVE A SORT OF CRITICAL THINKING about how an actuator is used, but if you want to compare it's important to consider when the transducer is.

KINDS OF ACTUATORS:

- Hydraulic jack
- Hydraulic brakes
- Hydraulic ram
- High precise positioning system for heavy load

On the slide are reported 2 examples of how actuators work (Costa Concordia and Morandi Bridge) → they are example of how hydraulic actuators can be used to displace or lift very high weights up. These systems are based on 2 brakes, and a free part is elongated and this telescopic part lifts cable up. So, by a sequentially activating the brakes and then the hydraulic you can have these cable lifted up. Step by step and time by time they do a very small changes, but you can see very heavy objects move. In the case of Costa Concordia, they used 36 spring jacks that are not small components that rotated the ship. For Morandi bridges, for lift the section up, they used spring jacks. In both the cases the hydraulic actuators worked **slow**.

PNEUMATIC SYSTEM



Rather than the hydraulic system, this system has only few differences. There is again the **filter**. In this case there is a compressor and a **motor that controls also the reservoir**. There is the **air treatment** because you don't want some dust to enter inside the cylinder. Then there is the **control valve** to control the

direction of the fluid. And again, inside the cylinder there is a stand that moves inside the cylinder according to the pressure on the upper side of lower side. Here the **control is usually not done**, in the previous case the control can be done with pressure (especially control in volume, that is more precise and easier because pressure increases extremely fast in hydraulic system). Instead, in pneumatic systems, volume control is still possible through integration of the flow, but it's not precise and not recommended. They usually are controlled in pressure. The main difference is that **the cylinder is not closed** → you have air in input and air in output, **exhausted air** (obviously unless you are underwater or in space). Of course, it has some implications:

1. Filtering is important because these systems use air from outside and they have to be “clear” before entering inside the reservoir.
2. Exhausted air is released in the environment, so we have to be careful in these cases in which, instead of motor or compressor, they rely on actions like combustion or decomposition (chemical reactions) so it's important also what we use to filter it. This chemical reaction can have implications with second products.

There is a table that summarizes what is the difference between the 2 actuators:

- The main difference is related to the **fluid**.
- The **pressure** rate is very much differentiating.
- Hydraulic system is **closed**, while the pneumatic is **open**.
- Hydraulic system with **leakages** slows down, while pneumatic systems have some effects but it's not a complete failure (we'll see on flexible fluidic part)
- Hydraulic **valves** are quite difficult to control, while pneumatic ones are easier to control. Pneumatic valves are also cheaper.
- The **weight**: the limit weight for the liquid is higher in hydraulic and also the total weight of the system is higher in hydraulic case.
- **Fire hazard**: hydraulic systems are unsafe. The pneumatic systems are safe, unless you use flammable gasses.
- **Lubrication** is automatic in hydraulic systems, while for pneumatic systems you need some special arrangements.
- The **cost** is higher for hydraulic systems and cheaper for pneumatic.
- The **speed** is relative, in hydraulic the speed is quite slow, but they are very powerful. While pneumatic systems are generally considered faster in reacting but with some limitation in applied forces.
- **Noise**: it's the same for both, because the pump makes much noise. Today there are some advanced technologies that try to reduce the noise. (low-cost compressors are very loud)
- **Positioning speed**: this is high for hydraulic systems and low for pneumatic, especially for the compressibility of the gas and this implies the following point.
- **Efficiency** is higher for hydraulic and less for pneumatic.

	HYDRAULIC	PNEUMATIC
FLUID	Pressurized liquid	Compressed gas
PRESSURE RANGE	Up to 700 bar	Up to 10 bar
SYSTEM	Closed system	Open system
LEAKAGE	Slows down when occurs	Does not affect the system much
VALVES	Operations are difficult	Operations are easy
WEIGHT	Heavier	Lighter
FLUIDIC SOURCE	Pumps	Compressors
FIRE HAZARDS	Unsafe	Free
LUBRICATION	Automatic	Special arrangements needed
COST	Expensive	Cheap
SPEED	Low	High
NOISE	High	High
POSITIONING SPEED	High	Poor
EFFICIENCY	Higher	Lower



TAKE HOME MESSAGE: as we saw these technologies can count on many years of evolution, optimization and advancements. They are very versatile, robust and reliable.

Whenever an application or a target where these technologies work well, there's no point in going into something new, that takes a lot of time to reach the same level that you can reach directly. But there are some cases in which these traditional technologies cannot work or cannot reach the level of performance you need. So, for those cases there are new technologies that we are going to see in next lesson.

Shape memory alloys.

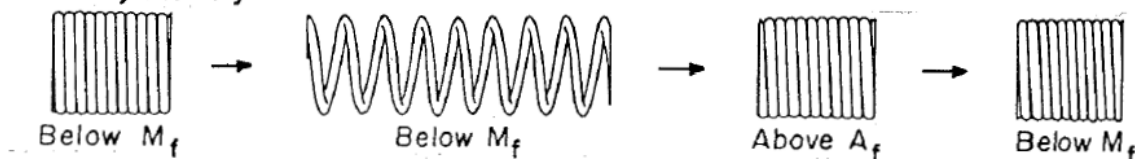
SHAPE MEMORY ALLOYS = This is an active technology. Basically, it's able to increase or decrease the energy level of a system it is integrated in. But it has also some features related to the semi-active principle but will see them at the end.

- Why is it called shape-memory alloy? Because it is based on metals, it's an alloy.
- And why shape-memory? Because it's exactly what this technology is about. This is based on the shape-memory effect that is based on the capability of this alloy to recover a deformed shape, that is induced at low temperature, by increasing the temperature of the alloy itself. (memorized shape $\rightarrow$ deformation at low temperature $\rightarrow$ increase in temperature $\rightarrow$ recovery of the memorized shape).
- At low temperature the material is deformable and plastically deformable.
- At high temperature application (through direct heat, diffusion effect of joule effect, since it's conductor) it's possible to recover the original shape.
(video: first step is the memorization process, first heating it up very much then water quenching, then deformation at room temperature and when you apply heat again you see it coming back to the original shape)

We need to distinguish 2 behaviours:

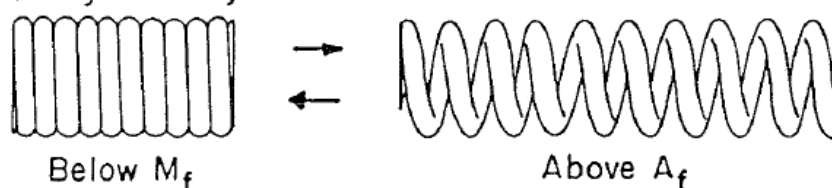
1. **ONE WAY MEMORY EFFECT** = in this case the effect of the temperature change is acting only when the temperature is increased. For example, you have a spring with a certain shape at low temperature (below M_f). At low temperature you deform it by applying an external load and you elongating the spring. And then, just applying the heat (above A_f) you have the recovery of the original shape. And then, when you return to the low temperature (M_f) you still have the same geometry. **There is no shape changing associated to the reduction of the temperature.** For this reason, it's called only one-way shape memory effect. In this case the opening part is by the external load, and the closing part is by the memory effect.

One-way memory



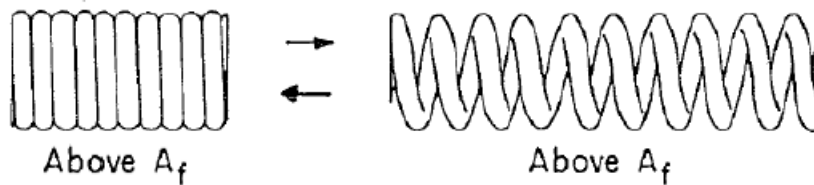
2. **TWO-WAY MEMORY EFFECT** = in this case you don't need basically the external load. The temperature itself is already enough to close the loop. So, increasing the temperature, the spring is opening and, by reducing temperature, spring is closing. In this case both the opening part and the closing part are related to temperature.

Two-way memory



One note → SHAPE MEMORY IS AN ACQUIRED PROPERTY. So, we need a specific part of the procedure, a specific thermomechanical treatment is applied to the alloy to make the memory effect appear.

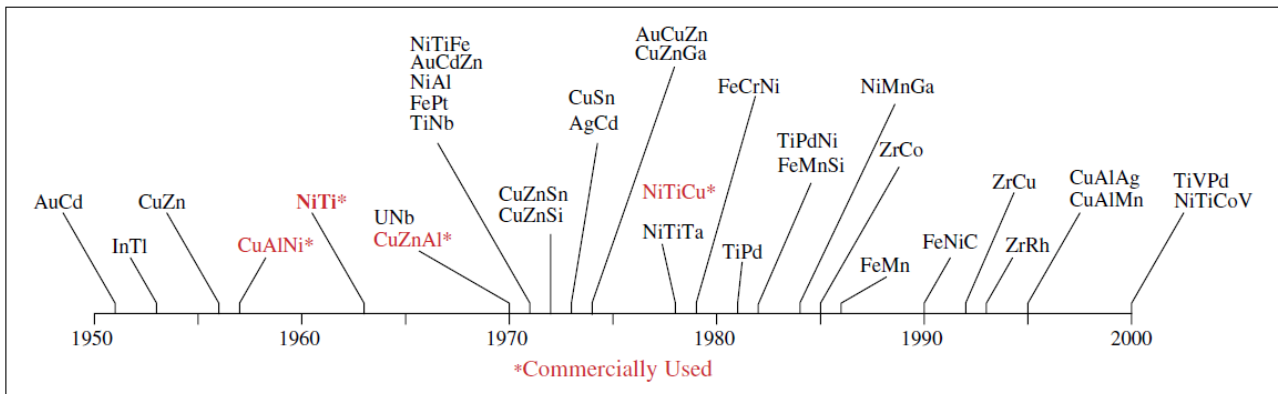
3. **SHAPE MEMORY ALLOYS WITH SE EFFECT** = these alloys have **super elastic effect**. On one side you have shape memory and the other side you have super elasticity properties. In 2 (two-way memory effect) there is temperature changing but, in this case, **temperature is always high**. (we'll see more in details what high means, now we are using relative terms, high means higher than something). If the alloy is maintained a certain temperature, the material can be subject to quite remarkable deformations and maintain elastic properties. This is a quite unique behaviour for a metal alloy. These materials, maintained at certain temperature, are able to behave in an elastic way on an extended range of deformation.



Historical notes:

- 1932 → this behaviour has been observed for the first time on an alloy based on cadmium and gold by a Swedish physicist.
- 1958 → the shape memory effect has been presented at Brussels World Fair.
- 1961 → 30 years later, a group from U.S Naval Ordnance Laboratory discovered that the same effect can be provided by a nickel-titanium alloys. This is an example of what the professor said one of previous lecture related to the fact that when you discover a transduction principle, the first phase to develop a mature technology is to master material properties and identify what are the materials that maximize the transduction effect. And this is what they did, because the gold-cadmium alloy was unable to transduce with a high efficiency, while nickel-titanium is probably the best kind of alloy that show this kind of behaviours. This alloy work really good that you can find easily in old literature NITINOL, that stands for Nickel-Titanium Naval Ordnance Laboratory, as a synonym of shape memory alloys.
- 1970 → First use of NITINOL in F-14 Tomcat (military application).

In commercial use, the easiest way to use this alloy is to directly heat to activate the alloy, because in this way you have full control of the temperature that you are using. Of course, **for robotic application this is not often the case and usually you are not allowed to use heat but electricity**, but this is much more complex. For the moment, in commercial all the available applications are direct activated by heat and not by electricity. The next picture is the chronological order of alloys that have been discovered with a shape memory effect. The first one was the cadmium-gold one, then NiTi and many others. The red alloys are commercially available, and the Nickel-Titanium is probably the most popular. NiTiCu has copper because it is responsible for the two-way memory effect. So, if you need one-way effect, nickel-titanium works perfectly. If you need something related to two-memory effects, copper is necessary.



ADVANTAGES AND DISADVANTAGES IF THIS TECHNOLOGY:

- ✓ We are considering an intrinsic property of the material; it's just heat that changes material properties. This means that you are allowed to use wires, or very easy and small elements as directly actuators that can be used in your system. So, this a **solution extremely lightweight**, because you don't need anything else than a wire.
- ✓ It has **high work density**. This is a consequence of the fact that the transduction efficiency is directly proportional to the deformation we are giving to the material. Higher is the deformation of the material, higher will be the force generated by the actuator itself. So, the density is basically considered to be the entire material. You just can rely on something that can contract.
- ✓ You can elaborate the actuation concept in a more complex way, introducing other mechanisms like gears or levers. But per se, **the basic mechanism is very simple**.
- ✓ **It's very clean**. There isn't secondary product, if you "want" the secondary product, is the heat.
- ✓ **It's silent**. There aren't no motion or rotatory parts. There is no relative motion between parts. It's just the deformation itself of the material. There is no loud at all.
- ✓ **Remotability(distributed)** the fact that you have a solution that can be placed wherever you like, and just with 2 wires you can activate it. It's very suitable for applications where you need distributed actuators for actuation system. (This is not case of cables tendons, you have the tendon that is addresses wherever you need, so the effect of the cable is from the motor itself to the insertion point. And you have this global effect on the overall system. In this case they are just wires)
- ✓ **Sensing capabilities**. The recovery of the shape is based on the deformation of the crystalline structure, and this crystalline structure is associated with a change of resistance of the material itself. For this reason, it has intrinsic sensing capabilities because the change in material structure, due to a deformation, can be also associated to a change of resistivity of the material and then of the resistance.
- ✓ **Scaling factor and miniaturization**. The low scaling of the material is quite well possible.
- ✓ **Positive scale effects**: indeed, while it is going down scale, the response speed increases as well as the power-ratio.
- ✓ Usually, it requires quite **low voltage** to be reactivated.

- ✓ It has **resistance to corrosion** because nickel and titanium are basically quite stable from the chemical point of view.
 **this is very good in resistance when it is passive, so it's not activated. But, when it is activated, especially in a wet environment, for electrolytes and high current, electrolysis that is induced in the material, is going to extract titanium from the alloy and release it in the environment. This makes the alloy not working anymore because there is not the combination, the alloy is not combined. **
- ✓ It is **biocompatible**. Not in term of activation, because heat is not something that humans don't tolerate very well, but in terms of material properties, titanium in something that is usually well used in bioengineering.

Shape memory alloy seems to be the best solution for all the problems in robotics, but of course there are some disadvantages that limit a lot the use of these alloys for general purpose actuation systems.

- × The **low energy efficiency**. In all cases where heat is included in the transduction principle, the efficiency is extremely low. Whenever heat is involved, you cannot expect to have high efficiency because the heat that you need is also dispersed in the environment. (1-10% maximum)
- × **Fatigue**, in terms of life cycles. (Remember the plot with 3 areas that were comparing stress, strain and lifetime). If you don't pay attention about how you are working with your alloy, it's easy to incur in fatigue issue, like the loss of memory and the capability of going back to restore the memorized shape.
- × Usually, the force generated is quite remarkable, but the **strain is usually limited**. (5-8% maximum strain and when you use two-way rarely it can exceed 2%). If you have this low strain capability, amplification mechanisms are needed and this is against the previous advances like remotability and light weight.

(Question from the class: "is it the super-elasticity in conflict with the low strain?")

Answer: the low strain we are talking about here is related to the memory effect, the capability of recovery after the application of the heat. Instead, in the super elastic temperature the temperature is not. By the way, we must notice that when we deal with the strain, we are considering a local effect. If you have a bending effect, you may have a remarkable displacement deflection, but with low strain. We need also to make a point on the difference between terms. *If you take an alloy and you bend it you can have displacement quite high, but the strain is distributed and it may be very low.*

- *Another question → they have a degenerative behaviour. You start looking at behaviour, maybe life cycles that are expected to have and also a certain stroke under a certain application of voltage, and time by time you see it decrease until you have no deformation at all)*
- × Although the driving voltage is usually quite slow, the **current is usually quite high**. This because the power that you need is related to the voltage resistance that is usually low because it's a metal. So, low driving voltage but high current to have the heat necessary to activate the transduction principles.
- × **Low bandwidth**. This is another important limitation. Since the effect of the behaviour is induced by thermal cycles, it takes time for thermal inertia. For example, dispersion of the

heat in the environment takes time, unless you have an active cooling. So, whenever you have a transduction principle that is related to heat, other than the low efficiency, you also have not very high velocity, high speed. This is something that you can deal with. (we'll see also an example of alloys that are used in the range of kHz).

It is also related with how much deformation you need. Velocity and displacement are coupled, and they are something that you can trade-off.

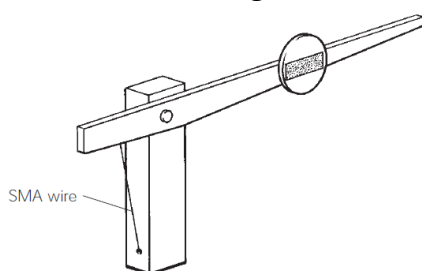
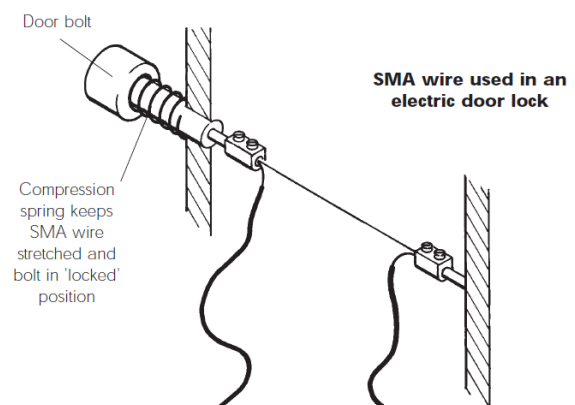
- × **Control issue**. It's another limiting factor, because these alloys have behaviours quite easy to observe. The cost is very low, and the observation of the principle is very easy, but the precise use of the phenomenon is not. This happens because there are no models that can really catch and completely describe the behaviour of the alloy. This is mainly related to the *number of parameters* that are involved in the process that are tightly coupled each other. This induced nonlinear behaviours. There is also a lot of hysteresis. There are some parameters that are not even easy to be defined, like for example the heat transfer coefficient.

All these advantages and disadvantages are what make this technology a very interesting approach, but they show also why it has limitations. Some features can be mitigated, for example, fabrication and process. You can trade off the low cost with the accuracy. Shape memory alloys have also versatility, because you can really shape them as you like. But, when you have too much freedom is maybe seen as disadvantages because they are no pre-configured/ pre-established shapes and ways to exploit this technology. Again, the material itself has a low cost, but when you start dealing with specific geometries and implementations, cost may rise significantly.

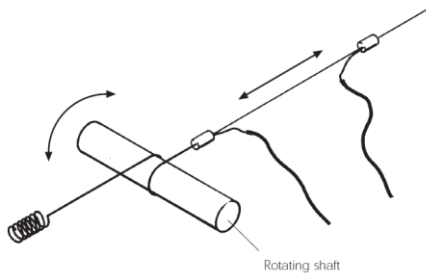
POSSIBLE IMPLEMENTATION/APPLICATIONS:

Let's see some possible basic implementations to see how alloys can be exploited directly.

- **Wires configured in a linear way**, a straight wire that can control a door bolt. It uses a one-way memory effect element, and the restoring force is provided by a compression spring. So, if you activate the alloy, it contracts, the door bolt is pulled, the compression spring compresses and you have the opening. When you deactivated it, the spring is able to make the door bolt go back into the original position and restoring also the elongated configuration of the wire.

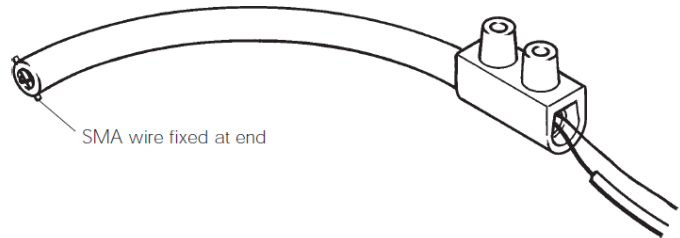


- **Angular actuation**. The same happens for angular actuation, where the wires can be used in combination with levers, just to have the possibility to displace the part.



- **Rotary actuations.** When the wires still used but the rotating shaft is connected to a spring, so the overall and the net action that you have from the system is a rotation of the shaft back and forth.

- **Bending actuation.** This can be related to bending actuation. It's not visible, but in this case the shape memory alloy is arranged eccentrically, so the alloy is on one side of this elastomeric or compliant tube, so when the shape memory alloy is activated, it generates this bending behaviour.



These are very basic possibilities of rotation. How can we make proper use of this technology in designing a shape memory alloy to an actuation system? **There are 6 points that you need to take care of.** Not all of them have the same difficulty, and not all of them are needed, so it's depending a lot on the application you have in mind, but the general workflow consists in 6 points:

1. **Derive your actuator specification:** you need to know and define very carefully what are the expected capabilities and properties of your actuator.
2. **Design the actuator.** If you are going to use a two-way memory effect, and you have intrinsically the capability of going back and forth, or if you want to exploit the one-way memory effect → at this point you need to take in consideration and design also the counter-actuator, so, what is the force that is restoring the deformed shape that the heat will recover ?
3. **Know your alloy:** you need to know all the alloys, or at least in general. You need to identify it and have a list of performances of the alloy, because each alloy has different behaviour and different performances. They have different activation temperature, elastic moduli, shapes and resistance to environment. There are many possibilities. So, you need to remember that every kind of alloy with shape memory effect has different behaviours and capabilities. So, you need to know what are searching for.

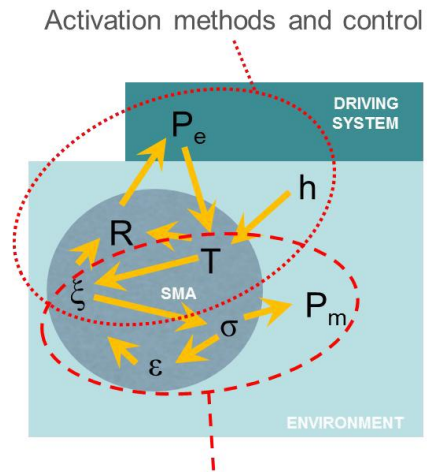
These first 3 are the main important, they have whatever you need to model mechanical performance of the actuator, but this is not the whole story. The performance of the actuator itself is very much depending also on what happening during the interaction with the environment since this is related with to heat exchange. For this reason, there are other 3 points.

4. **Environment:** you know where the actuation system is expected to work, and you have to understand the main parameters that describe the environment in terms of heat exchange to whatever (air, water, kind of flow...). You need to know everything that imposes conditions to the environment.
5. **Decide the activation and the cooling method:** you have to decide if you are going for back heat, joule effect, if you are going to use some specific cooling methodology, like water flowing very close to the wire, or maybe just air.

For example, in the octopus project they were exploiting the cooling method related to the environment itself and it works underwater. In this case the cooling method is not the issue but the heating method because they had water and cooling was so fast. For activation they had remarkable currents.

6. **The driving system:** especially if you are considering using current to activate the technology, you need to design the control current so you can achieve the working cycles as they are expected to act.

As the teacher said before, *the shape memory effect is a very easy effect to observe and obtain but it's a very difficult to use accurately.* And this happens because there are a lot of parameters that are involved in this process. In the image is reported only the driving system, that is basically your electric source, source of power. There is also the environment that can be air, water or whatever you want. In the environment, the alloy is immersed. So, in the image there are:

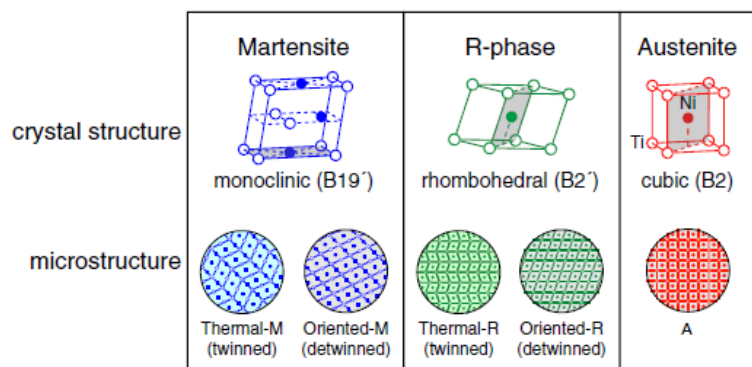


- the electric power (P_e) from the driven system to the alloy.
- the mechanical power (P_m) is given by the alloy to the environment.
- h is the heat exchange coefficient; it governs how the heat is exchange with the environment.
- And these other five parameters (R , T , ϵ , ζ , σ) describe all possible condition of the alloy. The condition is related with R (electric resistance), T (temperature), stress and strain. It's also related to ζ (it will be defined later) → this parameter is linked with the configuration state that is the phase/current state in which is the alloy.

SHAPE MEMORY ALLOY STRUCTURE

Now, let's go a little bit more into the structure of the alloy that is the heart of this technology. This kind of alloy can pass through 3 different phases, that are temperature dependent:

- Martensite phase (twinned and detwinned)
- R-phase
- Austenite phase (cubic)



They are mainly depending on temperature (we'll see an exception, but for the moment we consider those phases depending on temperature). We put more attention on Martensite and Austenite because, although R-phase is always present, it is not associated to a macroscopic and large

reconfiguration that induces deformation. From now on we study the passage from martensite to austenite and back. Again, they go always through the r-phase but it's not visible macroscopically.

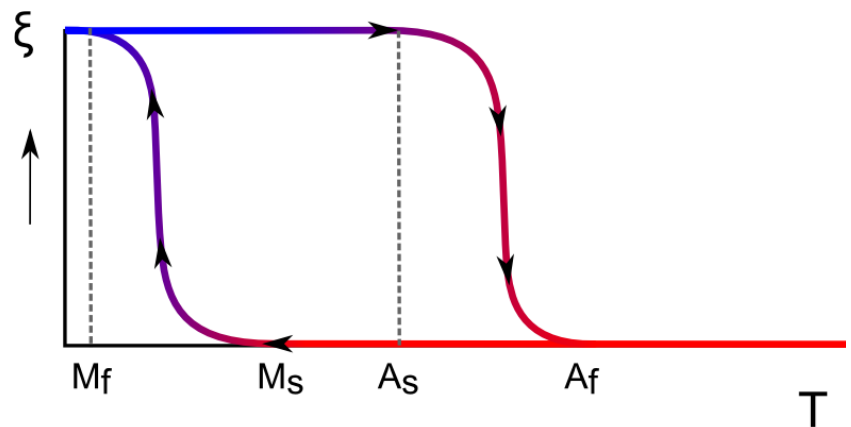
- *Austenite* is present in only one possible structure, the *cubic structure*.
- Martensite is present in 2 possible microstructures that are twinned and detwinned microstructure, they are both useful and important to describe the behaviour of the material. In martensite structure atoms are arranged in a monoclinic configuration.
In twinned configuration = unit cells are arranged in a zig-zag way.
In detwinned configuration = unit cells are aligned in the same direction. It's an oriented configuration.

One important thing to notice is that **from martensite to austenite, there is a remarkable variation of elastic moduli** and there is also where we have the possibility to use this alloy like an actuator. While **the passage between twinned and detwinned is associated with large deformation.**

The material microcrystalline structure can be both martensite and austenite, also a unit cell of alloy can be martensite or austenite. So, at any point in the transformation of the material, from martensite to austenite, you have in the material some parts of martensite and some parts of austenite. It starts at low temperature, martensite, to high temperature, austenite, and in the middle part you can have some austenite and some martensite (this is a proportional and progressive behaviour).

Volumetric Fraction $\xi_M = \frac{V_M}{V_{tot}}$

describes exactly this: how much of your alloy in that specific moment is in martensite state respect the total volume of the alloy. On the left there is the plot that describes the behaviour of volumetric fraction related to the temperature. Low temperature is usually between 0 and 1 but all alloys have different number, so now are reported transformation temperatures but without a specific number. The volumetric fraction is in a range from 0 to 1. When it is 1, the entire material is martensite, while when it is 0, everything is austenite.



- At low temperature → everything is martensite
- Then it moves towards higher temperature until a certain point there is A_s the point that is called **Austenite Start** → when the temperature is enough for the microcrystalline structures to transform some unit cells into austenite.
- Increasing temperature makes the number of unit cells in austenite to increase quite also abruptly.
- Then there is the point A_f , called austenite finish → moment when all unite cells have been transformed into austenite. If you still continue to increase the temperature there are no other phases unless the temperature arrives to liquefy the alloy.

When the process is reversed, so when it passes from high temperature to low temperature, it doesn't follow the same pathway, but it creates a hysteretic loop.

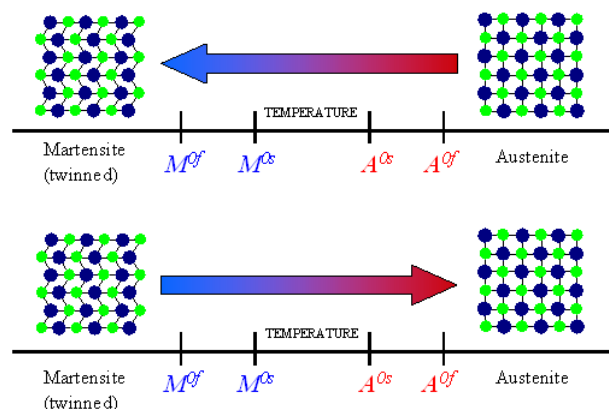
- the point M_s (martensite start) is much lower than the A_s .
- After that it goes to the point M_f (martensite finish) → when everything is martensite. Again, decreasing more the temperature, doesn't change the crystalline structure.

The definition of these 4-transition temperatures depends on the material composition and many ranges and combinations are available, but the sequence is always the same= A_s , A_f , M_s , M_f . And also, the hysteretic loop is always present.

Try to merge the concept of crystalline structure with the concept of transition phases:

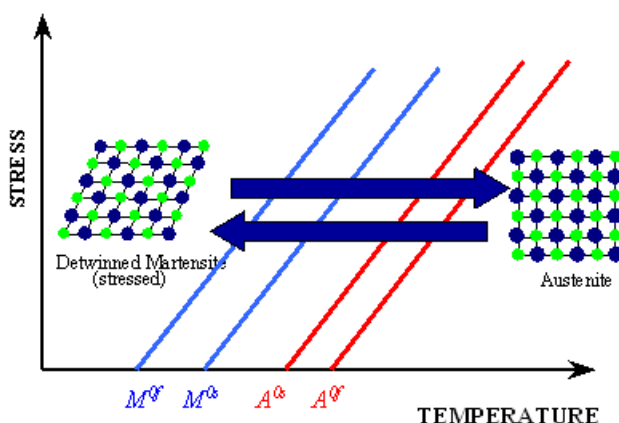
1. SHAPE MEMORY EFFECT WITHOUT MECHANICAL LOADING:

this is the easiest condition, when you have an alloy material that is subject to no-loading condition. So, if the material is at low temperature and without loading condition, it is martensite. Starting from martensite state: if you increase the temperature it goes to austenite. It goes through the 4 characteristic transformation temperature until austenite, without any macroscopic shape changing → although the microcrystalline structure is different, the overall shape is the same. **So, without mechanical loading or pre-strain on the alloy, increasing temperature affects the microcrystalline structure but it does not affect the shape.**



Going back, from austenite to martensite, is exactly the same thing.

2. SHAPE MEMORY EFFECT WITH MECHANICAL LOADING (I):

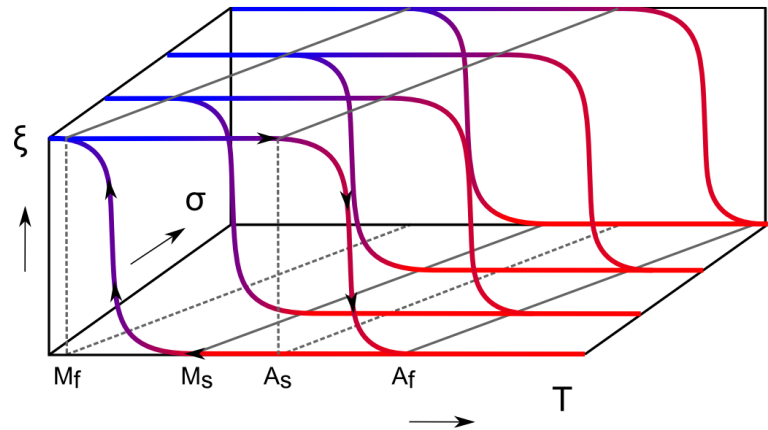


the situation is different when you have mechanical loading. In this case, the graph is 2D and it depends on the stress. This means that instead of starting from twin martensite, the mechanical load provides the detwinned martensite. All the unit cells are aligned in the same direction. After loading condition, there is the application of heat, and it goes through 4

transformation temperatures to arrive at austenite. If you go back to the room temperature, you pass the 4 phases and then you obtain again the deformed shape because the load is still applied. **With static loading → it's possible to cycle between this deformed shape to the recovered shape.**

It's also important to underline that **the 4 characteristic temperatures are not constant**, they depend on the stress. The fact that they are not constant it's bad news. But, **the good news, is that they are all parallel**, the dependency of these values with the stress, is the

same for each of them → they are defined through the equation $\frac{d\sigma}{dT} = \frac{\rho\Delta H}{T_0\Delta\varepsilon} = \frac{1}{cm}$. It's depended on the transformation latent heat (ΔH), on the null stress transformation temperature (original value T_0), on the strain and the stress gradient ($\Delta\varepsilon$, cm → basically the slope of those curve).



Usually this is a knowledge that the provider, who fabricates the alloy, should provide. If we take the 2D graph of the volume fraction/temperature and you plot it in the third dimension with an applied stress, we obtain something like the graph on the left. The hysteresis cycle migrates toward the right part of the plot because those lines describe the dependency of the characteristic temperatures on the stress and it translates the entire plot towards higher temperatures. This is direct consequence of the coupling between the 4 characteristic temperatures and the loading condition.

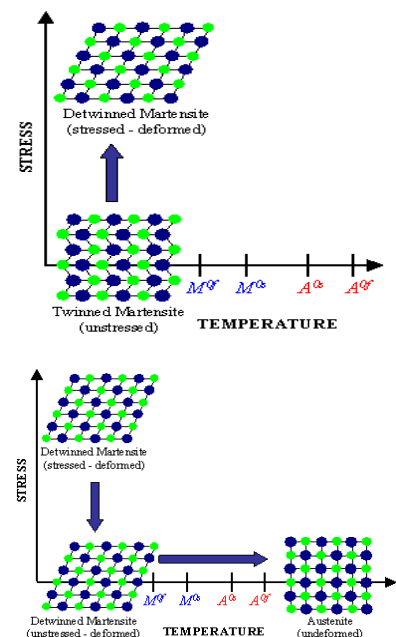
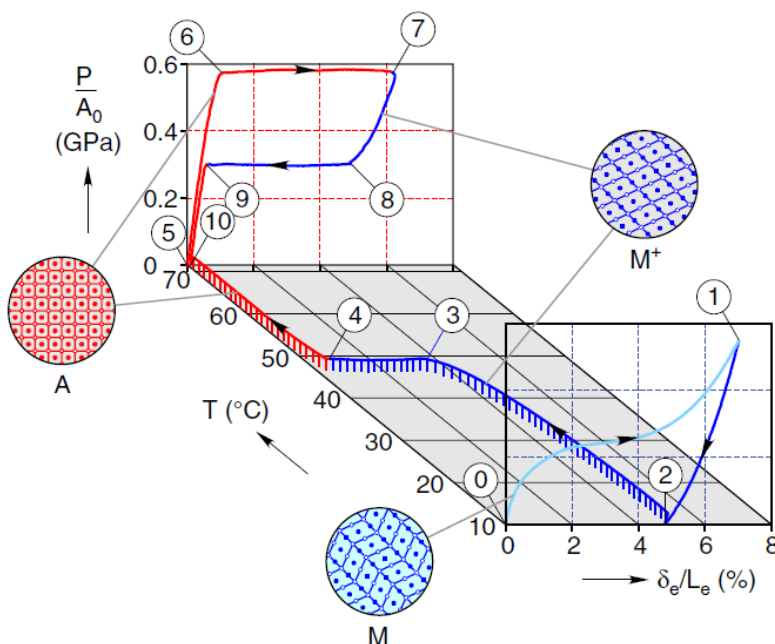
(remember → the characteristic temperatures have always a dependency on the stress)

3. SHAPE MEMORY EFFECT WITH MECHANICAL LOADING (II):

on the slide there is everything (14). This graph embedded everything you need to know and understand about shape memory alloys, at least the basic working principle and the relationship between stress, strain and temperature (**thermomechanical behaviour**). There are 2 plots:

- 3D plot of strain, stress and temperature.
- 2D plot that is connecting the stress and temperature but also the shape, so somehow, it's deformation.

These 2 plots are referred to the same thing, and there is also coherence with numbers.



- At zero there is low temperature, in the example is 10 degrees. At the beginning there is no stress and strain. In this condition there is for sure, martensite twinned with a crystalline structure.
- From 0 to 1 = Now, let's applied a load, that is imposing the deformation at low temperatures. So, **from twinned martensite (at zero) the load leads to a different shape that is detwinned martensite microcrystalline structure**. The load aligns the unit cells, so from twinned passes to detwinned.
- From 1 to 2 = what happens if you remove the loading condition? The alloy goes to the point 2. Here there is some elastic deformation that is recovered, but most of the deformation actually remains. This is the apparent deformation that the loading condition induces. In both point, 0 and 2, there is zero stress but in the second case the deformation is maintained. The difference between twinned and detwinned martensite is due to the fact that we applied enough stress to induce **an apparent deformation**.
- From point 2 to 5 = at point 2, everything is at low temperature. In this case no loading condition but increase in temperature. From point 2 the transition goes to point 3, point 4, when austenite starts, to point 5, where austenite finish. In this phase the alloy is transformed by heat from martensite to austenite (this is also coded by the colours blue and red). From detwinned martensite the material goes through austenite start and austenite finish. And at point 5 the full alloy is now in austenite phase. Increase more temperature doesn't change anything from the microcrystalline point of view. At point 5 there is high temperature, no stress and no strain and deformation. We recover the original shape but with **a completely different microcrystalline structure**.
- From point 5 to 7 = At point 5 there is a load with high temperature until point 7. This is a deformed configuration.
After point 7, there is a change in colour. It may be strange because we know that there are 2 stable phases that are martensite at low temperature and austenite at high temperature. **Why we have martensite at high temperature?** → this is called **stressed-induced martensite**. So, if stress is enough to make the unit cells collapse in martensite, it's possible to have martensite microstructure at high temperature. In terms of stability this is not normal. So, this configuration is not stable but it's martensite.
- From 7 to 10 = in the last part of the plot is possible to see that the material is not stable. Indeed, if the loading condition is removed, it's naturally moving back to austenite, that is the real stable phase at high temperature. From 7 to 10 the alloy goes back to the original condition in austenite. In absence of loading condition, the austenite is the target configuration that can be recovered at high temperature.

FEATURES IMPORTANT TO UNDERLINE:

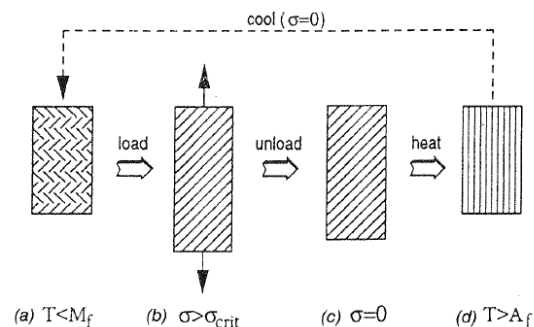


- first of **all are the hysteretic loops on the 2 plots**. In the first plot (low temperature) it's due to the plastic deformation. In the second plot there is also a plateau.
- **Plateau** → when you have the maximum value of the allowed stress at point 6, the unit cells are arranged in a cubic way. Unit cells, one by one, start to degenerate into martensite. There is the first unit cells that collapses and others resist, then the second one change configuration and others resist, then you have the third one and so and so... at the

end all the unit cells change into martensite. **This one-by-one process** is what generates plateau. The deformation is increasing without the increasing of stress because the energy is spent to deform one by one all the unit cell phases. This is what happens until the entire material collapses into detwinned martensite. This process is exactly the same when it is reversed → one by one-unit cells go back to austenite, inducing no different stress but remarkable deformation.

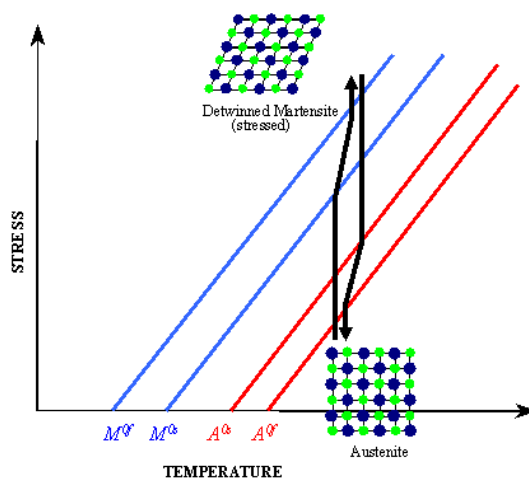
- **Slope** → **martensite is associated with a low elastic modulus**. In this case is not linear, so there isn't just a single value associated with that, but if you take a linearization, you have a slope like the one in the graph, associated with a simplification to the young modules of the material in martensite phase. While **austenite has a higher elastic modulus**. This can be used to the actuation, because the force generated by martensite is lower due to a lower modulus, otherwise the austenite is higher.
- **This graph is valid just for one-way SMA**. → this is the answer to a question from the class + usually it's asked at the exam. In the one way if you are at 10 and you decrease the temperature without loading conditions you go back to 0. If you have a two- memory effect alloy from 10, decreasing the temperature, you go back to 2. (NOT SURE, CHECK ON INTERNET). And also, this plot cannot be associated to two-ways memory effect because 5% of deformation is not possible for this type of SMA.

This is another simplify representation if this process. Remember at 0 and 10 you have the same overall shape but with different microcrystalline structures.



4. SUPER ELASTIC EFFECT (PSEUDOPLASTIC EFFECT)

In shape memory effect, when there is a loading condition applied, the alloys cycles between de-twinned martensite and austenite and basically there are these 2 configurations. In super



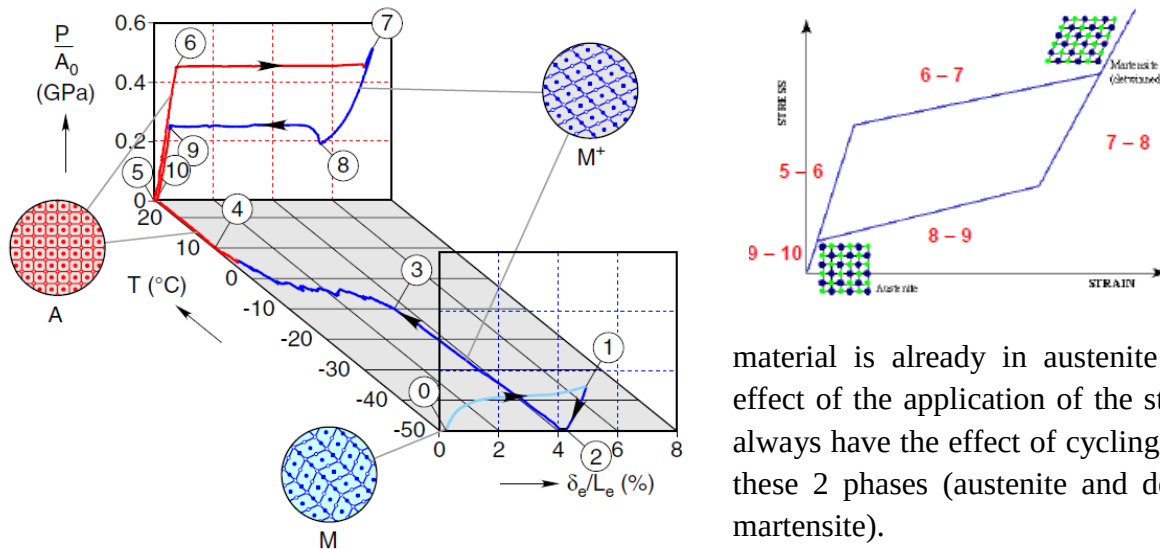
elasticity the macroscopically effect is the same, but the cycle is different, in this case the temperature is constant and the stress changes.

This passes from de-twinned martensite to austenite but without changing temperature.

This is an intrinsic property in the nature of the material → to maintain the possibility to go back and forth in the austenite and in the detwinned martensite at a specific level of temperature, just by changing the stress).

The following graph are similar with previous ones. **The overall temperature is different and**

also the stress is lower. But basically, the behaviour is the same, it's only related to the condition of the material. If the austenite phase is finished at environmental temperature, the



material is already in austenite and the effect of the application of the stress will always have the effect of cycling between these 2 phases (austenite and detwinned martensite).

So, basically the behaviour is the same, but according to material composition and the working environment, the alloy can present shape memory effect or super-elasticity. They are the same but working in different ranges! The thermos-mechanical behaviour of a shape memory alloy working in super-elasticity is the same, but what changes is the room temperature (point zero) that is minus 50 → if you have an alloy that in standard room temperature has super-elastic effect, it's mean that it has a shape memory effect at much lower temperature. (example: in space exploration temperatures are really different that here, so you should know how work your material → you must know the range of temperature).

MODELLING THE SHAPE MEMORY EFFECT

Now, let's try to understand why the modelling of the shape memory effect is very complex. This because SMA has **highly non-linear behaviour** with **pseudo-plastic response**. There is also an apparent plasticity, it's possible to say that there are many non-linear and strange behaviours. Whatever is the model that you are using, what you need is something that catch the overall behaviour that you really need.

- There are some models that are able to catch the microscopic behaviour, going into the level of molecular behaviour, that from our point of view is not really interesting. Maybe it's too much higher
- The macroscopic approach is focused on the phenomenological features and it's also the simplest from a computational point of view. It's very much simplified. This is the case of Landau-Devonshire theory.
- Models that start from the phenomenological features, but they have established mathematical function to describe the phase transformation mechanics. They are the most popular and used by engineers. The 4 models most used are Liang & Rogers, Brinson, Tanaka & Nagaki and Boyd and Lagoundas. (hysteresis loop and most of non-linearity sub theory)
- Model based on elastoplasticity theory (il prof nella realtà non l'ha mai visto usare)
- Model based on plasticity concept: Auricchio is the guy who developed the model, that is also already include in ANSYS and already evaluating and considering. It's good to know

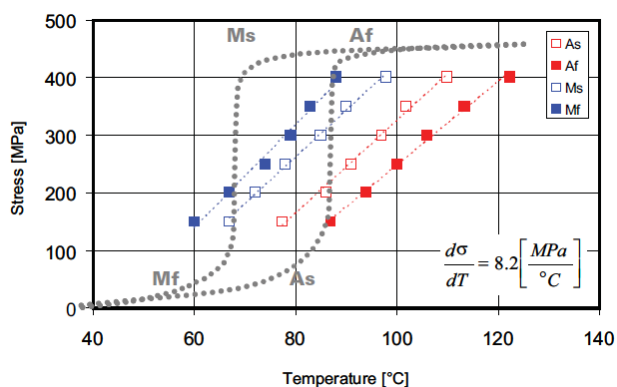
that there is an already in-built module in ANSYS that already implements the constitutive equations for shape memory algorithms.



INVITATION/ REMEMBER = accurately decide how much efforts you want to put in this modelling part. Usually, a model is something you would like to have as a general tool applicable to any kind of situation. The suggestion is to go for something that is able to catch most of the theory but also include some empirical parts that is working quite well. In any case, whatever is the model that you are going to use, there are some parameters that you will need in any case, and this is related to:

1. Characteristic transformation temperatures (the four temperatures)
2. Young modulus for the 2 phases (austenite and martensite)
3. Stress dependencies of the temperatures.

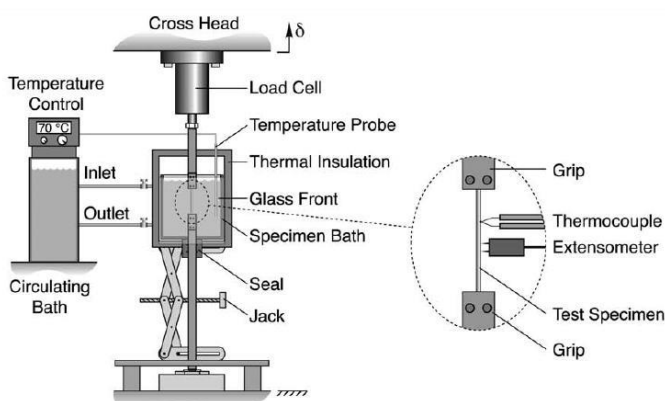
These are often provided by the manufacturer or the provider in this form:



Transition T	-200 ÷ 100°C
Transformation strain	8% (1 cycle), 6% (10 ² cycles), 4% (10 ⁵ cycles)
Thermal hysteresis	15 - 30°C
Bulk density	6,45 g/cm ³
Electrical resistivity	80 μΩcm (M) ; 100 μΩcm (A)
Corrosion resistance	Excellent
Melting point	1310°C
Elastic modulus	28 - 41 GPa (Martensite), 83 GPa (Austenite)
Breaking load	1500 (cold worked) MPa

The information is reported in a graph in temperature and stress, where are also reported the characteristic temperatures of transformation and the stress gradient (first image). The second image is another example of data, where there are a lot of properties. We are more interested in the elastic modulus → we are not considering materials that are changing 2 or 3 order of magnitude. If you are lucky you can double the elastic modulus, at maximum 3 time (in the example there are 21 and 83). There is a problem sometimes with these numbers → not all the time they are written, and they can also be wrong. The prof's suggestion is to test the alloy by yourself, if it's possible. In this case you are sure about properties.

In the image is shown the typical thermos-machinal testing machine and it doesn't involve anything extremely expensive. There are basically 2 components:



- one is the universal testing machine: it can be monoaxial or it's universal. In any case you need a machine that has a fixed support and then a crosshead that can be controlled to move up and down with a load cell, applied in the crosshead.

- In the middle there is the specimen, that is inside a controlled chamber, where

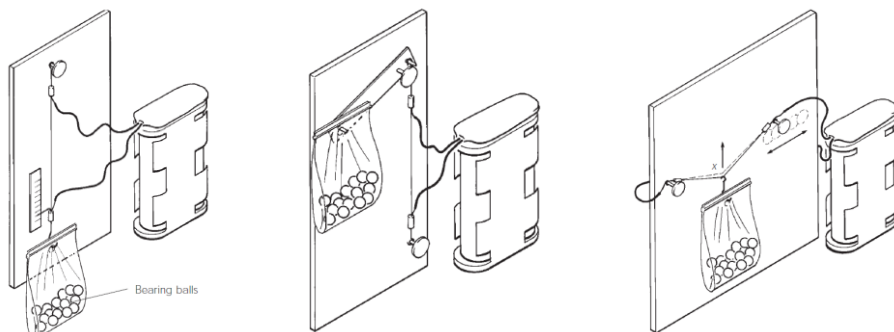
you can control the temperature inside it. → is fluidic driven, so there are an inlet and outlet.

What do you need id the capability of measuring deformation and stress applied to the alloy. The stress is direct read by the load cell, and the strain maybe can be derived by the motion of the crosshead directly. If you want to be extremely accurate, you can also add an extensometer on the wire or any kind of optical technology to track the deformation locally. If you want to rely on the external temperature control, you can use a thermocouple to measure the specific, the exact pressure on the alloy. Once you have this you can have a very precise texting procedure.

ACTUATOR MECHANICAL DESIGN PRINCIPLES

Let's move to see how to design an actuator that is based on shape memory alloys.

1. First of all, you need to look at the overall geometrical choice. The geometrical choice is your starting geometry, the geometry of the material that you are using. For example, if you have a bag with some spheres inside, you have also a battery (our supply), and you need to lift it up. This is a very simple problem, and you have several possibilities. The easier way is to have wires, they are set vertically, and the direct application of electricity will shorten the wires, and the bag will be lifted up. You can also have some leverage systems to increase the capability of the displacement. But wires are not the only possibility.

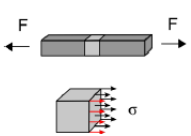
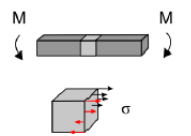
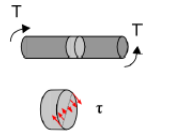


Wires are the most explored and the cheapest one but not the only one.

Wires can come in different shapes, diameter and you can also made spring out of them.

It's possible to have **also ribbons, foils and thin films** (they are extremely sensitive). You can have a lot of possibility and all of them are pretty low cost.

2. Then, there is the mechanical load concept, how do you want to exploit the recovery behaviour of the alloy?

	Tensile deformation	Bending deformation	Torsion deformation
			
stress	$\sigma = \frac{F}{A}$	$\sigma_{(x)} = \frac{M_b x}{I}$	$\tau_{(r)} = \frac{M_t r}{J}$
efficiency	$\frac{\dot{\sigma}}{\sigma_{\max}} = 1$	$\frac{\dot{\sigma}}{\sigma_{\max}} = \frac{1}{2}$	$\frac{\dot{\tau}}{\tau_{\max}} = \frac{2}{3}$

What is the way you are deforming your alloy that is fixing the way you will have your energy back. Now it should be clear that everything is related to the deformation of the material. But there are many ways to

deform the material: tensile, bending and torsion.

In tensile deformation there is the Hooke's law, and the stress is related to the force applied on the area. The entire area is considered, and efficiency is maximum because the average of the stress on the maximum stress is equal (maximum efficiency that you can obtain).

In bending deformation, the efficiency is not the maximum, and the stress is defined as $\sigma_{(x)} = \frac{M_b x}{I}$ and it depends on x, that is the distance between the mid-plane and the upper and the lower side of the beam. The stress has an average that is a half of the maximum, so the efficiency of mechanical loading is 0.5.

In torsion deformation, when there is the twisting the material from the 2 ends. In this case the torque, so the shear stress, is depending on the radius (it's not constant) and it cannot be maximum, indeed the maximum efficiency is 2/3

The order is tensile-torsion-bending.

Question from the professor: "if you have a wire, and you are using it in the tensile configuration. It's easy right? But if you have a spring? What is the modality/ mechanical load that you use with the spring? A spring is elongated and then heated up to be short" If you look of what happens in the coils and if you pretend to be inside them, you will see that there is **torsion in order to open the coil**. There is torsion locally, all around the coils. This is pretty good, not much as tensile, but really better than bending.

On slid e 31 there are typical formula and definitions that will be useful like Poisson's ratio, young modulus, Shear modulus and Bulk modulus.

SPRINGS:

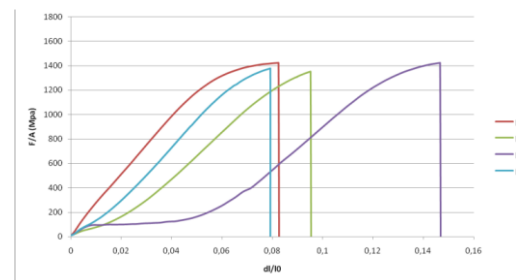
Helicoidal spring

Not, let's goes to couples of examples (actuator based on spring) → this is the simplest way to shape SME of a spring. It's important to have general knowledge on the behaviours of the spring, but there are much more accurate models. These are open parameters that are required to have a complete design of the spring:

- Wire diameter (d)
- Spring diameter (D)
- Number of coils (n)

And then, as an input, you have these parameters/material properties:

- Poisson's ratio = 0.33
- Austenite young modulus ($E_A = 55GPa$)
- Martensite young modulus ($E_M = 22GPa$) e
These are the 2 values chosen in the datasheet before.
- Ultimate stress ($\sigma_r = 1400 MPa$)
- Wahls's correction factor in an experimental factor.
It's an ignorance factor.



The breaking strength tests has been used to derive the ultimate stress.

The following one is the set of equation:

- Austenitic shear modulus $G_A = \frac{E_A}{2(1+\nu)}$ this is necessary to pass from material properties (E and ν) to the shear modulus
- Martensitic shear modulus $G_M = \frac{E_M}{2(1+\nu)}$
- Spring index $C = \frac{\hat{D}}{d}$ this index is really important. It's the ratio between the average of the diameter and the diameter of the wire. **This ratio should not be less than 6 !!** because according to an experimental approach, if the ratio is less than 6, it will more propense to lose memory. It happens because the curvature and strain concentration is very high also at very small elongation of the spring. So, C values high than 6 is very much suggested.
- Max shear stress $\tau_{max} = \frac{\sigma_r}{2c_f}$
- Austenitic spring rate $K_A = \frac{G_A d}{8nc^3}$ can be derive from the shear relate to austenite, this n appears and also c.
- Martensitic spring rate $K_M = \frac{G_M d}{8nc^3}$
- Spring free length $L_0 = dn$
- Maximum force $F_{Max} = \frac{\tau_{max} \pi d^3}{8D}$
- Maximum HOT Spring elongation $\Delta x_{A_{max}} = \frac{F_{Max}}{K_A}$ this is the maximum range of deformation that the spring should be able to sustain when it is at high temperature
- Maximum COLD Spring elongation $\Delta x_{M_{max}} = \frac{F_{Max}}{K_M}$ the same as before but at low temperature conditions.

If you combine all of those equations, and you have any constraints related to strain, stress or other things, you can design the specific spring for your specific application.

Planar spring

Another possibility is to use a planar spring. In this case, the twist is the main loading condition, so the mechanical loading concept is the bending. It's already known that it doesn't have the maximum efficiency, but there are some advantages in planar spring.

- In this case there are 4 parameters, instead of 3
- The set of equations is similar in terms of procedure, the flow is exactly the same but here there are other features, like the deflection of the single arm.
- There are again the spring rate and the maximum force and the 2 deformations in hot and cold condition.

DIFFERENCES BETWEEN PLANAR AND HELICOIDAL SPRING:

- The planar one has more parameters that can be adjusted and pack them better than in helicoidal case.

- Helicoidal spring is easier to be fabricated and the planar is much more difficult.

One-way vs two-way memory effect: if you want to use the 2-way memory effect, the first thing that you need to consider is the heat treatment that you need to execute in order to have 2-way memory effect. and the problem that is quite complex, and instable. This is the procedure proposed by Duerig:

- Overdeformation while in the martensite phase
- Shape memory cycling (cool -> deform -> heat -> repeat)
- SuperElastic cycling (load -> unload -> repeat)
- Constrained temperature cycling of deformed martensite

Duerig et al., 1990

You need to apply an over-deformation at martensite phase (low temperature), then you need to apply shape memory-cycling: cool-deform-heat and repeat (shape memory) or load-unload-repeat (super elastic cycling) and then you need a constrained temperature cycling. It's complex. For one shape memory effect you need:

- Constrain it mechanically
- Apply heating at certain temperature
- Then, water.

What are the downs and drawbacks? The recoverable strain rarely exceeds 2%, the long-term fatigue is low in any case and also dare are still missing. Stability is not documented because it's easy to have memory erasing. The prof suggestion is to use one-way memory effect and provide bias forces.

RESTORING FORCE:

In the 2-memory effect you have everything completely integrated. The full working cycle is completely implemented (deformation-restoring). In one-way case you need to provide an additional element to obtain the deformed shape that can be recovered by the shape memory effect. What is it the additional element that you can use? It's called restoring force that can be provide by:

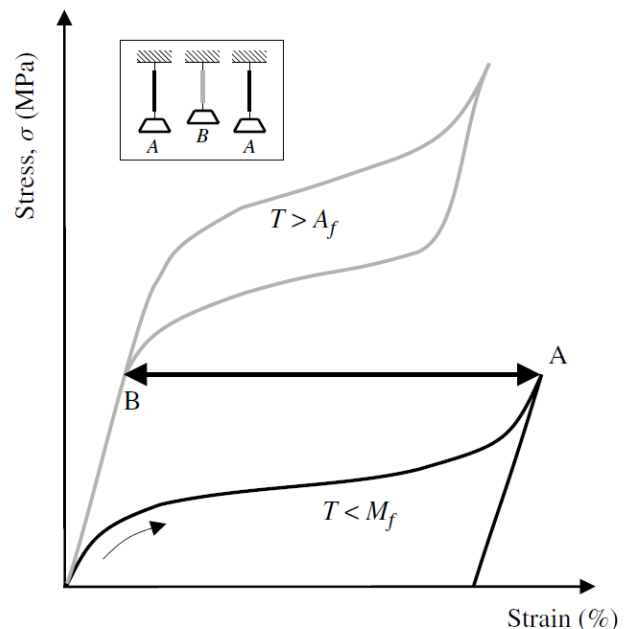
- A constant force.
- Variable passive force.
- Variable active force.

These are the 3 main possibilities:

Constant force:

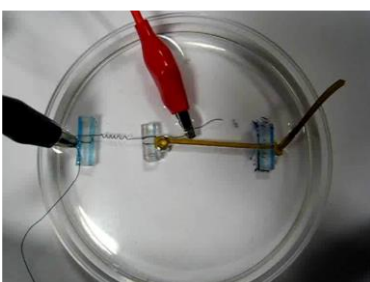
This is the example related to the constant force. There is a spring, made by shape memory alloy, and there is a stone at the end, so the weight is constant. There is also a current, that flows inside the alloy to activate it. In the video, you can see how the material lifts up the stone. Why?

1. The spring is in martensite condition, and the stone applies a constant load to the spring. The deformation is set at the equilibrium point between the martensite spring and the weight of the stone. (point A)
2. Activating the alloy through the flow of current ($\uparrow$ temperature), so it goes from the black behaviour (martensite) to the grey behaviour (austenite).
3. The second equilibrium point is B, when the force created by the stone is equilibrated.



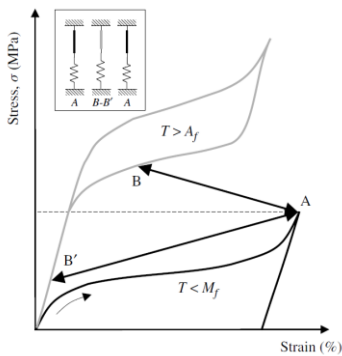
The force that the alloy is able to generate has changed according to the degree of activation (current/temperature). So, it goes from A (no current $\rightarrow$ martensite) to B (current $\rightarrow$ austenite) by a horizontal line because the stone cannot change its weight so the force is constant. The difference between the position in A and B is called **stroke = how much displacement the actuator is able to use**.

Variable passive force



The second possibility for restoring force is a variable passive force. In the video there is the spring that is activated through a low current and an elastic band which provides a variable but passive force. So, it's completely passive behaviour and not constant. Since the restoring element is not constant from A, it cannot move

horizontal but just upwards or downwards, according to the configuration of the coupling of the 2 elements.



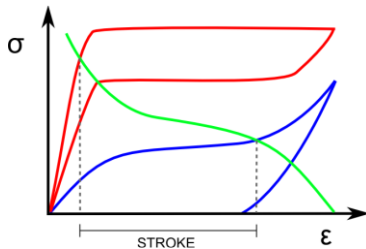
If they are in series like the sequence of the image, from A the graph goes to one point between B and B', if the spring and the shape memory alloy are inverted, from A it goes in the other point (in this case the spring is compressed by the alloy. The important thing is again, it moves from A (martensite maximum equilibrium point between elastic force of the passive element and the alloy) to another point, B or B', that is a new equilibrium point between the spring and the austenite configuration of the alloy. So, everything related to shape memory alloys is always related to equilibrium points. The

stroke in your actuation system is always given by 2 equilibrium points:

- One when the actuator is not active and the counter actuator or restoring element.
- The second is the actuator in active conditions and the same restoring element.

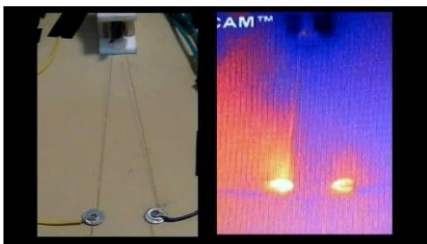
The stroke cannot be from zero to any point, because zero means that the strength of material is zero, and it can be zero if it is in equilibrium with something with zero forces and it is impossible. Any stroke comes from 2 equilibrium points that are higher than zero.

The alloy, unless it is completely disconnected, is always ranging between 2 points, with a strain different from zero!!!! (important)

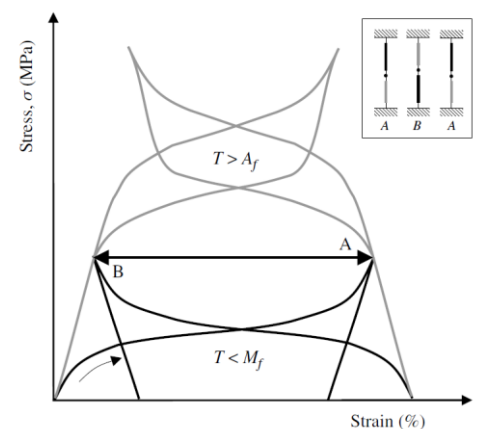


This is a more realistic representation of the same system. The austenite is red, the martensite is blue and in green there is the hyper-elastic behaviour of an elastomer that is reversed to visualize 2 equilibrium points, a visually you can define the stroke of the system

Variable active force:



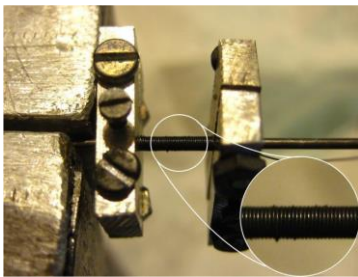
Finally, the restoring force is variable active force. In this example, there are 2 shape memory alloys which are acting antagonistically to rotate the shaft. By activating one and deactivating the other in an alternative way, the shaft is activated. There is the overlap the same behaviour in 2 wires. Also, in this case the 2 points of equilibrium are A and B. In A the first martensite alloy is in equilibrium with the austenite phase of the second alloy. In B there is the equilibrium between the martensite of the second alloy and the austenite of the first.



MEMORIZATION PROCESS:

Now let's see the memorization process of a shape memory alloy to set a certain shape/ an arbitrary shape into alloy and wire → that is the shape that will be recover every time you heat the alloy.

1. You need to constrain tightly into a specific geometry/specific shape.
2. Then, you need to put your system, very well tightened into a furnace at very high temperature (over 550 degrees for 30 minutes). It's very much suggested to have **flowing nitrogen** gas while doing it, because in presence of air sometimes one surface of the alloy reacts with oxygen and creates a thin layer of oxidated material. This is not a real problem, but if you like to connect the alloy with electric wires, this layer of non-conductive material can introduce a contact resistance. If you have nitrogen instead, this doesn't happen and the alloy is ready to be connected to electric connectors.
3. After the time is elapsing, there is the water-cast. The system is taken out from furnace and quickly put into fresh water, and this is setting the shape.



This seems easy, but if you want accurate results, it can be tricky, a lot of very small defects can be present. Some coils can be slightly bigger than others. This can be a medium-long term issue, because a coil with a slightly different geometrical definition has a weak point on a chain. So, if you have a coil slightly larger it's force and resistive force will be lower than others, and all the stresses will be concentrated and focused on that coil. In a medium-long term

condition it can caused over-stressing and lose momentum. If you don't want to do the alloy by yourself there are some commercial forms available → either you design your system in a way that those products are already suitable for you, or you design yourself your spring and the ask these companies to fabricate the spring exactly the way you want (in this way the cost is higher).

CONTROL METHODS:

The control method depends on the application and the way you want to control the alloy. There are 4 possible "conditions":

- **On-off activation** or **modulating** the alloy.
- **System in an open or closed loop**, so with feedback on a certain parameter.
- **Simple or complex geometry**, this depends a lot on the knowledge that you have on the capability of simplification that you can have on the device.
- The **environment**. The environment can be simple very well known or constrained or unknow.

Let's start with the simplest case: on-off, open loop, simple geometry and well-known environment.

This is the equation/the balance that you can use. When you have an input (electric power), some of the power is dissipated:

$$\rho c V \frac{dT}{dt} + \rho V \Delta H \frac{d\xi}{dt} = Ri^2(t) - hA(T - T_a)$$

- $hA(T - T_a)$ is the term related to the heat exchanged with the environment, when h is convection heat transfer coefficient, T is the temperature of the alloy and T_a is the ambient

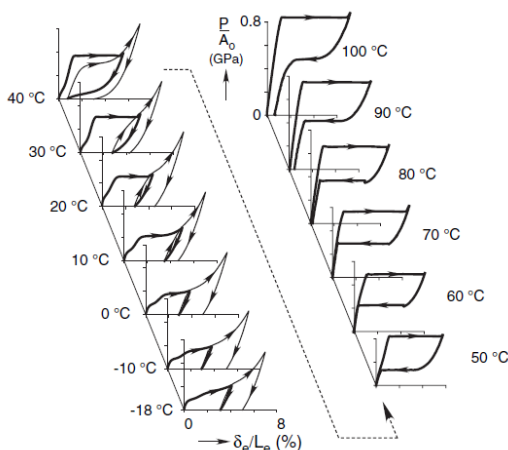
temperature (**heat exchange with the environment**). The resto of the power is divided in 2 terms:

- the first one is related to how much power (heat) is accumulated in the alloy $\rho c V \frac{dT}{dt}$, indeed is relate to density +, specific heat and volume. It's directly proportional to the gradient of temperature (**accumulated**)
- The other term is the energy used to transform the material, so to transform one phase to the other $\rho V \Delta H \frac{d\xi}{dt}$. Indeed, it's obtained by the latent transformation heat H, and the martensitic fraction (**spent in transformation**)

The problem with this kind of approach usually is related to h. This parameter is related to how much and how the heat is transferred to the environment and with the interaction with the environment. This parameter h is very well defined for very simple geometries, in other cases is not. It's almost impossible to have a very clear definition of the convection heat transfer coefficient.

If you have the case of on-off activation, open loop, well know environment and a complex geometry. In this case there is a coiled shape → you can assume that it is a cylinder, but this implies that the heat is constant all around and it's assumption doesn't work at all. Another possibility is to use complex solutions, like finite element analysis even just to simulate what happens from the

point of view of distribution of the temperature. In this case it's shown the stress-strain graph at different temperature.



- Until 50 degrees, it's also impossible to see the presence of austenite phase.

- A very particular phenomenon is that: when the loading condition is released at temperatures higher than 90, it doesn't go to zero. It's goes to something that is a little more than zero. Remember 2 things: the first one is that it's not precisely zero, and the second is that this phenomenon shows that the loading condition is exceeded.

In this case the loading condition for transforming the austenite in martensite by stress-induction is too much and the material remains with a permanent part deformed (this is not recoverable and it's visible on the plot).

This is the case of: on-off or modulated activation, closed loop, simple or complex geometry and well know or unknown environment. Now let's try to close the loop, what is the parameter that we can use to close the loop? We need to monitor one parameter that is indicative of what happens to the alloy. There are different possibilities:

- Use external variables feedback, like displacement or force. But it is likely to cause overheating. So, this is not a good idea.
- Internal state variable feedback can also be used. They are either temperature or electric resistance. Temperature is a good one, especially if you are inducing yourself a temperature change but if, instead you are using an electric input and you want to know

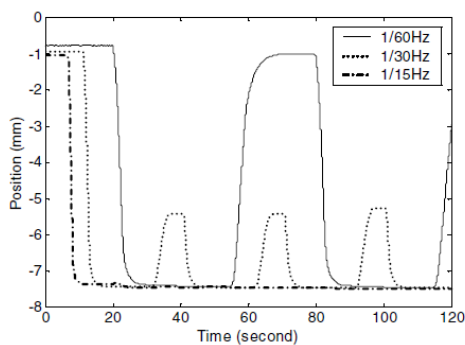
how much the temperature is, it's very complex to measure because it's difficult to do at micro-scale level.

So, **electric resistance, is a good possibility** and it's usually used. The only drawback is that the electric resistance is directly proportional to the microcrystalline structure → a problem is that we can have the same structure in 2 different moments, when it's a low temperature and at high temperature when austenite passes to martensite for the stress. There are 2 conditions with the same resistance that cannot be differentiated. The condition (low temperature and permanent deformation or high temperature but stress-induced) is something that you cannot know a priori unless you have the knowledge about other parameters. This depends a lot on your application- if you know the stress you can calibrate it. But in other systems, for example a multi-purpose system that should operate in unknown condition, the resistance is impossible to use unless you know stress conditions.

DRIVING METHODS:

There are basically 2 possibilities, one is the continuous PID controller and the other is the PWM PID controller. PWM is the same that we saw for servomotors, it works in modulation and the signal is modulated only in width: the amplitude is always the same, and the width changes every time. The parameters that are used to control the overall power are duty cycle and frequency.

- Duty cycle ($D = \frac{\tau}{T}$) is the duration of the signal active. How much time the signal is up respect the overall trigger. When the duration of activation is maximum, the D is equal to 1, it's a continuous signal. If the duration of activation is half, D is 0.5.
- Frequency. While duty cycle modulates the signal shapes, the frequency modulates how



much times in a certain period, the signal with this duty cycle is given to the alloy. In the image is possible to same the same alloy and the same signal with 3 different frequencies. The one with higher frequency is activated at the beginning, and then remains always active → so, the position (how much the alloy is deformed) is proportional to the power that is provided and the heat that the alloy experiences is continuous, so basically the

temperature is maintained constant by the signal provided every 15 seconds. Instead, increasing the time, so every 30 or 60 s, this increases also the position, because from one signal to the next, the alloy has enough time to cool down (lose capability to maintain position). PWM is indeed the standard to activate the memory effect because it's robust to disturbance, saves energy and it's easy to implement in any kind of processes.

PiezoElectric Actuators

Introduction

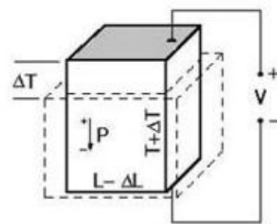
Piezoelectric are a type of active actuation that can increase or decrease the energy level of the system in which they are inserted.

History

- Piezoelectricity was first observed in some natural materials (quartz, tourmaline and Rochelle salt), which exhibited electrical polarization as the result of an applied mechanical load by Pierre and Jacques Curie in 1880.
- Lippmann predicts the inverse piezoelectric effect in 1881, and the Curies verify this experimentally.
- The first engineering application of piezoelectricity was devised by Langevin, who laid the foundations of ultrasonic submarine detection in 1916 (quartz-steel sandwich transducer).

Definition

- **Piezoelectric effect:** is the change in a material's polarization when it is subjected to mechanical stress, meaning the mechanical energy applied to the material is transformed into electrical polarization. This phenomenon also has an inverse effect, where applying polarization results in a physical deformation of the material.

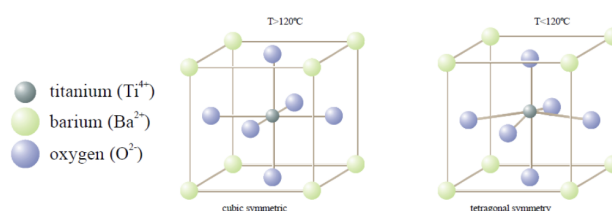


It is a material that cannot conduct electricity, meaning that applying an electric potential does not generate a current. However, it is distinct from a standard insulator. The key difference lies in the material's ability to be polarized by an applied electric field; dielectric materials exhibit a much stronger polarization than basic insulators. This capability to be polarized is quantitatively measured and defined by the electric displacement ($\vec{D}$), where ϵ is the dielectric constant.

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} = \epsilon \epsilon_0 \vec{E}$$

NOTE: that a material being dielectric does not automatically mean it is piezoelectric.

- **Ferroelectricity:** This is the capability of a material to be polarized spontaneously.

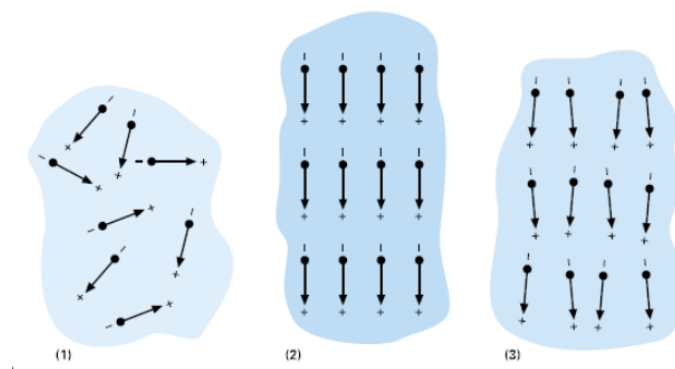


- Below 120°C , this material has a slightly asymmetric structure, which creates a polarization. The internal titanium atom is not aligned with the oxygen plane. We use 120°C here because it is specific to this material, but more generally, this threshold value is called the **CURIE POINT**.
- Above 120°C , the symmetry is recovered, so there is no longer any spontaneous polarization.

NOTE: that being ferroelectric does not necessarily mean being piezoelectric.

The core idea is to combine both dielectric and ferroelectric properties. This provides the ability to change the polarization by applying an external field (dielectricity), while also maintaining the spontaneous polarization provided by ferroelectricity.

- **Spontaneous polarization:** The crystal structure of the *dielectric* material is such that in the absence of external electric fields the centers of positive and negative electrical charges do not coincide. If this polarization can be modified when an external electric field is applied, the material is said to be *ferroelectric*. Normally a ferroelectric material have all the domain randomly positioned so the net effect is null.

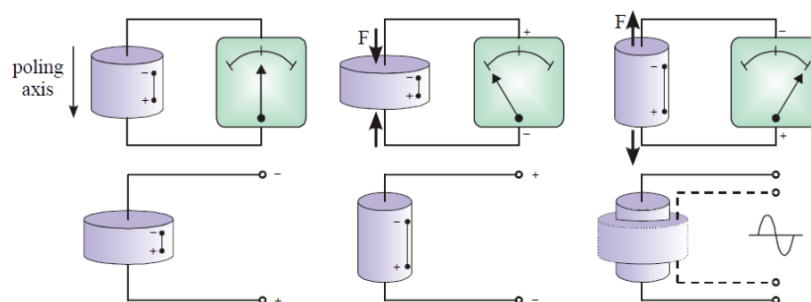


In order to act as a piezoelectric, we must use an external electric field called a **poling field**. This allows us to align the domains, giving us the ability to reorient them later through the application of an external electric field, which in turn deforms the material itself.

The *process of poling* involves heating the material to a relatively high temperature and applying an electric field. The high temperature essentially nullifies the initial polarization, allowing the applied field to easily reorient the domains. When the material is cooled back down to room temperature, the domains remain permanently aligned. From that moment on, applying an external electric field causes a variation in the material's shape.

NOTE final formula: Piezo = dial + ferro + polign process.

- **Dual effect:**



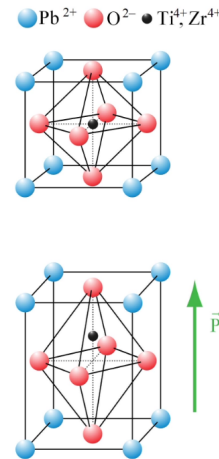
- **DIRECT:** On the right, you have the poling axis, which is the axis along which the domains are aligned. By applying a force in this direction, such as a compressive force, you change the orientation of the domains. This creates a measurable polarization within the material, a behavior typically associated with sensors.

- **INVERSE:** Depending on the direction of the applied electric field, you may induce deformation in one direction or the other, and naturally, this deformation can be cycled.

NOTE: We are more interested in this effect because it relates to an actuator. From now on, when we discuss the piezoelectric effect, we will be referring to the inverse one.

Materials

Structure	Material
Single Crystals	Quartz Lead Magnesium Niobate (PMN-PT and PZN-PT)
Ceramics	Lead Zirconate Titanate (PZT) Lead Metaniobate (LMN) Lead Titanate (LT) Lead Magnesium Niobate (PMN)
Polymers	Polyvinylidenedifluoride (PVDF)
Composites	Ceramic-polymer Ceramic-glass



The one on the right is a crystal of PZT, which stands for lead zirconate titanate. It is a ceramic material commonly used for piezoelectricity, similar to what we have seen with nitinol in SHA. In general, alongside different types of lead-based ceramics, there is also at least one piezoelectric polymer, which is PVDF.

Pros:

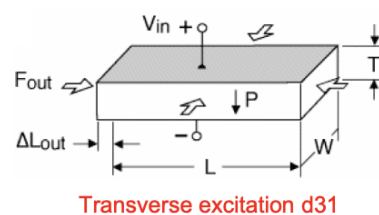
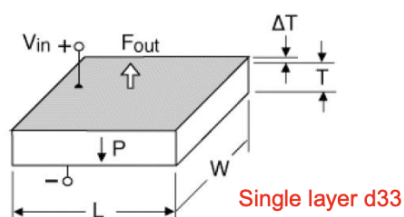
- High force and torque, resulting from the direct deformation of the material itself.
- Fast response time; the polarization effect is very quick. This is why these types of materials are commonly used at very high frequencies.
- No magnetic fields required or generated.
- Low inertia, because there are no moving parts.
- Very strong in compression, as they are generally made of ceramic materials.

Cons:

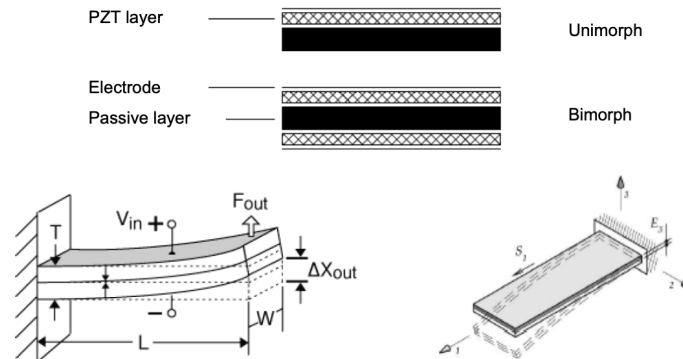
- Small deformation (0.1% to 0.18%), invisible to the naked eye.
- Relatively low stability (degrades gradually depending on the composition and manufacturing process).
- Electrical limitations (exceeding the breakdown voltage causes dielectric breakdown), mechanical limitations (very weak in tension), and thermal limitations (heating past the Curie point causes a loss of strong polarization).

Deformation modes

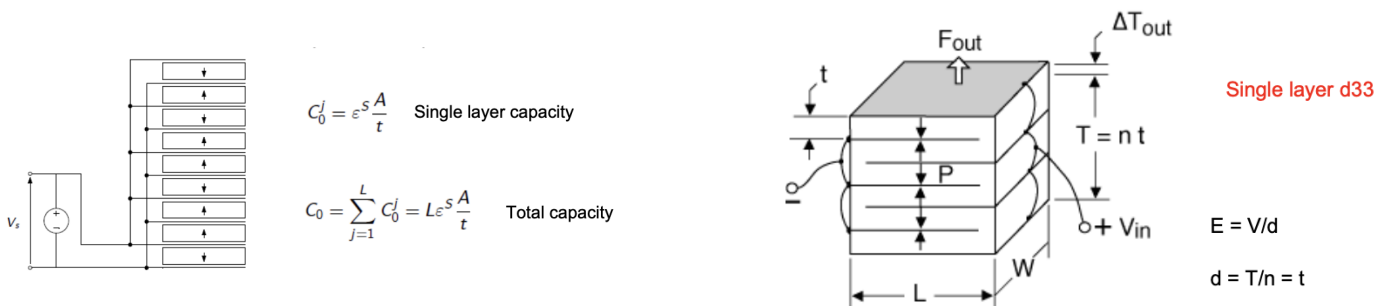
- **Longitudinal and transverse excitation:** First one is the simplest one, apply the field and on the same direction you have the deformation. Second one is when the polling field that would make the piezoelectric effect appear is moving to the other direction.



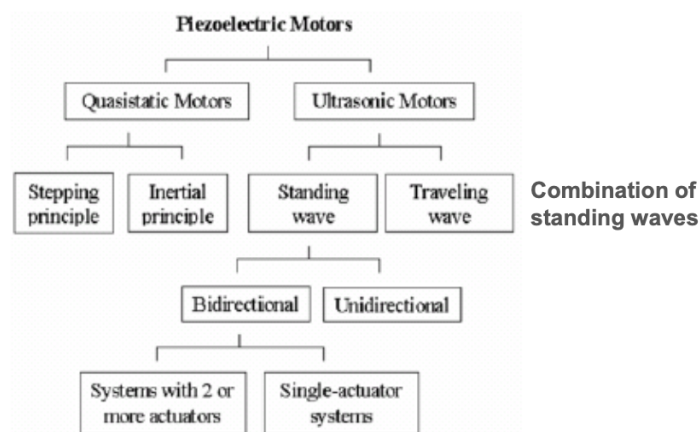
- **Bending:** Here you are using mono or multi-layer laminates that bends with an electric field. In this one you can transform the monoaxial deformation of materials into a bending motion.
 - **UNIMORPH:** where you have a piezoelectric layer combined with a passive layer that makes the deformation asymmetric.
 - **BIMORPH:** there are two piezoelectric layers and you can exploit this asymmetry in both direction.



- **Stack:** These are mono- or multi-layer laminates that deform (expand or contract) when an electric field is applied. The goal is to combine the displacement of multiple individual elements. Electrically, the layers are connected in parallel and driven by a single voltage source, which is highly advantageous for activation. Mechanically, however, the layers are arranged in series. Because of this mechanical series configuration, the absolute deformation (total stroke) increases by summing the displacement of each layer, but the relative deformation (strain percentage) remains the same. From a force perspective, there is no amplification; because they are mechanically in series, the maximum output force of the entire stack is equal to the force of a single layer.



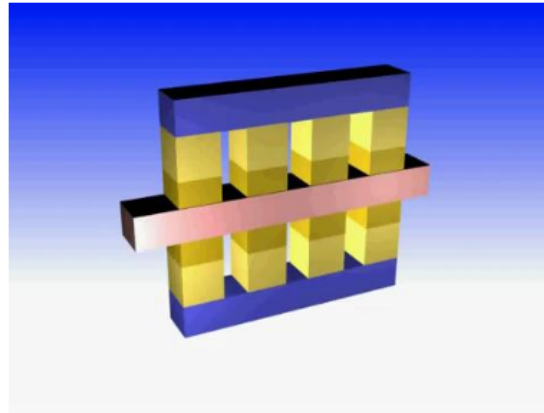
Piezo-motors



Non-resonant (quasi-static piezoelectric actuators)

Stepping

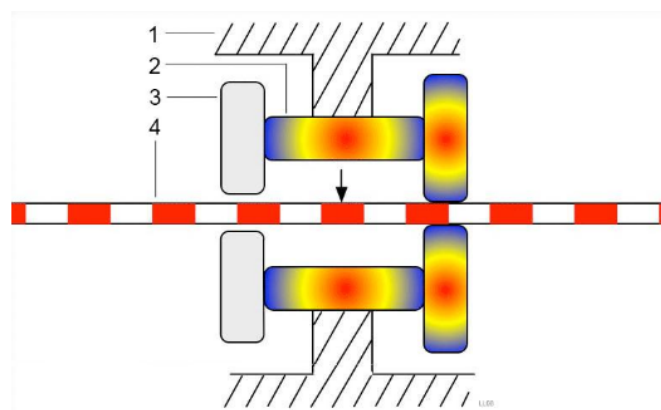
The rotor or slider is passed hand-over-hand between two sets of clamping and driving actuators. At any point in time, the set of actuators that is moving in the drive direction is the set that is clamped to the slider or rotor, while the other set is unclamped and moving backwards, preparatory to taking a new bite.



Inchworm

A similar implementation is the inchworm motor, which uses three independently driven piezoelectric ceramics: two of them clamp the shaft (the red and white pole), while the third elongates to move the shaft laterally. The operating cycle is as follows:

- The first ceramic (1) clamps the rotor.
- Ceramic (2) is actuated in axial mode.
- Ceramic (3) is actuated to clamp the rotor, similarly to (1).
- Ceramic (1) relaxes, releasing the rotor.
- Ceramic (2) relaxes, axially displacing the rotor.
- Ceramic (3) relaxes.



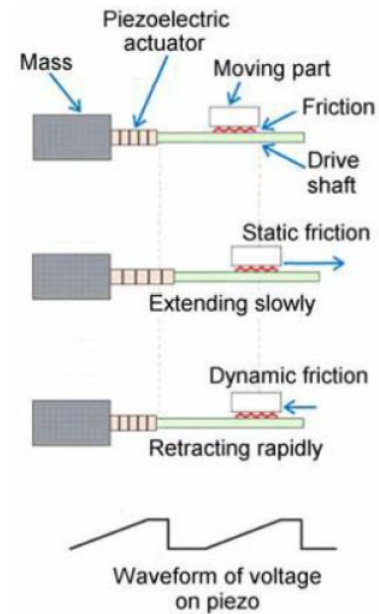
NOTE: This operates under quasi-static conditions, meaning it does not rely on dynamic time-domain effects. You can stop and resume the motion at any time regardless of velocity, relying solely on the intrinsic characteristics of the piezoelectric material.

Inertial

This exploits an inertial principle. There is a stable *mass* and a *piezoelectric actuator* (such as a stack capable of extending and retracting), meaning nothing is completely rigid. Attached to the tip of the stack is the *drive shaft*, which is in contact with the *moving part*.

PS: I am not entirely sure, but he referred to the mass and piezo assembly as the **STATOR** (defined as the stationary part of a machine that contains moving components).

Friction is crucial because it drives the movement. The stack can be actuated to extend very slowly, ensuring that static friction is sufficient to maintain contact between the drive shaft and the moving part. As a result, the moving part is displaced along with the piezo. Once the stack reaches its maximum deformation, it is retracted very quickly; this sudden movement overcomes the static friction, leaving the moving part in its new position.

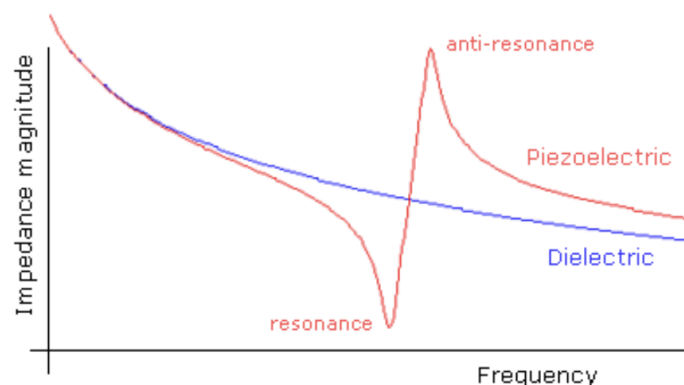


Repetition: By extending slowly and then abruptly retracting, this asymmetric waveform generates a net displacement cycle by cycle.

NOTE: The displacement per cycle is small, but the movement occurs at a very high frequency.

Resonant (ultrasonic) piezoelectric actuators

Impedance vs frequency

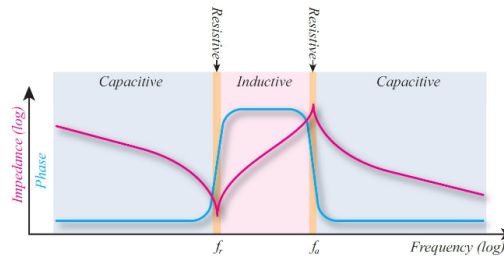
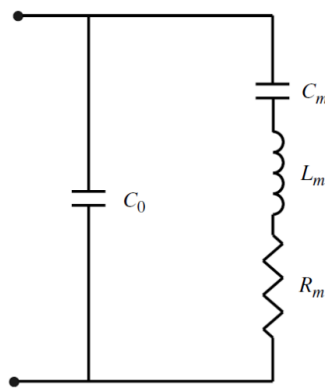


The main difference between classic dielectric materials and piezoelectric ones is how their impedance changes with the applied frequency of the electric field.

- **Dielectric:** The relationship is very smooth and single-valued.
- **Piezoelectric:** There is a lower peak corresponding to the minimum impedance, which is associated with the *resonance* frequency, and a maximum impedance peak corresponding to the *anti-resonance* frequency.

Equivalent circuit

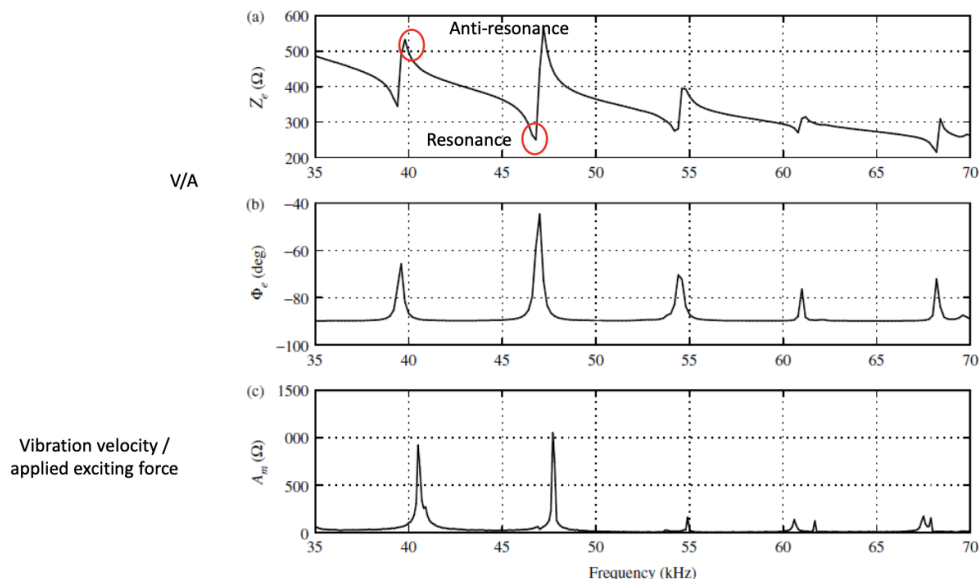
Equivalent circuit



- C_0 Clamped capacitance – capacitance of the piezoelectric ceramic - Stator
 - C_m Motional impedance
 - L_m Motional inductance
 - R_m Motional resistance
- } Rotor properties

We can also study this behavior using an equivalent circuit, where a clamped capacitance (C_0) represents what happens at the stator level, and a series branch ($C_m - L_m - R_m$) represents the rotor part. Around the resonance and anti-resonance frequencies, the phase shifts up and down. As a result, there is no phase delay in the regions outside the band between resonance and anti-resonance.

NOTE: I didn't understand the part above at all, but in short, the concept is simple. The system behaves like a *capacitor* (no phase delay) in the outside regions ($f < f_r \wedge f > f_a$), like an *inductor* in the middle region (high phase delay), and like a *resistor* exactly at the resonance and anti-resonance frequencies. This defines the electrical behavior, which we need to understand to choose the correct operating frequency based on how we want the actuator to perform.



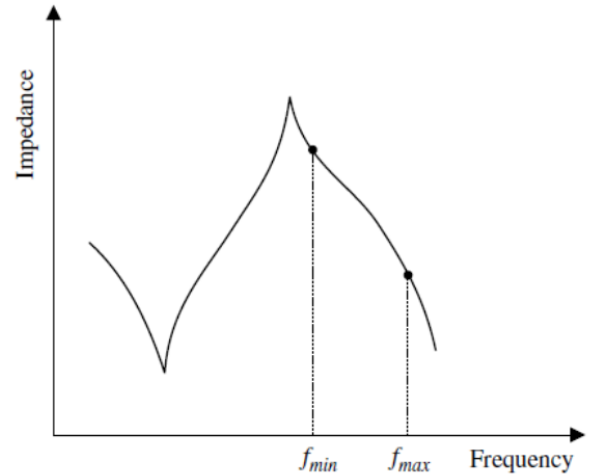
The peaks repeat cycle by cycle and in the graph you have the amplitude impedance ($Z_e (\Omega)$), the imp. in term of phase ($\Phi (\text{deg})$) and the mechanical impedance ($A_m (\Omega)$).

Operation point

Quality factor:

$$Q = \frac{\sqrt{L_m/C_m}}{R_m}$$

The material is frequency-dependent, and we want to find the optimal range and frequency to achieve maximum performance. Quantitatively, this factor is essentially a measure of the transduction efficiency from electrical to mechanical energy. Based on calculations from real data, the best frequency range for controllable and highly efficient operation lies between the two peaks. In this region, the slope is smooth—avoiding the exact resonance or anti-resonance frequencies—with optimal operation occurring very close to the anti-resonance.



This is interesting because the relationship between frequency and impedance directly determines how effectively the motion of the piezoelectric element is transmitted to the moving part of the rotor. By tuning the actuation frequency, we gain direct control over the speed of the motion.

NOTE: Using frequency modulation, we can control the motor speed.

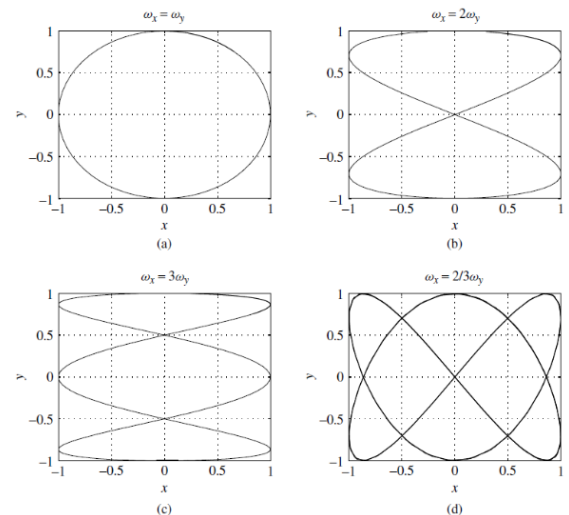
How to pass from resonance to macroscopic motion?

We want to exploit vibration and high-frequency motion to achieve the actual actuation.

In order for the actuator to convert resonance into macroscopic motion, two steps are required:

- Transform the vibration into an **elliptic** motion, rather than a linear one.
- The elliptic motion has to be **transmitted** to the moving part.

To create this *elliptic* motion, we can combine two linear motions along orthogonal axes. We need two vibrating elements with connected tips; this is exactly analogous to using sine and cosine functions to generate a circle.



The mathematical relationship describing the motion of these two elements along orthogonal axes is:

$$y = A_y \sin(\omega_y t + \varphi_y)$$

$$x = A_x \sin(\omega_x t + \varphi_x)$$

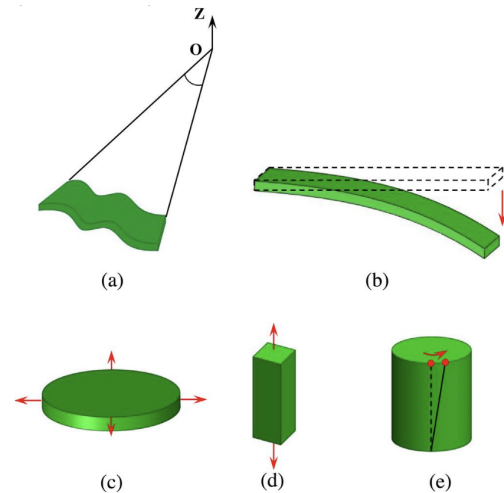
Once this is established, if you set the same amplitude and vary the pulsation/frequency ratios, you obtain what are known as **Lissajous curves**. We are specifically interested in the (1,1) case.

Standing waves

You can generate high-level deformation, by applying the field monodirectionally.

- *Flexural vibration mode(a)*: pure bending in circumferential direction.
- *Bending vibration mode (b)*
- *Radial vibration mode(c)*
- *Longitudinal vibration mode(d)*
- *Torsional vibration mode(e)*

These are the basic elements that you can create vibration with.

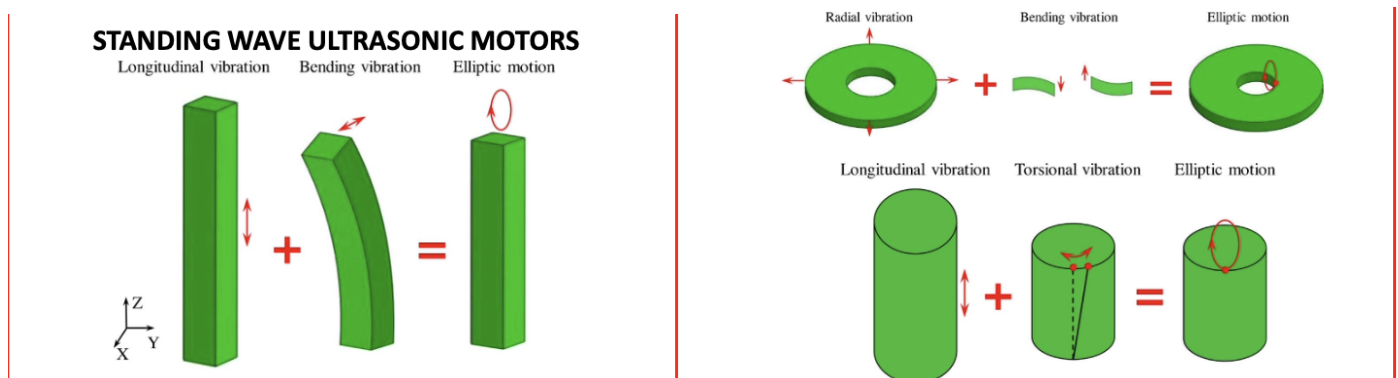


Combination of different modes to achieve elliptic motion

The figures below show these two combinations:

- Motors using resonance modes of different types at the same frequency.
- Motors using resonance modes of different types at different frequencies.

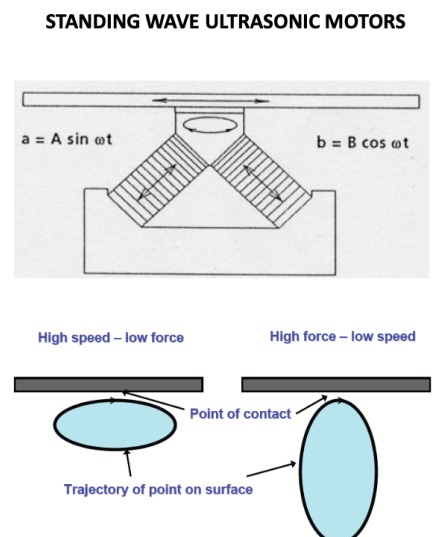
Basic resonance modes are combined at the same or at different frequencies to generate elliptic motion.



- Motors using mode separation (same modes, same frequencies, nonsymmetric geometry).

Two motions with the same amplitude and frequency generate an elliptic motion at the connection point between the two motors, which can then be transmitted to the plate.

NOTE: The generation of elliptic motion has different effects on the moving part depending on the shape of the ellipsoid (blue shape in the figure).

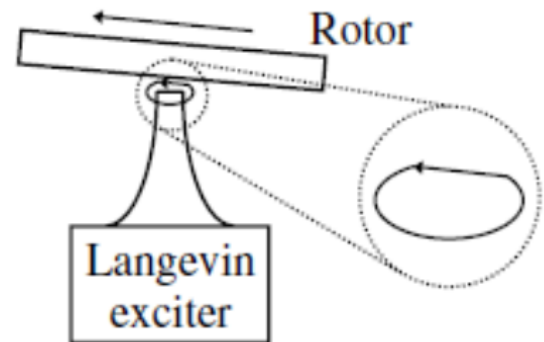


For example, if we have the ellipsoid on the left (x -axis $>$ y -axis), we obtain a higher speed because the area, and therefore the interaction between the actuator and the moving part, is larger than in the ellipsoid on the right.

NOTE 2.0: A larger interaction area means higher speed, but also lower force.

- Motor using mode conversion (single longitudinal vibration mode + impact).

It emerges from the interaction between the zoomed part, which is normally deformable, and the rotor itself. The exciter makes contact with the rotor and continues to push against it due to the inclination of the moving part. The zoomed part represents the *pathway in terms of vertical displacement*.



Explanation: The **Langevin** piezoelectric transducer vibrates by moving up and down, generating only longitudinal vibrations.

Asymmetry: The rotor and the exciter are not perfectly perpendicular to each other.

This combination creates an elliptic motion at the tip of the exciter, which drives the rotor and defines the displacement pathway.

Traveling Wave Ultrasonic Motor (TWUM)

The general idea is to combine single piezo elements to transform local vibrations into a global motion. A single vibrating piezo generates a **standing wave** (motion up and down without going laterally).

What happens if we combine multiple standing waves?

By combining standing waves with specific phase delays, the ultrasonic motor generates a **travelling wave**. As this wave propagates through the elastic substrate, it creates the **elliptical motion** on the surface needed to drive the rotor.

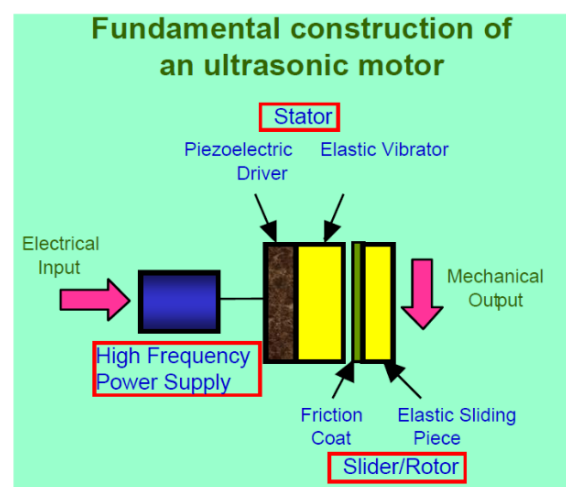
Let's start discussing the different components inside the TWUM. Obviously, there are two main parts: the Stator and the Slider/Rotor:

The first three compose the **stator**:

- **High-frequency power supply.**
- **Piezoelectric driver.**
- **Elastic vibrator.** This is important because the stator must be compliant to allow deformation.

After the stator, there is a gap followed by the different components of the **rotor**:

- **Friction coat.** It allows movement through the interaction between the stator and the rotor. This layer must be carefully engineered to ensure a precise and controlled interaction.
- **Elastic sliding piece** (the main body). It serves a similar compliance purpose as the stator's vibrator.



Focus on the Elastic vibrator

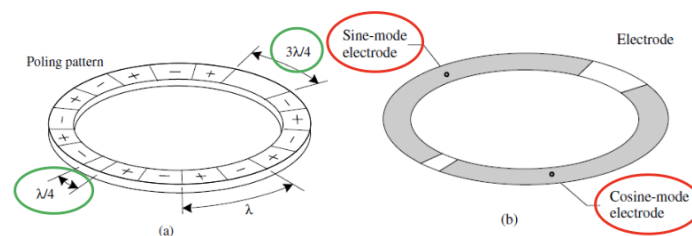
The two rings below illustrate the structure of the piezoelectric driver. They are not separate components, but rather two different views (internal and surface) of the same single ring.

Poling pattern: This shows the alternation of positive and negative polarization, indicating how the internal dipoles of the piezoelectric material are oriented. The single sectors are like the basic piezo elements studied previously, capable of generating standing waves. The key characteristic is that the pattern is not continuous; there are two gaps of $\frac{3\lambda}{4}$ and $\frac{\lambda}{4}$, which create a 90° phase delay **in space**.

The other schematic refers to the two electrodes deposited on top of the patterned material. They are grouped into a *sine-mode* electrode and a *cosine-mode* electrode to create a 90° phase delay **in time**.

NOTE: 'Sine' and 'cosine' modes refer to the type of AC current provided by the high-frequency power supply. It is literally the specific electrical signal sent to the two groups of electrodes (the two currents are 90° out of phase).

$$u(\phi, r, t) = AR(r) \cos(k\phi) \cos(\omega t) \mp AR(r) \sin(k\phi) \sin(\omega t)$$



NOTE: The crucial point is that we create a 90° phase delay **both in time and space**.

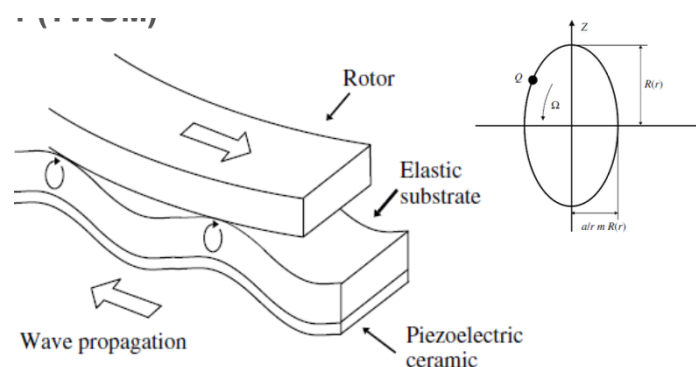
How it works?

The two patterns vibrate simultaneously. The physical connection (including the specific gaps) ensures they interact. Instead of behaving like isolated standing waves, they combine, and the wave starts to propagate, creating a **travelling wave**.

Transient time: When the current is first applied, the travelling wave doesn't appear instantaneously. The two standing waves start to vibrate and take time to synchronize. During this brief phase, the gaps do not move. However, once this transient time passes (reaching steady state), the gaps become impossible to distinguish, as the entire ring deforms continuously as part of the travelling wave.

KEY TAKEAWAY: The sectors are PHYSICALLY CONNECTED, AND THIS GENERATES THE TRAVELLING WAVE. Furthermore, IT IS THE EXACT VALUE OF THE GAPS ($\lambda/4$) THAT ALLOWS THE WAVES TO PROPERLY SYNCHRONIZE AND MERGE.

How is the motion transferred to the rotor?

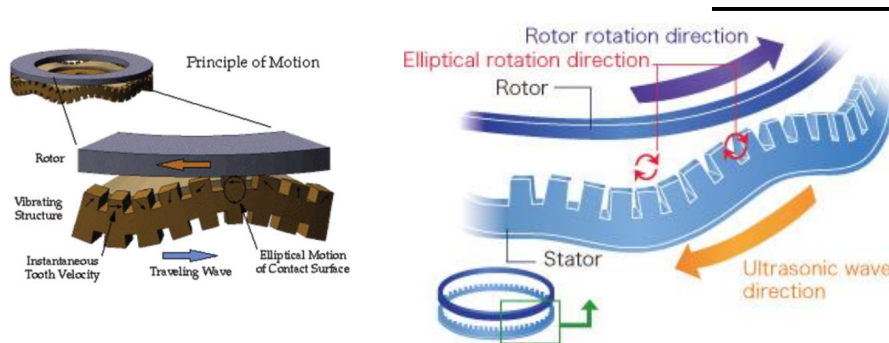


The rotor moves in the opposite direction of the wave propagation.

The piezoelectric ceramic deforms the elastic substrate, resulting in two orthogonal displacements:

- The **vertical movement** (z) comes directly from the wave's peaks and valleys.
- The **longitudinal movement** (φ) comes from the outermost surface of the material physically stretching at the peaks and compressing in the valleys.

Friction is the mechanism that transmits this combined elliptical deformation to the rotor, pushing it forward.



NOTE: It is the **thickness** of the elastic substrate that amplifies the rotation created by the travelling wave. The thicker the material, the greater the longitudinal amplification transmitted to the moving part.

Essentially, if you only had the bare piezoceramic, the longitudinal movement would be so negligible that you would only perceive the up-and-down motion typical of a standing wave.

Recommended video: <https://www.youtube.com/watch?v=8eiHTkralYY>

Figure scaling trends

Figures of merit	Nonresonant actuators			Resonant actuators
	Stack	Benders	Inchworm	TWUM
Force, F	$10^2 - 10^5$ N	$10^{-2} - 10^2$ N		Torque $\leq 10^4$ Nm
Displacement, S	0.1–0.6% (dynamic up to 1.7%)	$10^{-2} - 10$ mm	Unlimited	Unlimited
Work density, W_V	$\approx 10^{-2}$ J/cm ³	$10^{-4} - 10^{-2}$ J/cm ³	$\approx 10^{-3}$ J/cm ³	$10^{-3} - 10^{-2}$ J/cm ³
Power density, P_V	$10^2 - 10^4$ W/cm ³	$10^{-1} - 10^1$ W/cm ³	$\approx 10^{-4}$ W/cm ³	$\approx 10^{-1}$ W/cm ³
Bandwidth, f	$\leq 10^5$ Hz	$\leq 10^3$ Hz	$\leq 10^2$ Hz	$\leq 10^2$ Hz
Efficiency, η		70–90%		
Scaling trends				
Force		$F \propto L^2$		
Stroke		$S \propto L$		
Work per cycle		$W \propto L^3$		
Energy density		$W_V \propto L^0$		
Bandwidth		$f \propto L^{-1}$		
Power density		$P_V \propto L^{-1}$		

Applications

Tremor compensation

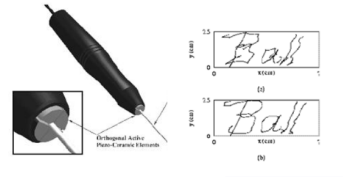
Consumer Electronics

Digital Camcorder
(Panasonic et al.)



Microsurgical device

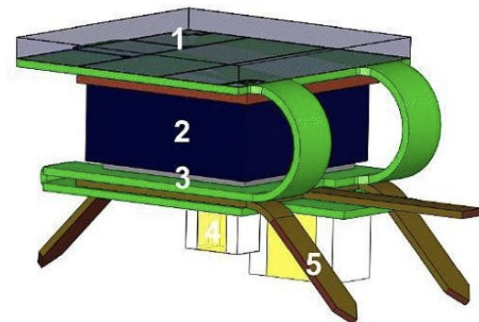
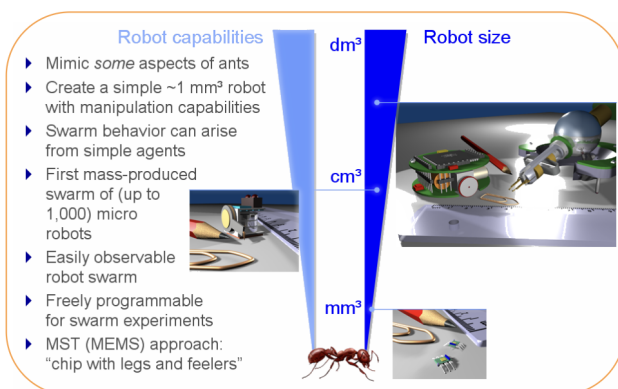
Intelligent hand-held instrument for ophthalmologic microsurgery
(MRCAS - Carnegie Mellon Univ. Robotics Institute - Pittsburgh, P.



Compensation strategy: The instrument detects its own motion, distinguishes between desired and undesired motion using active filtering and inertial sensors, and actively compensates for undesired motion by an equal but opposite deflection of its own tip.

I-SWARM PROJECT

The goal of this project is to demonstrate that a swarm of very simple elements, much like an ant colony, can exhibit highly complex behaviors. These behaviors emerge from simple rules at the individual level and the nonlinear, complex interactions between the agents.



(1) solar cell, (2) IR-communication module, (3) an ASIC, (4) capacitors, (5) locomotion module

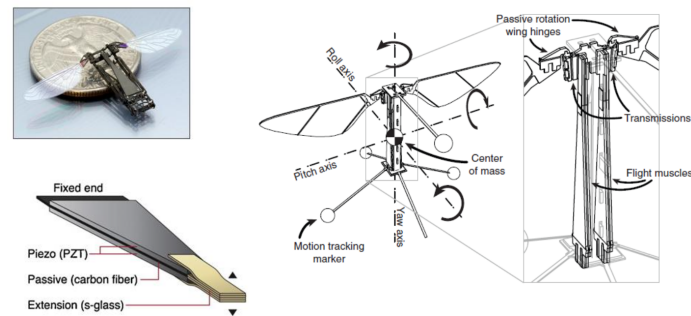
These units are extremely small and locomote entirely using piezoelectric film "microlegs". The high-frequency vibration of these legs generates a loud, high-pitched noise (**NOTE**: in fact, this is exactly why I couldn't understand a damn thing from the audio recording).

Flying insect

The core concept is relatively straightforward; the primary complexity lies in the design of the shoulder joint and the wings.

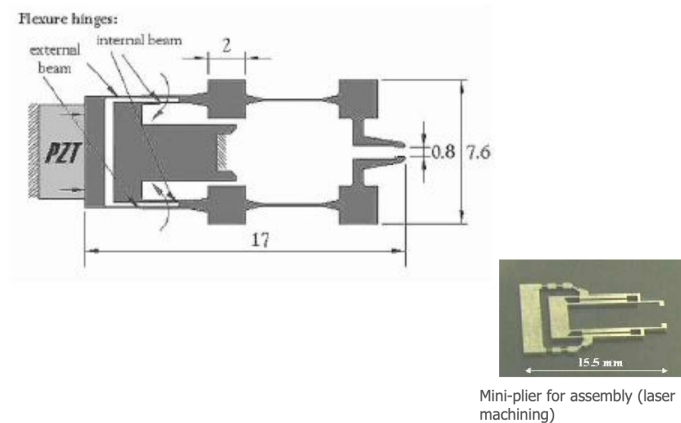
Two piezoelectric (PZT) actuators function together as a bimorph bender, causing the tip to oscillate up and down. The shoulder mechanism then transmits this vertical displacement to generate the complex flapping motion of the wings.

Flying insect



Displacement amplification mechanism

In this configuration, a PZT actuator is coupled with a kinematic chain composed of rigid links and compliant flexure joints. This transmission mechanism amplifies the initial piezoelectric displacement while ensuring highly precise actuation of the micro-gripper. Consequently, it enables the grasping of microscopic objects with exceptional accuracy.



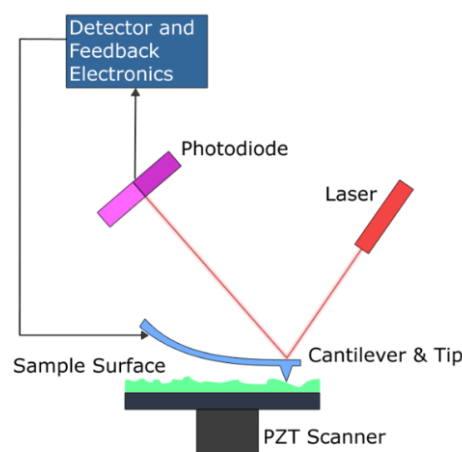
Scanning mechanism for AFM

This device is utilized to study material surfaces, specifically focusing on properties such as roughness and surface tension.

The system consists of a *cantilever* that physically interacts with the sample surface. A laser beam is focused on the back of the cantilever tip and is reflected onto a position-sensitive photodiode, which detects even the most minute topographical variations based on the cantilever's deflection.

Additionally, a *PZT scanner* is employed to translate the sample in multiple dimensions (X, Y) with extremely high positioning precision.

Atomic Force Microscope



Other Examples of Piezoelectric Applications

- **Nanomotors:** These represent the smallest class of motors currently in existence, relying on nanoscale piezoelectric actuation.
- **Auto-focus camera systems:** These systems integrate PZT elements within the stator to achieve rapid and highly precise lens positioning.
- **Optical zoom modules:** Employ miniature piezoelectric actuators for continuous and precise optical adjustments in compact devices.
- **Squiggle motor:** This type of motor is entirely non-magnetic, making it highly suitable for use near MRI systems. In medical applications, the motor's threaded screw can be used to drive a syringe plunger to precisely deliver contrast media during a scan. The driving mechanism consists of four piezoelectric actuators that bend in a coordinated, high-frequency sequence. Internally, the design utilizes either a combination of interlocking teeth or smooth, friction-based interfaces. By exploiting these mechanisms, the coordinated bending of the actuators induces rotation in the threaded shaft, which is consequently translated into linear movement.

PS:LA PARTE SULLE APPLICATION È ANDATO MOOOOOOOLTO FAST.

FFA - Flexible Fluidic Actuators

1. Introduction: what Flexible Fluidic Actuators are and why they matter

Flexible Fluidic Actuators, often abbreviated as **FFAs**, are one of the central technological families within soft robotics.

They are defined as **pneumatic or hydraulic actuators operating in a soft-bodied chamber**. This means that the active part of the actuator is not a rigid cylinder with rigid moving elements, but a compliant chamber, usually elastomeric, which changes shape when pressurized by a fluid. The fluid may be a gas or a liquid → the word **fluid** is used in the general sense, covering both pneumatic and hydraulic actuation.

This definition is important because it already contains the two essential ingredients of the technology:

- **fluidic actuation**, meaning that the input energy is delivered through pressure.
- **flexible structure**, meaning that the body of the actuator is soft and deformable rather than rigid.

These actuators should not be thought of as just “soft balloons.” A balloon expands because pressure acts normally to its membrane more or less uniformly. A flexible fluidic actuator starts from the same basic physical principle but then modifies it through design. The volumetric deformation of the chamber is **guided**, constrained, or redirected so that the expansion does not remain generic. Instead, it becomes a specific mechanical output: contraction, elongation, bending, curling, and in more advanced designs, even multi-directional or continuum motion, i.e. **the volumetric deformation of the chamber is usually driven to cause expansion toward specific directions so as to result in predefined deformation modalities**.

So, already at the most general level, FFAs are a perfect example of a soft-robotics idea: **using morphology, material properties, and structural constraints to embody part of the function directly into the body of the device**.

Before going deeply into the mechanics, let's make an important terminological clarification. In the literature, many names are used for devices that are very similar or partially overlapping. There are some terms such as:

- Pneumatic Muscle Actuators
- Fluid Actuators
- Fluid-Driven Tension Actuators
- Axially Contractible Actuators
- Tension Actuators
- McKibben
- PneuNets

- Fluidic Muscles
- Air Muscles
- Fluidic Elastomer Actuators
- PneuFlex
- Pneumatic Artificial Muscles (PAM)
- Flexible Fluidic Actuators (FFA)

This is why once the community started to explore soft elastomeric chambers coupled with fluidic actuation, many researchers and groups introduced their own terms. Sometimes these names emphasized a specific embodiment, sometimes a specific application, sometimes a branding choice, and sometimes a small design distinction. As a result, the literature became crowded with many labels → for the purpose of the course, these can be grouped under the broad family of **Flexible Fluidic Actuators**, and then specific distinctions can be introduced where useful.

2. General advantages of Flexible Fluidic Actuators

Let's begin with the overall and generic advantages of FFAs, and let's explain **why** soft fluidic actuation became so attractive in robotics and biomedical engineering.

1) Compatibility with magnetic fields and certain medical environments

One of the first listed advantages is compatibility with **magnetic fields**, and the transcript also mentions compatibility in settings relevant to the medical field. The point is that these actuators are typically based on elastomers and soft non-rigid bodies, and they avoid some of the problems associated with conventional rigid electromechanical systems. In areas where electromagnetic compatibility, safety, or low interference matters, soft fluidic systems can therefore be attractive.

2) No relative motion of rigid parts

This is one of the most conceptually important advantages. Traditional actuators (e.g. piston-cylinder actuator) often rely on rigid parts moving against other rigid parts: pistons sliding inside cylinders, gears meshing, bearings rotating, linkages pivoting around joints. In these cases, motion is produced through components that move relatively to one another while in rigid contact.

In FFAs, this is often absent or strongly reduced. The body itself deforms, instead of consisting of many hard parts moving against each other. This has several consequences. It reduces classical sources of friction, inertia concentration, heat generation, and mechanical wear. In conventional devices, repeated motion leads to friction, heat, possible particle release, lubrication needs, and degradation after many cycles. In flexible structures, these issues are usually much less dominant because the mechanism is based on distributed deformation rather than rigid contact mechanics.

3) No lubricants, no leakages of the classical kind, no wear particles

Closely related to the previous point, FFAs usually avoid some of the maintenance and contamination issues of rigid mechanisms. The slides explicitly mention **no lubricant, no leakages and no wear particles**. The spirit of this statement is not that every real soft actuator is magically free of every engineering issue, but that, compared with mechanisms involving rubbing rigid parts, FFAs can avoid

the classical lubrication–wear–particle chain. This is especially attractive in medical robotics, delicate interaction tasks, and environments where contamination is undesirable.

4) No friction, or at least much less relevant friction in the classical sense

The slide lists “no friction,” while the transcript speaks more carefully about the absence of friction due to rigid parts sliding against each other. Strictly speaking, some internal dissipative effects still exist in real soft systems, and later in the lecture we even mention advanced models that explicitly account for friction in McKibben actuators. So the right interpretation is that FFAs avoid **the dominant rigid-contact friction mechanisms of conventional actuators**, not that every physical dissipative phenomenon disappears. This is one of those points where the transcript helps interpret the slide in a more realistic way.

5) Smooth and complex motion

Traditional rigid robots often have motion defined by discrete joints and links. This gives excellent precision and predictability, but it also tends to constrain motion into a set of relatively simple geometric primitives: rotation around a joint, translation along a guide, etc. Flexible fluidic actuators instead can produce **smooth, continuous, and highly shape-adaptive deformation**.

We can contrast rigid systems, which often move mainly along one axis or one degree of freedom at a time, with flexible structures, where “the sky is the limit” in terms of possible deformation patterns, as long as the design is done properly. This is a key advantage in continuum manipulation, soft gripping, wearable robotics, and bioinspired locomotion. The motion is not necessarily better in every possible sense, but it is often **richer** and more naturally compatible with curved, compliant, deformable, and uncertain environments.

6) Mechanical, chemical, and electrical safety

The slides summarize this as **mechanically, chemically and electrically safe**. The transcript elaborates: many of these devices are built from insulating materials, so they reduce electrical hazard; they are often chemically stable; and mechanically they are compliant, so they interact in a less dangerous way with humans and delicate environments. This is particularly important in soft robotics because compliance is not just tolerated, but actively exploited as a functional property.

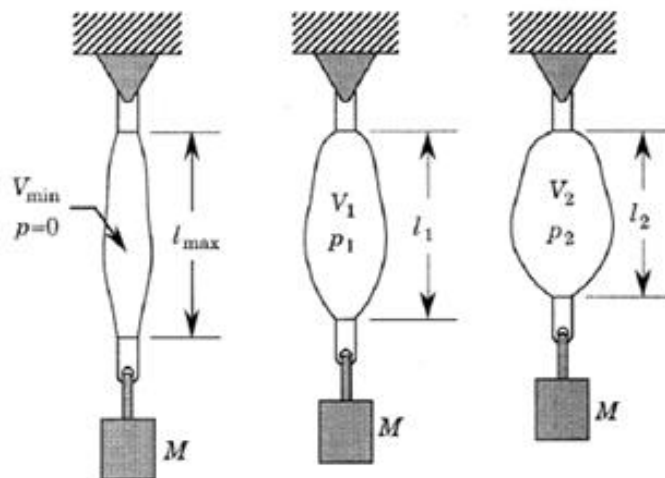
Overall, the general-advantages section serves to position FFAs not merely as a curiosity, but as a major actuation technology for domains where compliance, safety, distributed deformation, and smooth body–environment interaction matter.

3. The fundamental physical principle behind all FFAs

The starting point is extremely simple: if you have a **pressurized fluid source** and an **elastomeric chamber**, then pressurizing the chamber deforms it. That is the common physical basis of all FFAs. If the chamber were just a balloon, pressure would act normal to the membrane everywhere, and the result would simply be a general expansion. (the balloon example precisely is the easiest way to picture what happens when pressure has no preferred direction).

But that is only the starting point. The real engineering step is to **add features** that make pressure no longer equivalent everywhere in terms of global effect. These features may include:

- inextensible fibers,
- braided sleeves,
- wall patterns,
- corrugations,
- composite layers,
- eccentric chamber geometries,
- different wall thicknesses,
- asymmetries in material stiffness,
- multiple chambers with independent inflation.



Once such features are introduced, pressure still acts locally normal to the wall, but the **global deformation is no longer uniform**. Instead, the chamber's volumetric expansion is redirected toward selected directions. This means that the same input—fluid pressure—can produce radically different outputs depending on **how the body is designed**.

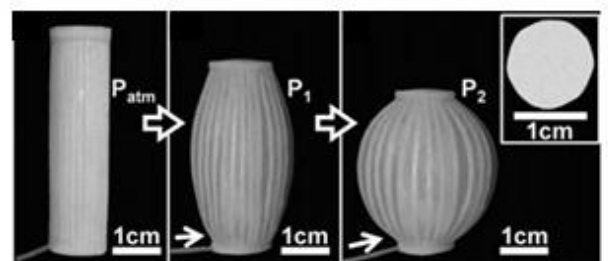
This is one of the most important messages of the lecture: the behavior is not “coded” only in the controller or in the pressure value; it is **embodied in the morphology** of the actuator. Pressure is the energy source, but morphology is what transforms that energy into a desired kinematic pattern.

4. First classification of motions: contraction and elongation

The first motion family considered in the lecture is axial deformation: either the actuator shortens or it lengthens.

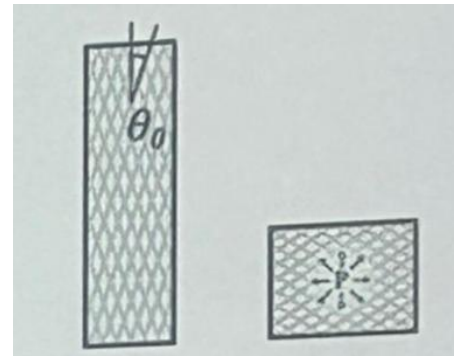
How is contraction obtained?

One way to obtain **contraction** is to use a **cylindrical elastomeric chamber** reinforced with **inextensible helical fibers**. When the chamber is **pressurized**, the internal pressure tends to make the cylinder expand **radially**. However, because the reinforcing fibers **cannot stretch**, this radial expansion is **geometrically coupled** with a change in the **axial length** of the actuator. In other words, as the **radius increases**, the fiber geometry forces the overall structure to **shorten along its longitudinal axis**.



The most emblematic example of this principle is the **McKibben pneumatic actuator**, which will be studied in detail later. In a McKibben actuator, contraction is **produced by an inextensible**

braided sleeve wrapped around the elastomeric chamber. The direction of motion depends on the **braid angle**, defined as the angle between the fibers and the **longitudinal axis** of the actuator. When this angle is **below approximately 54.7°**, pressurization produces **axial contraction**. At about 54.7°, the actuator is close to a **neutral configuration**, while for **larger angles** the same mechanism may lead to **axial extension** instead of contraction. Therefore, **McKibben-type contraction** is associated with **braided helical fibers** arranged at a suitable angle, rather than with **purely longitudinal reinforcement**.



The main idea is therefore:

- pressure tends to enlarge the chamber;
- geometric constraints prevent arbitrary free expansion;
- constrained expansion is converted into shortening.

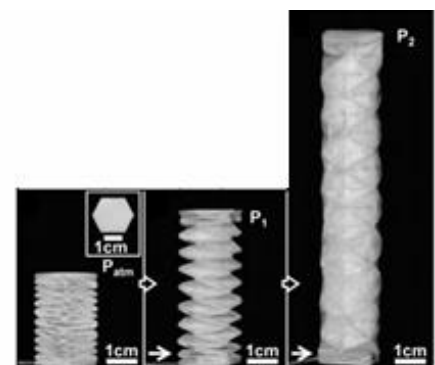
That is why these devices are often called “artificial muscles”: they can generate tension by contracting under pressure.

How elongation can be obtained?

At first sight, it may seem intuitive that internal pressure should always make a chamber become “fatter” and, if its radial expansion is constrained, also shorter. However, the opposite behaviour is also possible: pressure can produce **elongation** instead of contraction.

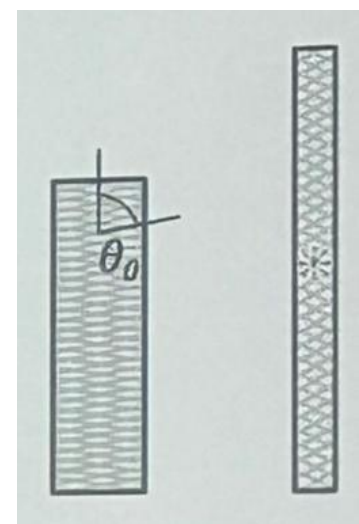
This can be achieved by designing the chamber and its reinforcing structure so that pressurization preferentially unfolds, deploys, or extends the actuator along its longitudinal axis, rather than mainly expanding it radially. Examples include

corrugated membranes, **folding structures**, and **origami-inspired designs**. In these cases, the pressurized chamber is shaped so that the dominant deformation mode is an **increase in axial length**.



A similar geometrical principle also appears in McKibben-type actuators. In the classical McKibben model, the direction of motion depends on the **braid angle**, defined as the angle between the braided fibers and the longitudinal axis. When the braid angle is below approximately 54.7°, pressurization produces **axial contraction**. When the braid angle is above approximately 54.7°, the same geometrical mechanism may instead favour **axial elongation**.

TO SUM UP: pressure alone does not determine whether an actuator contracts or elongates. The resulting motion is mainly determined by the **geometry of the chamber**, the **arrangement of the reinforcing**



fibers, and the **constraint structure**. This is why geometry and material arrangement are central in **fluidic flexible actuators (FFAs)**.

5. Bending: the importance of breaking symmetry

After contraction and elongation, let's talk about bending, which is one of the most common motions in soft robotics.

The key rule is simple, but really important: **to obtain bending, you must break symmetry** (e.g. if the actuator deforms equally on both sides, it will expand or elongate symmetrically → To produce curvature, one side must effectively extend more than the other).

There are several ways to achieve this:

- **Inextensible layer**

A very common design is to place an **inextensible but flexible layer** on one side of the actuator. When the actuator is pressurized, the unconstrained side can extend more, while the constrained side extends less. The result is a curved shape.



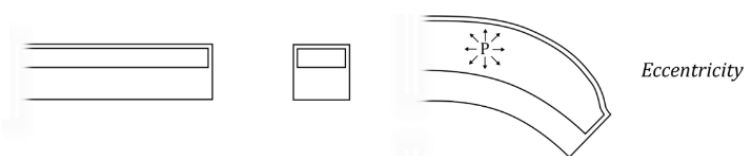
- **Oriented fibers**

Instead of a full inextensible layer, one can also use **fibers** arranged so that they are flexible in the transverse sense but not extensible along the axis. This produces a similar asymmetry: one side is structurally constrained more than the other, and a bend appears under pressure.



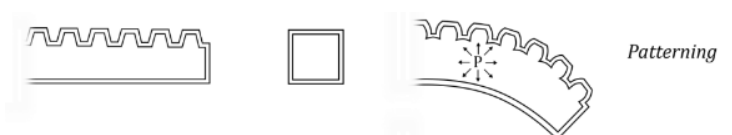
- **Eccentric chamber geometry**

Another possibility is to play only with **geometry**, without changing materials. If the internal cavity is **eccentric**—so the wall thickness is not symmetric—the thicker or stiffer side deforms less, while the thinner side deforms more. Again, the result is bending. → asymmetry can come from **material design** or purely from **shape design**.



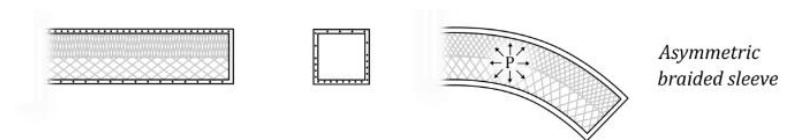
- **Patterning**

The chamber walls themselves can be patterned so that one side unfolds or expands more readily than the other. Corrugations, ribs, and repeated geometric features are examples of this. In such a case the asymmetry is not due to a second material, but to different structural response on the two sides.



- **Asymmetric braided sleeves**

The symmetry-breaking also extends logic to braided structures. If one side of a braided sleeve has a different braiding angle from the other, the two sides respond differently to pressure. One side may have greater elongation tendency than the other, and the interaction between them produces curvature.



The key conceptual conclusion is that all these are just different embodiments of the same rule: **bending is obtained whenever pressure-induced deformation is made asymmetric.**

6. McKibben actuators

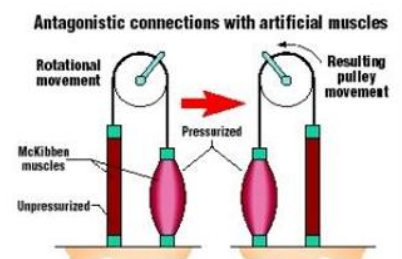
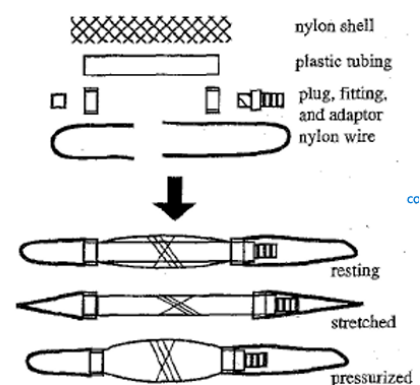
We now move our attention on the McKibben actuator, which is probably the most fundamental example of a flexible fluidic actuator.



History facts:

PAMs (pneumatic artificial muscle) were first developed in the 1950s as **McKibben Artificial Muscles** for artificial limbs, especially orthotic appliances for polio patients, and that Bridgestone commercialized the concept later, in the 80s, under the name **Rubbertuators**.

They were a discovery of Joseph Lawrence McKibben since his daughter's hand had been paralyzed by polio. He developed a device to help restore grasping ability. This historical note is relevant not just as background, but because it places McKibben actuators in a biomedical, assistive origin, which resonates strongly with modern soft robotics.



Structure of the actuator

A McKibben actuator is made of:

- an **inner elastic tube or bladder**, which is airtight and expands under pressure; this internal tube provides airtightness and elasticity;

- an **outer braided sheath**, typically made of inextensible fibers; this external braid determines the kinematics of transformation;
- end fittings that fix both the bladder and the braid so that they do not move independently.

NOTE: although the bladder is necessary because it contains the fluid and provides compliance, the **deformation kinematics are mainly dictated by the external braid**. The bladder supplies the pressurizable soft chamber; the braid decides how that chamber is allowed to deform.

When the chamber is pressurized, the internal tube tends to expand. The external braid constraints this expansion and converts it into axial motion. Depending on the braiding angle (we will see more in the details later in the lecture), the actuator may contract or elongate.

That is why, conceptually, the McKibben actuator is such a pure illustration of embodied mechanical design: the actuator behavior is literally encoded into the geometry of the fiber network.

Let's now develop a **geometrical model** to understand how the actuator behaves, before introducing force.

Assumptions of the model

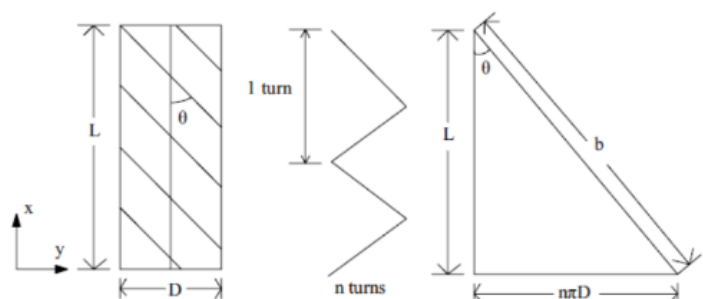
We have to list four strong assumptions:

1. the actuator is cylindrical in shape;
2. the sheath threads are inextensible and always in contact with the bladder;
3. friction between bladder and sheath and between fibers is negligible;
4. latex tubing forces are negligible.

All these assumptions are restrictive. The actuator is not perfectly cylindrical at the ends, friction is not really zero, and the bladder is not truly force-free. But this simplified model is still extremely useful because it captures the main geometrical law of the braid.

To derive the geometry, one fiber of the braid is conceptually “unwrapped” from the cylinder and drawn as a straight segment in a triangle. In this idealization:

- b is the length of the fiber,
- L is the actuator length,
- D is the diameter,
- n is the number of turns,
- θ is the braiding angle.



From this triangle, one can relate the fiber length b , the actuator length L , the diameter D , and the number of turns n .

A common geometric relation is:

$$b^2 = L^2 + (\pi Dn)^2$$

and

$$L = b \cos \theta$$

From this, the diameter can be expressed in terms of b , L , and n .

$$D = \frac{b \sin \theta}{n\pi}.$$

Then the actuator volume is written as

$$V = \frac{b^3 \cos \theta \sin^2 \theta}{4n^2\pi}.$$

These equations are central because they connect actuator length, diameter, and internal volume through one single design quantity: the **braiding angle**. This allows us to study how volume changes as the actuator changes shape.

IMP: when length changes, diameter changes too, but **not according to an arbitrary or linear law**. The braid imposes a specific geometric coupling. It's important to not confuse the behaviour of a braided actuator with that of a bulk incompressible elastomer. They are not the same thing. In an incompressible body, volume preservation would impose one law (related to a Poisson ratio close to 0.5); in a braid-constrained pressurized chamber, the relation is different because the inextensible fibers impose their own kinematic constraints.

This difference is important because if a solid silicone cylinder is inserted inside a braided structure, the two components may impose **incompatible** deformation laws. The braid wants to deform according to one relationship between length and diameter, while the elastomer wants to deform according to another. The system can become over-constrained.

Therefore, the chamber inside a McKibben actuator must be compliant enough to follow the kinematics imposed by the braid.

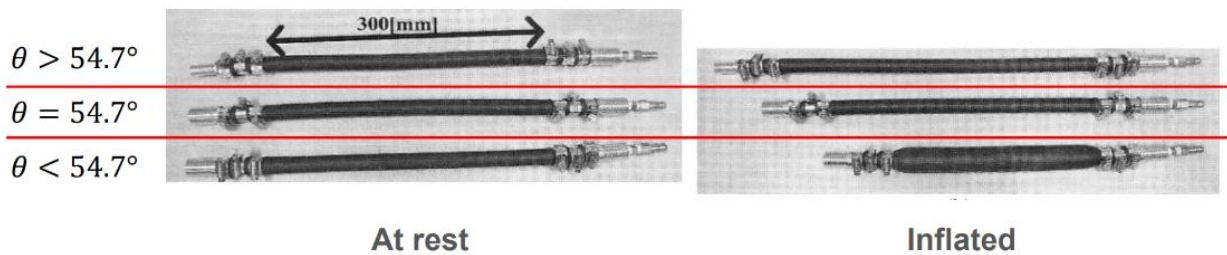
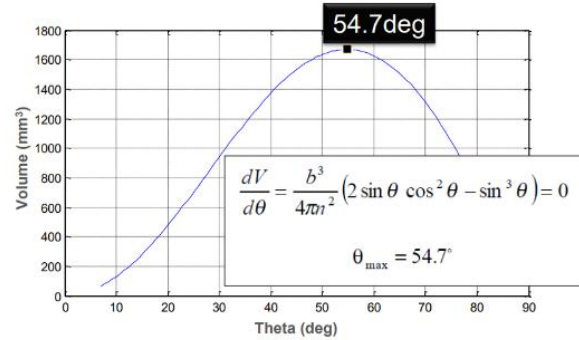
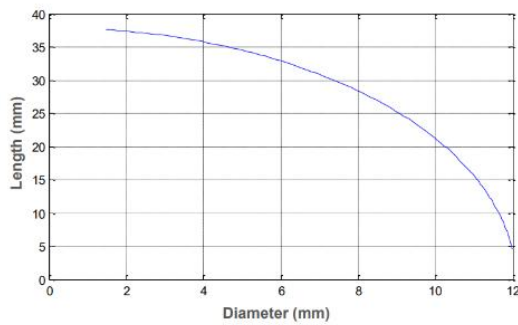
This is also important because it explains why the braid is **not just reinforcement**: it is a genuine geometric mechanism.

The importance of the angle of 54.7°

One of the most famous results is that the internal volume of the McKibben braid has a maximum when the braid angle is approximately **54.7°**. → This number is independent of the specific dimensions and proportions of the actuator.

If pressure is increased, the actuator tends to move toward a configuration with **larger internal volume**. Since the volume reaches its maximum at 54.7°, the system tends to evolve toward that angle.

From this, three cases follow.



If $\theta > 54.7^\circ$

If the initial braiding angle is larger than 54.7° (suppose is around 70°), then as pressure increases and the actuator tends toward the volume maximum, the angle decreases. A decrease in braiding angle corresponds geometrically to an **elongation** of the actuator. So a pressurized actuator with an initially large braiding angle will lengthen.

If $\theta < 54.7^\circ$

If the initial braiding angle is smaller than 54.7° (suppose is around 30°), pressure again drives the system toward the volume maximum, but now that means the angle increases. An increasing angle corresponds geometrically to **contraction, to an axial shortening**. So in this range, the actuator shortens when pressurized.

If $\theta = 54.7^\circ$

If the actuator is designed exactly at this critical angle, increasing pressure does not strongly change the equilibrium length. Instead, the main effect is a **stiffening** of the actuator. In other words, pressure loads the structure and makes it mechanically stiffer, but without producing much axial deformation.

This is one of the most important design results of the whole lecture. The braiding angle is not a secondary detail. It is the **main design parameter** that determines whether the actuator behaves as a contracting muscle, an elongating element, or mainly a stiffening element.

7. Static modeling: how force is estimated

After geometrical modeling, we move on adding static modeling to estimate the force produced by the actuator.

The idea is to compare:

- the **input work** done by the fluid pressure as the chamber volume changes,

$$dW_{\text{in}} = \int_{S_i} (P - P_0) d\mathbf{l}_i \cdot d\mathbf{s}_i = (P - P_0) \int_{S_i} d\mathbf{l}_i \cdot d\mathbf{s}_i = P' dV$$

- the **output work** done by the actuator as it exerts an axial force through an axial displacement.

$$dW_{\text{out}} = -F dL$$

Using the virtual work argument (and neglecting the losses) the input work can be equated to the output work and we can obtain a relation between force, pressure, and geometry. The key point is that the force can be expressed from the derivative of volume with respect to length. In a compact sense, the actuator force is related to how much pressure-driven volume increase is converted into axial deformation.

$$dW_{\text{out}} = dW_{\text{in}} \quad F = -P' \frac{dV}{dL} = \frac{P' b^2 (3 \cos^2 \theta - 1)}{4\pi n^2}$$

This is useful because it gives an engineering-level estimate of the force that a McKibben actuator can generate without requiring a full-blown nonlinear material simulation.

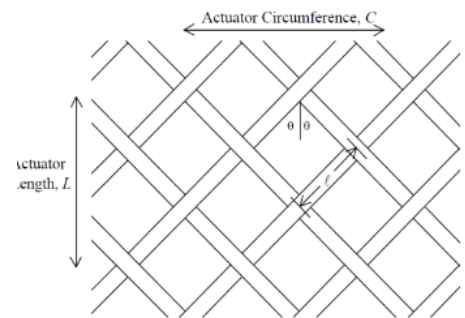
We need always to take in account that this model relies on strong assumptions; It ignores friction, fiber stretch, elastomer elasticity, end effects, and nonideal contact. Therefore, it is useful for a first estimate, but it is not a precise model.

More advanced models remove some of these assumptions. For example, some models account for internal friction, fiber stretch, rounded end effects, and the force contribution of the internal elastomeric tube. These more complex models are closer to experimental results but require more detailed parameters.

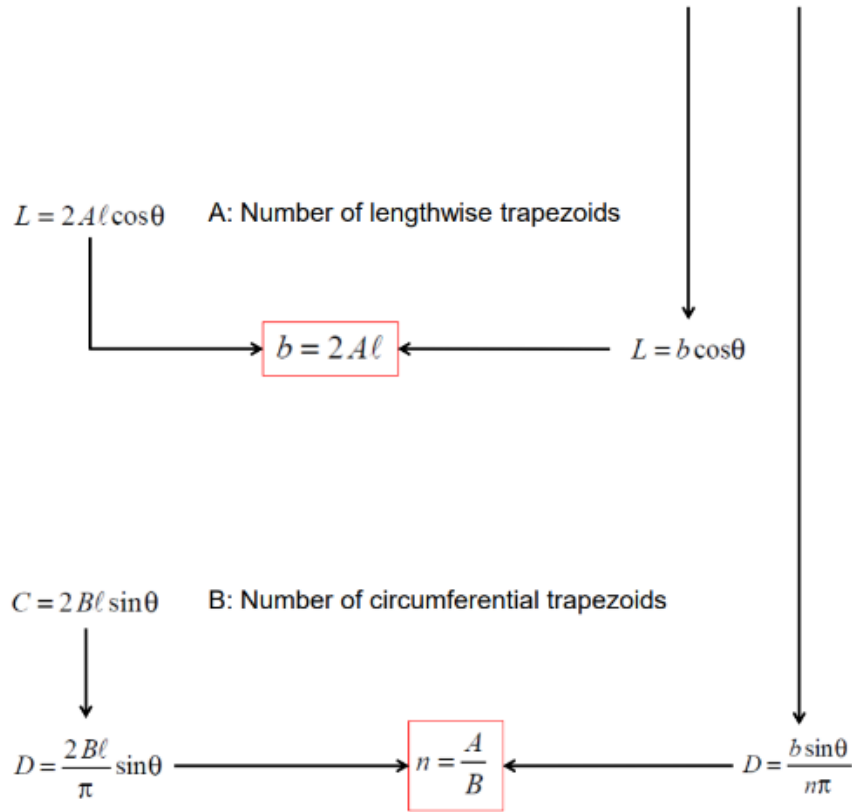
Practical problem: how to determine braid parameters

It's important to underline that some quantities in the ideal formulas, like the exact number of braid turns n , or the fiber length b , are not always easy to measure directly in practice. To make the model more usable, let's introduce:

- A : number of **lengthwise trapezoids** (cells along the actuator length)
- B : number of **circumferential trapezoids** (cells around the circumference)



These counts come from the mesh structure of the braid. With them, the formulas can be rewritten in a form that removes difficult-to-measure terms and replaces them with more practical geometric observables. Then the dependence on θ can also be eliminated with trigonometric substitutions, giving a control-oriented expression for force; also we simplify the work by removing θ because it's hard to measure.



So, we have defined:

A = number of lengthwise trapezoids
 B = number of circumferential trapezoids
 ℓ = length of one side of a single trapezoidal cell

Look at one diagonal fiber path in the braid.

Along the actuator length, the fiber passes through a sequence of repeated trapezoidal cells. Each complete lengthwise cell contributes two inclined segments of thread, each of length ℓ .

So, for one lengthwise trapezoid:

$$\text{thread length contribution} = 2\ell$$

If there are A trapezoids along the actuator length, the total thread length becomes:

$$b = 2A\ell$$

So b is not measured directly anymore. It is reconstructed from the number of lengthwise cells and from the length of one cell side.

Instead, for L each inclined thread segment of length ℓ has a vertical projection along the actuator axis.

From the triangle:

$$\cos \theta = \frac{\text{axial projection}}{\ell}$$

so the axial projection of one segment is:

$$\ell \cos \theta$$

Since each lengthwise trapezoid contains two such segments, the axial length contribution of one cell is:

$$2\ell \cos \theta$$

If there are A cells along the actuator length:

$$L = 2A\ell \cos \theta$$

This is coherent with:

$$b = 2A\ell$$

because if $b = 2A\ell$, then:

$$L = b \cos \theta$$

which is exactly the relation shown previously in the lecture

.Now consider the circumference of the actuator, called C .

Each inclined segment of length ℓ also has a horizontal/circumferential projection. From the triangle:

$$\sin \theta = \frac{\text{circumferential projection}}{\ell}$$

so the circumferential projection of one segment is:

$$\ell \sin \theta$$

Each circumferential trapezoid contributes two such projected segments, so:

$$2\ell \sin \theta$$

If there are B circumferential trapezoids around the actuator:

$$C = 2B\ell \sin \theta$$

Since the circumference is also:

$$C = \pi D$$

we obtain:

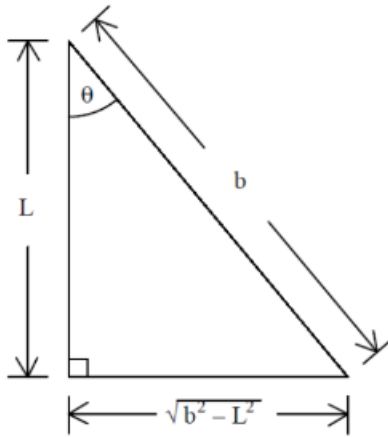
$$\pi D = 2B\ell \sin \theta$$

and therefore:

$$D = \frac{2B\ell}{\pi} \sin \theta$$

This explains the formula in the slide:

$$D = \frac{2B\ell}{\pi} \sin \theta$$



$$\sin \theta = \frac{\sqrt{b^2 - L^2}}{b}$$

$$\cos \theta = \frac{L}{b}$$

$$F = \frac{P_g b^2 (3 \cos^2 \theta - 1)}{4\pi n^2} = \frac{P_g b^2}{4\pi n^2} \left(\frac{3L^2}{b^2} - 1 \right) =$$

$$= \frac{P_g B^2 \ell^2}{\pi} \left(\frac{3L^2}{4A^2 \ell^2} - 1 \right)$$

The important point is this: the original force model depends on parameters such as b , n , and θ , which are natural from a theoretical point of view but inconvenient in practice.

The rewritten model depends on:

$$P_g, A, B, \ell, L$$

These are easier to access experimentally:

- P_g is the gauge pressure applied to the actuator;
- L is the actuator length;
- A is the number of cells along the length;
- B is the number of cells around the circumference;
- ℓ is the length of one braid segment.

So A , B and ℓ are simply **a more practical way to describe the same braided geometry**.

Again, this simplified model is only a **rough approximation**. (*è la quinta volta che lo ripeto ma ha detto che lo chiede all'esame*). It works reasonably well as a first estimate, but it neglects several real effects:

- internal friction,
- fiber stretch,
- non-cylindrical end effects,
- bladder elastic forces.

That is why the slides then mention more refined models from Tondur and Lopez, Davis et al., Tsagarakis and Caldwell, and Delson et al., each relaxing one of the simplifying assumptions. So the right way to understand the static model is:

- very useful for intuition and preliminary design,
- not sufficient for a full high-fidelity description of the real actuator.

8. Application examples of McKibben actuators

We now move to showing several examples to connect the theory to real robotic systems.

Shadow hand and FESTO fluidic muscles

Shadow air muscles and **FESTO fluidic muscles** are examples of McKibben-like actuation used in robotic systems. The point here is to show that the same physical principle can be used both in research and industrial contexts.

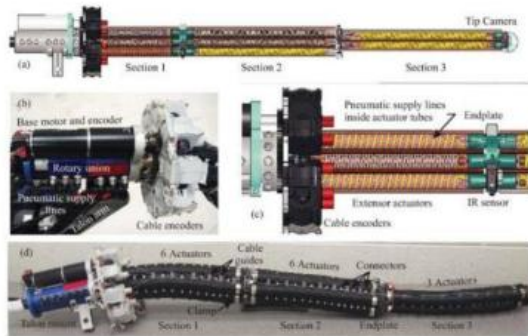
It's interesting also underlining the biological inspiration in the hand, pointing out that many finger-driving muscles are extrinsic, meaning located away from the fingers themselves. This makes muscle-like remote actuation a natural and bioinspired architectural choice.

OctArm

Another example is the **OctArm**, a continuum arm inspired by the octopus. → biologically inspired maneuvering, including bending in different directions, expansion and contraction, and using the



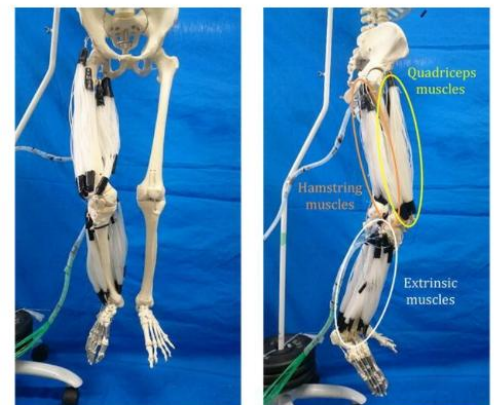
whole arm body to grasp objects. This is important because it shows that flexible fluidic actuation can be distributed over the body of a manipulator, not restricted to one joint or one end-effector.



Multifilament muscles and recruitment-like actuation

We have also seen **micro-McKibben-based multifilament muscles**, for example in musculoskeletal lower-limb robots. -
 → human muscles are not one giant single fiber, but collections of many fibers, and force can be modulated by **recruitment**, that is, by activating different numbers of fibers. Similarly, bundles of many small McKibben-type actuators can be activated in groups, giving more biologically plausible force modulation and more human-like motion.

This example is very meaningful in the context of bionics because it moves beyond “an actuator that contracts like a muscle” toward “an actuator architecture that also mimics the organization of biological muscles.”



Multifilament muscles based on micro-McKibben actuators

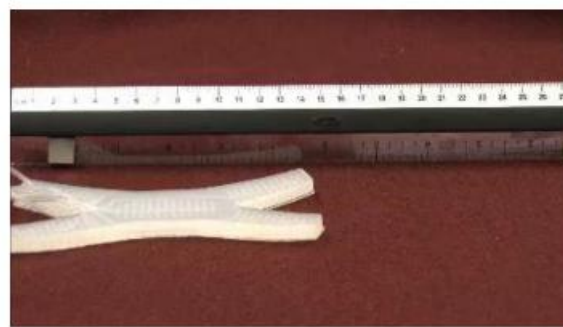
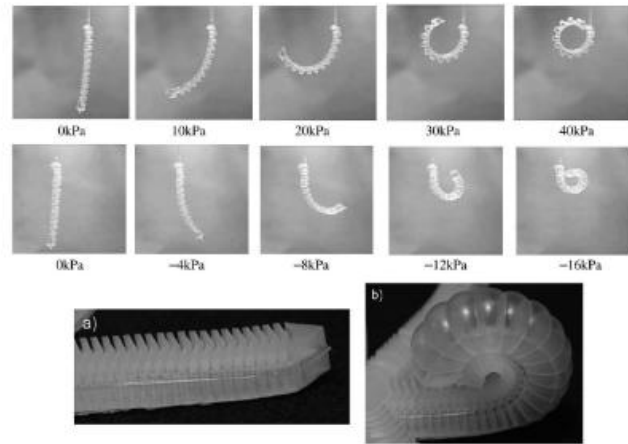
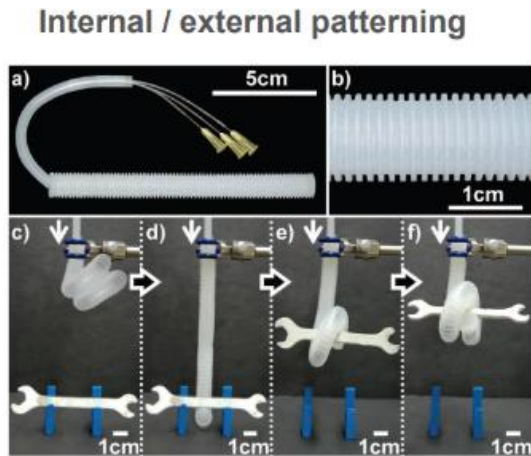
9. Other FFAs beyond McKibben

After the McKibben section let's see other actuator families use the same general principle but different constraint strategies.

Internal and external patterning

An important concept is the one that the **patterning** of the chamber walls determines the motion. Patterned geometries can cause curling, directional bending, locomotion, and more complex deformations (examples are work by Ilievski, Shepherd, Martinez, Whitesides, Wakimoto, and others cited in the slides).

A very interesting point is the bidirectional curling actuator where **positive pressure bends one way** and **vacuum bends the other way**. This is a beautiful example of how morphology can encode asymmetric response to opposite pressure signs. It also reinforces the central lesson of the entire lecture: the same chamber can support multiple deformation modes depending on how it is designed.



Soft locomotion and gripping based on patterning

There exist robots where patterned pneumatic chambers are arranged so that coordinated inflation generates locomotion. This includes the famous **multigait soft robot**. → the material properties like the surface stickiness of silicone can contribute functionally, for instance by helping traction or grip. This is an important soft-robotics concept: not only chamber geometry, but also passive material interaction with the world becomes part of the behaviour.

Different materials

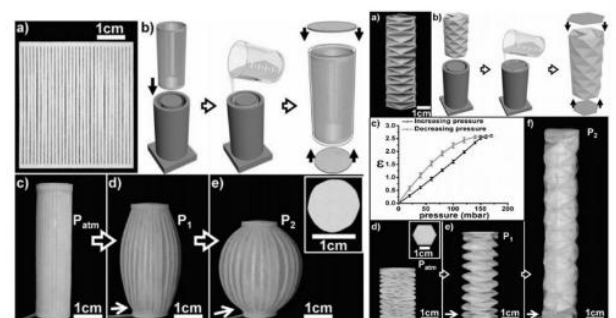
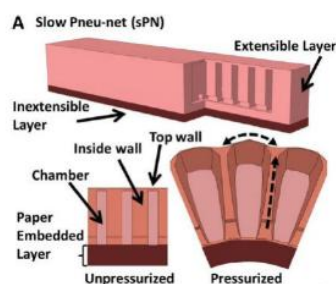
Another family of FFA actuators is based on **combining different materials**. For example, there exist systems where elastomeric chambers are cured over an inextensible layer, as in soft gloves and many PneuNet-derived actuators. The idea is to combine:

- an extensible chamber side,
- a less extensible or inextensible backing layer.

This directly creates asymmetry, and therefore bending.

We can connect this to wearable robotics. In rigid exoskeletons, one major challenge is joint alignment:

Different materials



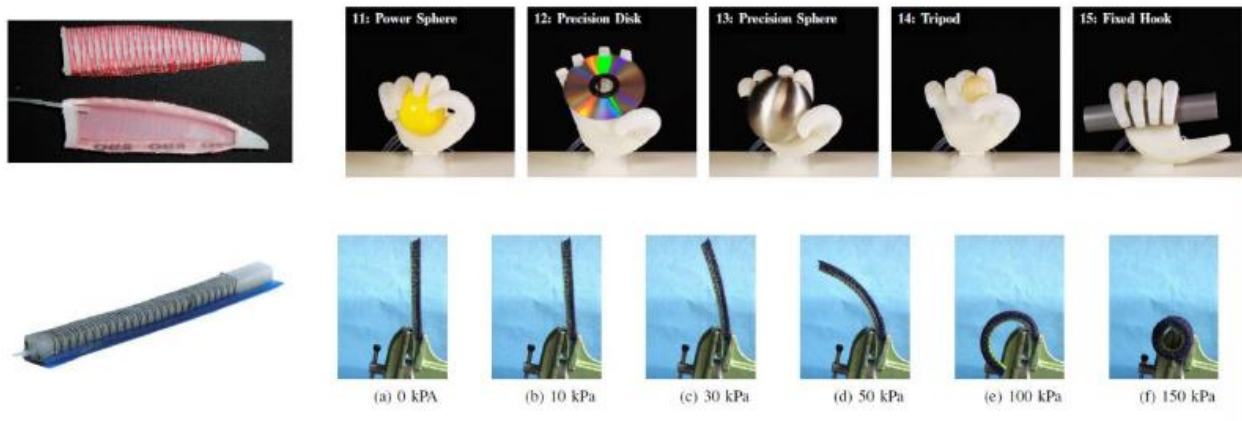
if the robot joint does not align perfectly with the human joint, unwanted torques and discomfort arise. In soft glove-like systems, there is no rigid artificial joint that must align. Instead, the soft actuator simply conforms to the finger and uses the human anatomy itself as the kinematic substrate. This is one of the strongest ergonomic arguments in favor of soft wearable actuation.



Crossed yarns

Then it is also necessary to introduce **crossed yarns**, citing work by Deimel and Brock. This approach can be seen as a lighter or simpler way of constraining expansion than a full braid. The crossed yarns limit radial expansion and help preserve desired bending behavior, while still allowing the actuator to remain compliant when bent. → they do not impose a rigid-joint kinematic chain, but just redirect deformation.

The resulting hands are highly adaptive. Because each finger is compliant, the grasp emerges from contact with the object rather than from precise pre-programmed trajectories. This is strongly connected to the ideas of underactuation and morphological computation.



Inextensible chambers and Baymax reference

Another approach shown in the lecture is based on **inextensible chambers**. In this case, the actuator is not made to work mainly through the stretching of an elastomeric wall. Instead, it uses thin flexible layers that are almost non-extensible. This means that the material can bend and fold, but it does not significantly elongate.

The key mechanism is therefore **deployment** rather than stretching. The chamber can start from a collapsed or folded configuration and, when pressurized, it unfolds and reaches a prescribed inflated shape. In this way, the deformation is mostly determined by the geometry of the chamber, by the position of folds, and by the way the flexible sheets are assembled.

This is different from highly stretchable elastomeric actuators. In stretchable actuators, pressure deforms the body by stretching the material. In inextensible-chamber designs, instead, pressure mainly changes the configuration of the structure: the material does not need to stretch a lot, but it deploys.

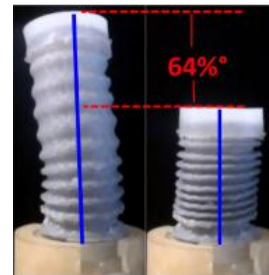
By designing folds and textile-like patterns in specific locations, it is possible to enable local deformations and obtain bending, opening, closing, or more complex soft robotic motions. This is why these systems are connected to textile-based soft robotics: the mechanical behavior can be programmed through the geometry and fabrication of the flexible chamber.

Different braided structures

Finally, the topic returns to braided structures, but in a different configuration from the classical McKibben actuator.

In a standard McKibben actuator, the braided sleeve is arranged around the whole cylindrical chamber in a relatively uniform way. Therefore, when the actuator is pressurized, the main effect is usually axial contraction or axial elongation, depending on the initial braiding angle.

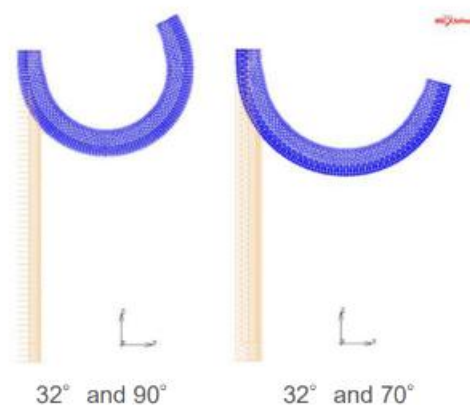
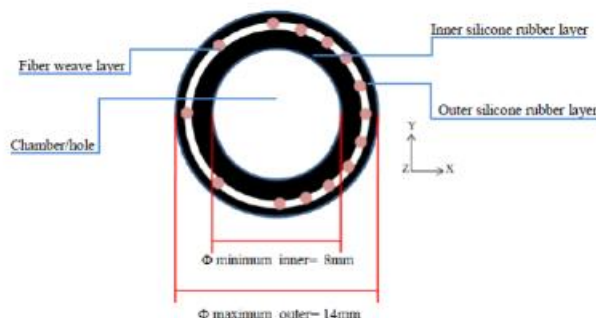
Here, instead, it is shown that braided structures can also be used to obtain **bending**. The idea is to break the symmetry of the braid by using different braided regions with different angles. For example, one side of the actuator can have a high braiding angle, while the opposite side can have a lower braiding angle.



Since the deformation of a braided structure depends on the braiding angle, the two sides do not respond in the same way to pressure. One side may tend to elongate more, while the other side may elongate less or even shorten. This mismatch between the two sides produces curvature, and the actuator bends.

So the important point is that the braid is not only a reinforcement layer. Its geometry actively determines the deformation of the actuator. By changing the braid pattern, the braid angle, or the arrangement of the braided regions, it is possible to program different mechanical responses.

This connects back to the McKibben model: in McKibben actuators, the braiding angle determines contraction or elongation; in asymmetric braided structures, different braiding angles on different sides can be combined to generate bending.



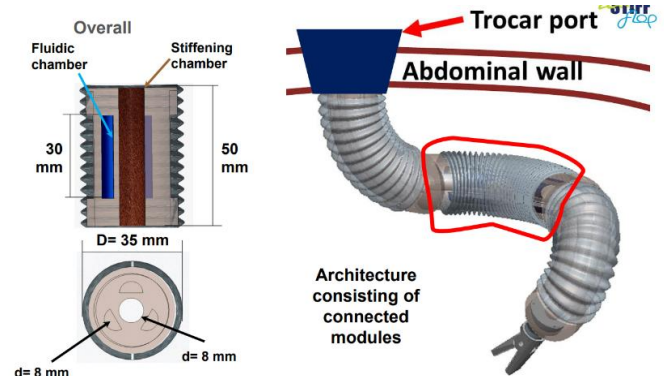
The STIFF-FLOP modular arm

A major applied example is the **STIFF-FLOP modular arm**, a soft manipulator developed for applications such as minimally invasive surgery.

Architecture

We can describe the module as a connected architecture consisting of:

- three embedded **fluidic chambers**,
- one **stiffening chamber**,
- an overall soft silicone body,
- an **external braided sheath** limiting lateral expansion.

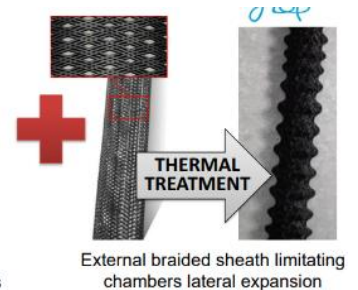
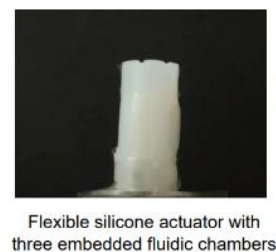


Dimensions shown on the slides are on the order of 30 mm and 50 mm, with the body diameter around 35 mm and embedded channels around 8 mm. The significance of this architecture is that different chambers can be pressurized selectively, producing continuum bending and elongation, while the stiffening chamber can modulate structural rigidity.

Material and fabrication aspects

We can also characterize the silicone actuator as:

- easy to fabricate,
- stable,
- able to withstand high strains without damage,
- not strongly susceptible to loading hysteresis,
- cheap.

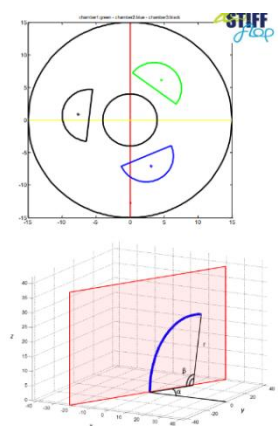


In the lecture context, this matters because the STIFF-FLOP arm is not only a proof of concept. It is intended as a practical soft manipulator, so manufacturability and mechanical robustness are part of its value.

Analytic model

The first model shown for the STIFF-FLOP module is an analytic model based on Euler–Bernoulli beam theory. The goal of this model is not to describe the full nonlinear behavior of the soft manipulator with high precision, but to obtain a fast relationship between the pressures applied in the three chambers and the resulting configuration of the module.

The model is based on several simplifying assumptions:



- the module can be approximated as a constant-curvature section of a circle;
- the deformation is assumed to be small;
- the base of the module is fixed at the origin;
- when no chamber is inflated, the module is aligned along the vertical axis;
- no external forces are applied to the module.

These are strong assumptions, especially the small-deformation assumption, because the real STIFF-FLOP module **can undergo large deformation**. However, the model is still useful because it gives a compact way to understand how the three chamber pressures influence bending, elongation, and curvature.

The inputs of the model are the three chamber pressures:

$$P_1, P_2, P_3$$

Each chamber is located in a different angular position around the cross-section of the module. Therefore, inflating different chambers produces bending in different directions. The model first computes the bending-plane angle α , which indicates the direction in which the module bends:

$$\tan \alpha = \frac{\sqrt{3}(2P_1 - P_2 - P_3)}{3(P_2 - P_3)}$$

This equation shows that the bending direction depends on the relative pressure difference among the three chambers. If the three pressures are equal, there is no preferential bending direction; if one chamber is more pressurized than the others, the bending plane shifts accordingly.

The model then estimates the radius of curvature r :

$$r = \frac{EI}{M}$$

where E is the elastic modulus, I is the second moment of area of the section, and M is the bending moment generated by the pressurized chambers. In the slide, this is rewritten as:

$$r = \frac{3EI}{A_p d \sum_{i=1}^3 (P_i \sin \alpha_i)}$$

This expression means that curvature depends on the balance between two contributions. On one side, EI represents the bending stiffness of the module: if the module is stiffer, it bends less, so the radius of curvature is larger. On the other side, the pressures in the chambers generate bending moments: if the pressure contribution increases, the module bends more, so the radius of curvature decreases.

The model also estimates the total length L of the module:

$$L = \frac{A_p L_0}{3A_0 E} \sum_{i=1}^3 P_i + L_0$$

Here, L_0 is the rest length of the module. The additional term depends on the sum of the three chamber pressures, so it represents the elongation produced by pressurization. If all chambers are inflated together, the main effect is elongation rather than bending.

Finally, the bending angle β is computed as:

$$\beta = \frac{L}{r}$$

This is consistent with the constant-curvature assumption: if the module is approximated as an arc of a circle, the bending angle is given by the arc length divided by the radius of curvature.

So, the analytic model provides a simplified description of the module behavior:

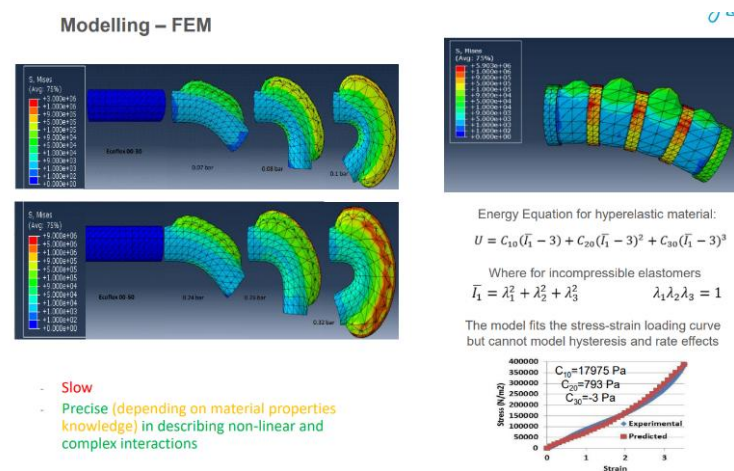
- α gives the bending direction;
- r gives the radius of curvature;
- L gives the elongated length of the module;
- β gives the total bending angle.

The advantage of this model is that it is fast and easy to compute, because it only requires the chamber pressures, the geometry, and the material properties. The limitation is that the real manipulator is soft, nonlinear, and can undergo large deformation, so the model is not highly precise. Nevertheless, it is useful for a first prediction and for understanding the qualitative relationship between pressure inputs and motion output-

FEM model

The second modelling approach shown in class is the finite element model, FEM. Compared with the analytic Euler–Bernoulli model, FEM is slower, but it can describe the behavior of the STIFF-FLOP module in a more realistic way, especially because the manipulator is made of soft silicone and undergoes large deformations.

This is important because soft robotic structures do not usually behave like linear elastic beams. In rigid or small-deformation problems, simple linear material models may be sufficient. Here, instead, the actuator experiences large strains, nonlinear deformation, and complex interactions between the pressurized chambers and the surrounding soft body. For this reason, the material must be described with a **hyperelastic model**.



In the slide, the material behavior is represented through an energy equation for a hyperelastic material:

$$U = C_{10}(\bar{I}_1 - 3) + C_{20}(\bar{I}_1 - 3)^2 + C_{30}(\bar{I}_1 - 3)^3$$

where, for incompressible elastomers:

$$\bar{I}_1 = \lambda_1^2 + \lambda_2^2 + \lambda_3^2$$

and:

$$\lambda_1 \lambda_2 \lambda_3 = 1$$

The λ_i terms are the stretch ratios, while the condition $\lambda_1 \lambda_2 \lambda_3 = 1$ expresses the incompressibility assumption: the material changes shape, but its volume is approximately preserved.

The idea is that, instead of using a simple linear stress-strain law, the model fits the experimental stress-strain loading curve of the silicone. This gives a more realistic description of the material response under large deformation. In the slide, the predicted curve is compared with the experimental one, showing that the hyperelastic model can fit the loading behavior reasonably well.

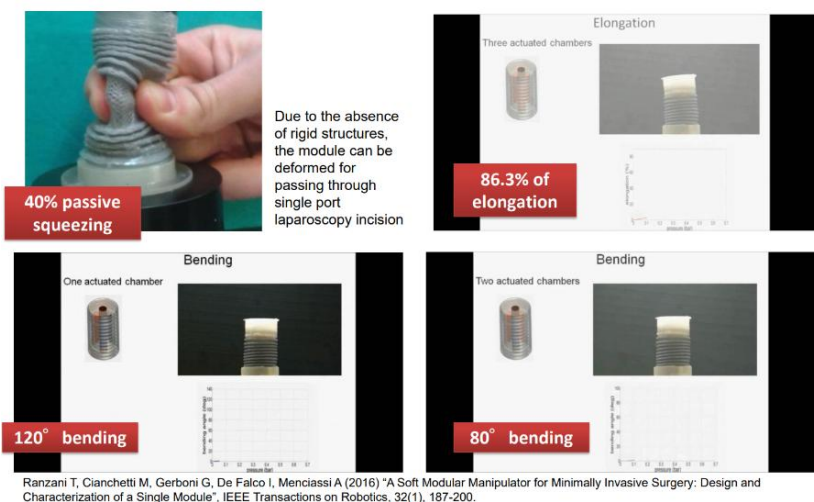
However, there is also an important limitation: even this model cannot fully describe hysteresis and rate effects. In other words, it can fit the stress-strain curve during loading, but it does not completely capture how the material behaves depending on loading history or deformation speed.

Therefore, FEM is more precise than the analytic model, but only if the material properties are known accurately. This is a key point in soft robotics: a detailed numerical model is useful only if the “virtual material” is close enough to the real material. Otherwise, even a complex simulation can give unreliable results.

So the meaning of this slide is that FEM offers a more powerful tool for predicting nonlinear deformation and chamber interaction, but it comes with two main costs: it is slower, and it depends strongly on correct material characterization.

The photo report large possible deformations, including about **120° of bending**, around **80° in another configuration**, and up to **86.3% elongation**. They also note that because the arm has no rigid skeleton, it can be passively deformed to fit through constrained access routes, such as single-port laparoscopic entry.

This is exactly the kind of application where soft actuation becomes compelling. A rigid robot might be precise but difficult to insert through confined curved anatomical paths. A

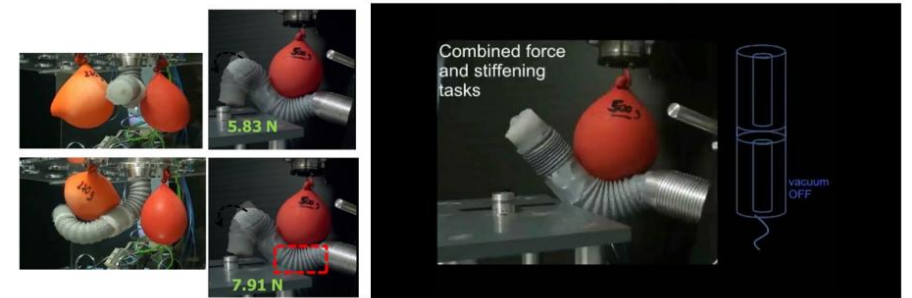


soft continuum manipulator can instead deform safely and adaptively, then later provide controlled motion at the target site. This is one of the clearest real-world motivations for the entire lecture.

Two-module and miniaturized STIFF-FLOP manipulator

After discussing the single STIFF-FLOP module, let's move to the two-module version of the manipulator. The important point is that the system is modular: each module can be actuated and stiffened independently.

This is useful in a surgical scenario because traditional surgery often requires different instruments for different tasks, such as grasping, retracting, dissecting, or visualizing. With a modular soft manipulator, different sections of the same body can potentially perform different tasks. For example, one module can stabilize or hold a structure, while the other module interacts with the surgical target.

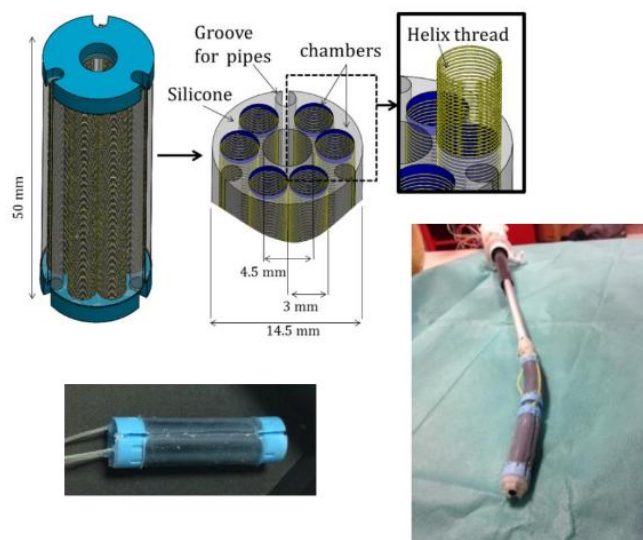


Ranzani T, Gerboni G, Cianchetti M, Menciassi A (2015) "A bioinspired soft manipulator for minimally invasive surgery" *Bioinspiration & Biomimetics*, 10, 03500

The first STIFF-FLOP prototype demonstrated the feasibility of the concept, but it was still too large for standard minimally invasive surgery. Moreover, even if the external braided sheath limited external ballooning, the internal chambers could still deform toward the inside of the module. This produced interaction between chambers, or cross-talk.

This cross-talk is problematic because the final shape of the manipulator should ideally depend only on the final pressure values, not on the order in which the chambers are activated. If inflating one chamber changes the mechanical conditions of the others, then the same final pressure combination may produce different final shapes depending on the activation sequence.

For this reason, a smaller version was developed. The new version had a diameter of about 14.5 mm, compatible with a standard trocar size of about 14.7 mm. The containment strategy was also changed: instead of relying only on a global external sheath, each chamber was locally constrained by helical threads. This helped reduce lateral expansion and chamber cross-talk.



The miniaturized version also allowed the integration of a camera. This is important because a soft bending camera can inspect the surgical field from different perspectives without requiring an additional access point.

Surgical advantages and Da Vinci integration

The main surgical advantages of the STIFF-FLOP system are related to field of view, dexterity, and safety.

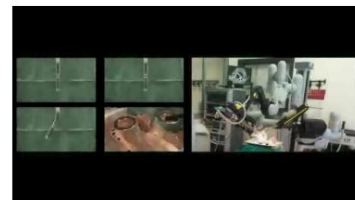
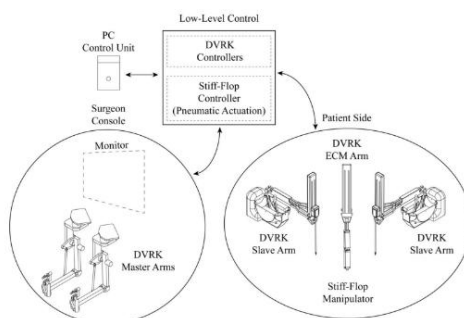
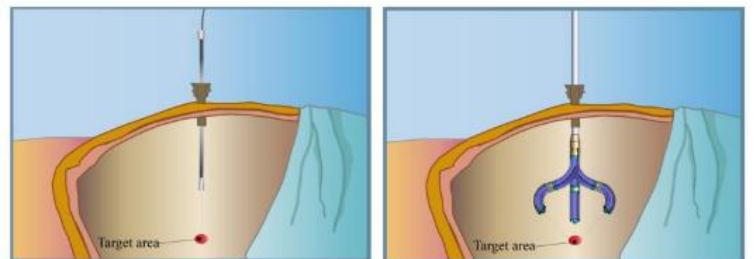
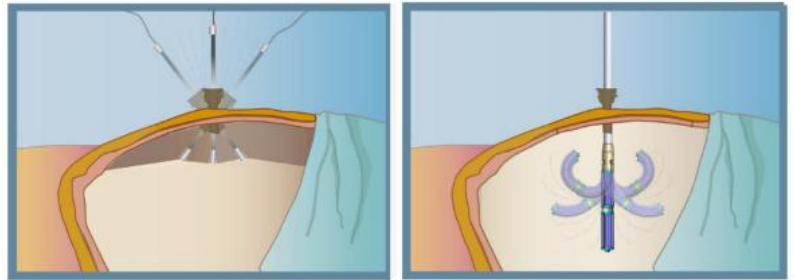
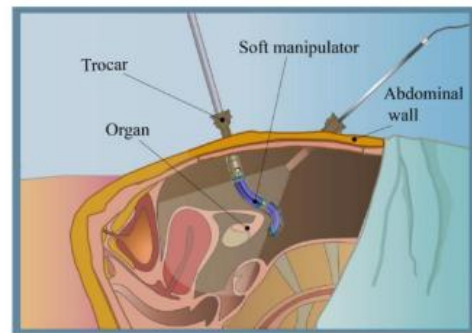
First, the bending capability increases the field of view. A rigid endoscopic camera usually observes the target from one main direction. A soft camera module, instead, can bend around the target and inspect it from different perspectives.

Second, the system improves maneuverability. Since the manipulator can bend and deform, it can reach regions that would be difficult to access with a straight rigid instrument.

Third, it allows safer navigation around organs and tissues. Since the manipulator is soft, if it touches an organ, it tends to deform rather than applying a large localized force. This makes interaction with delicate tissues safer during navigation.

However, as usual, there is also an important limitation: being soft is useful during navigation, but at the surgical target the robot may need to generate force. For this reason, stiffening mechanisms are important. The ideal behavior is to be soft while moving through the body, but stiffer when manipulation is required.

We also mentioned the integration of STIFF-FLOP with the Da Vinci Research Kit. This required hardware interfaces to connect the soft manipulator to the robotic platform and software integration to control it. The control system included a state-machine logic and allowed the surgeon to switch control modes, for example by using a pedal. This shows that a soft actuator is not useful only as a single component: to be clinically relevant, it must be integrated into a complete robotic system.



Hardware Integration: HW interfaces to connect Stiff-Flop to ECM arm



Software Integration:
 ○ State Machine Diagram
 ○ Stiff-Flop control

I-SUPPORT project: soft assistive robot for personal care

After the surgical example, at lesson we presented another application: the I-SUPPORT project, an EU-funded project for assistive robotics.

The goal was to develop a robotic shower assistant for frail older adults, especially for tasks such as washing the back or lower limbs. This is a completely different application from surgery, but the design requirements are similar in one important aspect: the robot must interact safely with the human body.

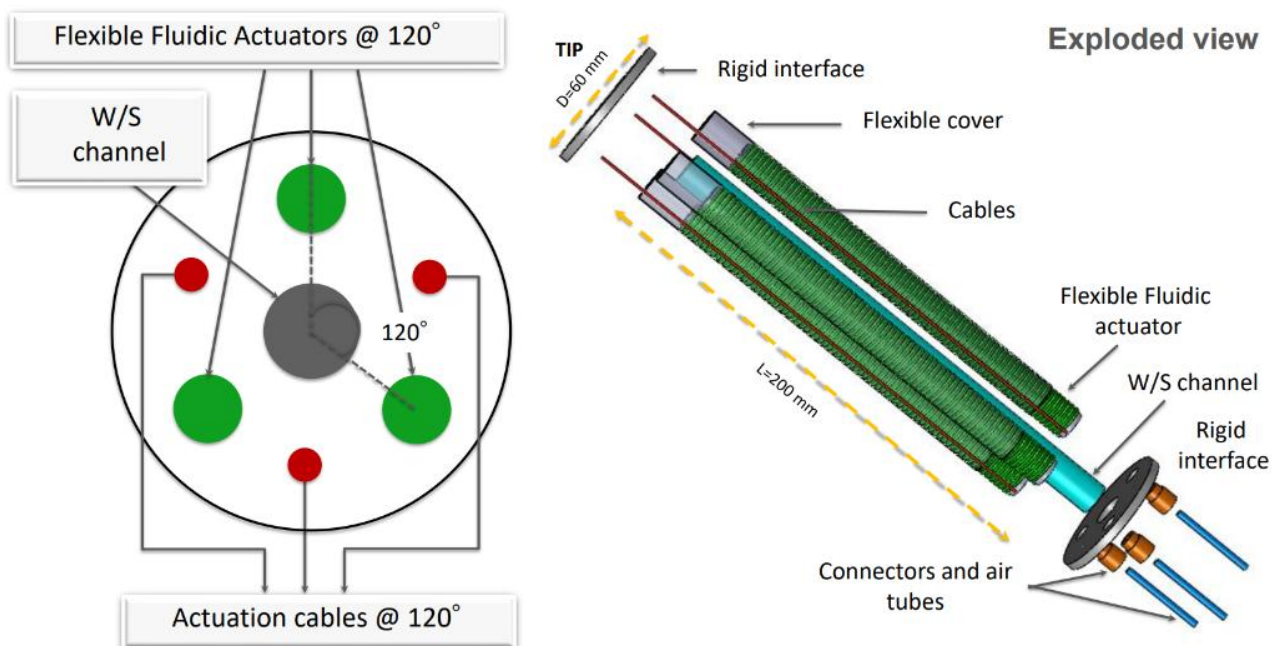
Since the robot operates in a shower environment, pneumatic actuation is particularly attractive because it avoids placing electrical actuation directly near water and the user. The robot must be soft, compliant, and safe, but also strong enough to perform useful actions such as moving a sponge or delivering water/soap.

The module combines flexible fluidic actuators, cables or tendons, soft materials, rigid interfaces, air tubes, and a water/soap channel. The fluidic actuators are arranged at 120° , and the cables are also arranged at 120° .

The fluidic actuators mainly provide elongation and bending. However, they are not naturally good at active shortening. For this reason, cables are added. The cables provide three functions:

- they allow shortening;
- they improve bending when combined with fluidic actuation;
- they contribute to stiffening when co-activated with the fluidic actuators.

So the I-SUPPORT module is a hybrid system: the pneumatic part gives soft deformation and compliance, while the cables add pulling capability and improve controllability.



But now an important question has to be done: Are flexible fluidic actuators really actuators?

According to the definition introduced earlier in the course, an actuator should convert energy from another domain into mechanical energy. For example, an electric motor converts electrical energy into mechanical energy.

In the case of flexible fluidic actuators, however, the input is often already a pressurized fluid. Pressure is already mechanical energy. Therefore, the soft chamber alone does not necessarily perform a complete energy-domain conversion. It mainly converts pressurized fluid into a specific mechanical deformation.

For this reason, the soft chamber can be seen more precisely as a mechanical converter or transducer. The complete actuation system must also include the pressure source, because the pressure must be generated somewhere.

This is an important point because, in soft robotics, it is easy to focus only on the soft chamber and forget the external hardware. But a pneumatic soft actuator is not just the silicone body: the complete system also includes compressors, pumps, tubes, valves, batteries, reservoirs, or chemical pressure sources.

Fluidic energy sources

The final part of the lecture discusses possible sources of pressurized fluid.

The first option is an off-board compressor. This is common in laboratories, hospitals, and research environments. It provides high pressure, high volume capacity, and fast actuation. However, it is bulky, heavy, often noisy, and requires tubes between the compressor and the robot. Therefore, it is powerful but not very portable.

A second option is to use onboard micro-compressors powered by lithium batteries. This makes the robot more autonomous and reduces external piping. However, micro-compressors usually provide lower pressure and lower flow rate. Multiple pumps in series may be needed to increase pressure, while multiple pumps in parallel may be needed to increase flow. They also add noise, mass, and battery limitations.

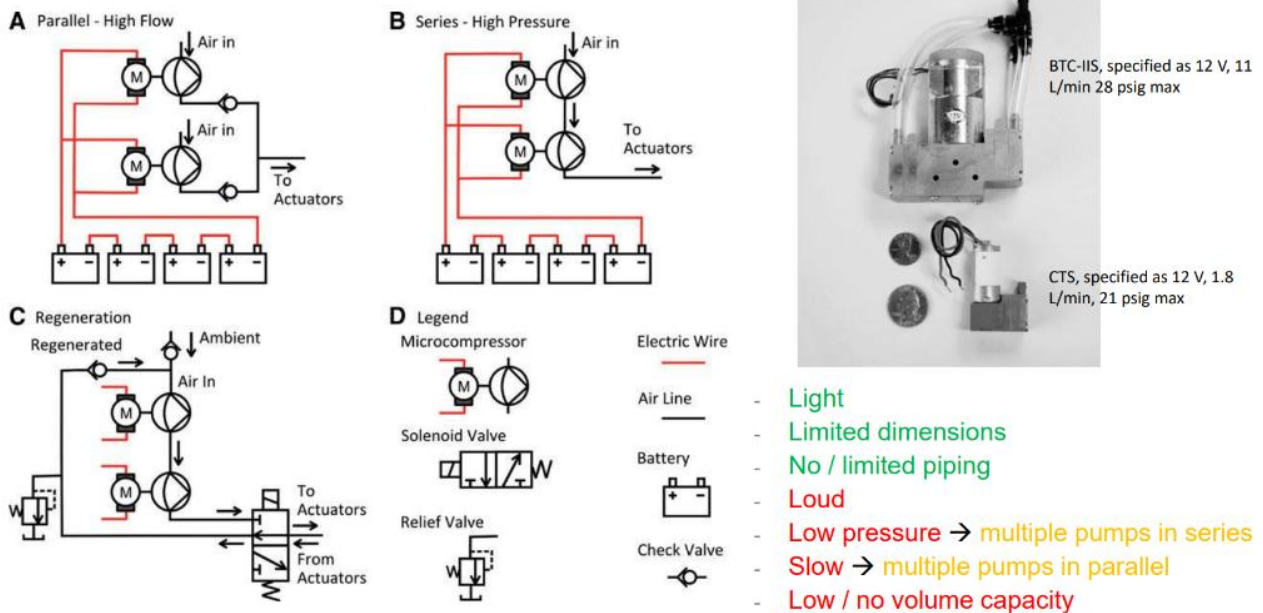
Another solution is compressed fluid stored in a cartridge or tank. This can provide high pressure and fast response in a relatively lightweight way. The main limitation is volume capacity: once the compressed gas is consumed, it must be replaced or refilled.

There is also the combustion of hydrocarbons. This solution has very high energy density and can produce very fast actuation, which is useful for impulsive motions such as jumping. However, it is loud, produces high local temperature, requires fuel, oxygen and ignition, and is difficult to control and model.

Finally, monopropellant decomposition is discussed. In this case, a chemical substance decomposes in the presence of a catalyst, generating gas and heat. Hydrogen peroxide is an example. This can be used for onboard pressure generation, but it introduces safety, temperature, and control issues.

The main idea of this final section is that the actuator body cannot be evaluated alone. A flexible fluidic actuator must always be considered together with its pressure source, because the source strongly affects portability, speed, safety, and practical usability.

On-board micro-compressors + Lithium batteries



Wehner M, Tooley MT, Mengüç Y, Park YL, Mozeika A, Ding Y, Onal C, Shepherd RF, Whitesides GM, Wood RJ, (2014) "Pneumatic Energy Sources for Autonomous and Wearable Soft Robotics", Soft Robotics, 1(4), 263-274

10. Final conceptual conclusion

The final message of the lecture is that flexible fluidic actuators are not simply soft chambers inflated by air or liquid. Their behavior is determined by how pressure-induced expansion is constrained and redirected by geometry, materials, fibers, patterns, braided structures, or tendons.

The McKibben actuator is the clearest theoretical example, because the braiding angle alone determines whether the actuator contracts, elongates, or mainly stiffens. The same general idea applies to all the other examples discussed in the lecture: patterned actuators, PneuNets, crossed-yarn systems, inextensible chambers, STIFF-FLOP, and I-SUPPORT.

The advantages of FFAs come from their softness, compliance, smooth deformation, safety, and adaptability. At the same time, these same properties make them difficult to model and control. This is why simplified models are useful for intuition, but real systems require attention to material nonlinearities, fabrication, pressure dynamics, chamber interaction, stiffening, and integration with the complete robotic platform.

Jamming Transition

In this lesson, we introduce a class of actuators based on the **jamming transition**. This is a **semi-active principle**, meaning that such actuators are not able to inject positive energy into the system; instead, they can only dissipate energy.

Jamming can be understood through the example of a flexible membrane filled with granular material. When the **pressure inside** and **outside** the membrane is **equal**, the particles are free to move and rearrange themselves, giving the system a soft, compliant, and flexible behavior. However, when air is removed from the membrane and a **vacuum** is created, the external atmospheric pressure causes the membrane to collapse onto the particles. The particles are compressed together, friction between them increases, and their motion becomes restricted. As a result, the system undergoes a solidification-like effect and becomes rigid. When air is reintroduced, the pressure difference disappears, the particles regain freedom of movement, and the system returns to its original floppy, liquid-like state.

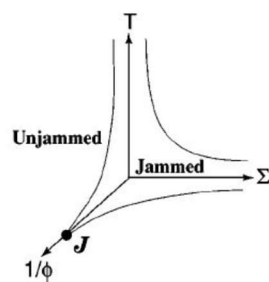
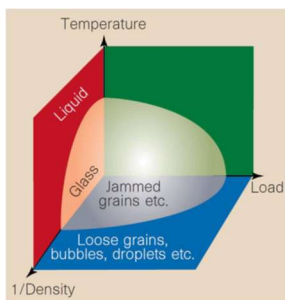
In general, any granular system enclosed within a flexible membrane behaves according to this principle. A common everyday example of this phenomenon is a **vacuum-sealed coffee bag**. While sealed, the bag is rigid because the coffee grounds are in a jammed state caused by the vacuum. As soon as the package is opened and air enters, the grains become free to move again, and the bag immediately becomes soft and flexible. This corresponds to the transition from the jammed to the unjammed state.

Definition

The jamming transition as a general phenomenon has been observed and studied for centuries, for example in situations involving the clogging or sticking of cereals and granular materials. Nevertheless, a complete theoretical and numerical understanding of the phenomenon remains elusive. The jamming transition is defined as:

→ a **critical slowing down of the dynamics of a system far from equilibrium**, mainly caused by the “overcrowding” of particles.

Initially, particles are free to move because interactions between them are negligible, and the system behaves in a flexible manner. As overcrowding increases—either because particles are compressed or because the particle density becomes sufficiently high—the number of available paths for particle movement decreases dramatically. Eventually, particle motion becomes arrested and the system behaves as a rigid structure. It is precisely this arrest of dynamics that is referred to as **jamming**.



ANALOGY: A similar type of state transition can also be observed in attractive liquids. In that case, overcrowding is not caused by mechanical loading or compression, but rather by a reduction in temperature accompanied by an increase in density. This phenomenon is known as the **glass transition**. Although extensively studied, the glass transition is still not completely understood. The analogy

between jamming and glass transition highlights how parameters such as load, density, and temperature can all contribute to transitions between fluid-like and solid-like states (see plots).

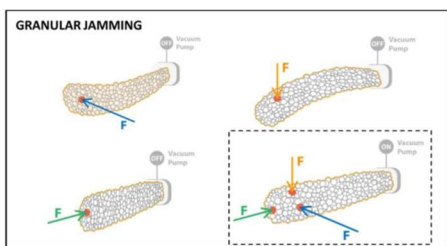
Jamming Possibilities

The same jamming principle can be applied to different types of filler materials, provided they are enclosed within a flexible membrane. Depending on the nature of the filler, three main categories of jamming systems can be identified:

- **Granular Jamming (GJ):** based on granular materials
- **Fiber Jamming (FJ):** based on flexible fibers
- **Layer or Laminar Jamming (LJ):** based on flexible sheets or layers

The dimensionality of the filler material strongly influences the mechanical behavior of the overall system. In particular, each type of jamming mechanism produces a characteristic variation of stiffness along different spatial dimensions. Although more complex behaviors can be achieved in practice, these systems can be understood through three fundamental cases.

1) Granular Jamming (0D → 3D)

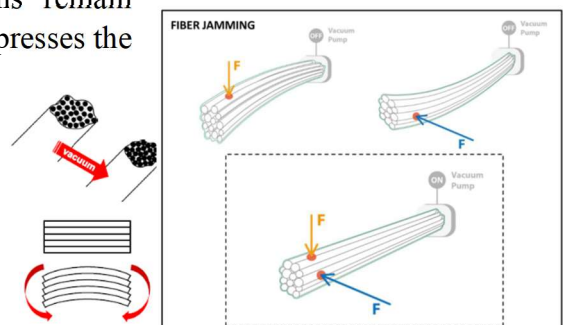


In granular jamming systems, the filler consists of small particles or grains. Since each grain is extremely small compared to the system dimensions, it can be approximated as a **zero-dimensional (0D)** element. If vacuum is applied, the membrane collapses onto the particles, increasing friction between them and restricting their movement. As a consequence, the stiffness of the structure increases simultaneously in all three spatial

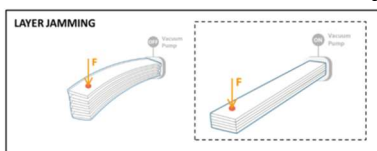
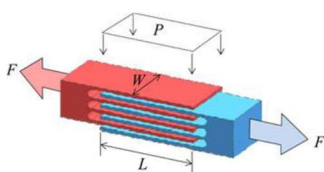
dimensions. For example, in a cylindrical structure filled with granular material, the initially soft response in every direction becomes rigid once the grains jam together. Therefore, granular jamming produces a stiffness variation that acts in **three dimensions (3D)**

2) Fiber Jamming (1D → 2D)

In fiber jamming systems, the filler consists of elongated flexible fibers, which can be considered **one-dimensional (1D)** elements. Before jamming, the fibers are naturally stiff along their longitudinal direction, while the transverse directions remain compliant. When vacuum is applied, the membrane compresses the fibers together, increasing friction between adjacent fibers and preventing them from sliding relative to one another. Consequently, the directions that were initially soft also become stiff. In this case, the jamming transition effectively converts a system with one dominant stiff direction into a structure with stiffness distributed over **two dimensions (2D)**.



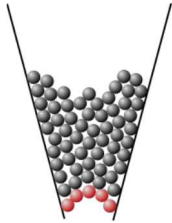
3) Layer Jamming (2D → 1D)



In layer or laminar jamming systems, the filler consists of thin flexible sheets, which represent **two-dimensional (2D)** structures. These layers can easily slide over one another, making the system highly compliant in bending and deformation. When vacuum is applied, friction between adjacent layers increases and sliding is inhibited. This produces a significant increase in stiffness, primarily along the direction associated with layer bending and relative motion. Therefore, the stiffness variation is concentrated mainly along a **single direction**, leading to a one-dimensional (1D) stiffening effect.

NOTE: These represent only the fundamental behaviors of jamming systems. In practice, much more sophisticated actuation concepts can be developed. As discussed previously, once a transduction principle is understood and the material properties are characterized, the next step is to design suitable actuation strategies. In the case of jamming technologies, combining multiple effects and exploiting geometric design enables the realization of highly complex and versatile behaviors.

Granular jamming



We now examine in greater detail the three different implementations of jamming systems, starting with **granular jamming**. Among the various jamming approaches, granular jamming is the most extensively studied. Historically, it was one of the earliest implementations to be investigated, and it has attracted significant interest because of the rich mechanical phenomena associated with the behavior of granular materials.

Modeling of Granular Jamming

The modeling of granular jamming remains a challenging topic in physics and engineering. As highlighted in the literature, *“the phenomenon of jamming in granular matter poses fundamental challenges in basic science because there is no known framework leading from the microscopic, grain-level interactions to the macroscopic properties that reflect collective behavior.”* Similarly, *“the theoretical description of granular systems remains largely a plethora of different, often contradictory concepts and approaches.”*

In this lecture, we focus on two simplified approaches that, although not exhaustive, are capable of capturing the overall behavior of granular jamming and identifying the key parameters governing the phenomenon. These approaches are:

- **SPM (Simple Probabilistic Modeling):** an analytical approach aimed at developing simplified design tools and performing virtual evaluations of system behavior.
- **DEM (Distinct Elements Modeling):** a more numerical and computational approach that explicitly models the interactions between particles.

a) Statistical Mechanics Approach

To understand granular jamming, it is necessary to introduce some fundamental parameters from statistical mechanics. One of the most important quantities is the **packing fraction**, also referred to as the **volume density**. This parameter is crucial for describing how the jamming transition occurs.

The packing fraction is defined as:

$$\phi = \frac{Nv_0}{V}$$

where:

- N is the number of particles,
- v_0 is the fixed volume of each particle,
- V is the total volume of the container or region in which the particles are confined.

The corresponding **free volume** of the system can be expressed as:

$$(1 - \phi)V$$

When the density ϕ is sufficiently low and there is no pressure difference between the inside and outside of the membrane ($\Delta P = 0$), the particles are free to move. Under external perturbations, the grains behave similarly to particles in a gas or liquid, continuously bouncing and rearranging themselves.

This behavior can be visualized using a simplified two-dimensional example. Consider a **box filled with particles**, where one side of the box is movable. If the movable wall leaves enough free space inside the box, the particles can move freely whenever an external disturbance is applied. Now suppose the movable wall is pushed inward, reducing the available free volume. As the free space decreases, the particles are forced into contact with one another, and eventually the outermost particles also come into contact with the boundaries of the box. In this condition, the pressure inside the system increases relative to the outside, meaning that $\Delta P > 0$. At the same time, the density ϕ increases until it reaches a critical value known as the **jamming threshold**, denoted by ϕ_J .



Once this threshold is reached, the particles form a rigid yet disordered structure. Except for a few isolated particles, often referred to as *rattlers*, the grains become locked in place and can no longer rearrange. Consequently, external perturbations no longer induce internal motion, because the entire system becomes mechanically arrested.

Random and Ordered Close Packing

To better analyze the density conditions of granular systems, it is useful to distinguish between two fundamental packing configurations that can both lead to jamming, although they differ significantly in the arrangement of the particles.

1) Random Close Packing (RCP)

In RCP, particles are distributed randomly while still being packed as densely as possible. The arrangement is disordered and lacks a regular geometric structure, yet the particles remain in contact with one another and can still reach a density high enough to produce a jammed state. This demonstrates that the jamming transition does not necessarily require an ordered particle arrangement.

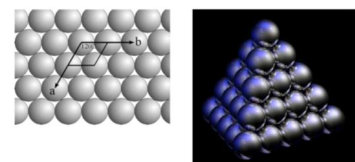
For systems composed of spherical, frictionless, and rigid particles—meaning incompressible and non-deformable particles—the density at which jamming occurs is commonly referred to as the **random close packing density**, denoted as ϕ_{RCP} .

2) Ordered Close Packing

If the particle arrangement is no longer random but instead perfectly organized, it is possible to achieve even higher density values. This configuration is known as **ordered close packing**. In this case, particles are arranged according to highly regular lattice structures that maximize the occupied volume fraction.

For spherical particles, the Bravais lattices capable of achieving the maximum possible density are:

- the **hexagonal lattice** in two dimensions (2D),
- the **face-centered cubic (FCC) lattice** in three dimensions (3D).



The corresponding maximum density values are:

$$\phi_{hex} = \frac{\pi}{2\sqrt{3}} \approx \mathbf{0.9069} \text{ (hexagonal 2D lattice)}$$

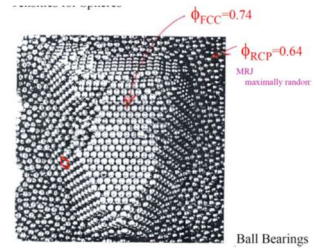
$$\phi_{fcc} = \frac{\sqrt{2}\pi}{6} \approx \mathbf{0.7405} \text{ (face-centered cubic 3D lattice)}$$

Any random close packing configuration necessarily exhibits a lower density than these ideal ordered packings.

Experimental Estimation of Random Close Packing Density

Because random close packing is, by definition, disordered, it is difficult to describe rigorously using a purely analytical mathematical model. For this reason, researchers have investigated experimentally generated random packings of spherical particles and analyzed them through numerical simulations.

These studies revealed that randomly packed systems often contain localized regions with ordered close-packing structures, while other regions remain disordered. By performing numerous experiments and simulations, researchers were able to estimate characteristic density values for random close packing:



$$\phi_{RCP} \approx 0.84 \text{ in 2D}$$

$$\phi_{RCP} \approx 0.64 \text{ in 3D}$$

Although these values are lower than those associated with ordered close packing, they are still relatively high. This explains why random close packed systems, despite being theoretically only **metastable**, exhibit very stable behavior in practice, especially for large numbers of particles in three-dimensional systems.

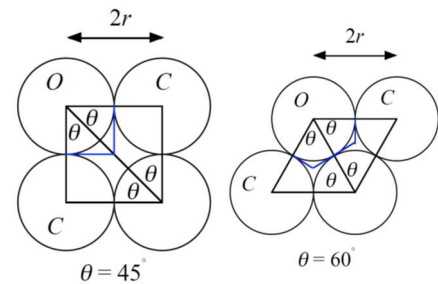
For this reason, random close packing configurations are widely exploited in soft robotics technologies based on granular jamming. Nevertheless, a precise mathematical definition of random close packing and an exact analytical determination of the values of ϕ_{RCP} remain unresolved problems. In fact, it is still debated whether a universally valid mathematical definition of RCP actually exists.

Model of Random Close Packing

Several theoretical approaches have been proposed to analytically describe the random close packing (RCP) state. One particularly simple and effective model is based on considering circles in 2D—or spheres in 3D—whose relative orientation can vary according to a single parameter, denoted by the **angle θ** .

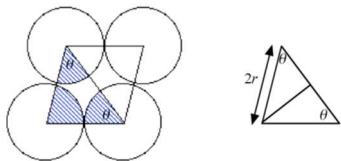
Two limiting configurations can be identified:

- a **square-like arrangement**, corresponding to $\theta = 45^\circ$, where the circles are fully aligned;
- a **hexagonal-like arrangement**, corresponding to $\theta = 60^\circ$, where one circle is positioned between the two circles below it.



Because of the geometric symmetry of the system, if the circles move beyond the second configuration the arrangement effectively returns to the first one. Consequently, in a random close packing configuration all possible orientations are assumed to lie within the interval:

$$45^\circ \leq \theta \leq 60^\circ$$



The density of the system can then be expressed as a function of θ . To derive this relationship, the packing density is first evaluated locally by considering a representative triangular cell. The density is obtained as the ratio between the area occupied by the circular sectors inside the triangle (blue area) and the total area of the triangle itself. This leads to the following expression:

$$\phi(\theta) = \frac{\frac{1}{2}\pi r^2}{2r^2 \sin(2\theta)} = \frac{\pi}{4 \sin(2\theta)}$$

Since all admissible values of θ within the interval are assumed to be equally probable, an estimate of the random close packing density can first be obtained by considering the average angle: $\bar{\theta} = 52.5^\circ$. Substituting this value into the density equation gives: $\phi_{RCP} \approx 0.813$ in two dimensions. This value is already very close to the experimentally observed density.

An even more accurate approximation can be obtained by averaging the entire density function $\phi(\theta)$ over the admissible interval rather than evaluating it only at the average angle. In this case:

$$\phi_{RCP} \approx \frac{3}{\pi} \int_{\pi/4}^{\pi/3} \frac{\pi}{\sin(2\theta)} d\theta = 0.824$$

which is extremely close to the experimentally observed value of approximately 0.84.

By extending the same reasoning to the three-dimensional case, the model predicts: $\phi_{RCP} \approx 0.6394$, which again matches very closely the experimentally measured value. This demonstrates that the model is a useful tool for relating the geometry and arrangement of grains to the overall density of the system.

Metastability of Random Close Packing

The random close packed state is considered only **metastable**. This means that mechanically stable configurations with higher density always exist, up to the limit of ordered close packing. In practice, one can trade randomness for density: introducing a certain degree of order into the arrangement allows the particles to achieve higher packing fractions.

As a consequence, the exact value of ϕ_{RCP} obtained in experiments or simulations may depend on the specific protocol used to generate the jammed state. Since the arrangement of particles strongly affects both the density and the effectiveness of the jamming transition, the performance of a granular jamming system is inherently influenced by the fabrication and loading procedure used to create it.

This variability is one of the main reasons why a complete and universally accepted theoretical description of jamming behavior is still lacking.

FOCUS: Role of the order

Although density is one of the most important parameters governing jamming behavior, the degree of order within the system also plays a fundamental role. For this reason, Torquato introduced an **order parameter**, denoted by ψ , to quantify how ordered a packing configuration is. In this formulation:

$$\psi = 1$$

corresponds to perfectly ordered close packing. Using this parameter, it is possible to represent jammed configurations in a $\psi - \phi$ plane, where:

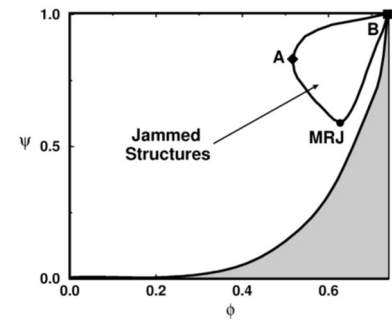
- the horizontal axis represents the density ϕ ,
- the vertical axis represents the order parameter ψ .

Different regions can then be identified in this plane:

- **low density and low order**, corresponding to loose disordered configurations;
- **high order and high density**, corresponding to ordered close packing;
- **high density with very low order**, which is theoretically inaccessible because achieving very high density necessarily requires some degree of structural organization.

Within this diagram, Torquato identified a region in which stable jammed packings can exist. The boundaries of this region are characterized by three important points:

- **Point A:** the jammed structure with the lowest achievable density;
- **Point B:** the densest possible ordered jammed structure, corresponding to ordered close packing configurations such as the FCC or hexagonal lattice;
- **MRJ (Maximally Random Jammed state):** the jammed configuration with the minimum possible order parameter ψ .



NOTE: The MRJ state is particularly important because it shows that jamming is not determined solely by density. A minimum degree of order is also necessary; below this threshold, the system can no longer sustain a jammed configuration.

Therefore, the **maximally random jammed state** is defined as the jammed structure that minimizes the order parameter ψ while still remaining mechanically stable. Numerical simulations performed by Torquato and collaborators found: $\phi_{MRJ} \approx 0.64$, which is in excellent agreement with the accepted experimental values for ϕ_{RCP} .

b) Distinct Elements Modeling (DEM)

We now move to a modeling approach that is more closely related to numerical simulations: Distinct Elements Modeling (DEM). DEM can be used as a virtual testing tool, allowing the study of granular systems without relying exclusively on physical experiments. The methodology is conceptually similar to the Finite Element Method (FEM), with the key difference that, instead of considering a continuous body divided into connected elements, DEM models systems composed of **discrete** and separated **particles**.

Today, DEM is widely recognized as an effective method for studying engineering problems involving granular and discontinuous materials, particularly in granular flows, powder mechanics, and rock mechanics (geomechanics). Its primary focus is not the deformation of individual grains, but rather the analysis of the **interactions** occurring **between particles**. In many applications, DEM and FEM can also be combined.

The DEM workflow follows steps similar to FEM:

1. **Generation of the model**, including particle geometry, spatial distribution, and initial conditions;
2. **Computation of forces**, using appropriate physical laws and contact models;
3. **Simulation**, including initialization, explicit time-stepping, and post-processing.

KEY ASPECT: Compared to FEM, a crucial aspect of DEM is the accurate description of particle interactions. In particular, the model must account for every mechanism capable of influencing particle interactions, such as:

- friction between contacting particles;
- plastic deformation or recoil during collisions;
- gravitational forces, relevant mainly in large-scale or astronomical simulations;
- attractive interactions such as cohesion, adhesion, liquid bridging, and electrostatic forces.

One of the main advantages of DEM is its ability to simulate a broad range of granular flow and rock mechanics problems with good agreement with experimental observations. Moreover, DEM enables

the investigation of the micro-dynamics of granular systems in ways that are often impossible experimentally. For example, force transmission networks inside granular media can be directly visualized through simulation.

The main limitation of DEM is its high computational cost. Real granular systems may contain billions of particles, while current DEM simulations are still constrained by available computational power and simulation time.

Granular Materials

Today, many research groups focus on the experimental comparison of granular jamming systems in order to identify the most suitable granular materials and to understand the relationship between particle properties and stiffness variation performance. Experimental campaigns typically compare different fillers under identical conditions to evaluate their effectiveness.

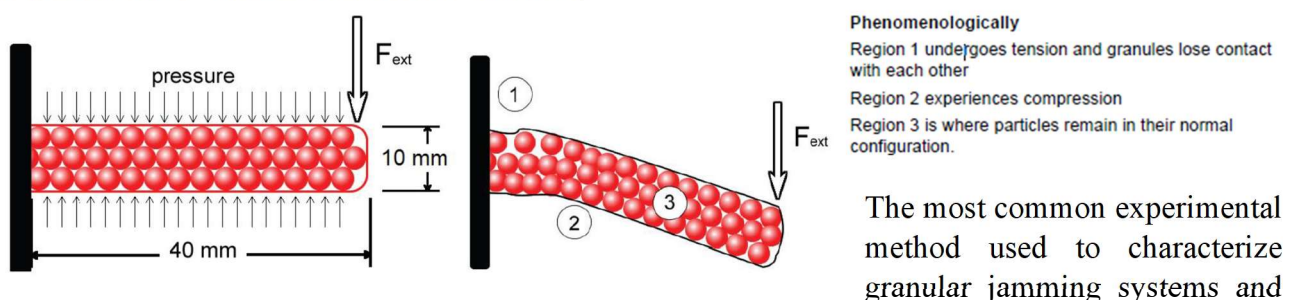
Among the most commonly used granular materials, **coffee grounds**—both fine and coarse—have shown particularly good performance. According to several studies, coarse coffee performs slightly better because of its irregular particle shape, which promotes stronger mechanical interlocking between particles during jamming. Other commercially available materials that provide good results include sand and rice.

In granular jamming systems, several parameters can be tuned, including:

- particle size,
- particle material,
- particle shape,
- particle texture and friction properties.

MAJOR ISSUES: However, these systems are highly **protocol dependent**. Since the particle arrangement is intrinsically random, achieving perfectly repeatable experiments is difficult. This randomness, combined with the large number of influencing parameters, represents one of the major challenges in the characterization and design of granular jamming systems.

Experimental Characterization of Stiffness



compare stiffness variations is the **cantilever beam test**, where a fixed-ended beam or cylindrical structure filled with granular material is subjected to a point load at its free end. By measuring the resulting deformation, it is possible to estimate the effective change in stiffness or elastic modulus of the system.

Typically, two testing conditions are compared:

- **ambient pressure**, where the structure remains soft and compliant;
- **vacuum conditions**, where the membrane compresses the particles and the structure stiffens.

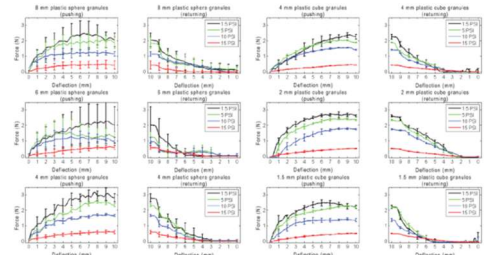
The applied vacuum level itself is another important parameter. However, stiffness modulation is often not linear with pressure. In many cases, there exists a pressure threshold below which the

structure remains relatively soft, followed by a sudden and sharp increase in stiffness once jamming is triggered. As a result, the usable stiffness modulation range is relatively narrow.

Experimental plots are generally presented in pairs, showing:

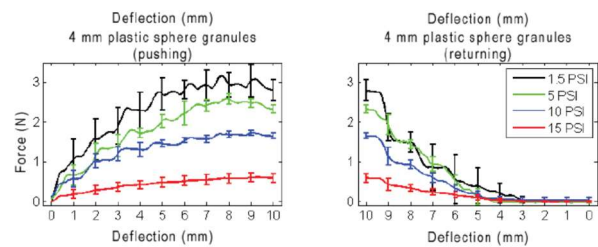
- a **loading curve** (during deflection increase),
- an **unloading curve** (during release).

These experiments are repeated for different particle geometries and sizes, such as spherical or cubic particles with dimensions ranging from 1.5 mm to 8 mm.



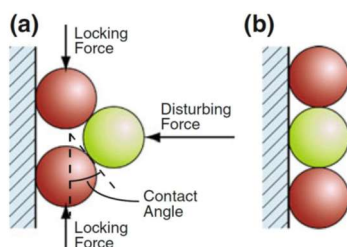
KEY INSIGHTS: Two important behaviors are consistently observed in these experiments:

- 1) **step-like increases in the loading curves**, characterized by sudden spikes in force during deformation;
- 2) **apparent plastic deformation during unloading**, where the force returns to zero before the structure fully recovers its original shape.



These effects arise from different particle-level mechanisms.

a) Particle Intrusion

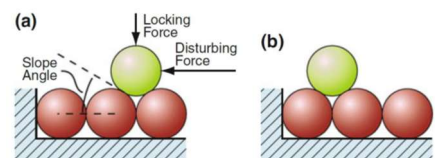


Particle intrusion occurs when a particle is forced between neighboring particles under external loading. For example, if a third sphere is pushed between two aligned spheres and the contact geometry allows it, the particle suddenly penetrates the structure. This rapid rearrangement produces abrupt jumps in the force-displacement curve, explaining the step-like behavior observed experimentally.

Because the particle arrangement changes during loading, the system may not immediately return to its original configuration after unloading. This produces an apparent plastic deformation, even though the deformation is not truly permanent. Once the vacuum is removed, the particles can rearrange again and the original configuration can largely be recovered.

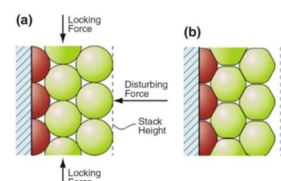
b) Particle Hopping

Another mechanism is **particle hopping**, in which particles change their relative position within the granular network without penetrating between neighboring particles. This rearrangement also contributes to the irregular steps observed in the loading curves and to the apparent residual deformation during unloading.



c) Particle Deformation

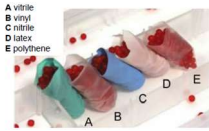
A third possibility is direct **particle deformation**. In this case, instead of rearranging, the particles themselves deform under load. The relative importance of deformation versus rearrangement depends strongly on the particle material and geometry.



FOCUS: Role of the Membrane

Although the granular filler is the primary element responsible for the jamming transition, the enclosing membrane also plays an important role in determining system performance.

Experimental studies comparing different membrane materials—including vitrile, vinyl, nitrile, latex, and polyethylene—have shown that membrane properties significantly affect the behavior of the system. Even without a complete analytical model, these studies helped identify the main parameters influencing granular jamming devices.



The relevant parameters can be divided into two groups:

Granular material parameters

- particle size,
- particle material and deformability,
- particle shape,
- particle texture and friction.

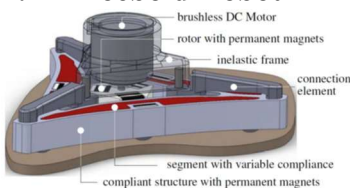
Membrane parameters

- membrane thickness,
- membrane elasticity,
- membrane surface friction and texture.

Applications of Granular Jamming

Although current modeling tools are still unable to fully describe the physics of granular jamming, the technology has already demonstrated remarkable effectiveness in many practical applications.

1. Amoeboid Robot

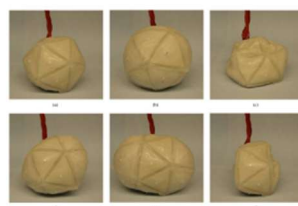
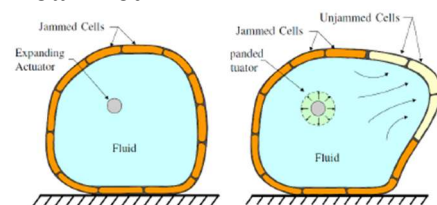


One of the earliest examples is the **Amoeboid robot**, a triangular locomotion system driven by a brushless DC motor that generates vibrations. The robot moves through the periodic deformation of an elastomeric structure containing three jamming-based segments with reversible variable stiffness. By selectively applying vacuum to these

chambers, the stiffness of individual sides can be modified, breaking the symmetry of the structure. As a result, the robot acquires a preferential direction of motion.

KEY-POINT: This application highlights an important concept: jamming can be exploited not only to increase stiffness, but also to intentionally introduce **asymmetries in material properties**, enabling directional locomotion and motion control.

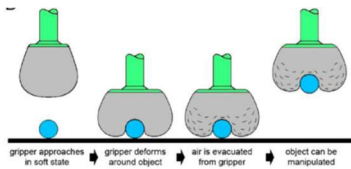
2. JamBot



Another important example is **JamBot**, developed by the research group led by H. M. Jaeger, E. Brown, and E. Steltz, who were among the first to introduce jamming systems into soft robotics around 2009–2010.

JamBot consists of a central expandable actuator surrounded by multiple jamming cells made of elastomeric membranes filled with granular material. Each cell can be independently controlled through dedicated vacuum lines. When selected cells are unjammed while others remain stiff, activation of the internal actuator causes expansion only in the compliant regions. Consequently, the initially spherical robot deforms into an asymmetric shape, shifting its center of mass and generating motion or rotation. In this way, jamming modulates both the direction and magnitude of the work produced by a single actuator.

3. Universal Gripper

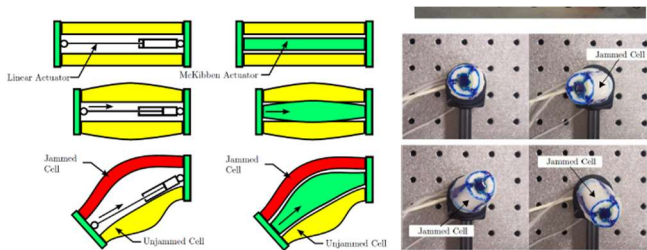


The most famous application of granular jamming is probably the **universal gripper**, developed by the same research group. The gripper consists of a flexible membrane filled with coarse coffee grounds. When the membrane is soft, the granular material flows around the target object and adapts to its shape. Once vacuum is applied, the coffee

grounds jam together, causing the membrane to contract and harden around the object. In this way, the gripper can securely grasp a wide variety of shapes without requiring complex sensing or control strategies.

This technology clearly demonstrates the practical potential of jamming systems. However, because predictive modeling tools are still limited, device design remains heavily dependent on experimental testing. This limitation also contributed to the difficulties faced by the startup company **Empire Robotics**, which attempted to industrialize the universal gripper concept at large scale.

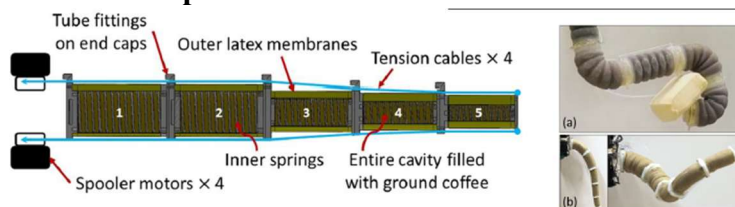
4. Jamming Modulated Unimorph (JMU)



The **Jamming Modulated Unimorph (JMU)** further illustrates the importance of symmetry breaking in jamming-based systems. The device combines a single linear actuator—independent of the specific actuation technology, for example a McKibben actuator—with multiple jamming cells.

By selectively jamming different regions, a one-degree-of-freedom linear actuator can be transformed into a multi-degree-of-freedom bending actuator. This approach effectively changes the kinematics of the system, introducing the concept of **variable kinematics**, which is particularly useful in soft grippers and adaptive manipulators.

5. MIT Manipulator

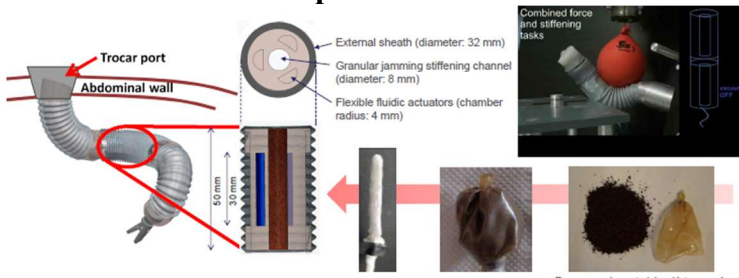


Another example is the **MIT manipulator**, developed by Cheng and collaborators. In this design, tunable stiffness obtained through granular jamming is combined with

off-board spooler motors and tension cables to generate complex manipulator configurations.

The manipulator consists of multiple segments filled with granular material. By controlling the jamming state of each segment, the structure can transition reversibly between soft and stiff configurations, enabling both flexibility and load-bearing capability.

6. STIFF-FLOP Manipulator



The **STIFF-FLOP manipulator**, previously introduced in earlier lectures, represents one of the most advanced medical applications of jamming technology. The manipulator is composed of a series of identical modules, each consisting of:

- a silicone tube with pneumatic chambers enabling 3D motion;
- a central channel dedicated to the jamming mechanism.

The central channel contains granular material that can be jammed to locally increase stiffness and force transmission when needed.

	WEIGHT	WORKING CHANNEL	DoM	VARIABLE STIFFNESS	STIFFENING/ STABILITY	WORKSPACE
STIFF-FLOP	8.5 g	YES	4 (3 bending + elongation)	NO	-	***
MODULE A	9.2 g	NO	4 (3 bending + elongation)	YES	+	**
MODULE B	7.6 g	YES	1 (bending)	YES	**	+
MODULE C	9.1 g	YES	4 (3 bending + elongation)	YES (+716%)	**	**

FIBER JAMMING: Initially, coffee grounds were used as the granular filler. However, because the manipulator is long and slender, granular jamming alone provided limited stiffening performance. This led researchers to explore alternative solutions, particularly **fiber jamming**, which proved more effective for elongated structures.

More recent versions of the STIFF-FLOP design further improved performance by relocating the stiffening chambers externally. Instead of fully surrounding the manipulator with fluidic actuators, the new architecture alternates pairs of pneumatic actuators with dedicated fiber-jamming chambers, producing stronger and more efficient stiffening behavior.

Layer Jamming

Layer jamming is based on the same physical principle as granular and fiber jamming, but with a different filler dimensionality. Instead of particles or fibers, the system uses thin flexible layers enclosed within a deformable membrane.

A layer jamming system typically requires:

- a **cover membrane** that is deformable and resistant to failure;
- a **filler material** characterized by low bending stiffness and high tensile strength.

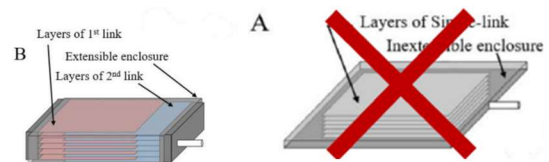
When vacuum is applied, atmospheric pressure compresses the layers together, increasing friction between adjacent surfaces and preventing sliding. This transforms the system from a compliant structure into a much stiffer one.

NOTE: Unlike granular jamming, layer jamming is easier to model analytically because the geometry and layer arrangement are more controllable and repeatable. Thin layers can be distributed with high accuracy, making the system less protocol-dependent. For this reason, layer jamming has enabled the development of more detailed analytical models capable of relating design parameters to stiffness variation performance.

Axial Forces: System Behavior

When studying axial forces in layer jamming systems, the configuration of the layers is critical. Two possible arrangements can be considered:

- **independent layers (A)**, which are not properly constrained and provide poor performance;
- **grouped layers (B)**, connected to opposite sides of the structure, which represent the preferred configuration.



Once vacuum is applied, a pressure difference ΔP is generated between the inside and outside of the membrane. The membrane collapses onto the layers, increasing friction and producing a stiff behavior. Initially, the response is approximately linear, but beyond a critical load the layers begin to slip relative to one another.

The **resulting** axial resisting **force** has been modeled as:

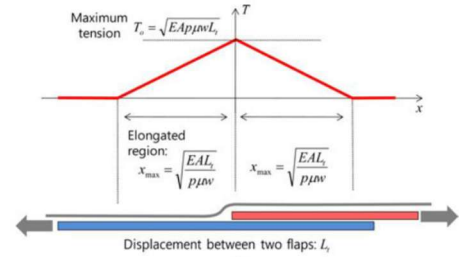
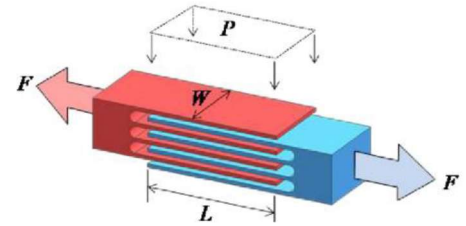
$$F = \mu npwL$$

where:

- μ is the friction coefficient,
- n is the number of layers,
- p is the applied pressure,
- w is the overlap width,
- L is the overlap length.

Therefore, the resisting force increases with friction, pressure, overlap size, and number of layers.

An additional contribution comes from the **membrane tension**, which is not negligible. The highest tension occurs in the elongation region of the membrane, which consequently becomes the area most prone to deformation and sliding initiation.



Static Conditions

The static behavior of a layer jamming system can be described using a one-dimensional free-body model. In the schematic representation, one side of the layered structure is fixed while the opposite side is pulled by an external tensile force $F_{tensile}$. Two resisting effects oppose this motion:

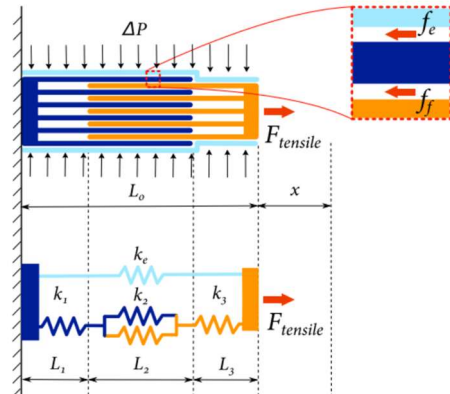
- the friction force between layers, F_f ;
- the elastic interaction force between the layers and the membrane housing, F_e .

Phase I: Pre-Slip Regime

Before sliding begins, the tensile force is smaller than the maximum static friction force between layers. In this condition, the system behaves like a stiff spring and stores mechanical energy.

Three distinct layer regions can be identified:

1. layers connected to the left side,
2. layers connected to the right side,
3. the overlapping central region where both layer sets interact.



Since each region has different geometric properties, the system can be modeled as **three springs in series**. The effective tensile response is therefore:

$$F_{tensile} = k_{effective}x$$

with

$$k_{effective} = \left(\frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3} \right)^{-1}$$

and

$$k_i = \frac{E_i A_i}{L_i} (i = 1, 2, 3)$$

where:

- E_l is the tensile modulus of the layer material,
- A_l and L_l are the cross-sectional area and length of each layer region.

Phase 2: Post-Slip Regime

When the applied tensile force exceeds the static friction threshold, the layers begin to slide relative to one another. In this regime, the system dissipates mechanical energy through friction.

The equilibrium condition becomes:

$$F = F_f + F_e$$

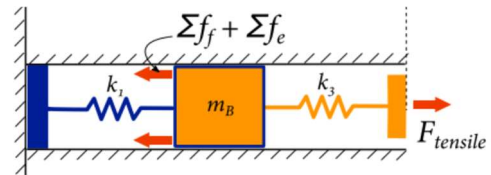
where the friction contribution is:

$$F_f = \mu(n - 1)W(L_2 - x)\Delta P$$

and the elastic interaction with the membrane housing is:

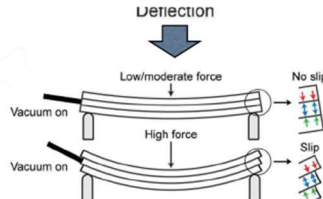
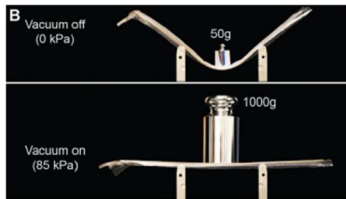
$$F_e = 2\sqrt{E_e A_e \Delta P \mu_e W x}$$

In this phase, the motion of the layers can be represented dynamically as a sliding block connected to springs and subjected to resisting frictional forces.



Transversal Forces:

The behavior of layer jamming systems under transversal loads is similar to the axial case and is strongly influenced by the applied vacuum pressure. This is clearly visible in the classical **three-point bending configuration**, where the structure exhibits a dramatic increase in stiffness when vacuum is applied.

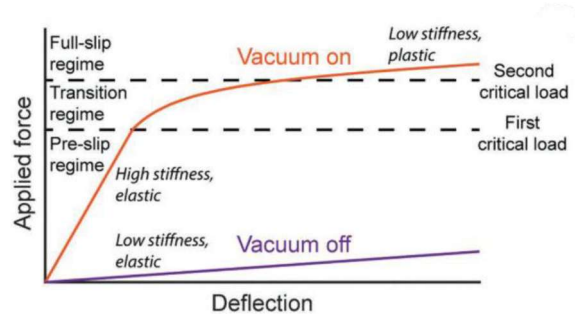


Unlike axial loading, transversal loading does not require the layers to be alternately

connected to opposite ends. Since the applied force is perpendicular to the layers, simpler layer arrangements can still produce effective stiffening behavior.

When vacuum is applied, the force–deflection response can be divided into three main regimes:

1. **Pre-slip regime:** the layers remain locked together by friction, and the structure behaves with very high stiffness and elastic response.
2. **Transition regime:** once the applied load exceeds a critical threshold, partial sliding between layers begins.
3. **Full-slip regime:** all layers slide relative to one another, and the structure loses much of its stiffening capability.



Simplified Analytical Model

A widely used simplified analytical framework for transversal layer jamming was proposed by Aktas et al. The model compares two idealized configurations:

- a system composed of n independent layers, representing the **unjammed state**;
- a fully bonded single block of material, representing the **jammed state**.

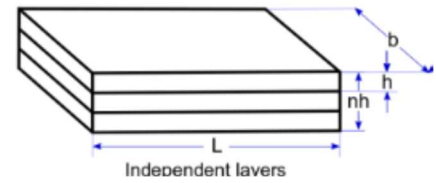
Unjammed State

For n independent layers, the total moment of inertia is obtained as the sum of the inertia of each individual layer:

$$I = n \frac{bh^3}{12}$$

where:

- b is the layer width,
- h is the thickness of a single layer.



The corresponding bending stiffness is:

$$k_0 = \frac{4Ebnh^3}{L^3}$$

where E is the Young's modulus and L is the beam length.

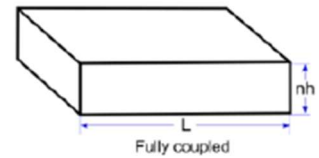
Jammed State

In the jammed condition, friction fully prevents relative sliding, and the layered structure behaves as a single solid block with total thickness nh . The moment of inertia becomes:

$$I = \frac{b(nh)^3}{12}$$

and the stiffness is:

$$k_j = \frac{4Ebn^3h^3}{L^3}$$



Stiffness Ratio

The ratio between jammed and unjammed stiffness is therefore:

$$r = \frac{k_j}{k_0} = n^2$$

KEY-POINT: This result shows that the **number of layers** n is the dominant parameter controlling stiffness variation. Doubling the number of layers leads to a fourfold increase in stiffness ratio. Consequently, for a fixed overall thickness, using thinner layers is advantageous because it increases the number of interfaces and amplifies the stiffening effect.

Force Regimes

Using this framework, the forces acting during the different operating conditions can be approximated as follows.

Unjammed State

In the unjammed configuration, the force–deflection relationship is:

$$F = k_0\delta$$

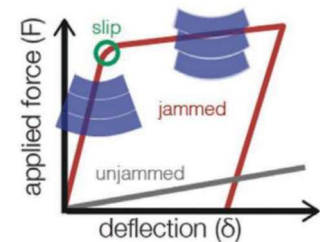
where δ is the deflection.

Jammed State: Pre-Slip

Before sliding begins, the jammed structure behaves as a rigid block:

$$F_{pre-slip} = k_j\delta$$

Slip Threshold: The transition between elastic and slipping behavior occurs when the applied force reaches the slip threshold:



$$F_{slip} = \frac{2\mu PA}{3}$$

where:

- μ is the friction coefficient,
- P is the applied pressure,
- A is the contact area.

This force separates the elastic regime from the sliding regime.

Post-Slip Regime

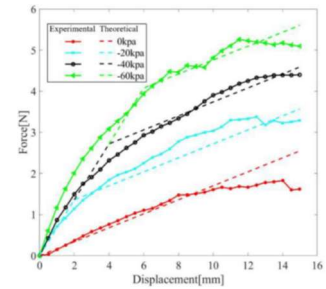
After complete sliding begins, the layers again behave approximately as independent elements, giving:

$$F_{post-slip} = k_0 \delta$$

LIMITATIONS: Although this analytical framework is useful for understanding the dominant parameters, it provides only a simplified description of the real behavior. The actual experimental response differs significantly from the idealized piecewise-linear model.

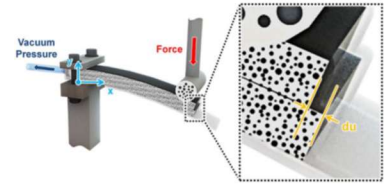
In practice:

- stiffness increases continuously with vacuum level;
- post-slip behavior is strongly nonlinear;
- a sharp slip threshold is rarely observed;
- the transition regime is always present, producing a gradual change in behavior rather than abrupt phase transitions



Hybrid Analytical–Experimental Model

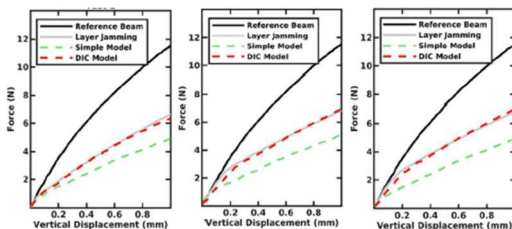
To obtain a more accurate description of transversal layer-jamming behavior, hybrid analytical–experimental models have been developed. These approaches combine analytical formulations with experimental measurements obtained through **Digital Image Correlation (DIC)** techniques.



Using DIC, researchers were able to measure more precisely the relationship between:

- the curvature of the structure during bending,
- the relative displacement and discontinuity between adjacent layers.

As curvature increases, the relative separation generated between layers also increases, strongly influencing the onset of slip and the overall stiffness behavior.



Validation results show that this hybrid model predicts the mechanical response much more accurately than the earlier simplified analytical framework. In particular, it successfully captures the nonlinear behavior observed experimentally and provides significantly better agreement with measured force–deflection curves

Recent Developments: Slip Evolution

More recent studies, including work by Y. Narang during his PhD at Harvard, investigated the spatial evolution of slip along the beam length. Analytical models initially suggested that slip could start near the center of the structure and progressively propagate toward the tip.

However, both intuition and experimental evidence indicate that the critical resisting force is largest near the fixed end of the beam and lower near the free end. Consequently, sliding is more likely to initiate near the tip and then propagate toward the clamped region. Experimental measurements confirmed this behavior by showing lower critical forces near the free end of the structure.

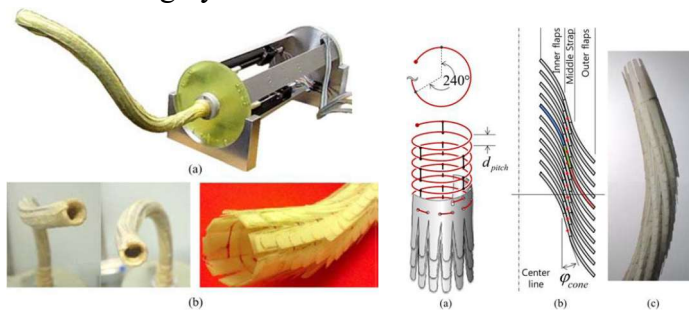
These studies also led to the development of systematic design methodologies for jamming-based technologies. The resulting procedures can be applied to systems subjected to axial forces, transversal forces, bending moments, and torsional moments.

Additional Applications of Layer Jamming

Layer jamming has also been explored in several other application domains, including:

- wearable devices,
- adaptive grippers,
- medical and surgical manipulators.

FOCUS: One particularly important example is the **snake-like manipulator** designed for minimally invasive surgery.



In this system, the layer-jamming mechanism exploits friction between thin flexible layers controlled through vacuum pressure. The structure is lightweight, compact, and hollow, making it particularly suitable for endoscopic and surgical applications where controllable stiffness is required.

The layers are arranged in a helical configuration using thin sheet-like materials with flaps wrapped around the structure. The entire assembly is enclosed within a membrane:

- when no vacuum is applied, the layers can freely slide and the manipulator remains highly flexible;
- when vacuum is applied, friction locks the layers together and the structure behaves as a rigid rod.

By integrating cable-driven actuation, the manipulator can also actively bend and steer, making it suitable for advanced endoscopic navigation and minimally invasive surgical procedures.

Electro-Active Polymers

HISTORY:

Electro-Active polymers is a huge class of technologies. Related to this technique, everything begins when some researchers started to play with elastomers, watching the effect of electricity/electric field on elastomers. The first effect that was observed and studied in depth was the piezoelectric effect related to elastomers, but it wasn't very successful (1925), for this reason there are just few applications. Then, some researchers, especially in Japan, started to see the effect of adding some conductive materials to elastomers in order to have conductive polymers. And this is the base of what happens later on, especially in 1990s, when a couple of more successful technologies were discovered, like ionic polymer composite and the use of dielectric elastomers (this technology was used also in 2003 in a spin-off called Artificial Muscle)

DEFINITION:

There are a lot of technologies, but the list of elements that are shown in this lesson, are less or more exhaustive. There are a lot of classes because, usually researchers introduce new technologies that are slightly different and try to define new classes and names. Despite of this, it's possible to divide materials in 2 main groups: **ELECTRONIC (DRY) EAP** and **IONIC (WET) EAP**.

- The common feature they all have is that they are electrically activated, and this is an ACTIVE TECHNOLOGY (as we saw at the beginning this technology is able to increase and decrease the level of energy of the system where it's installed into).
- The second common feature is that they are based on elastomers and whatever is added to make them reactive to electricity, power, current or electric field, doesn't change their mechanical properties, which is really soft and close to what we have in our muscle. This is one of the first class of artificial muscles associated to artificial muscles, especially for passive characteristics, but in some cases also for active.

Now let's start with the 3 groups

ELECTRONIC (DRY) EAP

This is the list of electronic EAPS:

1. Electrostrictive Graft Elastomers
2. Ferroelectric Polymers
3. Liquid Crystal Elastomer (LCE) Materials
4. Dielectric EAP

Electronic vs ionic:

Why electronic? Because the triggering of the actuation principle is due to the migration of electrons, while in ionic EAP, migration of ions is the triggering of the actuation mechanism. Why dry? Because usually these kinds of actuators can work in dry conditions, they don't need wet condition or solution. They don't need specific condition of the environment; in contrast ionic EAP works in solution and exchange ions with the environment (their trigger mechanism is based on the

exchange of inside and outside the main matrix of the actuator). For this reason, usually ionic EAP **encapsulated**, if you need it in a general purpose outside a wet environment.

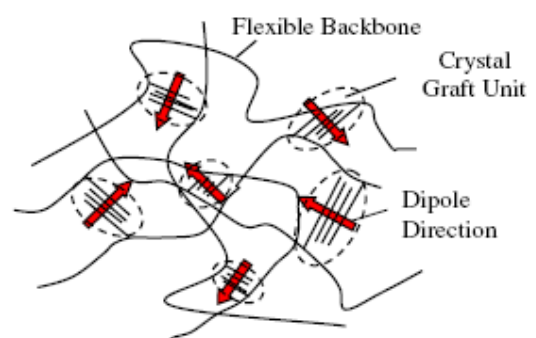
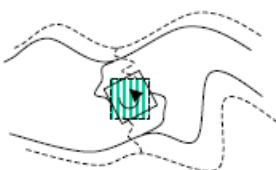
Going back to electronic EAP, what are the **main advantages and disadvantages of this class:**

- ✓ In general, this entire class, **can operate at room temperature and conditions**. It doesn't need specific conditions of the environment.
- ✓ These materials are quite **fast in reacting**. This implies that the response time is quite low (ms level). This is linked to the fact that the trigger mechanism is based on electrons migration, that is very quick.
- ✓ They can **hold strains under DC activation**.
- ✓ They can induce **relatively large actuation forces**. Force is usually quite high, but this implies that the deformation (strain) is quite limited → whenever there is force, there isn't deformation and whenever there is strain, there isn't force. Every time you see an advantage related to stress or force, it's because there is a disadvantage related to strain. In any case you need a leverage system or amplification mechanism or whatever you like to have a different trade-off.
- × This technology requires a very **high voltage**. The field is usually about 150 Mv/m. of course you have to pay attention of this. But, although electric elastomers require a very high electric field, usually they **require a very low amount of current**. Indeed, you don't need an extremely powerful sources, but you need a DC-DC converter, it's converting the power that you have (the source) in two products between the potential and the current. This technology is based on a capacity principle, so, if you have high voltage and low current it is not going to kill you. It's not dangerous, it's also not advised touching it with your finger, but the power that you need to activate the actuator, is not very high (although a high voltage, the current is extremely small).
- × **Strains are usually low**, so you need some compromise
- × You need to pay attention to **glass transition**, but for very low temperature tasks.

Electrostrictive Graft Elastomers

In this technology, there are elastomers composed by chains of polymers. The usual elastomeric matrix is formed by chains of polymers, as main backbones, and residuals that are attached to other polymeric chains. The backbones and the residuals, that are connecting several other chains, are what composes an elastomer.

In the electrostrictive graft, there are graft → they are inclusions of crystal units with a dipole that is attached



between flexible chains, composing the matrix. Applying an electric field it's possible to reorient grafts, depending on the dipole direction. Since the grafts are connected to flexible chains, the reorientation makes the grafts pull the backbones. This reorientation causes a re-adaptation of the chains and overall makes the entire material deformed.

Application of external electric field → reorientation of grafts attached to chains → chains are also reorientated → overall deformation of the material.

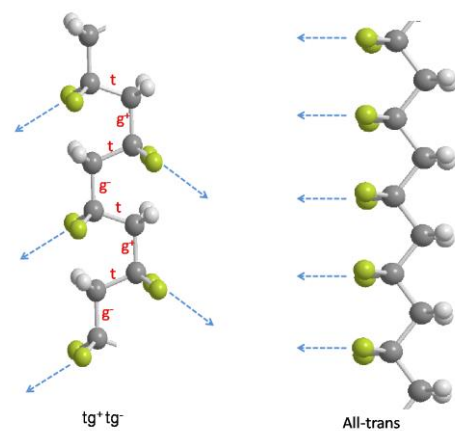
On the slide are also reported some characteristics that are used to fabricate these materials. The voltage required is quite high, but the energy density is also quite high. The strain is about 5%, as it was anticipated it's not too good.

CHARACTERISTICS

- > High stiffness ($E=550$ MPa)
- > Strain above 5 %
- > Relatively high force (22 MPa)
- > High voltage required (150 MV/m)
- > Energy density: 247 kJ/m³
- > Dielectric constant: 11

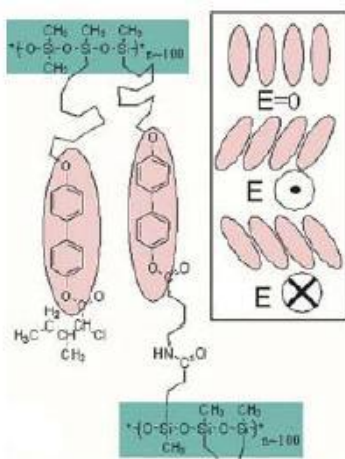
Ferroelectric Polymers:

In this case the polymer itself is made of ferroelectric materials like PDVF (Poly-vinyl-difluoride). In this case these polymers are in an amorphous matrix with dipoles randomly oriented because this is the configuration with the lower internal energy. So, what happens applying the external field? There is the reorientation of all the domains, and the polymer passes from the first image configuration to the second configuration where there is also dimensional difference. This is basically how you can use ferroelectric domains with an external electric field.



- Also, in this case the stress is relatively high.
- The required electric field is very high.
- There are some limitations related to the fatigue.
- There are also limitations on the operational range, because the ferroelectric effect is constrained by the Curie temperature, after which the material loses ferroelectric effect.

Liquid Crystal Elastomer materials:



Then there are liquid crystals elastomers.

These materials work inside the display → there is the source that emits light, then there are 2 filters at 90 degrees of orientation difference, in the middle there is the liquid crystal material. The light passes through the first filter is process by liquid crystal. According to the orientation of the liquid crystal a certain amount of light passes to the second filter and then, the second filter distributes the light into the single pixel and sub-pixel that through RGB composition generates the colour.

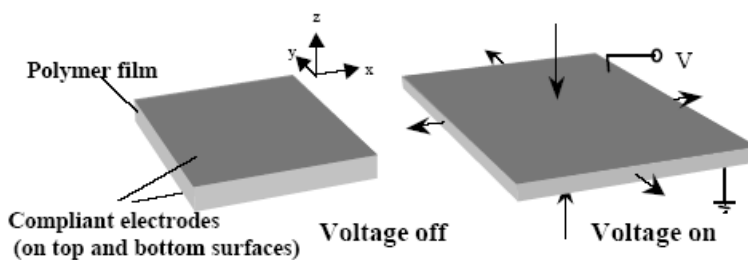
So basically, a liquid crystal is a liquid made of crystals that **can be oriented thought the application of an electric field**. In this case,

you have elastomers, so more solid phase, but the principles are exactly the same→ these crystals, called mesogens, are connected between polymer backbone and, by applying an electric field, the mesogens are reorientated. The orientation and rotation of these domains is able to deform the material.

- Response time is very fast.
- Stress is quite remarkable.
- Stiffness is low.
- The reverse process requires cooling, because the effect is also associated with an increase of temperature. Indeed, you can also activate the process also just applying a thermal input.
- Quite remarkable electric field.

Dielectric EAP:

The last class of electronic dry electroactive polymers is dielectric elastomers. The principle is extremely simple and for this reason this kind of activator are very famous and the most studied also in terms of publications.



Main principles:

- **Structure** → the polymer film is between 2 compliant electrodes on top and bottom of surfaces. What are the characteristics of these 3 layers? The film inside is elastomeric in terms of mechanical properties (flexible and deformable) but it's not conductive, so it's a dielectric material. the 2 compliant electrodes must bring current and charges. They are also very thin respect to the elastomeric film, but in any case, extremely flexible and deformable. From the mechanical point of view, they are very close, so both are very deformable, but the internal layer must be dielectric so not conductive.
- **Dimensions** → numerically, it's better to have the polymer film with as low as possible about hundreds of microns/ microns at maximum. At the same time, the main idea is to have compliant electrons to be almost negligible in the thickness of the entire actuator. Usually, the thickness of compliant electrodes is less than 1/10 of the thickness of polymer film.
- **Actuation idea** → when you have a film like this (a polymer between 2 electrodes) and you use an external generator to apply a field between the 2 electrodes, charges distributed on the 2 surfaces with opposite sign tend to attract each other, for the Columbian attraction. So, there are + and – on the 2 sides. If they are enough close and charges high, this Columbian attraction is enough high to squeeze the polymeric film that is in between. When there is enough charge, the Columbian attraction is able to generate what is called **Maxwell stress**

effect. for this reason, the polymer should be as soft and deformable as possible, and it shouldn't be conductive.

Why the 2 electrodes must be deformable to? The elastomer is very close to the assumption of constant volume deformation and, for this reason, whenever it's deformed in the vertical direction, it has also a deformation in the plane. This happens especially when the actuator is thin and large. So, it's really important to have electrodes compliant enough to follow this deformation and not obstacle lateral deformation of the actuator.

Actually, it's a capacitor. It's exactly the same equations to evaluate how much is the charge that is accumulated in the actuator. The only difference is that capacitors are usually made by ceramics (not deformable), that are able to maintain entirely that is given to the capacitor

This a set of equations can be used to grasp behaviour and have a rough evaluation of the force and the stress that these actuators are able to generate and what are the main parameters that affect the generation of the force and stress.

- Energy density of an electric field in a capacitor $u = \frac{1}{2}DE = \frac{1}{2}\epsilon_r\epsilon_0E^2$
- Entire energy that is accumulated that is just the density multiplied by the volume $U = \frac{1}{2}uAZ = \frac{1}{2}CV^2$
- By combining the previous two and considering the variation of the energy in a capacitor with the geometry, so how much energy varies with geometry $dU = -\frac{1}{2}\epsilon_r\epsilon_0V^22\frac{A}{z^2}dz$.
- from the previous equation, it's possible to define the force as the relationship between the variation of energy and variation of thickness. This equation is a good approximation of force generated $F = -\frac{dU}{dz} = \epsilon_r\epsilon_0V^2\frac{A}{z^2}$. ϵ_r is the material relative dielectric constant and ϵ_0 is the absolute dielectric constant. V voltage is constant; A is the area and z the thickness.
- For evaluating the stress, this is the formula $S = \frac{F}{A} = \frac{\epsilon_r\epsilon_0V^2}{z^2}$

The polymer film

The polymer film must have low stiffness, high deformability and a dielectric constant high than possible with a breakdown strength higher than possible.

Breakdown voltage = any kind of material can be conductive if the electric field is sufficient high. **There is a voltage that makes dielectric material to become conductive.** This breakdown voltage depends a lot on the chemistry of the material and varies a lot according to the composition. (For example, this is the phenomenon of thunders, indeed air is not usually conductive, but when there is enough different potential between the ground and the clouds, air becomes conductive and thunders are formed.) the same principle is valid for silicon. Usually silicon is dielectric, because it's not prone to bring charges, but if the electric field is high, also elastomers can be conductive. This is the concept of breakdown voltage. Usually, the breakdown voltage for elastomer is not very high but

also, it's not very low. But, in this case the applied electric field is really high, so it's not rare that the material can also reach this breakdown voltage

The breakdown voltage is also related to the equation of the stress, because it's true that increasing the voltage, the force is increased by the square, but you cannot use the voltage high as you like because at some point you will hit the breakdown voltage and the dielectric becomes conductive. And once, you hit the breakdown voltage, the breakdown value of the entire material lowers, and it also determines the end of the actuator → of course there is a discharge of the 2 plates, the actuator doesn't work anymore.

Electrodes

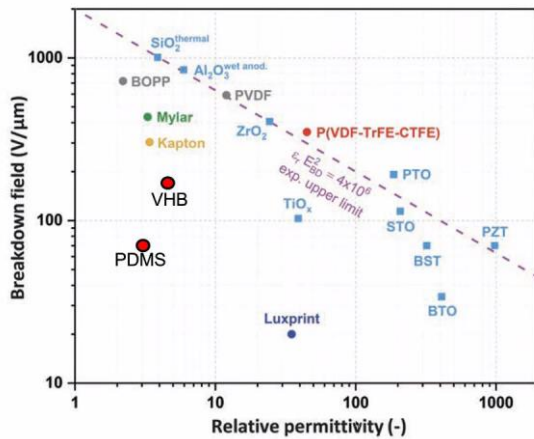
Electrodes usually are made of high-compliance materials and high electrical conductivity, and they provide **a really good contribution to stiffness (negligible)** → this is really important because if you want to make an elastomer conductive, you need to drive it with many ferroelectric material or materials that are conductive, and this usually make the elastomer quite stiff, but this is not what you need, because **stiffer is the electrode, lower will be the deformation that the dielectric material will be able to achieve**. The electrode should be this as possible with a low thickness, but **why it should be thin as possible?** → basically, the electrodes are the passive part of the actuator. They are not counted into deformation. You need electrodes, but the part that contributes to the deformation is the elastomer. The elastomer is between the electrodes and for this reason you need them are this as possible. This is why usually graphite spray, carbon grease, Graphite powder, electrolyte solution are usually used.

What are the parameters that affect this technology? The 3 main parameters are the dielectric constant, the voltage applied and thickness.

$$S = \frac{F}{A} = \frac{\epsilon_r \epsilon_0 V^2}{z^2}$$

In the force or stress equation, two out of three are on the square. Let's start from the voltage. It's possible to increase the voltage as much as you like, but it's important to pay attention to the breakdown voltage. Also, the dielectric constant can be increased. The “native” dielectric constant of an elastomer, for example a silicon, is about 8, and it's not high (for ceramics is about 1000). It's possible to increase the dielectric constant, but you need a lot of dielectric material, like ceramics powder (barium titanate). You need a lot of ceramics material just to have a small increase of the dielectric constant and this also makes the entire material much stiffer. This is the limitation: *you can find your own compromise between the dielectric constant and the stiffness of the material*, but, usually, the acceptability limit on the stiffness of the material does not justify the increase of the dielectric constant that you can obtain. There are more specific studies on material science that are trying to develop specific materials that are already very flexible, but at the same time, with a very high dielectric constant.

what we can say at the moment is that for the Maxwell stress effect, there is a sort of threshold that has been empirically derived. It seems there are some limitations about the maximum in the product between the breakdown voltage and the dielectric constant of the material. the limitation is

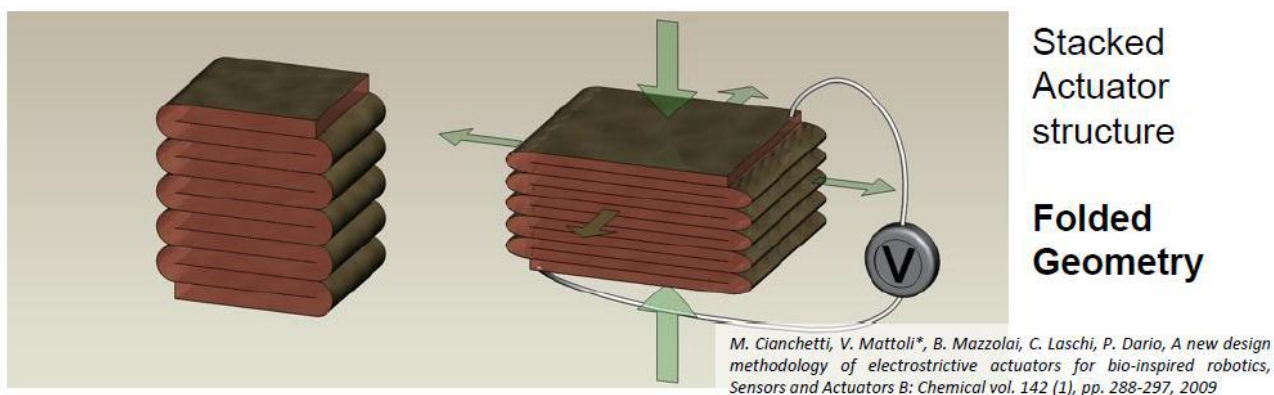


$$\epsilon_r E_{BD}^2$$

This is the experimentally value that we cannot exceed. Whatever in the lower side is possible by combining the breakdown field and the relative permittivity.

Since we are considering actuators that are not able to generate very high deformations, solutions like stacked actuator structures are very much used. In this case is possible exploit the fact that the material is an elastomer, so the stack is not made of single layers but

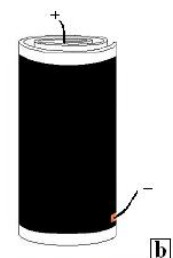
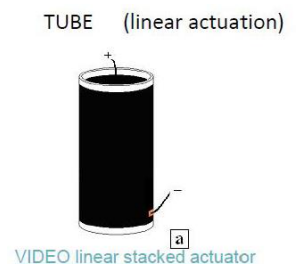
is formed by just one single layer folded on itself.



APPLICATIONS: different geometries

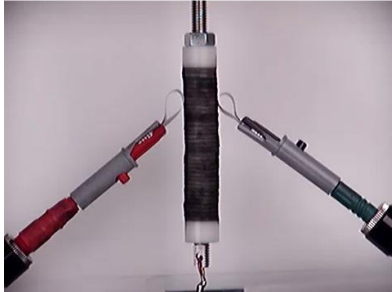
Another possibility, even simpler are reported at slide 13.

1. **The tube** is one of the possibilities, it's a way to exploit the highest deformation dielectric elastomer usually generate (it's not a real amplification). The application of the voltage is in the direction, of the image and the main effect (the squeezing effect) is in the same direction, but the small deformation on the axial direction is associated with a large deformation of the area, because of volumetric constrains. **So, the idea is to use directly the increase of the area as the output deformation.** This is the idea of linear actuator, the tube. The field, between the inner black electrode and the outer black layer, generates squeezing effect, that makes the wall of the tube thinner. **This is not a very big deformation, but if you consider how much this small radial deformation is translated into the increase of the area, you can understand how the deformation of the elongation of the tube may be.** Instead of having a contraction, there is an elongation, so basically the opposite. The effect is amplified but with the cost of having the opposite way of deforming.
2. **Spring-rolled**: building upon the previous approach, the spring rolled actuator is another option. This is the same concept but, instead of having a single



layer to create a cylinder, you have a spiral shape. What is the advantage in this case? The deformation will be exactly the same, but the force will increase, because it's basically like you have forces in parallel related to the entire spiral of the actuator.

3. Another possibility is the circular plate. There are also other possible implementations, like stacks of the elastomer fixed to a diaphragm that is moved up and down through the activation of one, the upper or the lower actuator, to generate the vertical motion.



4. On the left image is shown what is an artificial muscle. In this case the vertical deformation (axial contraction of each element) is really considered. In this case the stack is formed around by 20 layers, different layers. The good thing is that there is just only one electric source for the field, because externally all the electrodes are connected together. It's the same principle of piezoelectric materials. You need just one source

and then connecting all the layers in parallel to have the same electric field distributed in all the layers and to have the amplification of the deformation. In percentage it will remain the same, but overall, in absolute value, it's possible to see a quite remarkable contraction of the actuator.

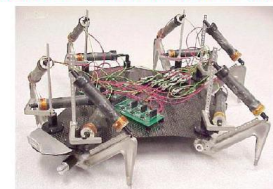
5. The other 2 cases of the slide 14 show how the opposite modality, the increase of area, is used. In the first video it's showed hoe to adjust the bending of the entire structure by having the length of the 2 layers to change. In the other case, there is a combination of 2 elastomers: one is enlarged and the other is a kind of resisting layer. In this case the usual concept of bending by the break of the symmetry, is used. One side increases the area, and the other side instead resists.

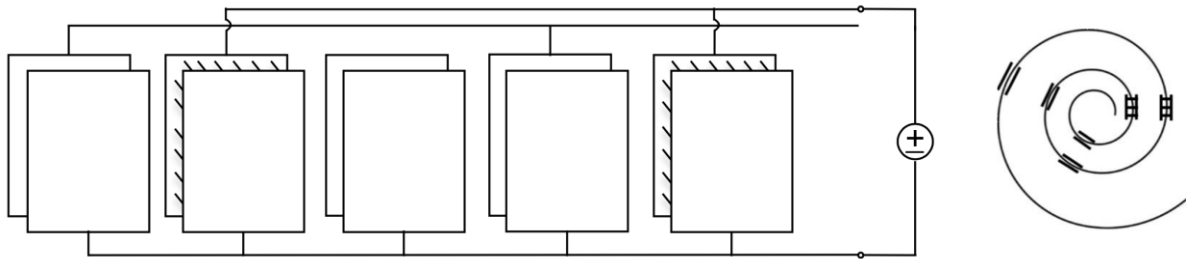
The simplicity of the principle also allows the possibility to model and study the electric elastomer in simulation trough MULTI-PHYSICS SIMULATION. Multi-physics it's a finite element method in which it's possible to combine 2 different domains. And this is exactly what you need here, the 2 domains are electric domain and mechanical domain. This is also possible on ANSYS, but we are not doing that.

Example of Dielectric EAP actuators, there are some soft actuators already produced:

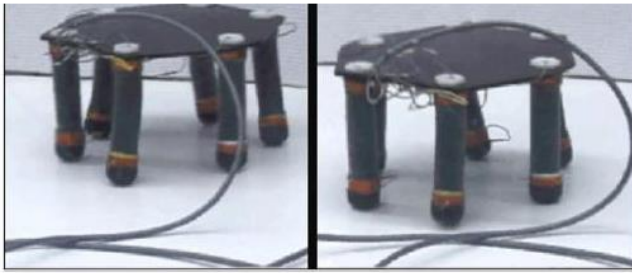
- Spring roll actuator = it's created by guys from EMPA and ETH. This is the version with spiral version. The free strain is remarkable, around 35%, but it's in elongation, and the block forces, or capability of pushing of the robot, is 7 N.
- This is another example of how this basic technology can be used also in a more complex hexapod robot. Every leg is controlled by a pair of actuators. One if for controlling the elevation and other is for controlling the planar angle. By combining them, each leg can generate a step.
- Another implementation of the same group is the MERbot, where they used a slightly different concept. The dielectric elastomer is still a spring roll, but in a little different way. Indeed, it's able to perform **omnidirectional banding**. It would not be possible with just 2 electrodes that are activated at the same time, so how did they reach it? the electrodes are not uniformly, there are pair of electrodes (1-4, 2-5, 3-6)

FLEX2 (SRI, International USA)



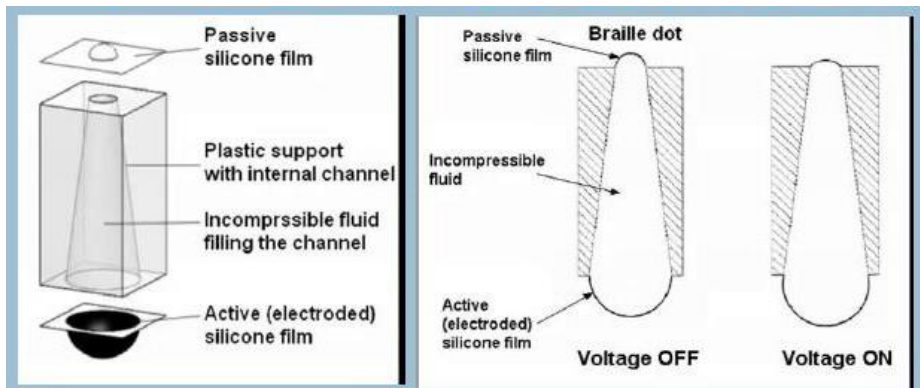


So, along the polymeric films, electrodes are in pattern, are not continuous. And then you need **to feed electrode pairs in different ways**. So, the first electrode is activated together with fourth electrode and so on, the second is fed together with the fifth. By connecting electrically



the right pairs, you are feeding electricity the same side of the actuator, and this generates the bending, of course in the opposite side because they are elongated. By combining these actuators in a more complex way, like in MERbot, that probably it's omnidirectional.

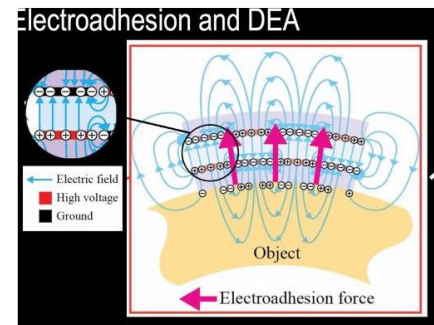
- Another possibility is the Dirigible by EMPA or Bubble EAP, when membrane is able to generate bubbles.
- **Hydrostatically coupled dielectric elastomer actuator** → this device has been developed at University of Pisa, by Centro Piaggio. This is a display for blind people, so it's a brail display. So how does it work?



So, basically, it's a display when there are areas with 6 pins and depending on if the pin is up or down, the combination of the pins means something for a blind person. Going through these areas with fingers and identify what are the pins up and what are the pins down, they can understand, read and communicate information. On the image there is a classical display. Usually, these kinds of display are quite bulky because they need to have actuators for each of these dots. So, they come up with the idea of having electroactive points on the electric elastomers to take care of these dots → of course they cannot be directly touched by the user because it wouldn't work if a finger is touching the electrode itself (there aren't conductivity risks). So, they used the **reverse principle**. It's a sort of triangle shape; in the top there is just the passive silicon and in the lower side there is the electroactive polymer that has this kind of dome shape already. Internally there is an incompressible and non-conductive fluid inside. When the **voltage is off**, the membrane is sustained the internal

pressure and **the dot is up**. By applying the voltage, the area of the dome increases. So, increasing the area means that the actuator is able to lodge more liquid, so the liquid goes in the lower side. It's a very simple approach. In this case you don't need remarkable deformations.

- Another application came up by the idea of combining electric elastomer with electro- adhesion. This is a genius idea, because basically it's free. If you have the actuator that already combines 2 electrodes, the positive and the negative, and you make it in contact with another material with a layer in between, you are recalling the opposite charges again. Whatever is the property of the material, you are able to obtain this effect of re-calling charges on the surface that are attracted by the outer layer of the electric elastomer. And this basically generates adhesion. You can take electric elastomer, deform it and conform it to whatever shape, and use electro-adhesion to make sure that the adhesion between the actuator and the object is higher. This concept is usually used to develop sort of grippers. The layer between the object and the electrodes is just some millimetres in thickness.



Below PROs and CONs are reported for dielectric elastomers, but they are very close to the entire class. Low cost and low current. Efficiency is usually quite high, because it's an electrostatic actuator, and electrostatic as principle is quite high (It doesn't involve thermal effect). What it is lost in efficiency is related to parasite current that is making the capacitor discharge on time, viscoelasticity of the material and maybe some heat, but it again related to parasitic current (extremely low).

Benefits	Drawbacks
Large strain (10% - 380%)	High voltages (> 1 KV) and fields (~150MV/m)
Moderate stress (several MPa peak)	Typically required DC-DC converters
Large work density (10k to 3.4 MJ/m ³)	Compliant (E ~ 1 MPa)
Moderate to high bandwidth (10 Hz to > 1 KHz)	Pre-stretching mechanisms currently add substantial mass and volume, reducing actual work density and stress
Low cost	
Low current	
Good efficiency (> 15% typ., 90 % Max)	

IONIC (WET) EAP

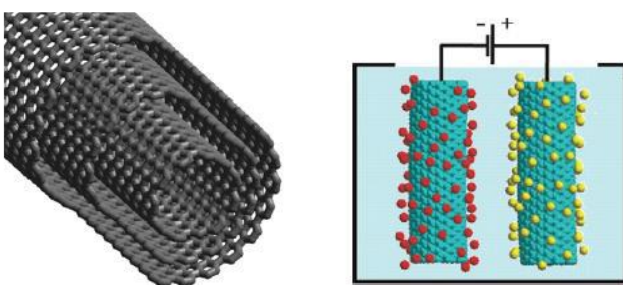
The second main group of electroactive polymers is related to ionic, wet electroactive polymers. Ionic because there is a migration of ions and wet because the triggering event is related to the migration of ions outside the material between the wet and the environment. The 4 main groups are:

1. Carbon Nano Tubes
2. Ionic Polymers Gel
3. Ionomeric Polymer-Metal composite
4. Conductive Polymers

Advantages and disadvantages of this class:

- ✓ Respect to the previous class, they need **low voltage** (2/3 V)
- ✓ They mainly provide **bending** motion than axial motion, and this bending is usually quite remarkable.
- ✓ **Large bending displacement.**
- ✓ They **don't hold strain under DC voltage**
- × **Slow response** in term of time. This is intrinsically due to the fact that ions migrate inside and outside the material, and this migration of big, bulky ions needs time. That's why the response time is about fraction of second but not millisecond. The response time is not high like in electronic case.
- × **Low forces** due to a remarkable displacement. If the displacement is high, usually forces are low.
- × Since they work in aqueous environment, there is also the **risk of hydrolysis** if the voltage exceeds the hydrolysis level.

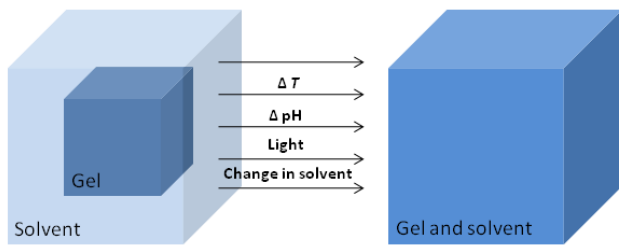
Carbon Nano Tubes (CNT):



How does it work? The quantum or chemical mechanisms are not important, instead we need to know the main principle that can make a carbon nanotube works as actuators. When you have this network of carbon nanotubes and you apply a different voltage in between 2 layers or 2 nanotubes made of these carbon layers, you can obtain the migration of ions that are

contained in the solution. They start to go into the matrix, into this net of the carbon nanotubes. The point is that this **migration and attraction of ions make them interact with the bond between the carbon atoms and basically deform them**. So, the carbon atoms, arranged in a certain way inside the specific net, start to interact with ions attracted by the carbon atoms themselves and these interaction changes the bonds and deform the carbon nanotubes, this is obviously due to the quantum mechanical effect (but we are not interested in it). We need to know that the activation of electrons attracts ions and ions act against the bond in between carbon atoms, deforming the structure.

Ionic Polymers Gel (IPG):



Hydrogels are a very huge class. There is a kind of matrix inside a solvent. **By the application of electric field** (more interesting), temperature, light, pH, or other chemicals in the solvent, **it's possible to change the osmotic interaction between the hydrogel and the solvent.**

Usually, this means that you can increase the volume of the hydrogel, induce the uptake of the solvent or dissolve the solvent. The change of osmotic pressure is caused by 3 main mechanisms

$$\Pi = \Pi_M + \Pi_{el} + \Pi_{ion}$$

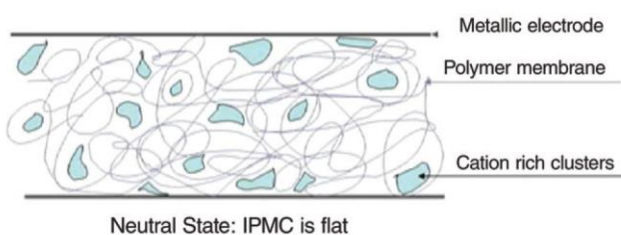
The first one is mechanical with the polymer-polymer affinity, when you have involved Van der Waals forces, hydrophobic interactions and hydrogels bonding. Elastic forces are also included in the mechanism, and the last one is related to electrostatic interaction. The main limitation of this technology is related with the time and with the reaction time.

$$t_{gel} = \frac{x^2}{2D}$$

The reaction time is inversely scales the diffusion coefficient (D). High diffusion coefficient → lower response in time (fast).

Ionomeric Polymer-Metal Composites (IPMC):

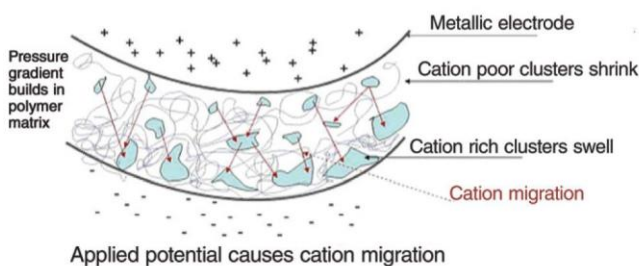
Another cases much studied is the IPMC (Ionomeric Polymer-Metal Composites). Here the trigger mechanism is a little bit different. In this case there is ions migration, but the migration is inside.



The IMPC is composed by 2 main parts:

- The internal part is an exchange membrane that acts as a polyelectrolyte. So inside there is a **polymer membrane** that has some ions clusters.

- The membrane is **between 2 electrodes**, metallic electrodes or conductive electrodes.



When the current is applied and then also the field is applied into the material, the charge accumulated on the 2 sides attracts the opposite kind of ions that are inside the polyelectrolyte. If there are cation-rich clusters and the voltage is applied in the way of the left

image, they are attracted on the lower part of the membrane. **This means that there is a volumetric excess of the material in the lower side of the membrane. The overall effect**

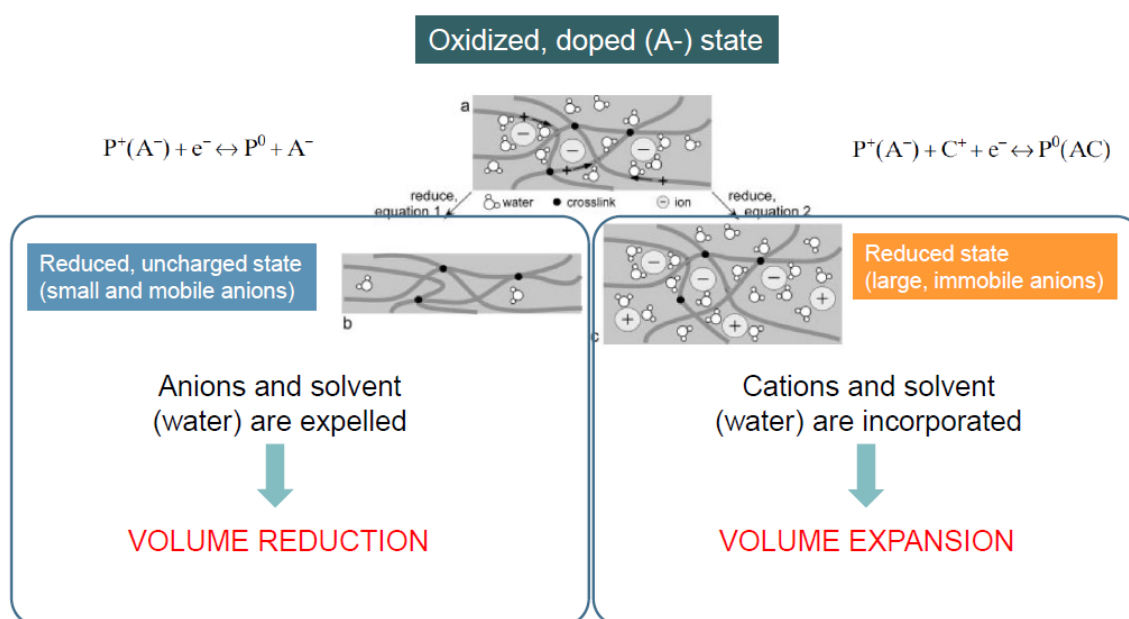
is bending. The bending is basically the main and the only way of deformation. In the video of slide 30 on the left there is a kind of swimming robot, and even though the velocity is a little bit up, it has a good performance, and the voltage is quite low.

Benefits and Drawbacks of IMPF:

- ✓ The voltage is quite low.
- ✓ Large displacement, but only in bending direction.
- × Low efficiency
- × It requires encapsulation

Conductive Polymers (CP):

Now, let's talk about conductive polymers. Conductive Polymers are polymers made conductive with addition of something else from a mechanical point of view (polypyrrole, polyaniline, polythiophene). By applying an electric field, we can change their capability of uptake or expulsion ions that are present in the solvent layer. **The capability of uptake or expulsion ions depends on the way the elastomers have been grown.** If the elastomers have been grown in presence of **bulky anions**, that are included in the polymer matrix, the way of reaction when there is the electric field is to **uptake cations that are in the solvent**. Instead, if they have been grown in presence of **small anions**, they are induced to **release anions** when are subject to electric field.

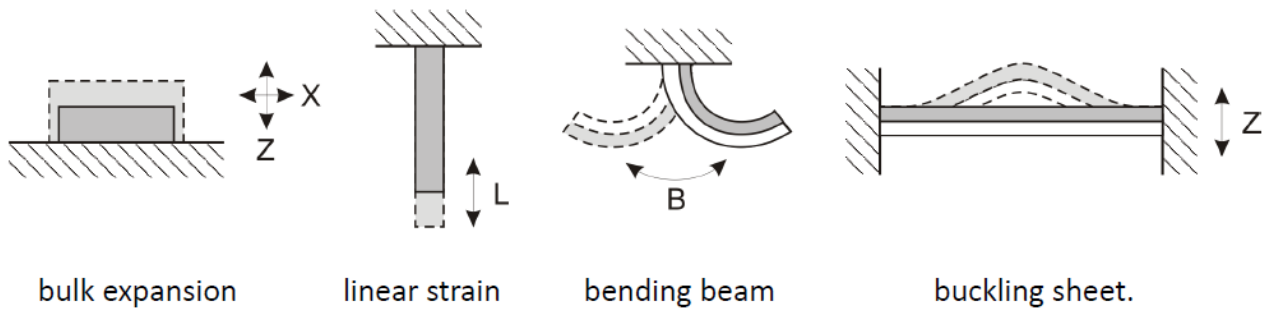


1. Left case = the polymer is produced in order to have inside small and mobile anions. if anions are small enough, they are free to move and for them it's not difficult migrate outside or inside the matrix. At beginning, the polymer and anions are in equilibrium. Then, if you add electrons, what happens to create a new equilibrium? Electrons are up taken and for the new equilibrium anion are release. The equilibrium is found by substituting anions with electrons. This generates a contradiction and reduction of the volume due to the release of anions.

2. Right case = if the polymer has been grown in presence of big anions, so they are trapped inside the polymer matrix and cannot escape. If you add electrons, to find a new equilibrium, the elastomer absorbs external cations for compensate the excess of electrons. In this case there is an increase of volume due to the adding more material inside the matrix.

Of course, it's possible to combine these 2 mechanisms, you can have different kind of actuation modes, like linear expansion, bending, asymmetries.

Actuation modes



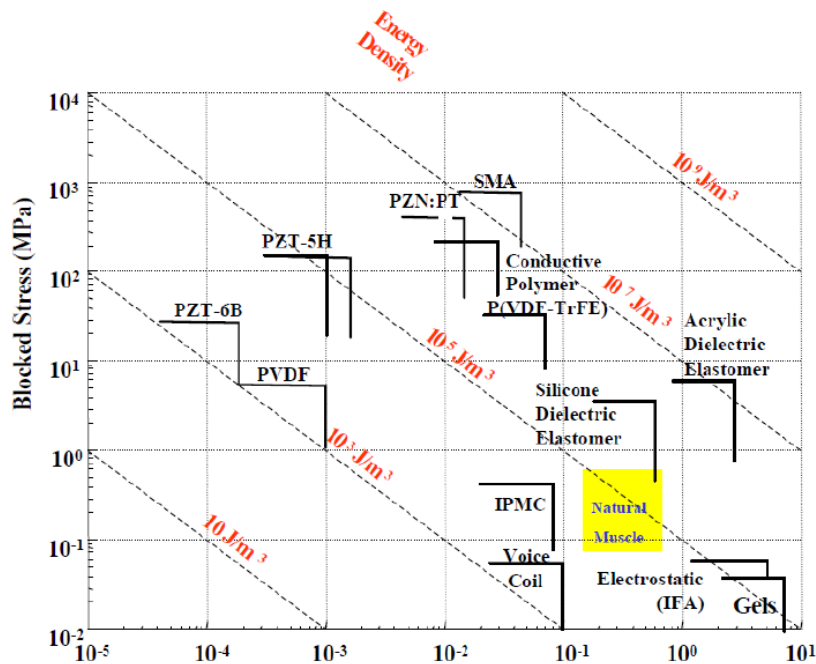
On slide 34 there are 2 examples of conductive polymers in action. The scale is very small; materials are very small.

Benefits	Drawbacks
Moderate to large strains (~3% in plane; >30% out of plane)	Typically need encapsulation
High stress (34MPa max; 5MPa typ.)	Low electromechanical coupling
Large work density (100 kJ/m ³)	Slow (several Hz maximum to obtain full strain)
Stiff polymers (~1GPa)	
Low voltage (1-2V)	

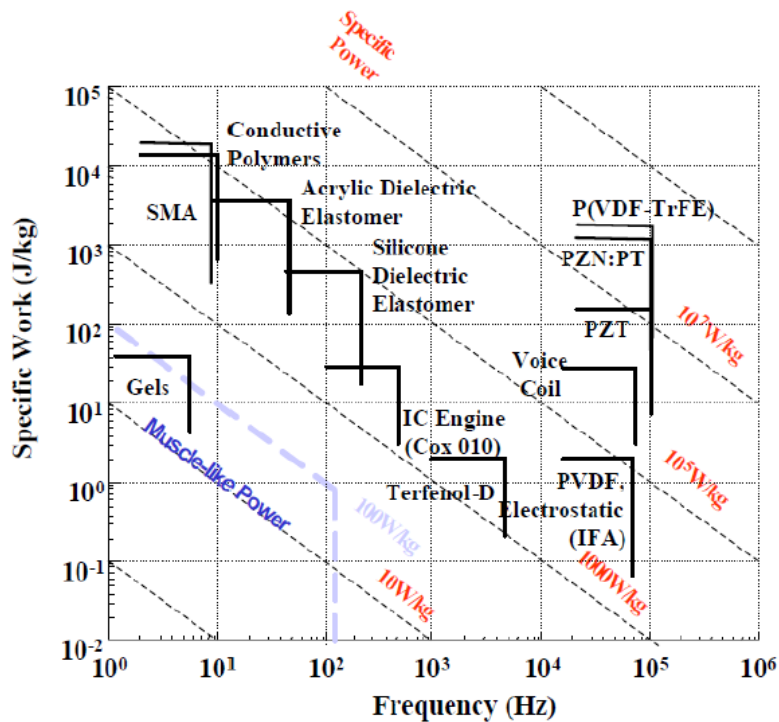
In the graph next table are reported typical values of all the technologies seen today and also the skeletal muscle is reported.

Actuator type	Strain (max) (%)	Stress (max) (MPa)	Work density (max) (J/cm ³)	Specific power (max) (W/g)	Max strain rate (%/s)	Elastic module (MPa)	Max efficiency (%)
Skeletal Muscle	20 (40)	0.1 (0.35)	0.008	0.05	500	10 - 60	40
Dielectric Elastomers	25 (>300)	1.0 (7.0)	0.01 (3.4)	0.5	>450	0.1 - 10	80
Ferroelectric Polymer	3.5 (10)	20 (45)	0.32 (1)	NA	>2000	800	40
Electrostrictive Graft Elastomers	2.5	NA	0.5	NA	NA	600	NA
Liquid Crystal Elastomers	2 (4)	0.12 (0.45)	0.02	NA	1000	100	75
Carbon Nanotubes	<2	>10	1	0.01 (0.27)	19	1000 (sheet) 10000 (fibre)	<1
Conductive Polymers	2 (20)	5 (34)	0.1 (1.0)	(0.1)	10	800 - 3000	1
Ionic Polymer - Metal Composites	0.5 (3.3)	3 (30)	(0.006)	0.003	3.3	50 - 100	1.5 - 3
Active Gels	30	0.2 (1.2)	NA	0.001	NA	<10	NA

There are also other ways to compare technologies, like the next one. In this case, you can see that silicon dielectric elastomer and acrylic dielectric elastomer are covering an area really close to natural muscle as well as IPMC



There is another table to compare these elements in terms of power, there are Dielectric elastomers, conductive polymers and hydrogels.



Electro-Magneto-Rheological Fluids

This is the last, but not least, technology. This is also the last technology related to semi-active actuation technology (technology only able to dissipate energy, it cannot increase the energy of the system that is installed into).

The history of this technology is short because both the electro-rheological and magneto-rheological principles have been discovered almost simultaneously in 1949/49.

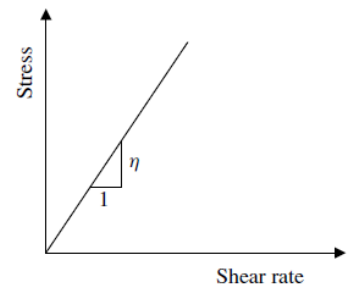
DEFINITIONS:

What is rheology? The terms rheology defines the science of the deformation and flow of matter. Basically, with these materials we can control the properties related to deformation and flow. **Rheology aims to define stress-strain relations through the rheological equations of state to study, again, deformation and flow of materials.**

Usually, fluids are divided in 2 main groups (Newtonian and Non-Newtonian)

- **Newtonian fluids** = they are fluids in which the stress and the shear rate (don't confuse it with shear) are related by the viscosity of the material. this is the simplest behaviour and simplest kind of fluid. An example of Newtonian fluid is water.

$$\tau = \eta \dot{\gamma}$$

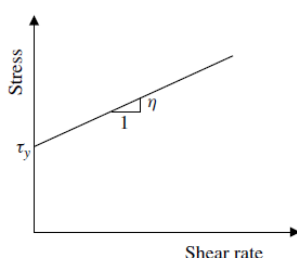


- **Non-Newtonian fluids** = there are many kinds of fluids that don't behave in the Newtonian way. They don't have a direct relation between stress and shear rate. There are mainly 3 kinds of non-Newtonian fluids:

Time dependent fluids → the state of the material depends a lot on the past/ on the previous state.

Elastoviscous fluids → they show a very limited part of elastic recovery. We are not interested in these fluids.

Time independent fluids → the relationship between stress and shear rate is not linear but still independent from time. We are interested in this group. In this group there are 3 main sub-groups (they only depend on viscosity and shear rate):

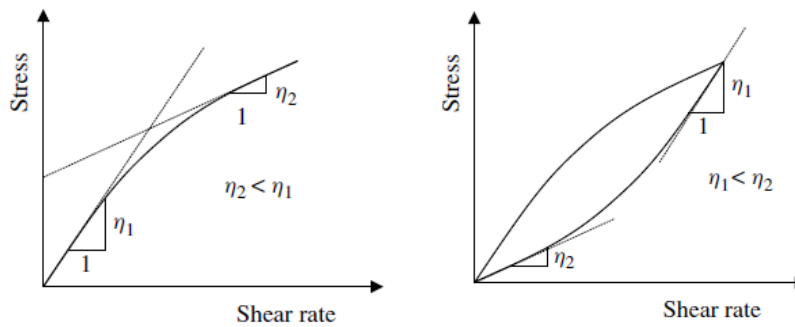


1. **BINGHAM FLUIDS:** basically, it's linear, linear respect to viscosity, but there is a threshold (τ_y). There is a threshold of stress, after that the material starts to behave linear.

$$\tau = \tau_y + \eta \dot{\gamma}$$

Before the threshold the material, the velocity it's too low, so the material acts like a solid. If the shear rate is enough high, the material behaves like a liquid, so it starts to flow.

2. PSEUDOPLASTIC $\rightarrow \tau = \eta \dot{\gamma}^n \quad n < 1$

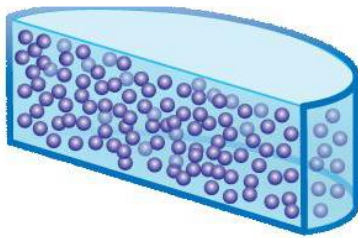


3. DILATANT FLUIDS $\rightarrow \tau = \eta \dot{\gamma}^n \quad n > 1$

Both don't behave linearly, and the non-linearity is due to the exponential terms. In the first case n is less than 1. While, for dilatant fluid, the behaviour is the opposite, so n is more than 1 and the slope of the graph increases.

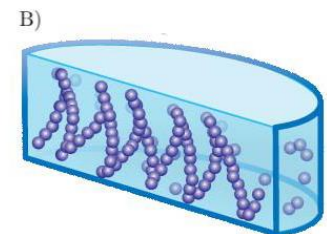
But we are more interested in Bingham fluids.

What is making this behaviour and this material interesting from an actuation point of view?



Imagine to have a liquid with particles inside, like in the image. These particles are electro- or magneto sensitive, for example soft magnets or ferroelectric materials. If you don't apply anything to the fluid, the entire material behaves homogeneously, and particles are free to flow from one side to others with very low viscosity and with more or less soft behaviours.

Now, if you apply an electric or magnetic field $\rightarrow$ in the location of the field lines, **particles, that form the magneto-/electrorheological fluid, align to the lines of the field**. It's like they are trapped inside the lines of the field. They tend to migrate and align on the lines, and this provides the creation of chains. Arranged between the 2 electrodes. This is not only a matter of redistributing particles, because they tend to remain aligned in this way. So, if any kind of load of deformation is applied to the sample, training to deform the chains, chains will resist as long as the field is applied. The resistance between the magnetic or electric field is what we can use as a mechanism of variable stiffness. The difference between the two images is that in the second case there is a field that decrease the deformability of the material.



Ferrofluids:



Now let's see more specifically what these materials are. The ferrofluids material is a non-example. Ferrofluids are materials able to react to the application of magnetic field. Ferrofluids are composed of extremely small particles $\rightarrow$ **ferrofluids are colloidal suspension of single-domain magnetic particles**. The size of particles is about 5-10

nanometers. They are very similar to magnetorheological fluids, but they are not considered inside the class because the size of the particles is too small and the application of the field changes a little the viscosity of the material. **The change between the particles is so small and it's usually not associated with a change of the resistance of the material.** The τ_y of Bingham fluids, here is not generated. This happens because the particles are too small and also the forces are so small that are comparable to Brownian forces that makes particles vibrate. This vibration in this scale is enough to prevent the generation of chains. Ferrofluids are sensitive to magnetic field but are not used as magnetorheological fluids.

Magnetorheological fluids:

Instead, magnetorheological fluids are suspension of non-colloidal, multidomain and magnetically soft particles that usually are organic or aqueous materials. It's a suspension and the size of particles is high than the previous case, and the range is **0.1 to 10 μm** . When the magnetic field is applied, the fluid undergoes a very drastic change of viscosity that can reach up to 5/6 orders of magnitudes.

Electrorheological fluids:

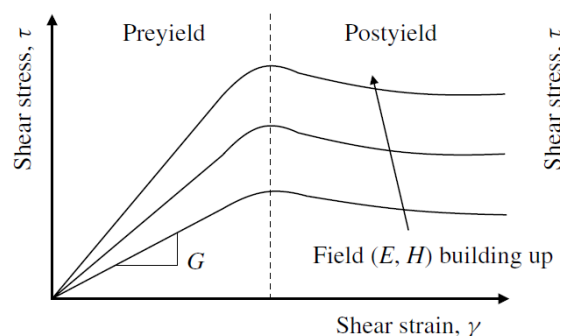
In this case the principle is the same. But, in this case, since the principle is the same, the liquid should be insulating. The particles are electro-active, so sensitive to the electric field. The size of particles is high than magnetorheological fluids (**1-100 μm**), but the maximum yield stress is 2 orders of magnitude less than the magnetorheological fluids. But the sensitive is very similar, also because the principle is similar. **There are always these particles that are suspended and trapped into field lines, increasing the viscosity of the material until a certain limit that is the yield stress. And then, removing the electric field, the fluid restarts to flow without any resistance.**

Models

Now let's see some basic models that show how this principle works. They are the same both for electric and magnetorheological fluids. We know that the material has solid-like behaviour until a certain stress threshold and after which, it starts flowing. We can study the material in the pre-yield and post-yield regions.

In the pre-yield condition, the shear stress is lower than the threshold and in the post-yield region the shear stress is higher and the material starts to flow.

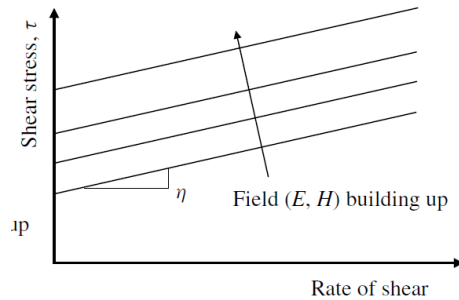
- When the material is in the pre-yield region, it's like a solid. The axis are the shear stress and the shear strain. In this case we use the "usual" constitutive equation of the material $\tau = G\gamma$ and $\dot{\gamma} = 0$. And G is just the shear modulus. This is what we can use to study the behaviour before the threshold. After τ_e you pass the elastic limit and then you find τ_y the yield limit. After τ_y



the material enters the dynamic domain and behaves like a liquid. In this graph it's important to see, that if you increase the field (E, H) there is also an increase of the shear modulus. And this is what happens in the first region. And the shear rate is zero.

- In the post-yield region, the material starts to flow. In this case the condition is described by the shear rate, the yield stress and viscosity. In this case, **increasing the field doesn't change the static viscosity of the material.**

$$\tau = \tau_y + \eta \dot{\gamma} \quad \tau > \tau_y$$



Mechatronic design concept:

How can we use this concept in terms of semi-active actuation system? We can use this mechatronic design principle that we can use to evaluate how much a system based on electro-magnetorheological tools, their effectiveness in changing their status. We can use this mechanical control lambda.

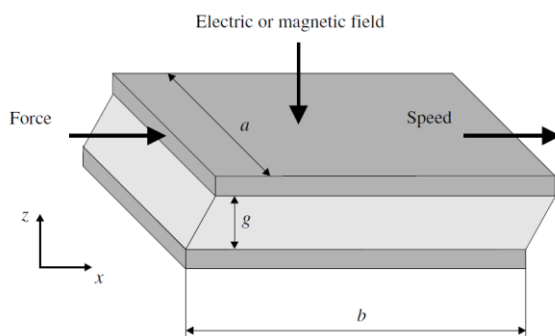
$$\lambda_E = \frac{X_\tau(E_{max})}{X_\eta(Y)}$$

$$\lambda_H = \frac{X_\tau(H_{max})}{X_\eta(Y)}$$

Lambda is the ratio between the performance and the dissipative force, when the system is subject to the maximum field → dissipative force when the field is maximum / dissipative force when the field is zero. **This ratio evaluates the dissipative capability of the system.**

How can we use in reality this concept?

1. SHEAR OR CLUTCH MODE DEVICE:



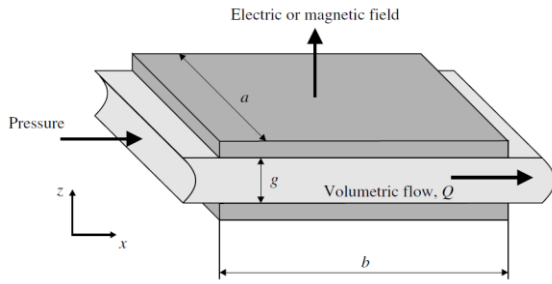
In this example there are 2 layers and between there is the fluid. The field is applied vertically, and the system is subject to a lateral force. The layer on the upper side slides on the fluids and the lower layer. There is a force, applied like in the image, that makes the layer moves laterally. If the field is applied vertically, the chains are generated between the 2 layers in the same direction of

the field. So, any external input that tries to deform these lines is obstacle and prevented. In conclusion: **if the field is not applied, the upper layer is free to slide because nothing prevents it to slide. If the field is applied, the creation of chains makes the sliding of the**

upper layer more difficult, because the layer has to break the chains that are created by the field. The 2 equation that describe the system and allow us to evaluate the mechanical control ratio, are the following:

- ✓ $F_\eta = \frac{\eta \dot{x} b a}{g}$ the upper part of the ratio describes the independent term. The viscosity term depends on geometrical conditions (a, b), velocity, so the motion of the upper plate, and by the viscosity of the material.
- ✓ $F_\tau(E, H) = \tau_y(E, H) b a$ the yield stress depends on the applied field and b and b are geometrical values.

2. FLOW OR VALVE MODE DEVICE:



In this case the geometry is the same, but here the layers are blocked and only the fluid moves. In this case the effect of the field blocks the fluid itself. In this case, the chains between the 2 plates prevent the flow of the solution between chains (for this reason it's a sort of valve). Also in this case, there are 2

equations that are the 2 terms of the control ration. One term is related to the field and the one independent.

$$P_\eta(S) = \frac{12\eta Q b}{g^3 a}$$

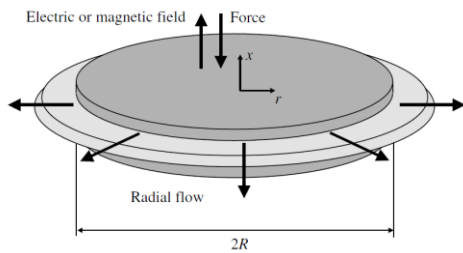
$$c = 2 \quad \text{for} \quad \frac{\Delta p_\tau}{\Delta p_\eta} \leq 1,$$

$$P_\tau(E, H) = \frac{c \tau_y(E, H) b}{g}$$

$$c = 3 \quad \text{for} \quad \frac{\Delta p_\tau}{\Delta p_\eta} > 100.$$

- in the term dependent to the field there is an additional term c, that depends on delta p, used to pressurize the liquid, and the rest is related to geometrical feature.
- The viscosity term depends on volumetric flow, geometrical features and viscosity.

3. SQUEEZE MODE DESIGN



In this case the field and the generation of the chains is in the same direction, but the force is applied vertically, and plates move close and not laterally. In this case there is a vertical motion and for this, is squeeze mode.

If you move the 2 plates closer, you

$$F_\eta = \frac{3\pi \eta R^4 \dot{x}}{2(g_0 - x)^3}$$

are squeezing down the fluid because the chains are generated and stabilized by the field. Also, in this case there are 2 terms:

- The fluid term is only related to the geometrical features and the velocity of the upper plate.
- The field dependent term depends on the geometry and stress.

$$F_\tau(E, H) = \frac{4\pi \tau_y(E, H) R^3}{3(g_0 - x)}$$

The following is a table for direct comparison:

Property	Units	MR Fluids	ER Fluids	Ferrofluids
<i>Physical properties and constituents</i>				
Particle type	–	Fe, ferrites	Zeolites, polymers, $BaTiO_3$	Ceramics, Fe, ferrites
Particle size	μm	0.05–10	0.5–100	$2–10 \cdot 10^{-3}$
Base liquid	–	nonpolar oils, polar liquids	Oils	Oils, water
Density	kg/m^3	$3–5 \cdot 10^3$	$1–2 \cdot 10^3$	$1–2 \cdot 10^3$
Max. yield stress	kPa	50–100	2–5	–
Operating temp.	C	–40 to +150	–25 to +125	–
No field viscosity	mPa·s	100–1000	100–1000	2–500
<i>Electromagnetic properties</i>				
Maximum field	–	$\approx 250 \text{ kA/m}$	$\approx 4 \text{ kV/mm}$	–
Field-induced change	kPa	$\tau_y(H) \approx 10^2$	$\tau_y(H) \approx 5$	$\frac{\eta(H)}{\eta(0)} \approx 2$
Max. energy density	kJ/m^3	100	1	–
Response time	s	$< 10^{-3}$	$< 10^{-3}$	–

Video.